Asymmetries, Covert Wh-Movement, and Nominality in Japanese Wh-Questions

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This dissertation explores the syntax of Japanese wh-questions from a number of new perspectives. It is still an open question whether Japanese wh-questions involve covert wh-movement. In this respect, the dissertation argues, based on novel diagnostics, that Japanese wh-questions are derived in a non-uniform manner depending on their types. More specifically, it is shown that covert wh-movement is involved in embedded wh-questions selected by predicates that can select concealed questions (embedded CQ wh-questions), but not in the other types of wh-questions. This claim builds on the observation that different types of wh-questions in Japanese show different behavior with respect to intervention effects and cleft formation, which is captured under the analysis proposed in the dissertation. An important implication of the analysis is that assuming covert wh-movement is necessary in analyzing (a subset of) Japanese wh-questions. The dissertation also investigates why Japanese wh-questions differ regarding the involvement of covert wh-movement. To this end, the dissertation proposes a novel analysis of Japanese wh-questions where a selectional feature of the matrix predicate in the numeration probes into an already formed syntactic object and undergoes Agree with a categorial feature of a wh-phrase, which induces movement of that wh-phrase. It is also shown that the proposed analysis captures the nominal nature of embedded CQ wh-questions; in particular, it ascribes their nominality to such questions having the structure of free relatives. The discussion in the dissertation has a number of additional consequences. One concerns sluicing: the dissertation proposes a novel analysis of Japanese sluicing which involves null operator movement associated with the semantic property of exhaustivity. Additionally, the dissertation provides novel arguments regarding two long-standing and debated issues concerning wh-questions in Japanese; in particular, the dissertation provides new evidence for the island sensitivity of covert wh-movement as well as for the availability of large-scale pied-piping with respect to covert wh-movement in Japanese.

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Asymmetries, Covert Wh-Movement, and Nominality in Japanese Wh-Questions

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Contents

Chapter 1

Introduction

1.1 Syntax of wh-questions: Focusing on Japanese

In languages like English, a wh-phrase must appear sentence-initially in wh-questions. In the English wh-question in (1), for example, the object wh-phrase, which is underlined, occurs in the sentence initial position, rather than in the object position.

(1) What did John buy?

The standard syntactic view is that the wh-phrase in (1) originates in the object position and then undergoes wh-movement to the sentence-initial position, i.e. Spec,CP. This derivation can be roughly represented as in (2), where the trace with an index t_i indicates that the wh-phrase with the same index, *what*$_i$, was base-generated in the position of the trace and has then undergone wh-movement.

(2) $[_{CP}$ what$_i$ did $[_{TP}$ John buy $t_i]]$

From a cross-linguistic perspective, this analysis gives rise to a significant research question: what kind of syntactic analysis should be given to wh-questions in wh-in-situ languages, i.e. languages where wh-phrases in wh-questions superficially stay in-situ, such as Chinese and Japanese? (3) illustrates wh-questions from these two languages.

(3) a. Chinese

Ni	xihuan	shei.
you	like	who

'Who do you like?' (Huang 1982a: 370)

b. Japanese

Ken-wa nani-o kai-masi-ta ka.

Ken-TOP what-ACC buy-POL-PAST Q

'What did Ken buy?'

The object wh-phrases in (3) have not undergone wh-movement, but rather stay in their original object positions, as underlined, unlike (1). One might then suggest that wh-questions in those languages involve no wh-movement. However, Huang (1982a,b) proposes a different view; according to him, for example, the Chinese wh-question (3a) involves wh-movement that takes place not in overt syntax but in LF (i.e. covert syntax). This derivation for (3a) can be represented as in (4).

(4) Overt Syntax: $[_{CP}$ C $[_{TP}$ you like what$]]$

LF: $[_{CP}$ what$_i$ C $[_{TP}$ you like $t_i]]$ (based on Huang 1982a: 370)

Though his discussion revolves around Chinese, Huang claims that the derivation shown in (4) holds for wh-questions in wh-in-situ languages in general. More specifically, he claims that wh-movement is a universal property of language, but languages differ regarding what level wh-movement takes place in; it occurs in overt syntax in languages like English, while it does so in LF in languages like Chinese and Japanese.

Let us now turn our attention to the main concern of this dissertation, i.e. Japanese wh-questions like (3b). Japanese wh-questions are characterized by in-situ wh-phrases and the question particle (henceforth, Q-particle) *ka* in the sentence-final position, as seen in (3b). If we follow Huang's claim noted above, (3b) is derived in the same manner as the Chinese wh-question

(3a) (see (4)), as represented in (5). (I assume that the Q-particle *ka* is the head of CP, following the standard view.)

(5) Overt Syntax: $[_{CP}$ $[_{TP}$ Ken what bought] Q]
LF: $[_{CP}$ what$_i$ $[_{TP}$ Ken t_i bought] Q]

This type of analysis for Japanese wh-questions, i.e. the LF/covert wh-movement analysis, has been adopted and argued for in a number of previous works (Lasnik and Saito 1984, 1992, Nishiguachi 1990, Saito 2017, among many others). However, other types of analyses have been proposed as well. Three representative analyses among them are the null operator movement analysis (Watanabe 1992), the Q-particle movement analysis (e.g. Hagstrom 1998, Cable 2007, 2010, Miyagawa 2010), and the non-movement analysis (e.g. Shimoyama 2001, 2006). Under those analyses, the wh-question in (3b) is derived as represented below:

(6) a. Null operator movement analysis
Overt Syntax/LF: $[_{CP}$ Op$_i$ $[_{TP}$ Ken [t_i what] bought] Q]

b. Q-particle movement analysis
Overt Syntax/LF: $[_{CP}$ $[_{TP}$ Ken [what t_i] bought] Q$_i$]

c. Non-movement analysis
Overt Syntax/LF: $[_{CP}$ $[_{TP}$ Ken what bought] Q]

The first two analyses assume that a certain element associated with a wh-phrase undergoes movement in overt syntax; that element is a null operator originally attached to the wh-phrase in (6a), while it is the Q-particle *ka*, assumed to be originally attached to the wh-phrase, in (6b). In contrast, (6c) does not involve any movement either in overt syntax or LF. What these three analyses have in common is that they do not assume the movement of the wh-phrase itself, in contrast to the LF/covert wh-movement analysis (see (5)). Importantly, it is still an open issue which analysis best captures properties of Japanese wh-questions.

1.2 Goals and contributions

In order to shed new light on the issue discussed above, namely how Japanese wh-questions should be analyzed, this dissertation will explore the syntax of wh-questions in Japanese from new perspectives. As a result, I will argue that Japanese wh-questions are derived in different manners depending on the type of the question. More specifically, I will argue that wh-questions in Japanese differ in whether they involve covert wh-movement depending on what syntactic environment they appear in. To illustrate this claim, I will briefly introduce the classification of Japanese wh-questions that this thesis will draw on; the classification will be spelled out in more detail in Chapter 3. The three basic types of wh-questions the thesis will be concerned with are: (i) *matrix wh-questions*, (ii) *embedded wh-questions* and (iii) to-*embedded wh-questions* (where the wh-question is followed by the reportative complementizer *to*; see Saito 2010, 2012, 2013, 2015); they are exemplified in (7a-c) respectively.

(7) a. Matrix wh-questions
Ken-wa nani-o kai-masi-ta ka.
Ken-TOP what-ACC buy-POL-PAST Q
'What did Ken buy?' (= (3b))

b. Embedded wh-questions
Mai-wa [Ken-ga nani-o katta ka] {sitteiru / sinpaisiteiru}.
Mai-TOP Ken-NOM what-ACC bought Q know / is.anxious
'Mai {knows / is anxious} [what Ken bought].'

c. *To*-embedded wh-questions

Mai-wa [[Ken-ga nani-o katta ka] to] {tazuneta / sinpaisiteiru}

Mai-TOP Ken-NOM what-ACC bought Q REP asked / is.anxious

Lit. 'Mai {asked / is anxious} [that what Ken bought].'

Furthermore, I will take *concealed questions* into consideration to divide embedded wh-questions into two subclasses. Concealed questions refer to nominal phrases that are selected by question-selecting predicates and are interpreted as wh-questions (Baker 1968, Grimshaw 1979, Heim 1979, Romero 2005, Nathan 2006, Schwager 2008, Frana 2017, 2020, among many others). In the English example (8), for example, the object nominal phrase counts as a concealed question in that it is construed as a wh-question, as the paraphrase shows.

(8) Mary knows Joan's phone number.

≈ Mary knows what Joan's phone number is. (Frana 2020: 1-2)

Concealed questions are observed in Japanese as well (e.g. Nishiyama 2003, Nishigauchi 2020); see, e.g., (9).[1]

(9) Mai-wa [sono taikai-no tyanpion]-o sitteiru.

Mai-TOP that tournament-GEN champion-ACC know

'Mai knows the champion of the tournament.'

≈ 'Mai knows who the champion of the tournament is.'

It (cross-linguistically) holds, however, that question-selecting predicates are different in whether they can select concealed questions. In Japanese, for instance, the two question-selecting predicates in (7b), *sitteiru* 'know' and *sinpaisiteiru* 'is anxious', differ in this respect; while the

[1] (9) also has an "acquaintance" reading, being interpreted as *Mai is acquainted with the champion of the tournament*.

former can select concealed questions as in (9), this is not the case with the latter, as suggested by (10) (see its translation, in particular).

(10) Mai-wa [sono taikai-no tyanpion]-o sinpaisiteiru.
Mai-TOP that tournament-GEN champion-ACC is.anxious
'Mai is anxious about the champion of the tournament.'
≉ 'Mai is anxious who the champion of the tournament is.'

Taking into consideration this dichotomy of question-selecting predicates regarding concealed questions, I will refer to embedded wh-questions selected by predicates that can select concealed questions, as in (11a), as *embedded CQ wh-questions* (where CQ stands for concealed questions). By contrast, I will dub embedded wh-questions selected by predicates that cannot select concealed questions, as in (11b), *embedded non-CQ wh-questions*.

(11) a. Embedded CQ wh-questions
Mai-wa [Ken-ga nani-o katta ka] sitteiru.
Mai-TOP Ken-NOM what-ACC bought Q know
'Mai knows [what Ken bought].'

b. Embedded non-CQ wh-questions
Mai-wa [Ken-ga nani-o katta ka] sinpaisiteiru.
Mai-TOP Ken-NOM what-ACC bought Q is.anxious
'Mai is anxious [what Ken bought].'

Based on this classification, I will show that the above four types of wh-questions (i.e. matrix wh-questions, embedded CQ wh-questions, embedded non-CQ wh-questions, and *to*-embedded wh-questions) exhibit asymmetric syntactic behavior with respect to intervention effects and the formation of cleft wh-questions. Importantly, it will be observed that matrix wh-questions, embedded non-CQ wh-questions, and *to*-embedded wh-questions behave in a parallel manner with regard to the two syntactic phenomena mentioned above, as opposed to embedded CQ wh-questions. Based on this, I will argue that covert wh-movement is involved in embedded CQ wh-questions, while it is not in the other types of wh-questions. This conclusion will thus provide support for the necessity of covert wh-movement in analyzing (at least a subset of) Japanese wh-questions.

I will further examine why Japanese wh-questions differ with respect to the involvement of covert wh-movement in the way described above. For this exploration, I will focus on the fact that the selectors of embedded CQ wh-questions can select a wider range of syntactic categories as questions compared with those of other types of wh-questions. More specifically, selectors of embedded CQ wh-questions (e.g. *sitteiru* 'know'; see (11a)) can select CP (i.e. interrogative clauses) and DP (i.e. concealed questions) as questions, while selectors of embedded non-CQ questions (e.g. *sinpaisiteiru* 'is anxious'; see (11b)) and *to*-embedded wh-questions (i.e. the reportative complementizer *to*) can select only CP as questions; compare, e.g., (12) with (13-14).[2]

(12) Embedded CQ wh-questions

a. Mai-wa [$_{CP}$ Ken-ga nani-o katta ka] sitteiru.
Mai-TOP Ken-NOM what-ACC bought Q know
'Mai knows [what Ken bought].' (= (7b))

b. Mai-wa [$_{DP}$ sono taikai-no tyanpion]-o sitteiru.
Mai-TOP that tournament-GEN champion-ACC know
'Mai knows the champion of the tournament.'
≈ 'Mai knows who the champion of the tournament is.' (= (9))

[2] Regarding *to*-embedded wh-questions, I here consider only the (local) selectional relationship between the interrogative clause headed by *ka* and the reportative complementizer *to*; I leave open how the matrix predicate is involved with selection (see Ishii 2013 for relevant discussion).

(13) Embedded non-CQ wh-questions

a. Mai-wa [$_{\text{CP}}$ Ken-ga nani-o katta ka] sinpaisiteiru.
Mai-TOP Ken-NOM what-ACC bought Q is.anxious
'Mai is anxious [what Ken bought].' (= (7b))

b. Mai-wa [$_{\text{DP}}$ sono taikai-no tyanpion]-o sinpaisiteiru.
Mai-TOP that tournament-GEN champion-ACC is.anxious
'Mai is anxious about the champion of the tournament.'
≉ 'Mai is anxious who the champion of the tournament is.' (= (10))

(14) *To*-embedded wh-questions

a. Mai-wa [[$_{\text{CP}}$ Ken-ga nani-o katta ka] to] {tazuneta / sinpaisiteiru}
Mai-TOP Ken-NOM what-ACC bought Q REP asked / is.anxious
Lit. 'Mai {asked / is anxious} [that what Ken bought].' (= (7c))

b. *Mai-wa [[$_{\text{DP}}$ sono taikai-no tyanpion] to] {tazuneta / sinpaisiteiru}.
Mai-TOP that tournament-GEN champion-ACC REP asked / is.anxious
Lit. 'Mai {asked / is anxious} [that the champion of the tournament].'

I will argue that this is what matters here under a particular way of implementing selection, which takes into consideration the proposal by Cecchetto and Donati that moved elements can project (Checcetto and Donati 2010, 2015, Donati and Chechetto 2011). I will then propose an analysis of Japanese wh-questions that incorporates the selection mechanism in question, and show that it can capture the different behavior of the four types of wh-questions regarding the involvement of covert wh-movement.

The investigation in this dissertation will have consequences regarding a number of additional aspects of wh-questions in Japanese. Below I will briefly illustrate three main consequences. The first consequence concerns sluicing in Japanese. Sluicing constructions refer to (embedded) wh-questions which lack their propositional portion. In the English example (15a), e.g., the underlined sluicing construction has the same interpretation as the underlined embedded wh-question in (15b).

(15) a. Jack bought something, but I don't know [what].

b. Jack bought something, but I don't know [what he bought].

According to the standard assumption, the sluicing construction in (15a) involves the ellipsis of its propositional portion *he bought*. Japanese also has similar constructions (Inoue 1978, Takahashi 1994, Fukaya and Hoji 1999, Hiraiwa and Ishihara 2002, 2012, Saito 2004, Abe 2015, 2016, among many others); see the underlined part in (16b), which is uttered after (16a).

(16) a. Ken-wa dokoka-ni ryokooniitta sooda.
Ken-TOP somewhere-to traveled I.hear
'I hear that Ken traveled to somewhere.'

b. (Demo,) boku-wa [doko-ni (da) ka] siranai.
but I-TOP where-to COP Q know.not
'(But) I don't know [to where].'

(16b) is interpreted in the same way as (17), suggesting that the underlined sluicing construction in (16b) involves the ellipsis of its propositional part (i.e. *Ken traveled*).

(17) *After (16a),*
(Demo,) boku-wa [Ken-ga doko-ni ryokooniitta ka] siranai.
but I-TOP Ken-NOM where-to traveled Q know.not
'(But) I don't know [to where Ken traveled].'

I will discuss two representative analyses of sluicing constructions in Japanese, which I will call the focus movement/cleft analysis (e.g. Hiraiwa and Ishihara 2002, 2012) and the in-situ analysis (e.g. Kimura and Takahashi 2011), by taking into consideration the claim of this dissertation that embedded CQ wh-questions involve covert wh-movement. This will turn out to make both analyses problematic. Given that, I will propose a new analysis of Japanese sluicing. In particular, through comparing sluicing with other elliptical constructions (i.e. *ka*(*dooka*)-stripping, *to*-stripping and fragment answers) with respect to several tests, I will propose that sluicing involves null operator movement motivated by exhaustivity.

Another consequence concerns the nominal nature of embedded CQ wh-questions, which has been recently argued for by Tomioka (2020). He makes a proposal that Japanese embedded (CQ) wh-questions have a nominal structure and thus are interpreted as concealed questions. For example, (18a), involving an embedded CQ wh-question, is construed in the same way as (18b), involving a (corresponding) concealed question.

(18) a. Mai-wa [Ken-ga nani-o katta ka] sitteiru.
Mai-TOP Ken-NOM what-ACC bought Q know
'Mai knows [what Ken bought].' (= (11a))

b. Mai-wa [[Ken-ga e_i katta] mono$_i$]-o sitteiru.
Mai-TOP Ken-NOM bought thing-ACC know
'Mai knows the thing that Ken bought.'
≈ 'Mai knows what the thing that Ken bought is.'

The thesis will investigate the potential nominality of embedded CQ wh-questions in Japanese, implementing it in terms of treating such questions as free relatives. More importantly, the thesis will investigate the possibility of ascribing the covert wh-movement in embedded CQ wh-questions to the nominality of those wh-questions. While I will argue that embedded CQ wh-questions do not always show nominality, its distribution being restricted, I will ultimately propose an analysis of Japanese wh-questions that captures not only their different behavior regarding the involvement of covert wh-movement (as described above) but also the nominality of embedded CQ wh-questions and its restricted distribution.

Another consequence of the discussion in the dissertation concerns properties of covert wh-movement in Japanese. One notable syntactic property of movement is island-sensitivity: movement cannot take place across islands (e.g. Ross 1967, Chomsky 1973). (19) illustrates one such island, namely, the complex NP island. In particular, (19) shows that while overt wh-movement in English can cross a declarative complement clause (19a), it cannot cross a complex NP island (19b), where the complex NP is an NP whose head noun takes a clausal complement.

(19) a. Who$_i$ does Mary believe [that John praised t_i]?

b. *Who$_i$ does Mary believe [the rumor [that John praised t_i]]?

In contrast, it has been observed that in-situ wh-phrases in Japanese can generally appear in islands. In the Japanese sentence (20), for example, the in-situ wh-phrase *dare* 'who' appears in the complex NP island without inducing any degradation (e.g. Kuno 1973).

(20) Mai-wa [[Ken-ga dare-o hometa toyuu] uwasa]-o sinzitei-mas-u ka.
Mai-TOP Ken-NOM who-ACC praise-PAST that rumor-ACC believe-POL-PRES Q
Lit. 'Mai believes [the rumor [that Ken praised <u>who</u>]]?'
(≈ 'Who is the person such that Mai believes the rumor that Ken praised him/her?')

Under the LF/covert wh-movement analysis, there have been two debated views regarding this observation. The first view is that LF/covert wh-movement in examples such as those discussed above is not sensitive to islands, in contrast to overt wh-movement (e.g. Huang 1982b). The second view is that LF/covert wh-movement is sensitive to islands, and (20) does not show any island effect because what undergoes movement there is the whole island, which is pied-piped by the wh-phrase inside of it (e.g. Choe 1987, Nishigauchi 1990). More specifically, according to this view, the LF structure of (20) can be roughly represented as in (21), where the complex NP containing the wh-phrase has undergone wh-movement (see Chapter 2 for more details on this type of pied-piping).[3]

(21) [$_{CP}$ [[Ken-ga dare-o hometa toyuu] uwasa]-o$_i$ [$_{TP}$ Mai-wa *t*$_i$ sinzitei-mas-u] ka]
Ken-NOM who-ACC praise-PAST that rumor-ACC Mai-TOP believe-POL-PRES Q

I will show that the discussion in this dissertation lends support to the second view. In particular, I will provide novel evidence, from a new perspective, that LF/covert wh-movement is sensitive to islands and that covert large-scale pied-piping is available in Japanese wh-questions.

More generally, the discussion in this dissertation will be shown to have consequences for a number of phenomena, including wh-movement, concealed questions, intervention effects, clefts, inverse scope, freezing effects, peculiarities of reason wh-phrases (e.g. *naze* 'why' in Japanese), nominality of embedded wh-questions, free relatives, selection, island effects, and pied-piping.

1.3 Outline

The dissertation is organized as follows:

In Chapter 2, I will discuss in more detail different analyses of Japanese wh-questions proposed in the literature. In particular, it will be shown that the main argument for covert wh-movement in Japanese wh-questions provided by Nishigauchi (1990), i.e. the observation that Japanese wh-

[3] In (21), for expository purposes, I assume that the topicalized subject phrase *Mai-wa* is in Spec,TP, rather than in Spec,Top(ic)P in the left periphery (e.g. Kishimoto 2009).

questions show wh-island effects while they do not show any other type of island effects (see, e.g., (20)), has been doubted in the literature, which has undermined its solidity. This chapter thus establishes a motivation for one goal of this dissertation, namely a new defense of the necessity of covert wh-movement for (some of) Japanese wh-questions.

In Chapter 3, I will discuss asymmetries in Japanese wh-questions that have gone unnoticed so far in the literature. Building on the classification of Japanese wh-questions that has been briefly presented above, I will show that the four classes of wh-questions (i.e. matrix wh-questions, embedded CQ wh-questions, embedded non-CQ wh-questions, and *to*-embedded wh-questions) do not behave uniformly with respect to intervention effects and the formation of cleft wh-questions. Importantly, I will show that the two asymmetries make the same cut regarding Japanese wh-questions; matrix wh-questions, embedded non-CQ wh-questions and *to*-embedded wh-questions exhibit parallel behavior, as opposed to embedded CQ wh-questions. Given this demarcation, I will propose that Japanese wh-questions are derived in a non-uniform manner. More specifically, I will propose that matrix wh-questions, embedded non-CQ wh-questions and *to*-embedded CQ wh-questions do not involve covert wh-movement, while embedded CQ wh-questions are derived through covert wh-movement. I will show that this proposal explains the two asymmetries in question in a uniform manner.

In Chapter 4, I will discuss a consequence of the proposal presented in Chapter 3 regarding how sluicing in Japanese should be analyzed. More specifically, taking that proposal into consideration, I will discuss two representative analyses of Japanese sluicing, i.e. the focus movement/cleft analysis and the in-situ analysis, and uncover several problems with those analyses. I will then pursue a new analysis of sluicing in Japanese, through comparing sluicing with other elliptical constructions with respect to several relevant properties. Consequently, I will propose a novel analysis of sluicing which crucially involves null operator movement that is associated with the property of exhaustivity.

In Chapter 5, I will examine why different types of Japanese wh-questions are derived differently, as proposed in Chapter 3. In particular, I will zoom in on two specific properties of embedded CQ wh-questions that distinguish them from the other types of wh-questions. First, I will focus on a distinct "internal" property of embedded CQ wh-questions, namely their nominal nature, argued for by Tomioka (2020). Specifically, adopting the analysis of free relatives proposed by Cecchetto and Donati (Cecchetto and Donati 2011, 2015, Donati and Cecchetto 2011), I will scrutinize the hypothesis that covert wh-movement in embedded CQ wh-questions is triggered in order to build the structure of free relatives, which makes them nominal. This investigation will uncover an important fact regarding the nominality of embedded CQ wh-questions: those wh-questions do not always show nominality, its distribution being restricted. Second, I will highlight a distinct "external" property of embedded CQ wh-questions, namely, the fact that their selectors (i.e. predicates that can select concealed questions) can select a broader range of syntactic categories as questions. To syntactically implement this, I will adopt the syntactic mechanism of selection proposed by Donati and Cecchetto (2011). I will then propose an analysis of Japanese wh-questions which incorporates that syntactic selection mechanism and that pays special attention to labeling of syntactic objects. It will be shown that this analysis can capture in a uniform manner the asymmetric derivations of Japanese wh-questions (i.e. embedded CQ wh-questions involve covert wh-movement, while the other types of wh-questions do not), as well as the nominality of embedded CQ wh-questions and its restricted distribution.

In Chapter 6, I will explore some consequences of the discussion in the previous chapters. More specifically, going back to Nishigauchi (1990), I will consider what the findings and proposals in this dissertation can tell us about his claims regarding the syntax of Japanese wh-questions. In particular, I will show that they provide novel evidence for the island sensitivity of covert wh-movement and the availability of covert large-scale pied-piping in Japanese wh-questions.

Chapter 7 concludes the dissertation.

Chapter 2

Background on Covert Wh-Movement in Japanese

2.1 Introduction

To begin with, this chapter provides a brief background for one of the main concerns of this dissertation, i.e. covert wh-movement in Japanese wh-questions (e.g. Lasnik and Saito 1984, 1992, Nishigauchi 1990, Saito 2017). More specifically, I will illustrate the empirical motivation for covert wh-movement for Japanese wh-questions presented by Nishigauchi (1990), and discuss why the analysis has been challenged.

The rest of this chapter is organized as follows: Section 2.2 introduces the main argument for assuming covert wh-movement for Japanese wh-questions, i.e. island effects, provided by Nishigauchi (1990). Section 2.3 presents two views against that assumption given in the literature, one offering a semantic alternative analysis (Shimoyama 2006) and the other taking into consideration the prosodic traits of Japanese wh-questions (e.g. Deguchi and Kitagawa 2002, Ishihara 2003, Kitagawa 2005, 2017). Section 2.4 summarizes this chapter.

2.2 Main argument for covert wh-movement: Nishigauchi (1990)

Nishigauchi (1990) crucially observes an asymmetry in island effects in Japanese wh-questions and argues that it provides an argument for the covert wh-movement analysis of Japanese wh-questions. As in other wh-in-situ languages, wh-questions in Japanese usually do not show island effects. As (1) shows, for example, wh-phrases can occur within complex NPs without any problems, yielding no complex NP island effect (Kuno 1973).[1]

(1) Ken-wa [[e_i nani-o kaita] hito$_i$]-o home-masi-ta ka.
Ken-TOP what-ACC wrote person-ACC praise-POL-PAST Q
Lit. 'Ken praised [the person who wrote what]?'
(≈ 'What is the thing such that Ken praised the person who wrote it?')

This suggests that Japanese wh-questions do not involve either overt or covert wh-movement, or that if they involve covert wh-movement, such movement is not sensitive to islands (Huang 1982b, Lasnik and Saito 1984). However, Nishigauchi argues that Japanese wh-questions do show island effects; in particular, he argues that wh-island effects are observed in Japanese. To understand his claim, it is important, in the first place, to bear in mind the fact that the Q-particle *ka* need not co-occur with a wh-phrase; matrix and embedded clauses that contain the Q-particle but no wh-phrases are interpreted as *yes*/*no*-questions and *whether*-questions respectively, as exemplified in (2).

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[1] Contrary to this generalization, however, it has been observed that the adjunct wh-phrase *naze* 'why' cannot appear in complex NP islands, or islands in general; e.g., compare (i) with (1). ((i) ends with the particle *no*, instead of *ka*. This is because matrix wh-questions with *naze* 'why' must contain *no* in the sentence-final part; see, e.g., Kuwabara 2008, 2013, and references therein.)

(i) *Ken-wa [[e_i naze syoosetu-o kaita] hito$_i$]-o hometa no.
Ken-TOP why novel-ACC wrote person-ACC praised Q
Lit. 'Ken praised [the person who wrote a novel why]?'

Assuming covert wh-movement for in-situ wh-phrases, Huang (1982b) argues that the ungrammaticality of (i) follows from the Empty Category Principle (ECP), according to which traces must be properly governed, i.e. either lexically governed (governed by a lexical category) or antecedent-governed (governed by an element with the same index). According to Huang, (i) is ungrammatical because the trace of the covert wh-movement of the adjunct wh-phrase *naze* can be neither lexically governed (given that the wh-phrase is an adjunct) nor antecedent-governed (given that in LF the wh-phrase is in the matrix clause, where it cannot govern its trace within the island), thus not satisfying the ECP. (See also Lasnik and Saito 1984 for a more refined analysis.) This account is also compatible with the grammaticality of (1); since the trace of the covert wh-movement is in the object position, it is lexically governed by the verb, the ECP thus being satisfied. If this is on the right track, this ECP-based account of the contrast between (i) and (1) would support the hypothesis that LF/covert wh-movement occurs in Japanese wh-questions.

However, some problems regarding the degraded status of (i) and the ECP account of it have been noted in the literature; see fn.7 for one such problem noted by Nishgiauchi (1990) and fn.18 for another by Kitagawa (2006b, 2017).

(2) a. Ken-wa ringo-o kai-masi-ta ka.
Ken-TOP apple-ACC buy-POL-PAST Q
'Did Ken buy an apple?'

b. Mai-wa [Ken-ga ringo-o katta ka] sitteiru.
Mai-TOP Ken-NOM apple-ACC bought Q know
'Mai knows [whether Ken bought an apple].'

With this fact in mind, let us now observe two examples that lead Nishigauchi to argue for the existence of wh-island effects in Japanese. The first example is shown in (3).[2]

(3) Mai-wa [Ken-ga nani-o katta ka] sittei-mas-u ka.
Mai-TOP Ken-NOM what-ACC bought Q know-POL-PRES Q
(i) ✓'Does Mai know [what Ken bought]?'
(ii) * Intended: 'What does Mai know [whether Ken bought (*t*)]?'
(≈ 'For which *x*, *x* a thing, does Mai know [whether Ken bought *x*]?')
(based on Nishigauchi 1990: 30)

(3) is characterized particularly by two traits: (i) the wh-phrase *nani* 'what' appears in the embedded clause, and (ii) two Q-particles are involved, one in the matrix clause (i.e. matrix Q-particle) and the other in the embedded clause (i.e. embedded Q-particle). It follows from this configuration that (3) is expected to yield two interpretations. In the first expected reading, the wh-phrase *nani* 'what' is associated with the embedded Q-particle and thus takes an embedded scope, rendering the embedded clause a wh-question. The matrix Q-particle then makes the whole sentence a *yes*/*no*-question, yielding the reading (i) in (3). This interpretation for the example does obtain. In the second expected reading, the wh-phrase is associated with the matrix Q-particle

[2] The grammaticality judgements on (3) and (4) are based on Nishigauchi's (1990) statements regarding those examples. See also fn.3.

beyond the clause boundary. This makes the matrix clause a wh-question. The embedded Q-particle, then, is not associated with any wh-phrase and thus renders the embedded clause a *whether*-question. This results in the reading (ii) in (3). Nishigauchi claims, however, that "it is completely impossible to interpret [(3)] as [(ii)]" (Nishigauchi 1990: 31). Note crucially that in (ii), the long-distance dependency between the wh-phrase *nani* 'what' and the matrix Q-particle holds across the embedded *whether*-clause, which is a type of wh-question. The unavailability of the reading (ii) in (3) thus suggests that Japanese wh-questions show wh-island effects.

Observe now the second example in (4), which involves a more complex pattern.

(4) Mai-wa [dare-ga nani-o katta ka] sittei-mas-u ka.
Mai-TOP who-NOM what-ACC bought Q know-POL-PRES Q
(i) ✓'Does Mai know [who bought what]?'
(ii) ?* Intended: 'Who does Mai know [what (*t*) bought]?'
(≈ 'For which *x*, *x* a person, does Mai know what *x* bought?')
(iii) * Intended: 'What does Mai know [who bought (*t*)]?'
(≈ 'For which *y*, *y* a thing, does Mai know who bought *y*?')
(iv) * Intended: 'Who what does Mai know [whether (*t*) (*t*) bought]?'
(≈ 'For which *x y*, *x* a person, *y* a thing, does Mai know whether *x* bought *y*?')
(based on ibid: 28)

(4) is interesting in two regards: (i) two wh-phrases, *dare* 'who' and *nani* 'what', appear in the embedded clause and (ii) it contains two Q-particles, i.e. the matrix and embedded Q-particle. This configuration yields four logically possible interpretations. First, both wh-phrases in the embedded clause are associated with the embedded Q-particle, rendering the embedded clause a multiple wh-question. The matrix Q-particle then makes the whole sentence a *yes*/*no*-question. This results in the reading (i) in (4). Second, the embedded subject wh-phrase *dare* 'who' is associated with the

matrix Q-particle, which makes the matrix clause a wh-question. Meanwhile, the embedded object wh-phrase *nani* 'what' is associated with the embedded Q-particle, which yields an embedded wh-question. This gives rise to the reading (ii) in (4). The third reading is the same as the second one except for which wh-phrase takes which scope; in this reading, the embedded object wh-phrase *nani* 'what' takes a matrix scope (hence, the matrix wh-question) while the embedded subject wh-phrase *dare* 'who' takes an embedded scope (hence, the embedded wh-question). This corresponds to the reading (iii) in (4). Finally, both embedded wh-phrases are associated with the matrix Q-particle, which results in a matrix multiple wh-question. The embedded Q-particle, then, has no wh-phrase to be associated with and thus renders the embedded clause a *whether*-question. The reading (iv) in (4) obtains this way. According to Nishigauchi, (i) is the only possible reading for (4), the other three readings being unavailable.[3] Crucially, those three readings differ from (i) in that they contain long-distance dependencies between embedded wh-phrases and the matrix Q-particle which cross the boundary of the embedded wh-questions. The unavailability of those three interpretations thus indicates the existence of wh-island effects in Japanese.

Building on those observations, Nishigauchi (1990) argues that island effects are observed in Japanese questions, despite Japanese being a wh-in-situ language. Therefore, based on the general idea that island effects arise due to movement, he concludes that covert wh-movement is involved in Japanese wh-questions and that it is sensitive to islands, *pace* Huang (1982b) and Lasnik and Saito (1984).

[3] Nishigauchi (1990) notes, however, that the reading (ii) in (4) becomes more readily available for some speakers, including him, if a heavy stress is put on the embedded subject wh-phrase *dare* 'who'. In contrast, he further notes, this kind of phonological amendment does not salvage the other readings, i.e. (iii) and (iv) in (4). (This is why in (4) I put "?*" for the reading (ii) but "*" for the readings (iii) and (iv).) While he treats this phonological effect merely as a matter of focus which is not directly related to island constraints or wh-movement *per se*, some researchers take it more seriously and argue that it has much to do with the nature of wh-questions in Japanese; see Section 2.3.2.

The immediate question, then, is why covert wh-movement in Japanese is apparently selective regarding types of islands. More specifically, why is it that covert wh-movement in Japanese exhibits wh-island effects (e.g. (3-4)) while it does not show, e.g., complex NP island effects (e.g. (1))? To capture this island asymmetry, Nishigauchi (1990) argues that when a complex NP contains a wh-phrase, what undergoes covert wh-movement is not that wh-phrase but the whole complex NP; more specifically, the wh-phrase covertly pied-pipes the complex NP. This operation is widely known as *large-scale pied-piping*.[4] For example, consider (1), repeated below:

(1) Ken-wa [[e_i nani-o kaita] hito$_i$]-o home-masi-ta ka.
Ken-TOP what-ACC wrote person-ACC praise-POL-PAST Q
Lit. 'Ken praised [the person who wrote what]?'
(≈ 'What is the thing such that Ken praised the person who wrote it?')

According to Nishigauchi, what undergoes covert wh-movement in (1) is the whole complex NP *nani-o kaita hito* '(Lit.) the person who wrote what', rather than the wh-phrase inside of it *nani* 'what'. Consequently, the wh-phrase does not cross the complex NP island and thus no island effect arises. On this view, it should follow that wh-island effects are observed in Japanese because large-scale pied-piping is not available for wh-islands. I have to note here, however, that Nishigauchi (1990) does not explain its unavailability for wh-islands. I believe, however, that the difference between the two islands regarding the availability of large-scale pied-piping actually falls out from his implementation of large-scale pied-piping, as illustrated below.

Nishigauchi proposes to implement large-scale pied-piping through feature percolation. To understand the mechanism he assumes, consider how the wh-question in (1) (see above), which involves a complex NP island, is derived. According to his assumption, the complex NP in (1) has

[4] See also Choe (1987), who argues for covert large-scale pied-piping for Japanese and Korean wh-questions.

the structure in (5) (in overt syntax), where the wh-phrase *nani* 'what' has a [+WH] feature, the relative clause is a CP, and it occurs in the Spec,NP of the noun.[5]

(5) [$_{\text{NP}}$ [$_{\text{CP}}$ [$_{\text{TP}}$ e_{i} nani-o$_{\text{[+WH]}}$ kaita]] hito$_{\text{i}}$] (in overt syntax)
what-ACC wrote person
Lit. 'the person who wrote what'

He argues that the wh-phrase in (5) moves to the Sepc,CP of the relative clause in LF, and there, its [+WH] feature is percolated up to the immediately dominating node, i.e. CP, as represented in (6).

(6) [$_{\text{NP}}$ [$_{\text{CP[+WH]}}$ nani-o$_{\text{j[+WH]}}$ [$_{\text{TP}}$ e_{i} t_{j} kaita]] hito$_{\text{i}}$] (in LF)

The percolated [+WH] feature on the CP is then further percolated up to the immediately dominating node, i.e. NP, as (7) shows.

(7) [$_{\text{NP[+WH]}}$ [$_{\text{CP[+WH]}}$ nani-o$_{\text{j[+WH]}}$ [$_{\text{TP}}$ e_{i} t_{j} kaita]] hito$_{\text{i}}$] (in LF)

As a result, the whole complex NP in (7) qualifies as one large wh-phrase by virtue of the [+WH] feature on the NP and thus undergoes covert wh-movement, which results in the LF structure of (1) shown in (8) (where the moved complex NP is underlined).[6]

(8) [$_{\text{CP}}$ $\underline{\text{[}_{\text{NP[+WH]}}\text{ [}_{\text{CP[+WH]}}\text{ nani-o}_{\text{j[+WH]}}\text{ [}_{\text{TP}}\ e_{\text{i}}\ \text{t}_{\text{j}}\text{ kaita]] hito}_{\text{i}}\text{]-o}_{\text{k}}}$ [$_{\text{TP}}$ Ken-wa t_{k} home-masi-ta]
what-ACC wrote person-ACC Ken-TOP praise-POL-PAST

[5] Nishigauchi (1990) labels the projection below CP as S. Instead, I will label it as TP for expository purposes.
[6] In (8), for expository purposes, I assume that the topicalized subject phrase *Ken-wa* is in Spec,TP, rather than in Spec,Top(ic)P in the left periphery (e.g. Kishimoto 2009).

ka] (in LF)
Q

Crucially, the wh-phrase in (8) has undergone movement only within the complex NP island, not crossing it; hence, no island effect obtains.[7]

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[7] Recall the observation that in-situ adjunct wh-phrases such as *naze* 'why' cannot appear in islands (Huang 1982b, Lasnik and Saito 1984), in contrast to in-situ argument wh-phrases, as in the contrast between (1) and (i).

(i) *Ken-wa [[e_i naze syoosetu-o kaita] hito$_i$]-o hometa no.
Ken-TOP why novel-ACC wrote person-ACC praised Q
Lit. 'Ken praised [the person who wrote a novel why]?'

Huang (1982b) and Lasnik and Saito (1984) argue for an ECP account of the contrast in question; see fn.1 for its details. The account is, however, problematic. In particular, a problem is raised by (ii), where the wh-phrase *dare* 'who' appears as the subject in the island.

(ii) Ken-wa [[dare-ga e_i kaita] syoosetu$_i$]-o home-masi-ta ka.
Ken-TOP who-NOM wrote novel-ACC praise-POL-PAST Q
Lit. 'Ken praised [the novel which who wrote]?'
(≈ 'Who is the person such that Ken praised the novel which (s)he wrote?')

Assuming that the subject wh-phrase *dare* in (ii) appears in the canonical subject position, i.e. the position immediately dominated by TP or S, and undergoes LF/covert wh-movement, its trace could not be properly governed and thus (ii) is incorrectly predicted to be ungrammatical. To get around this issue, Huang (1982b) suggests that the governor of the subject wh-phrase in (ii) (or more precisely, in Chinese examples like (ii)), namely T or Infl, is actually a lexical category; the trace of the subject wh-phrase in LF is then lexically governed by the embedded T/Infl. Lasnik and Saito (1984) note, however, that this solution constitutes a problem for the ungrammaticality of (i); if T/Infl were a lexical category, the trace of the adjunct wh-phrase *naze* 'why' in LF would be lexically governed by T/Infl, satisfying the ECP. Lasnik and Saito then suggest that 'lexical government of the subject by Infl is made possible by a special relation between these two elements' (Lasnik and Saito 1984: 243). However, Nishigauchi (1990) notes the stipulatory nature of this solution.

This motivates him to suggest an alternative account of (i) based on the mechanism of feature percolation that makes large-scale pied-piping possible, as described in the text. In particular, he suggests the condition that in order for the [+WH] feature of a wh-phrase to be percolated up to the dominating node, the category feature of the wh-phrase must be identical with that of the dominating node. Consider how this condition works with the complex NP containing an object wh-phrase in (1), whose structure in overt syntax is shown in (5). I here adopt Nishigauchi's assumption that CP, or S', is neutral regarding its categorial status; it is determined based on the category feature of the element in its specifier. Under this assumption, when the wh-phrase undergoes covert movement to Spec,CP in the relative clause, as in (6), that CP obtains the category feature [+N], as a result of the wh-phrase *nani* 'what', which has the category feature [+N], occupying its specifier. The feature percolation of [+WH] in (7) thus successfully takes place, since the category feature of CP, [+N], is identical with that of the dominating node, NP. Consider now the complex NP in (i). Suppose that the adjunct wh-phrase *naze* in (i) moves to Spec,CP of the relative clause in LF, as in (iii).

(iii) [$_{NP}$ [$_{CP}$ naze$_j$ [$_{TP}$ e_i t_j syoosetu-o kaita]] hito$_i$] (in LF)
why novel-ACC wrote person

Nishigauchi crucially assumes that *naze* 'why' has the feature [-N], given that it is an adjunct. Then, under the aforementioned assumption, CP in (iii) obtains the category feature [-N], resulting in (iv).

(iv) [$_{NP}$ [$_{CP[+WH, -N]}$ naze$_{j[+WH, -N]}$ [$_{TP}$ e_i t_j syoosetu-o kaita]] hito$_i$] (in LF)
why novel-ACC wrote person

(iv) also shows that the [+WH] feature of *naze* is percolated up to CP. Notice that in (iv), the category feature of CP, [-N], does not match that of the dominating node NP, i.e. [+N]. Hence, the [+WH] feature of CP cannot percolate up to NP. As a result, large-scale pied-piping is unavailable in (i). This means that in (i), *naze* 'why' itself must undergo

Let us next turn to wh-islands. As noted above, Nishigauchi (1990) does not discuss wh-islands in terms of feature percolation, thus giving no explanation why covert pied-piping is unavailable for them. The reader should thus keep in mind that the following is not the account presented in Nishigauchi (1990), though it is based on the assumptions from that work.

Let us consider how (4) could get the reading in (iii), for example. The data is repeated below with the relevant reading:

(4) Mai-wa [dare-ga nani-o katta ka] sittei-mas-u ka.
Mai-TOP who-NOM what-ACC bought Q know-POL-PRES Q
(iii) * Intended: 'What does Mai know [who bought (*t*)]?'
(≈ 'For which *y*, *y* a thing, does Mai know who bought *y*?')

With the reading (iii), (4) should involve the embedded wh-question whose Q-particle is associated with the subject wh-phrase *dare* 'who'. Given Nishigauchi's claim that wh-phrases in Japanese undergo covert wh-movement, the LF structure of that embedded wh-question can then be represented as in (9), where *dare* 'who' has moved to Spec,CP (assumed to be headed by the Q-particle).

(9) [$_{CP}$ dare-ga$_{i[+WH]}$ [$_{TP}$ t_i nani-o$_{[+WH]}$ katta] ka] (in LF)
who-NOM what-ACC bought Q

Recall that under the current mechanism of feature percolation, a wh-phrase must move to a specifier of the topmost projection of an island to enable its [+WH] feature to be percolated to that projection. In (9), however, the specifier of the wh-island, i.e. Spec,CP, is already occupied by the wh-phrase *dare* 'what'. This means that the object wh-phrase *nani* 'what' cannot move there so

covert wh-movement, crossing the island. With Nishigauchi's claim that LF/covert wh-movement is sensitive to islands, the movement under consideration induces an island effect, hence the ungrammaticality of (i).

that its [+WH] feature can be percolated to the wh-island, or CP. Consequently, it has no choice but to undergo covert wh-movement that crosses the wh-island, thus leading to an island effect.

To recap, while Japanese wh-questions do not show complex NP island effects, Nishigauchi (1990) argues that they exhibit wh-island effects. He takes this to be a straightforward argument for covert wh-movement, which he argues is sensitive to island constraints (*pace* Huang 1982b and Lasnik and Saito 1984). Concerning the asymmetry of island effects between complex NP islands and wh-islands, he argues (or more precisely, *would* argue) that it follows from the difference between the two regarding the availability of covert large-scale pied-piping, which is implemented by feature percolation. I finally add that assuming covert large-scale pied-piping is not unreasonable, given that its overt counterpart is observed in other languages such as Basque (Ortiz de Urbina 1989, 1990) and Imabura Quechua (Cole 1982); see, e.g., (10), where in each example, the wh-phrase has undergone movement inside the complex NP and then pied-piped it to the Spec,CP of the matrix clause.

(10) a. Basque:

[Nork idatzi zuen liburua] irakurri du Peruk.
what.ERG write AUX book read AUX Peter.ERG
Lit. '[The book that who wrote] did Peter read?'
(≈ 'Who is the person such that Peter read the book that (s)he wrote?')

(Ortiz de Urbina 1990: 249)

b. Imbabura Quechua:

[Ima-ta randi-shka runa-ta taj] riku-rka-ngui.
what-ACC buy-NMLZ man-ACC Q see-PAST-2P
Lit. '[The man that bought what] did you see?'
(≈ 'What is the thing such that you saw the man that bought it?') (Cole 1982: 24)

2.3 Views against Nishigauchi (1990)

Since Nishigauchi (1990), however, his argument for covert wh-movement in Japanese wh-questions has been challenged in the literature. Below I will introduce two positions against his argument, one concerning the semantics of wh-questions (Shimoyama 2006) and the other the prosody of Japanese wh-questions (e.g. Deguchi and Kitagawa 2002, Ishihara 2003, Kitagawa 2005, 2017). (Note that the main discussion of this dissertation, which will start form the next chapter, will not depend on the two analyses of Japanese wh-questions introduced below; in this thesis, I will be agnostic regarding the validity of those analyses.)

2.3.1 Semantic solution: Shimoyama (2006)

Shimoyama (2006) argues that the asymmetry regarding island effects in Japanese wh-questions described above can follow from her semantic analysis of general wh-constructions in Japanese based on Hamblin (1973), which crucially does not assume covert wh-movement. The analysis she proposes builds on the traditional idea that the elements that are apparently wh-phrases in Japanese are actually so-called *indeterminates* (Kuroda 1965).[8] That is, those elements do not have any quantificational force in themselves, and instead, their quantificational interpretations depend on other quantificational items. For example, wh-items (i.e. indeterminates) obtain the interpretation of wh-phrases by virtue of the Q-particle *ka*, as the above examples have suggested. In addition, they can also be interpreted as universal quantifiers when they are followed by the (quantificational) particle *mo*, as exemplified in (11).

[8] Nishigauchi (1990) in fact follows this idea as well. For a recent discussion of indeterminates in Japanese, see Oda (2021).

(11) Dare-mo-ga odotta.
who-MO-NOM danced
'Everyone danced.'

Following Shimoyama, I will refer to sentences involving the association between wh-items and the particle *mo* as *universal constructions*. This association can hold even in a long-distance manner; a wh-item can be associated with *mo* across a (single) complex NP, to which *mo* is attached; see, e.g., (12).

(12) [[<u>Dare</u>-ga e_i syootaisita] sensee$_i$]-<u>mo</u> odotta.
who-NOM invited teacher-MO danced
'For every person *x*, the teacher(s) that *x* had invited danced.'
(based on Shimoyama 2006: 139)

Shimoyama (2006) crucially shows that the asymmetry regarding islands that is observed with respect to wh-questions holds as well with universal constructions.[9] First, universal constructions do not show complex NP island effects; a wh-phrase can be associated with the particle *mo* across complex NP islands. This has in fact already been indicated by (12). Shimoyama further shows that the association in question can hold even when a wh-item appears in another complex NP, as exemplified in (13).[10]

[9] To be fair, Nishigauchi (1990) makes the same observation with respect to *unconditionals* (or concessive conditionals, like *wh-ever* clauses in English), where (i) wh-items occur and (ii) the particle *mo* appears sentence-finally, as exemplified in (i).

(i) [Dare-ga ki-te mo], boku-wa kinisi-nai.
who-NOM come-TE MO I-TOP care-not
'Whoever comes, I don't care.'

More specifically, he observes that unconditionals do not show complex NP island effects but they exhibit wh-island effects (i.e. they show the patterns in (14) and (19)). This section will not deal with his analysis of unconditionals.

[10] I find this type of universal construction to be a bit marginal, while Shimoyama does not. This is why I mark (13) with "(?)".

(13) (?)[[[Dare-ga e_i osieta] gakusee$_i$]-ga e_j syootaisita] sensee$_j$]-mo kita.
who-NOM taught student-NOM invited teacher-MO came
'For every *x*, the teacher(s) that the student(s) that *x* had taught invited came.'
(based on ibid: 146)

Universal constructions parallel wh-questions in this respect; recall that wh-questions do not exhibit complex NP island effects either, as noted in Section 2.2. The relevant example (1) is repeated below:

(1) Ken-wa [[e_i nani-o kaita] hito$_i$]-o home-masi-ta ka.
Ken-TOP what-ACC wrote person-ACC praise-POL-PAST Q
Lit. 'Ken praised [the person who wrote what]?'
(≈ 'What is the thing such that Ken praised the person who wrote it?')

Under the assumption that the two particles *ka* and *mo* belong to the same class of particles, say quantificational particles, the above parallel can be generalized as follows: a wh-item can be associated with a quantificational particle across a complex NP. This general idea can be schematized as in (14).

(14) [… [$_{\text{Complex NP}}$ … wh-item(/indeterminate) …] …]-*ka*/*mo*
(based on ibid: 148)

Second, in contrast to complex NP island effects, universal constructions show wh-island effects; wh-items cannot be associated with the particle *mo* across an embedded wh-question. Before observing the relevant data, it is important to note that when *mo* is not associated with any wh-phrase, it is interpreted as *also*, as exemplified in (15).

(15) Ken-mo odotta.
Ken-MO danced
'Ken also danced.'

Bearing this fact in mind, observe now the relevant example below:[11]

(16) [[*e*$_i$ [Ken-ga dare-ni nani-o okutta ka] sitteiru] syoonin$_i$]-mo damatteita.
Ken-NOM who-to what-ACC sent Q know witness-MO was.silent
(i) ✓ 'The witness that knew what Ken sent to whom was also silent.'
(ii) * 'For every person *x*, the witness that knew what Ken sent to *x* was silent.'
(iii) ** 'For every thing *x*, the witness that knew to whom Ken sent *x* was silent.'
(iv) * 'For every person *x*, for every thing *y*, the witness who knew whether Ken sent *y* to *x* was silent.'

(based on ibid: 145)

In (16), two wh-items, *dare* 'who' and *nani* 'what', occur within an embedded clause headed by the Q-particle *ka*, and the particle *mo* is located above the Q-particle. With this configuration in mind, consider the four logically possible readings of (16) shown above. In the reading (i), both wh-items are associated with the embedded Q-particle, yielding an embedded multiple wh-question. The outer particle *mo* is then interpreted as *also*. In the readings (ii) and (iii), one wh-item is associated with the embedded Q-particle *ka*, and the other with *mo*, crossing the Q-particle. In the reading (iv), both wh-items are associated with *mo*, skipping *ka*. Shimoyama observes that (i) is the only possible reading for (16). Importantly, what the other three unavailable readings have in common is that they involve long-distance dependencies between wh-items and *mo* that cross the Q-particle *ka*. Note here that this reading pattern is parallel to that of (4), repeated below, which indicates that wh-questions exhibit wh-island effects (see Section 2.2). The only relevant

[11] The grammaticality judgements on (16-18) are Shimoyama's (2006).

difference between (16) and (4) is which particle, *mo* (in (16)) or *ka* (in (4)), is used as the higher particle.

(4) Mai-wa [dare-ga nani-o katta ka] sittei-mas-u ka.
Mai-TOP who-NOM what-ACC bought Q know-POL-PRES Q
(i) ✓'Does Mai know [who bought what]?'
(ii) ?* Intended: 'Who does Mai know [what (*t*) bought]?'
(≈ 'For which *x*, *x* a person, does Mai know what *x* bought?')
(iii) * Intended: 'What does Mai know [who bought (*t*)]?'
(≈ 'For which *y*, *y* a thing, does Mai know who bought *y*?')
(iv) * Intended: 'Who what does Mai know [whether (*t*) (*t*) bought]?'
(≈ 'For which *x y*, *x* a person, *y* a thing, does Mai know whether *x* bought *y*?')
(based on Nishigauchi 1990: 28)

Note that in the above examples, (16) and (4), the lower particle is the Q-particle *ka*, forming embedded (wh-)questions; this is why these two examples instantiate wh-island effects. Related to this point, Shimoyama makes another significant observation: (almost) the same reading pattern as (16) and (4) obtains when *mo* is used as the lower particle, regardless of whether the higher particle is *mo* or *ka*; compare (17-18) with (16) and (4).[12]

(17) [[e_i [[Ken-ga <u>nan-nen</u>-ni <u>nani</u>-nituite e_j kaita] ronbun$_j$]-<u>mo</u> yonda] sensee$_i$]-<u>mo</u>
Ken-NOM what-year-in what-about wrote paper-MO read teacher-MO

[12] The only difference between (16) and (4), on the one hand, and (17-18), on the other, is whether they can yield the reading (iv), where both wh-items are associated with the higher particle. In the reading (iv) in (16) and (4), the lower particle *ka* is interpreted as *whether*; this reading is unavailable, as with the readings (ii) and (iii), where the lower *ka* is the Q-particle associated with a wh-item. It can then be suggested that the particle *ka* interpreted as *whether* is identical to *ka* associated with a wh-item. In the reading (iv) in (17-18), the lower particle *mo* is interpreted as *also*; this reading is (at least marginally) available, unlike the readings (ii) and (iii), where the lower *mo* is associated with a wh-item. This consideration then suggests that the particle *mo* interpreted as *also* is a lexically different item from *mo* associated with a wh-item (Shimoyama 2006: 147). I will not consider further this issue since it is not relevant to the current discussion.

totemo tukareta.
very got.tired

(i) ✓'The teacher who read, for every topic *x*, every year *y*, the paper that Ken wrote on *x* in *y* also got very tired.'

(ii) * 'For every year *y*, the teacher who read, for every topic *x*, the paper that Ken wrote on *x* in *y* got very tired.'

(iii) ** 'For every topic *x*, the teacher who read, for every year *y*, the paper that Ken wrote on *x* in *y* got very tired.'

(iv) (?)'For every topic *x*, every year *y*, the teacher who also read the paper that Ken wrote on *x* in *y* got very tired.'

(based on Shimoyama 2006: 147)

(18) Mai-wa [[[Ken-ga <u>nan-nen</u>-ni <u>nani</u>-nituite e_i kaita] ronbun$_i$]-<u>mo</u> yuu datta <u>ka</u>]
Mai-TOP Ken-NOM what-year-in what-about wrote paper-MO A COP.PAST Q
siritagatteiru.
want.to.know

(i) ✓ 'Mai wonders whether for every topic *x*, every year *y*, the paper that Ken wrote on *x* in *y* got an A.'

(ii) ?* 'Mai wonders for which year *y*, for every topic *x*, the paper that Ken wrote on *x* in *y* got an A.'

(iii) * 'Mai wonders for which topic *x*, for every year *y*, the paper that Ken write on *x* in *y* got an A.'

(iv) (?) 'Mai wonders for which topic *x* and for which year *y*, the paper that Ken wrote on *x* in *y* also got an A.'

(based on ibid: 146-147)

Building on these observations, Shimoyama presents the following generalization: a wh-item cannot be associated with a quantificational particle (i.e. *ka* or *mo*) across another quantificational particle. This can be schematically represented as in (19).

(19) *[… [… wh-item(/indeterminate) …] -*ka*/*mo* …]-*ka*/*mo*

(based on ibid: 148)

With the two generalizations (14) and (19) established, Shimoyama then aims to propose a semantic analysis that would uniformly capture those generalizations, both of which concern the locality (or island asymmetry) of Japanese wh-constructions.

Her analysis builds on Hamblin's (1973) analysis of wh-questions, according to which English wh-pronouns denote sets of individuals. On this view, for instance, *who* and *what* denote sets of (all) persons and things, respectively. Adopting this idea, Shimoyama assumes that wh-items, or indeterminates, in Japanese denote sets of individuals. Thus, *dare* 'who' denotes a set of human individuals, as shown in (20). (In the following semantic representations, I will ignore tense and possible worlds for expository purposes.)

(20) $[\![\text{dare}]\!]^g = \{x \in D_e : \text{person}(x)\}$

She further assumes, following Hamblin (1973), that lexical items that are not indeterminate phrases denote singleton sets whose sole members are their normal denotations. For example, the predicate *odoru* 'dance' denotes a singleton set whose only member is its ordinary denotation, as (21) shows.

(21) $[\![\text{odoru}]\!]^g = \{\lambda x[\text{dance}(x)]\}$

Consider then how the (incomplete) wh-question (22) is semantically derived. ((22) is incomplete in the sense that it lacks the Q-particle *ka*. The semantic contribution of the Q-particle will be taken into consideration in the later discussion.)

(22) dare-ga odoru
who-NOM dance
Intended: 'Who dances?'

With a Hamblin-style analysis, the semantic derivation of (22) proceeds via pointwise functional application, i.e. by applying the function "$\lambda x[\text{dance}(x)]$" to each member of the set of human individuals. As a result, (22) obtains the denotation in (23).

(23) $[\![(22)]\!]^g = \{f(x): f \in [\![\text{odoru}]\!]^g, x \in [\![\text{dare}]\!]^g\}$
$= \{[\text{dance}(x)]: \text{person}(x)\}$

According to (23), (22) denotes a set of propositions (or answers to it), which is the standard denotation for interrogative sentences (e.g. Hamblin 1973, Karttunen 1977).

Note here, crucially, that a Hamblin-style analysis of wh-questions succeeds in deriving the standard denotation of wh-questions (i.e. sets of propositions) without assuming any (wh-)movement; the set of individuals denoted by an in-situ wh-item "expands" up to the point of becoming the set of propositions, by virtue of pointwise functional application.

Bearing this in mind, let us turn to the particles *mo* and *ka* and consider how they work with the above assumptions. Regarding *mo*, Shimoyama considers it to be a generalized quantifier; after it has combined with a set of individuals (as a restrictor), it takes a property (i.e. a predicate) and returns a singleton set whose sole member is a universally quantified proposition; see (24).

(24) For $[\![\alpha]\!]^g \subseteq D_e$,
$[\![\alpha \text{ mo}]\!]^g = \{\lambda P \forall x[x \in [\![\alpha]\!]^g \rightarrow P(x) = 1]\}$ (ibid: 155)

With (24), consider the basic case (25) (= (11)), for example.

(25) Dare-mo-ga odotta.
who-MO-NOM danced
'Everyone danced.'

In (25), *mo* first takes as its input the wh-item *dare* 'who', which denotes a set of human individuals, rendering it a restrictor. After that, the predicate *odotta* is combined. The resulting semantic representation of (25) is shown in (26).

(26) $[\![(25)]\!]^{g} = \{\forall x[x \in \{y: \text{person}(y)\} \rightarrow \text{dance}(x)]\}$ (based on ibid: 155)

Consider next a bit more complicated example in (27).

(27) [[<u>Dare</u>-ga e_i teisyutusita] syukudai$_i$]-<u>mo</u> yuu datta.
who-NOM submitted homework-MO A COP.PAST
'Every homework assignment that a person had handed in got A.' (based on ibid: 154)

(27) crucially differs from (25) in that *mo* does not directly take wh-item *dare* 'who' as its input; rather, it combines with the complex NP that contains that wh-item, i.e. *dare-ga teisyutusita syukudai* '(Lit.) homework assignments that who submitted'.[13] Notice, however, that this is not compositionally problematic; in that complex NP, the set of alternatives denoted by the wh-item has expanded by virtue of pointwise functional application, and consequently, that complex NP

[13] With the assumption that a wh-item itself serves as a restrictor of *mo* (which Shimoyama 2006 calls the *embedded restrictor view*), it has been proposed in the literature that even when these two elements are superficially non-locally positioned as in (27), they actually establish the association in a local manner in syntax (e.g. Takahashi 2002) or in LF (e.g. Nishigauchi 1990). Under this view, some type of movement needs to be posited to guarantee the association under consideration. However, Shimoyama (2006) departs from this view and proposes, as noted in the main text, that *mo* directly takes its sister constituent as its restrictor (i.e. the *direct restrictor view*), which thus makes movement unnecessary (see the discussion in the main text).

denotes a set of individuals. The complex NP under consideration can be semantically represented as in (28), which refers to the set of (unique) homework assignments that each person submitted.[14]

(28) $[\![$ [[1 [dare-ga t_1 teisyutusita]] syukudai] $]\!]^g$
= {ιy[homework.assignment(y) & submit(y)(z)]: person(z)} (based on ibid: 154)

Then, (27) eventually obtains the denotation in (29).

(29) $[\![(27)]\!]^g$ = {∀x[x∈{ιy[homework.assignment(y) & submit(y)(z)]: person(z)} → was.A(x)]}
(based on ibid: 155)

What about the Q-particle *ka*, then? While her main focus is on the particle *mo*, Shimoyama notes in a footnote that "*ka* can be considered to [take a question and] return a singleton set whose sole member is a question denotation as proposed in Groenendijk and Stokhof (1982)" (Shimoyama 2006: 154). I here adopt this assumption since it is crucial to the account of the locality puzzles under consideration. Notice here that what the two particles, *mo* and *ka*, have in common is that they take sets of alternatives (i.e. a set of individuals for *mo*, a set of propositions for *ka*) and return singleton sets.

With the above illustration in mind, let us finally turn back to Shimoyama's goal, which is to give a uniform account to the two generalizations about the locality in Japanese wh-constructions, namely (14) and (19), repeated below as (30) and (31) respectively.

(30) [… [$_{\text{Complex NP}}$ … wh-item(/indeterminate) …] …]-*ka*/*mo*
(based on ibid: 148)

[14] Following Shimoyama (2006), I here consider only the case where the bare noun *syukudai* 'homework assignment' in (27) is interpreted as definite and singular. This is why the iota operator ι is used in (28).

(31) *[… [… wh-item(/indeterminate) …] -*ka*/*mo* …]-*ka*/*mo*

(based on ibid: 148)

Let us first consider (30), which indicates that wh-items can be associated with a quantificational particle (i.e. *ka* or *mo*) across a complex NP island. The two instances of (30), (13) and (1), are repeated below as (32) and (33) respectively.

(32) (?)[[[Dare-ga e_i osieta] gakusee$_i$]-ga e_j syootaisita] sensee$_j$]-mo kita.
who-NOM taught student-NOM invited teacher-MO came
'For every *x*, the teacher(s) that the student(s) that *x* had taught invited came.'

(based on ibid: 146)

(33) Ken-wa [[e_i nani-o kaita] hito$_i$]-o home-masi-ta ka.
Ken-TOP what-ACC wrote person-ACC praise-POL-PAST Q
Lit. 'Ken praised [the person who wrote what]?'
(≈ 'What is the thing such that Ken praised the person who wrote it?')

Recall that under Hamblin's analysis, adopted by Shimoyama, wh-items do not undergo wh-movement, staying in-situ even in LF. Therefore, the fact that wh-items appear within complex NP islands in (32-33) should not raise any syntactic problems. In addition, according to Hamblin's analysis, the set of alternatives denoted by in-situ wh-items expands up by virtue of pointwise functional application. Importantly, there is no semantic or compositional problem in this expansion crossing complex NP islands. Therefore, (32-33) do not have any problems in their semantic derivations either. Their acceptability thus follows.[15]

[15] Shimoyama (2006) does not consider how her semantic analysis of Japanese wh-questions is compatible with the fact that adjunct wh-phrases such as *naze* 'why' cannot appear in complex NP islands, as in (i) (= (i) in fn.1) (see fn.1 and fn.7 for relevant discussion); I leave this issue open here.

(i) *Ken-wa [[e_i naze syoosetu-o kaita] hito$_i$]-o hometa no.
Ken-TOP why novel-ACC wrote person-ACC praised Q

Consider next the generalization (31), according to which wh-items cannot be associated with a quantificational particle (i.e. *ka* or *mo*) across another quantificational particle. Two of the four relevant examples, (4) and (17) are repeated below as (34) and (35) respectively; recall particularly that they do not yield the readings where wh-items are associated with the higher particle crossing the lower one (except for the reading (iv) in (35); see fn.12).

(34) Mai-wa [dare-ga nani-o katta ka] sittei-mas-u ka.
Mai-TOP who-NOM what-ACC bought Q know-POL-PRES Q

(i) ✓'Does Mai know [who bought what]?'

(ii) ?* Intended: 'Who does Mai know [what (*t*) bought]?'
(≈ 'For which *x*, *x* a person, does Mai know what *x* bought?')

(iii) * Intended: 'What does Mai know [who bought (*t*)]?'
(≈ 'For which *y*, *y* a thing, does Mai know who bought *y*?')

(iv) * Intended: 'Who what does Mai know [whether (*t*) (*t*) bought]?'
(≈ 'For which *x y*, *x* a person, *y* a thing, does Mai know whether *x* bought *y*?')

(based on Nishigauchi 1990: 28)

(35) [[e_i [[Ken-ga nan-nen-ni nani-nituite e_j kaita] ronbun_j]-mo yonda] sensee_i]-mo
Ken-NOM what-year-in what-about wrote paper-MO read teacher-MO
totemo tukareta.
very got.tired

(i) ✓'The teacher who read, for every topic *x*, every year *y*, the paper that Ken wrote on *x* in *y* also got very tired.'

(ii) * 'For every year *y*, the teacher who read, for every topic *x*, the paper that Ken wrote on *x* in *y* got very tired.'

(iii) ** 'For every topic *x*, the teacher who read, for every year *y*, the paper that Ken wrote on *x* in *y* got very tired.'

Lit. 'Ken praised [the person who wrote a novel why]?'

(iv) (?)‘For every topic *x*, every year *y*, the teacher who also read the paper that Ken wrote on *x* in *y* got very tired.’

(based on Shimoyama 2006: 147)

First, as noted above, under the assumptions under conssideration, the wh-islands in (34-35) should not raise any syntactic problems, since the wh-items stay in-situ even in LF. It is then reasonable to consider that their unacceptability should arise due to a semantic reason. Of relevance here is the semantic role of the particles *ka* and *mo*; they take sets of alternatives and return singleton sets. This means that if their sister constituent involves wh-items, the expansion of the set of alternatives denoted by them will be trapped (or “absorbed” in Shimoyama’s term) by those particles. As a result, the expansion under consideration can never go over those particles, making those wh-items inaccessible from the higher particle. The generalization (31) thus follows.

To recap, Shimoyama’s (2006) semantic analysis of Japanese wh-questions, or Japanese wh-constructions in general, captures the asymmetry regarding islands under consideration without assuming covert wh-movement. Therefore, if Shimoyama’s semantic analysis is a viable option, the island asymmetry of Japanese wh-questions would not be as strong an argument for covert wh-movement in Japanese wh-questions as Nishigauchi (1990) argues.

2.3.2 Prosody of Japanese wh-questions

Another view against Nishigauchi’s (1990) argument originates from the status of his grammatical judgements on the data that suggest the existence of wh-island effects in Japanese. As a case study, consider again (3), repeated below as (36). (Here, I will not discuss the other relevant example, i.e. (4), to simplify the discussion; see Deguchi and Kitagawa 2002 and Kitagawa 2006a, 2017 for relevant discussion.)

(36) Mai-wa [Ken-ga nani-o katta ka] sittei-mas-u ka.
Mai-TOP Ken-NOM what-ACC bought Q know-POL-PRES Q
(i) ✓‘Does Mai know [what Ken bought]?’
(ii) * Intended: ‘What does Mai know [whether Ken bought (*t*)]?’
(≈ ‘For which *x*, *x* a thing, does Mai know [whether Ken bought *x*]?’)
(based on Nishigauchi 1990: 30)

As described in Section 2.2, according to Nishigauchi, (36) can yield only the reading (i), not (ii). More specifically, in (36), the wh-phrase *nani* ‘what’ in the embedded clause can be associated only with the embedded Q-particle, which forms an embedded wh-question. Concerning the reading (ii), he notes that “it is completely impossible to interpret [(39)] as [(ii)]” (Nishigauchi 1990: 31). Given that in the reading (ii) the embedded wh-phrase is associated with the matrix Q-particle across the *whether*-question, the unavailability of that reading indicates, he argues, that Japanese wh-questions show wh-island effects.

Since Nishigauchi (1990), however, the grammaticality judgement on (36) has been cast doubt on. For example, Watanabe (1992) notes that there is speaker variation regarding the availability of the reading (ii) and, taking that into consideration, judges his relevant examples to be at least marginal (i.e. “?” or “??”). However, Takahashi (1993) judges wh-questions with the reading like (ii) in (36) to be fully acceptable. How could this puzzling status be resolved?

An answer to this question has been suggested in the literature; the key to this issue may lie in the prosodic facet of Japanese wh-questions (e.g. Deguchi and Kitagawa 2002, Ishihara 2003, Kitagawa 2005, 2017). Japanese wh-questions are prosodically characterized by the focus prominence on wh-phrases and the post-focal reduction, i.e. the fact that the pitch of the string that follows a wh-phrase (with focus prominence) is compressed. To see these prosodic traits, consider the matrix wh-question in (37).

(37) Ken-wa nani-o kai-masi-ta ka.
Ken-TOP what-ACC buy-POL-PAST Q
'What did Ken buy?'

(38) shows the prosodic representation of (37), where the aforementioned prosodic properties are reflected.

(38) Ken-wa **NA**ni-o kai-masi-ta ka↑
Ken-TOP what-ACC buy-POL-PAST Q

In (38), the accented mora of the wh-phrase *nani* 'what' obtains the pitch prominence, represented by bold capitals. Then, the pitch of the following string is compressed until the sentence-final Q-particle *ka*. This pitch compression, or *post-focal reduction*, is represented by underlined reduced fonts. The upward arrow on the Q-particle indicates the interrogative rising intonation. Following Kitagawa (2017), I will refer to the combination of the focus prominence on wh-phrases and the post-focal reduction as the *focus prosody*.

Importantly, it has been observed in the literature that the domain of the focus prosody corresponds to the scope that wh-phrases take (e.g. Deguchi and Kitagawa 2002, Ishihara 2003, Kitagawa 2005, 2017). E.g., consider the wh-question in (39).

(39) Mai-wa [Ken-ga nani-o katta to] omottei-mas-u ka.
Mai-TOP Ken-NOM what-ACC bought REP think-POL-PRES Q
'What does Mai think that Ken bought?'

In (39), the wh-phrase *nani* 'what' appears in the embedded declarative clause. Since the embedded clause is a declarative one, so there is no Q-particle to be associated with in it, the wh-phrase must take a matrix scope, being associated with the matrix Q-particle. As a result, (39) must be interpreted as a matrix wh-question. Let us now observe the focus prosody for (39). (40) shows a prosodic representation of (39); with this, (39) sounds natural, thus being acceptable.

(40) Mai-wa [Ken-ga **NA**ni-o katta to] omottei-mas-u ka↑
Mai-TOP Ken-NOM what-ACC bought REP think-POL-PRES Q

In (40), importantly, the domain of the focus prosody crosses the embedded clause and ranges over the matrix clause; that is, the post-focus reduction continues up to the matrix Q-particle. Following Kitagawa (2017), I will call such focus prosody *global focus prosody*. This is consistent with the fact that the wh-phrase in (39) must take a matrix scope; hence, the naturalness of (43). Next, observe (41), another prosodic representation for (39); with this, (39) sounds strange, thus being unacceptable.

(41) # Mai-wa [Ken-ga **NA**ni-o katta to] omottei-mas-u ka↑
Mai-TOP Ken-NOM what-ACC bought REP think-POL-PRES Q

(41) crucially differs from (40) in that the domain of the focus prosody resides in the embedded clause, not stretching over the matrix clause; i.e., the post-focal reduction stops at the edge of the embedded clause, not breaking into the matrix clause. Following Kitagawa (2017), I will call such focus prosody *local focus prosody*. This is incongruent with the fact that the wh-phrase in (39) has to take a matrix scope; hence, the strangeness of (41). Those observations thus indicate that the domain of the focus prosody is in harmony with the scope of wh-phrases.

Bearing this syntax-prosody synchronization in mind, let us now turn back to the example under consideration, i.e. (36). Specifically, let us re-examine (36) taking the focus prosody into consideration. Observe first its prosodic representation in (42).

(42) Mai-wa [Ken-ga **NA**<u>ni-o katta ka</u>] sittei-mas-u ka↑

Mai-TOP Ken-NOM what-ACC bought Q know-POL-PRES Q

(i) ✓'Does Mai know [what Ken bought]?'

(ii) * Intended: 'What does Mai know [whether Ken bought (*t*)]?'

(≈ 'For which *x*, *x* a thing, does Mai know [whether Ken bought *x*]?')

(42) involves local focus prosody; the domain of the focus prosody sits within the embedded clause, not ranging over the matrix clause. It is then expected that the wh-phrase in (36) will take an embedded scope, not a matrix scope; i.e., the reading (i) in (36) is expected to arise, but not (ii). This expectation is borne out, as indicated in (42). Note that this parallels Nishigauchi's judgement pattern for (36).

Let us now consider the crucial case (43), another prosodic representation for (36).

(43) Mai-wa [Ken-ga **NA**<u>ni-o katta ka] sittei-mas-u</u> ka↑

Mai-TOP Ken-NOM what-ACC bought Q know-POL-PRES Q

(i) *'Does Mai know [what Ken bought]?'

(ii) ✓/ ? Intended: 'What does Mai know [whether Ken bought (*t*)]?'

(≈ 'For which *x*, *x* a thing, does Mai know [whether Ken bought *x*]?')

In contrast to (42), (43) involves global focus prosody; the domain of the focus prosody in (43) breaks into the matrix clause. It is then predicted that the wh-phrase in (36) will take a matrix scope, rather than an embedded scope; hence, the reading (ii), not (i), is predicted to obtain. Note here that if the wh-island constraint holds in Japanese, as Nishigauchi argues, the reading (ii) should be unavailable and thus (43) should sound strange. Crucially, however, (43) is observed to be acceptable for many native speakers with the reading (ii) in the literature (e.g. Deguchi and

Kitagawa 2002, Ishihara 2003, Kitagawa 2005, 2017).[16] It has now turned out that the puzzling status of the grammaticality judgement in (36) can be resolved by taking into consideration the prosodic aspect of Japanese wh-questions; uttering (36) with a global focus prosody makes the reading (ii) readily available for many native speakers. The availability of the reading (ii), then, suggests that the wh-island constraint does not hold in Japanese.[17] This raises a problem for Nishigauchi (1990), given that his observation on wh-island effects, i.e. the unavailability of the reading (ii) in (36), serves as a crucial argument for the covert wh-movement analysis of Japanese wh-questions.

While it has been shown that the reading (ii) for (36) is actually available, there still remains one important question: why does the reading (ii) tend to be judged to be unavailable, or at least marginal? For example, the reading (ii) of (36) is unavailable for Nishigauchi (1990) (i.e. *) and marginally available for Watanabe (1992) (i.e. ?/??), while it is available for Takahashi (1993) (i.e. ✓). Recall that (43) shows that the reading in question obtains more easily if (36) is assigned global focus prosody. It should then follow that global focus prosody for (36) is not readily available for many speakers, including the first two authors above. Given that, the following generalization should follow: when a wh-phrase occurs in a wh-island, global focus prosody tends to be unavailable. An account of this generalization will then constitute an answer to the question raised above.

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[16] While the reading (ii) in (36) is certainly more readily available with (43) than with (42), one might still find it to be marginal (as suggested by "✓/ ?" in (43)). This could be because interpretations like the reading (ii), where a wh-phrase takes a matrix scope from within an embedded wh-question, requires a very specific (and sometimes unusually elaborated) pragmatic context for their presuppositions to be satisfied, as Kitagawa and Fodor (2003) observe.

[17] One might assume that a wh-island effect does occur in (43) but it is merely "alleviated" by a "special" prosodic amendment, i.e. global focus prosody; Nishigauchi (1990) in fact holds such a view (see fn.3). Under this view, global focus prosody is regarded as some extra-grammatical strategy to "hide" wh-island effects. However, it is not clear why that would be the case.

It has been argued in the literature that the generalization under consideration results from several extra-syntactic/grammatical factors (e.g. Kitagawa and Fodor 2003, 2006). I refer the reader to Kitagawa (2017) for a summary of those factors. Here, I illustrate only one of them, which concerns (un)markedness of focus prosody.

According to Kitagawa and Fodor (2003), global focus prosody is phonologically more marked than local focus prosody. This is because, compared with local focus prosody, global focus prosody creates a long string of rhythmically and tonally undifferentiated material by virtue of post-focal reduction, but such a string is dispreferred in natural language (Selkirk 1984). Then, when one silently reads a sentence where a wh-phrase appears in a wh-island to examine its grammaticality, one tends to choose the unmarked option, i.e. local focus prosody (i.e. the *Implicit Prosody Hypothesis*; Fodor 2002) and thus the more marked option, namely global focus prosody, tends to be excluded.

To recap, we have considered how the grammaticality judgement of (36) depends on the prosodic aspect of Japanese wh-questions. In particular, it has been observed in the literature that assigning global focus prosody to (36) makes the reading (ii) available for many native speakers (see (43)). This may then suggest that there are no wh-island effects in Japanese. This observation, in turn, serves as a potential counterargument to Nishigauchi's (1990) argument for covert wh-movement in Japanese wh-questions, since his argument crucially depends on his observation regarding the presence of wh-island effects. Concerning the question why the reading (ii) of (36) tends to be unavailable, it has been suggested in the literature that global focus prosody is less accessible than local focus prosody.[18]

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[18] I finally note Kitagawa's (2006b, 2017) remark on the fact that the adjunct wh-phrase *naze* 'why' cannot appear in islands. Witness again the relevant example in (i), where *naze* 'why' occurs in a complex NP island.

(i) *Ken-wa [[e_i naze syoosetu-o kaita] hito$_i$]-o hometa no.
Ken-TOP why novel-ACC wrote person-ACC praised Q

2.4 Summary of the chapter

This chapter has provided brief background regarding covert wh-movement in Japanese. I have first illustrated the main argument for it, i.e. Nishigauchi's (1990) observation that while Japanese

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Lit. 'Ken praised [the person who wrote a novel why]?'

It has also been claimed that when *naze* 'why' appears in an embedded (wh-)question, it cannot take a matrix scope (i.e. wh-island effects), as exemplified in (ii).

(ii) Kimi-wa [Ken-ga naze sono hon-o katta ka] siritai no.
you-TOP Ken-NOM why that book-ACC bought Q want.to.know Q
(A) ✓ Do you want to know [why Ken bought that book]?
(B) * Intended: Why$_i$ do you want to know [whether Ken bought that book t_i]? (based on Saito 1994: 205)

As noted in fn.1, Huang (1982b) and Lasnik and Saito (1984) ascribe the ungrammaticality of (i-ii) to the ECP, assuming LF wh-movement of *naze*. However, Kitagawa (2006b, 2017) argues that (i-ii) are unacceptable for pragmatic reasons. More specifically, he argues that they are unacceptable because they require elaborate and specific types of pragmatic contexts to be felicitously interpreted (cf. fn.16). For example, (i) and (ii) (with the reading (B)) are expected to be uttered in situations such as (iiia-b), which may be hard for one to imagine when s/he is asked to judge the grammaticality of (i-ii) with no specific context given.

(iii) a. The speaker believes that there is some specific reason that Ken praised the person who wrote a novel for it (= that reason), and wants the hearer to identify the reason for which this is true.
b. The speaker believes that there is some specific reason such that the hearer wants to know if Mai bought the book for it (= that reason), and wants the hearer to identify the reason for which this is true.
(based on Kitagawa 2006b: 108-109)

If this is correct, it is expected that sentences where *naze* 'why' appears in islands will be ameliorated if they can allow one to easily imagine situations like (iii), and they are assigned appropriate prosody, especially in the case of wh-islands. Kitagawa claims that this expectation is borne out, giving examples like (iv-v), which involve a complex NP island and a wh-island respectively. (A few notes are in order: Regarding (iv), the wh-phrase *naze* is intended to take an embedded rather than matrix scope. Regarding (v), it involves the phonological representation used in the main text, where (v) is pronounced with global focus prosody so that the scope of the wh-phrase stretches across the embedded clause. Note also that the grammaticality judgements in (iv-v) are Kitagawa's.)

(iv) [Every year, numbers of players leave professional baseball.]
[[[*e*$_i$ naze yameteiku] sensyu$_i$]-ga itiban ooi ka] sitte-mas-u ka. Kega desu yo, kega.
why quit player-NOM most many Q know-POL-PRES Q injury COP.POL SFP injury
'Do you know for what reason the most number of players quit the team? It is injury!' (ibid: 109)

(v) [A law professor lecturing on court cases says: "Generally speaking, in any trial, why the assailant committed a crime becomes a very important point, though what kind of motive is considered to be the most important differs depending on the type of the trial. For instance, in criminal cases, …" The lecture on criminal cases continues for a while:]
Dewa, minzisaiban-de-wa [[kagaisya-ga **NA**ze tumi-o okasitesimatta ka]-ga mottomo
then civil.trial-in-TOP defendant-NOM why crime-ACC have.committed Q-NOM most
zyuuyoosis-areru ka] to ii-mas-u to, …
view.important-PASS Q that say-POL-PRES if
'Then, what reason is regarded as most important if the defendant committed a crime for that reason? I would say …'
(ibid: 110)

The ECP-based account would not accommodate the acceptability of those examples. Kitagawa adds, however, that when he asked over 40 speakers to judge (iv-v), the most popular reaction was that "interpreting [(iv)] and [(v)] requires some pondering" (Kitagawa 2006b: 110).

At any rate, I will leave for future research to clarify the status of adjunct wh-phrases within islands (both empirically and theoretically).

wh-questions do not show complex NP island effects, they exhibit wh-island effects; the existence of island effects constitutes a straightforward argument for wh-movement in Japanese. Concerning the asymmetry with respect to island effects, Nishigauchi argues (or more precisely, *would* argue) that covert large-scale pied-piping is available to complex NP islands, which voids the island effect in this case, while it is not to wh-islands, due to the mechanism of feature percolation.

I have then introduced two views that challenge Nishigauchi's argument. First, Shimoyama (2006) argues that the island asymmetry can be accounted for by her semantic analysis of wh-constructions in Japanese without assuming wh-movement. Second, the data that Nishigauchi (1990) crucially judges to be unacceptable due to wh-island effects (e.g. (3-4)) turn out to be acceptable if the prosodic properties of Japanese wh-questions are taken into consideration, as observed in the literature (e.g. Deguchi and Kitagawa 2002, Ishihara 2003, Kitagawa 2005, 2017).

What has been shown thus far is that the main argument for covert wh-movement in Japanese provided by Nishigauchi (1990), i.e. (asymmetric) island effects, has lost some of its original solidity, due to the challenging views discussed above (though the alternative accounts discussed above also face some issues). Against this backdrop, in Chapter 3, I will investigate the syntax of Japanese wh-questions from new perspectives, and as a result, provide brand-new arguments for covert wh-movement in (at least a subset of) Japanese wh-questions.

Chapter 3

Asymmetries and Covert Wh-Movement in Japanese Wh-Questions

3.1 Introduction

As discussed in Chapter 1, it is still controversial how to syntactically analyze Japanese wh-questions, which are characterized by in-situ wh-phrases and the sentence-final Q-particle *ka*, as exemplified in (1).

(1) Ken-wa <u>nani</u>-o kai-masi-ta ka.
Ken-TOP what-ACC buy-POL-PAST Q
'What did Ken buy?'

In particular, (2) shows four representative analyses of Japanese wh-questions and the derivation of (1) in each of these analyses.

(2) a. LF/covert wh-movement analysis (e.g. Lasnik and Saito 1984, 1992, Nishigauchi 1990, Saito 2017)
Overt Syntax: $[_{CP}$ $[_{TP}$ Ken what bought] Q]
LF: $[_{CP}$ what$_i$ $[_{TP}$ Ken t_i bought] Q]

b. Null operator movement analysis (e.g. Watanabe 1992)
Overt Syntax/LF: $[_{CP}$ Op$_i$ $[_{TP}$ Ken [t_i what] bought] Q]

c. Q-particle movement analysis (e.g. Hagstrom 1998, Cable 2007, 2010, Miyagawa 2010)
Overt Syntax/LF: $[_{CP}$ $[_{TP}$ Ken [what t_i] bought] Q$_i$]

d. Non-movement analysis (e.g. Shimoyama 2001, 2006):
Overt Syntax/LF: $[_{CP}$ $[_{TP}$ Ken what bought] Q]

The LF/covert wh-movement analysis (2a) differs from the other three analyses (2b-d) in that it involves the movement of the wh-phrase itself. As illustrated in Chapter 2, however, the main argument for the LF/covert wh-movement analysis presented by Nishigauchi (1990), i.e. (asymmetric) island effects, has been challenged in the literature.

Against this background, this chapter aims to shed a new light on the syntax of Japanese wh-questions through investigating it from novel perspectives. This investigation will crucially build on a particular classification of Japanese wh-questions; in particular, I will categorize Japanese wh-questions into four types, namely matrix wh-questions, embedded CQ wh-questions, embedded non-CQ wh-questions, and *to*-embedded wh-questions (as briefly introduced in Chapter 1). With this classification, I will examine what syntactic behavior the above four types of wh-questions show with respect to intervention effects and the formation of clefts. I will crucially show that these different types of wh-questions exhibit asymmetric behavior with respect to the two syntactic phenomena in question. In particular, it will be shown that those wh-questions can be divided into two groups based on their asymmetric behavior in question.

Based on this finding, I will argue that Japanese wh-questions should be analyzed in a non-uniform manner. More specifically, I will propose that the involvement of covert wh-movement differs depending on the type of wh-questions. I will demonstrate that this proposal can capture in a uniform manner the aforementioned asymmetric behavior of the four types of wh-questions with respect to intervention effects and cleft formation. The proposal in question will thus serve as a novel argument for the necessity of covert wh-movement in analyzing (a subset of) Japanese wh-questions.

The rest of this chapter is structured as follows: Section 3.2 establishes the classification of wh-questions in Japanese which consists of matrix wh-questions, embedded CQ wh-questions, embedded non-CQ wh-questions, and *to*-embedded wh-questions. In Section 3.3, it will be shown that those four types of wh-questions show different behavior with respect to intervention effects and cleft formation. Section 3.4 presents a proposal regarding the syntax of wh-questions in Japanese and illustrates how this proposal gives an account of the asymmetries in Japanese wh-questions observed in Section 3.3. Section 3.5 concludes by giving a summary of this chapter and discussing some implications of the proposals presented in this chapter.

3.2 Classification of Japanese wh-questions

This section provides the classification of Japanese wh-questions that this dissertation will crucially draw on. I will first introduce three basic classes of wh-questions: matrix wh-questions, embedded wh-questions, and *to*-embedded wh-questions. I will then further divide embedded wh-questions into two subclasses: embedded CQ wh-questions and embedded non-CQ wh-questions.

3.2.1 Three basic classes

The three basic classes of Japanese wh-questions that this dissertation revolves around will be referred to as *matrix wh-questions*, *embedded wh-questions*, and to-*embedded wh-questions*. They are roughly schematized in (3) and are exemplified in (4).

(3) a. Matrix wh-questions: [… wh …] *ka*

b. Embedded wh-questions: [… wh … *ka*] V

c. *To*-embedded wh-questions: [[… wh … *ka*] *to*] V

(4) a. Matrix wh-questions

Ken-wa nani-o kai-masi-ta ka.

Ken-TOP what-ACC buy-POL-PAST Q

‘What did Ken buy?’ (= (1))

b. Embedded wh-questions

Mai-wa [Ken-ga nani-o katta ka] {sitteiru / tazuneta / sinpaisiteiru /
Mai-TOP Ken-NOM what-ACC bought Q know / asked / is.anxious /
gironsita / …}.
discussed
‘Mai {knows / asked / is anxious / discussed / …} [what Ken bought].’

c. *To*-embedded wh-questions

Mai-wa [[Ken-ga nani-o katta ka] to] {*sitteiru / tazuneta / sinpaisiteiru /
Mai-TOP Ken-NOM what-ACC bought Q REP know / asked / is.anxious
*gironsita / …}.
discussed
Lit. ‘Mai {knows / asked / is anxious / discussed / …} [that what Ken bought].’

A few notes regarding those types of wh-questions are in order. First, matrix wh-questions in Japanese are stylistically constrained in that their predicates obligatorily contain a politeness affix such as -*mas*, as (4a) shows (e.g. Miyagawa 1987). Without that affix, they are unacceptable; compare (5) with (4a).[1]

(5) *Ken-wa nani-o katta ka.
Ken-TOP what-ACC bought Q
‘What did Ken buy?’

[1] Matrix wh-questions in Japanese can also end with another particle *no* (see, e.g., fn.1 in Chapter 2). In this case, the politeness affix does not appear, as exemplified in (i).

(i) Ken-wa nani-o katta no.
Ken-TOP what-ACC bought Q
‘What did Ken bought?’

The stylistic constraint under consideration thus applies only to matrix wh-questions that end with the Q-particle *ka*.

This property is not shared by the other classes of wh-questions; observe that the embedded wh-question in (4b) and *to*-embedded wh-question in (4c) do not involve the politeness affix in their predicates.

Second, *to*-embedded wh-questions (3/4c) differ from (standard) embedded wh-questions (3/4b) in that they are followed by the complementizer *to*. This complementizer is viewed as a reportative complementizer, introducing a report (or paraphrase) of direct discourse (e.g. Saito 2010, 2012, 2013, 2015).[2] Thus, it can take as its complement not only an interrogative clause (as in (4c)) but also a declarative clause and even an imperative clause (cf. Kuno 1988; see also Schwager 2006 and Kaufmann 2012), as exemplified in (6). (See Saito 2010, 2015 for other clause types that *to* can take as its complement.)

(6) a. Mai$_i$-wa [[Ken-ga kanozyo$_i$-no heya-o soojisuru] to] omotteiru.
Mai-TOP Ken-NOM she-GEN room-ACC clean REP think
'Mai thinks [that Ken will clean her room].'

b. Mai$_i$-wa Ken-ni [[kanozyo$_i$-no heya-o soojisi-ro] to] itta.
Mai-TOP Ken-to she-GEN room-ACC clean-IMP REP said
Lit. 'Mai said to Ken [that clean her room].'

Another difference between the two types of wh-questions concerns the predicates that select them; the comparison between (4b) and (4c) reveals that only a restricted range of predicates can select *to*-embedded wh-questions, compared with embedded wh-questions. This property of *to*-embedded wh-questions follows from the semantic role of the reportative complementizer *to*; since it plays a role of introducing a report/paraphrase of direct discourse, *to*-embedded wh-questions

[2] Saito's (2010, 2012, 2013, 2015) claim that *to* is a reportative complementizer is (at least partly) motivated by the fact that *to* parallels Spanish complementizer *que*, which Plann (1982) argues can serve as a complementizer introducing a paraphrase of direct discourse (see also Rivero 1994).

can be selected only by verbs that can take a direct quote as their complement (i.e. verbs of saying and thinking), which include *tazuneta* 'asked' and *sinpaisiteiru* 'is anxious', but not *sitteiru* 'know' and *gironsita* 'discussed' as (7) shows.[3]

(7) a. Mai-wa ["Ken-wa, eeto, nani-o kai-masi-ta ka?" to] {tazuneta / *sitteiru /
Mai-TOP Ken-TOP well what-ACC buy-POL-PAST Q TO asked / know /
*gironsita / …}.
discussed
Lit. 'Mai {asked / knows / discussed /…} "Well, what did Ken buy?" '

b. Mai-wa ["Ken-wa, nnn, nani-o katta ndaroo ka?" to] {sinpaisiteiru /
Mai-TOP Ken-TOP hmm what-ACC bought MOD Q TO is.anxious /
*sitteiru / *gironsita / …}.
know / discussed
Lit. 'Mai {is anxious / knows / discussed / …} "Hmm, what did Ken buy?" '

As one might have noticed, in Japanese, direct quotes are introduced by the particle that is homophonic to the reportative complementizer *to*, as (7) shows. One might then suspect that the *to*-embedded wh-question in (4c) is not reportative but a direct quote. This is not the case, however, because it does not involve the politeness affix *-mas*, which matrix wh-questions must contain, as described above (e.g. (4a) vs. (5)). In examples in this thesis, *to*-embedded wh-questions will not

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[3] Saito (2010, 2012, 2013) presents a partial list of predicates that can select *to*-embedded wh-questions and those that cannot (i). Note particularly that the predicates in (ia) can select a direct quote as well while those in (ib) cannot.

(i) a. matrix predicates that allow *ka-to*: *kiku* 'ask', *situmonsuru* 'question', *yuu* 'say', *sakebu* 'scream', *omou* 'think'

b. matrix predicates that do not allow *ka-to*: *tyoosasuru* 'investigate', *hakkensuru* 'discover', *rikaisuru* 'understand', *siranai* 'don't know'

(Saito 2010: 88)

include the politeness affix and can thus be safely regarded as genuinely embedded clauses, rather than direct quotes.[4]

3.2.2 Concealed questions and two subclasses of embedded wh-questions

In this subsection, I will further divide embedded wh-questions like (4b) into two subclasses. The notion that is relevant to this subclassification is *concealed questions* (Baker 1968, Grimshaw 1979, Heim 1979, Romero 2005, Nathan 2006, Schwager 2008, Frana 2017, 2020, among many others). They refer to noun phrases that are selected by question-selecting predicates and are interpreted like corresponding wh-questions (or more specifically, copula/specificational wh-questions; see fn.5). E.g., the objects of the English sentences in (8) are concealed questions and are thus interpreted like wh-questions, as their paraphrases show.

(8) a. Mary knows <u>Joan's phone number</u>.
≈ Mary knows <u>what Joan's phone number is</u>.

b. The X-ray revealed <u>the gender of the baby</u>.
≈ The X-ray revealed <u>what the gender of the baby was</u>.

c. Sam asked me <u>the outcome of the election</u>.
≈ Sam asked me <u>what the outcome of the election is</u>. (Frana 2020: 1-2)

Concealed questions are observed in Japanese as well (e.g. Nishiyama 2003, Nishigauchi 2020). For example, the object in (9a) is a concealed question and the sentence is thus construed as (9b), which contains the corresponding embedded wh-question.[5]

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[4] One might also wonder if the embedded clauses in (6) are direct quotes, rather than reportative clauses. Notice, however, that the embedded clauses in (6) contain a pronominal element *kanozyo* 'she' which refers to the matrix subject *Mai*, a reading which would not be possible if they are direct quotes.

[5] Note that the embedded copula wh-question in (9b) itself is ambiguously interpreted either as a specificational sentence or a predicational sentence (e.g. Higgins 1973). With the former interpretation, Ken knows (or

(9) <u>(9a) ≈ (9b)</u> (with the embedded wh-question in (9b) being interpreted as specificational; see fn.5)

a. Ken-wa [sono taikai-no tyanpion]-o {sitteiru / akasita / tazuneta}.
Ken-TOP that tournament-GEN champion-ACC know / revealed / asked
'Ken {knows / revealed / asked} the champion of the tournament.'

b. Ken-wa [sono taikai-no tyanpion-ga dare (da) ka] {sitteiru / akasita / tazuneta}.
Ken-TOP that tournament-GEN champion-NOM who COP Q know / revealed / asked
'Ken {knows / revealed / asked} [who the champion of the tournament is].'

It is (cross-linguistically) not the case, however, that concealed questions can appear in the complement position of all question-selecting predicates; some of them cannot select concealed questions. For instance, predicates such as *sitteiru* 'know', *akasita* 'revealed' and *tazuneta* 'asked' can select concealed questions, as (9) indicates. However, question-selecting verbs such as *sinpaisiteiru* 'is anxious', *gironsita* 'discussed' and *kaisetusita* 'explained' cannot take concealed questions as their complements. For example, while (10a) is itself grammatical (but sounds infelicitous with *gironsita* 'discussed'), they cannot have the interpretation of (10b), which contains the (intended) corresponding embedded wh-question.[6]

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revealed/asked) who won the championship of the tournament. With the latter interpretation, Ken knows some essential fact of the person referred to as the champion of the tournament (e.g. s/he used to be his roommate); with this reading, in particular, Ken need not know that that person won the championship of the tournament. Importantly, Frana (2010) argues that, even though concealed questions are paraphrased as copula wh-questions (Nathan 2006), their interpretation is equivalent to specificaitonal sentences, not to predicational sentences. In fact, (9a), which contains a concealed question, must imply that Ken knows (or revealed/asked) who won the championship. That is, (9a) is interpreted in the same way as (9b) with the specificational reading of the embedded wh-question.

[6] Regarding (10), two notes are in order here. First, (10b) is degraded with the matrix predicate *sinpaisiteiru* 'is anxious', even though it can select non-copula wh-questions (e.g. (4b)). This thus suggests that this verb is generally incompatible with copula wh-questions like the one in (10b). Second, one might wonder whether the non-CQ predicates in (10) (except for *sinpaisiteiru* 'is anxious') can select copula wh-questions with specificational interpretations; if they cannot, that property might be able to explain the fact that the object noun phrase in (10a) cannot be construed as a concealed question, given that concealed questions are construed as specificational copula wh-questions (Frana 2010), as noted in fn.5. Note, however, that the embedded copula wh-question in (10b) can be

(10) <u>(10a) ≉ (10b)</u>

a. Ken-wa [sono taikai-no tyanpion]-o {sinpaisiteiru / #gironsita /
Ken-TOP that tournament-GEN champion-ACC is.anxious / discussed /
kaisetusita}.
explained
'Ken {is anxious about / discussed / explained (about)} the champion of the tournament.'

b. Ken-wa [sono taikai-no tyanpion-ga dare (da) ka] {??sinpaisiteiru /
Ken-TOP that tournament-GEN champion-NOM who COP Q is.anxious /
gironsita / kaisetusita}
discussed / explained
'Ken {is anxious / discussed / explained} [who the champion of the tournament is].'

I will refer to question-selecting verbs that can select concealed questions as *CQ predicates*, and those that cannot select concealed questions as *non-CQ predicates* (where *CQ* stands for *concealed questions*). (11) gives examples of these two types of predicates.

(11) a. an incomplete list of CQ-predicates:
akasu 'reveal', *kiku* 'ask', *kirokusuru* 'record', *kotaeru* 'answer', *oboeteiru* 'remember', *osieru* 'teach/tell', *sitteiru* 'know', *tazuneru* 'ask', etc.

b. an incomplete list of non-CQ-predicates:
gironsuru 'discuss', *kaisetusuru* 'explain', *kansatusuru* 'observe', *kansin-ga aru* 'be interested', *sinpaisiteiru* 'be anxious', *situmonsuru* 'question', etc.

interpreted as a specificational sentence, as well as a predicational sentence (as in the case of (9b); see fn.5); e.g., (10b) with *gironsita* 'discussed' will be true if Ken discussed who won the championship of the tournament (i.e. a specificational reading), or if he discussed some essential fact about the person referred to as the champion of the tournament (i.e. a predicational reading). Hence, the observation that the object noun phrase in (10a) cannot serve as a concealed question should not be ascribed to the possible interpretations of copula wh-questions selected by the matrix predicates in (10a).

With this distinction, I now add two subclasses of embedded wh-questions, which differ in which type of predicate they are selected by. I dub embedded wh-questions selected by CQ-predicates *embedded CQ wh-questions*, and those selected by non-CQ-predicates *embedded non-CQ wh-questions*. Those are roughly schematized in (12) and are exemplified in (13).

(12) a. *Embedded CQ wh-questions*: [… wh … *ka*] CQV

b. *Embedded non-CQ wh-questions*: [… wh … *ka*] non-CQV

(13) a. Embedded CQ wh-questions

Mai-wa [Ken-ga nani-o katta ka] {sitteiru / akasita / tazuneta}.
Mai-TOP Ken-NOM what-ACC bought Q know / revealed / asked
'Mai {knows / revealed / asked} [what Ken bought].'

b. Embedded non-CQ wh-questions

Mai-wa [Ken-ga nani-o katta ka] {sinpaisiteiru / gisonsita / kaisetusita}.
Mai-TOP Ken-NOM what-ACC bought Q is.anxious / discussed / explained
'Mai {is anxious / discussed / explained} [what Ken bought].'

Note that the two embedded wh-questions in (13) apparently look the same. In the following discussion, however, it will turn out that they crucially show different properties.

In summary, this section has established the classification of wh-questions in Japanese that constitutes the basis for the following discussion, as summarized in (14).

(14) Complete list of categories of Japanese wh-questions

a. Matrix wh-questions: [… wh … *ka*] (e.g. (4a))

b. Embedded wh-questions: [… wh … *ka*] V

A. Embedded CQ wh-questions: [… wh … *ka*] CQV (e.g. (13a))

B. Embedded non-CQ wh-questions: [… wh … *ka*] non-CQV (e.g. (13b))

c. *To*-embedded wh-questions: [[… wh … *ka*] *to*] V (e.g. (4c))

3.3 Two asymmetries in Japanese wh-questions

Building on the classification of Japanese wh-questions established in the last section (see (14)), this section will present new observations regarding Japanese wh-questions. More specifically, I will show that there exist asymmetries among those classes of wh-questions with respect to intervention effects and cleft formation. I will then conclude this section with the observation that these two asymmetries reveal a demarcation between two groups of wh-questions.

3.3.1 First asymmetry: Intervention effects

The first asymmetry to be discussed concerns the phenomenon that is widely known as *intervention effects*, the gist of which will be provided below.

3.3.1.1 An overview of intervention effects

It has been cross-linguistically observed that wh-questions become marginal when a certain phrase (i.e. an intervenor) is placed between an in-situ wh-phrase and a complementizer that determines the scope of that wh-phrase (e.g., Beck 1996, 2006, Beck and Kim 1997, Pesetsky 2000, Kim 2002, 2006, Kotek 2017, 2018).[7] This phenomenon is widely referred to as *intervention effects*. (15a) and (16a) show examples of intervention effects in German and Korean, respectively.[8]

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[7] Beck (2006) argues that focusing and quantificational elements, as in (i), typically serve as intervenors.

(i) only, even, also, not, (almost) every, no, most, few (and other nominal quantifiers), always, often, never (and other adverbial quantifiers)

(Beck 2006: 4)

[8] Not all speakers of German find (15a) ungrammatical (Magdalena Kaufmann p.c.).

(15) German

a. *Wen hat niemand wo gesehen?
   whom has nobody where seen
   'Where did nobody see whom?'

b. Wen had wo niemand gesehen?
   whom has where nobody seen (Beck 2006: 4)

(16) Korean

a. *Minsu-man nuku-lûl po-ss-ni?
   Minsu-only who-ACC see-PAST-Q
   'Who did only Minsu see?'

b. nuku-lûl Minsu-man po-ss-ni?
   who-ACC Minsu-only see-PAST-Q (ibid: 3)

In (15a) and (16a), the intervenors *niemand* 'nobody' and *Minsu-man* 'only Minsu', respectively, are located above the in-situ wh-phrases, thus inducing intervention effects. In contrast, (15b) and (16b) show no intervention effect because the in-situ wh-phrases are located above the intervenors as a result of scrambling.

Intervention effects are also observed in Japanese (Hoji 1985, Takahashi 1990, Tanaka 1997, Hagstrom 1998, Tomioka 2007, Miyagawa 2010, Morita 2019, among others). Previous studies agree with the view that the intervenors in Japanese include the negative polarity item (NPI) *X-sika* 'only X', the NPI wh-*mo* (e.g. *dare-mo* '(not) anyone'), the universal quantifier wh-*mo* (e.g.

da're-mo 'everyone'), and the disjunction *X ka Y (ka)* 'X or Y'.[9] Relevant examples are shown below:[10]

(17) NPI: *X-sika* (… NEG) 'only X'

a. ?*$\boxed{\text{Ken-sika}}$ nani-o {kaw-anakat-ta desu ka / kai-masen desita ka}.
Ken-SIKA what-ACC buy-not-PAST COP.POL Q buy-not.POL COP.POL.PAST Q
'What did only Ken buy?'

b. Nani-o$_i$ $\boxed{\text{Ken-sika}}$ t_i {kaw-anakat-ta desu ka / kai-masen desita ka}.
what-ACC Ken-SIKA buy-not-PAST COP.POL Q buy-not.POL COP.POL.PAST Q

(18) NPI: wh-*mo* (… NEG) (e.g. *dare-mo* (who-MO) … NEG 'no one …')

a. ??$\boxed{\text{Dare-mo}}$ nani-o {kaw-anakat-ta desu ka / kai-masen desita ka}.
who-MO what-ACC buy-not-PAST COP.POL Q buy-not.POL COP.POL.PAST Q
'What did no one buy?'

b. Nani-o$_i$ $\boxed{\text{dare-mo}}$ t_i {kaw-anakat-ta desu ka / kai-masen desita ka}.
what-ACC who-MO buy-not-PAST COP.POL Q buy-not.POL COP.POL.PAST Q

(19) Universal quantifier: wh-*mo* (e.g. *da're-mo* (who-MO) 'everyone')

a. ?? $\boxed{\text{(Hotondo) da're-mo}}$-ga nani-o kai-masi-ta ka.
almost who-MO-NOM what-ACC buy-POL-PAST Q
'What did (almost) everyone buy?'

b. Nani-o$_i$ $\boxed{\text{(hotondo) dar'e-mo}}$-ga t_i kai-masi-ta ka.
what-ACC almost who-MO-NOM buy-POL-PAST Q

[9] Although the universal quantifier *da're-mo* 'everyone' looks identical to the NPI *dare-mo* '(not) anyone', they are disambiguated in some regards. For example, the former has a pitch accent on the first syllable (as indicated by the apostrophe in *da're-mo*), while the latter does not. In addition, the former is usually followed by a Case particle, which is not the case with the latter.

[10] The degree of the degradedness induced by intervention effects in Japanese seems to be different depending both on different speakers and on different intervenors; for the author, for example, the NPI *X-sika* shows the strongest degradation, and the NPI wh-*mo* results in stronger intervention effects than the other two intervenors.

(20) Disjunctions: *X ka Y (ka)* 'X or Y'

a. ??Ken-ka Kooji(-ka)-ga nani-o kai-masi-ta ka.
   Ken-or Koji-or-NOM what-ACC buy-POL-PAST Q
   'What did Ken or Koji buy?'

b. Nani-o$_i$ Ken-ka Kooji(-ka)-ga t_i kai-masi-ta ka.
   what-ACC Ken-or Koji-or-NOM buy-POL-PAST Q

Notice that the contrast between the a- and b-examples above parallels that in (15-16); the a-examples show that placing an intervenor between an in-situ wh-phrase and its licensing complementizer (or the Q-particle *ka*) degrades the sentence, while the b-examples show that scrambling the in-situ wh-phrase across the intervenor salvages the marginal sentence. This observation indicates that Japanese wh-questions give rise to intervention effects, like German and Korean wh-questions.

I will show below that Japanese wh-questions exhibit different properties with respect to intervention effects depending on their types, by reconsidering Morita's (2019) generalization regarding the presence/absence of intervention effects in Japanese wh-questions.

3.3.1.2 Morita's (2019) generalization

Morita (2019) importantly observes that intervention effects in Japanese show an asymmetry; they are obviated in some embedded wh-questions. To illustrate this asymmetry, he first shows, following Ishii (2013), that the semantic classes of question-selecting predicates (Lahiri 1991) can be divided into two groups depending on whether the selected questions can be followed by the reportative complementizer *to* (cf. Section 3.2.1); considering three classes of verbs, namely *know*-class verbs (e.g. *sitteita* 'knew' in (21a)), *investigate*-class verbs (e.g. *sirabeta* 'investigated' in (21a)), and *wonder*-class verbs (e.g. *tazuneta* 'asked' in (21b)), he notes that *to* can be added only to questions selected by *wonder*-class verbs (see also the discussion in Section 3.2.1), as shown in (21).

(21) a. Ken-wa [John-ga nani-o kaita ka (*to)] {sitteita / sirabeta}.
Ken-TOP John-NOM what-ACC drew Q REP knew / investigated
'Ken {knew / investigated} [what John drew].'

b. Ken-wa (Susan-ni) [John-ga nani-o kaita ka (to)] tazuneta.
Ken-TOP Susan-to John-NOM what-ACC drew Q REP asked
Lit. 'Ken asked (Susan) [(that) what John drew].'

(Morita 2019: 319)

With this dichotomy, Morita observes the data in (22), where the intervenor occurs above the in-situ wh-phrase within each embedded wh-question and thus an intervention effect is expected to arise.[11]

(22) a. Ken-wa [Mary-sika nani-o kak-anakat-ta ka] {sitteita / sirabeta}.
Ken-TOP Mary-SIKA what-ACC draw-not-PAST Q knew / investigated
'Ken {knew / investigated} [what only Mary drew].'

b. ?*Ken-wa (Susan-ni) [Mary-sika nani-o kak-anakat-ta ka to] tazuneta.
Ken-TOP Susan-to Mary-SIKA what-ACC draw-not-PAST Q REP asked
Lit. 'Ken asked (Susan) [that what only Mary drew].'

(ibid: 320)

[11] In most examples where intervention effects are concerned, I use only the NPI *X-sika* as an intervenor. Note that the same result obtains with other intervenors, unless noted otherwise.

Crucially, (22a) is unexpectedly judged as grammatical, in contrast to (22b). This means that an intervention effect occurs in the embedded wh-question in (22b) but not in (22a).[12] To capture this difference, Morita pays attention to the fact that the matrix predicates in (22) are different in their semantic classes; the matrix predicates in (22a) belong to the *know*/*investigate*-class, while the one in (22b) to the *wonder*-class. Given this difference, he argues that it is the semantic class of question-selecting predicates that matters for the difference in (22), and thus concludes that intervention effects are obviated in wh-questions selected by *know*/*investigate*-class verbs, while they still occur in those selected by *wonder*-class verbs.

3.3.1.3 Reconsidering Morita's generalization: A new paradigm

Notice, however, that the semantic class of the selecting verb is not the only difference between (22a) and (22b). They are also different in whether the embedded wh-questions are followed by the additional complementizer *to*; it is present in (22b) but not in (22a). Recall here that the attachment of *to* to a wh-question selected by a *wonder*-class verb is optional (see (21b)). Morita does not discuss what happens when *to* does not appear in (22b). Consider now this case:

(23) Ken-wa (Susan-ni) [Mary-sika nani-o kak-anakat-ta ka] tazuneta.
Ken-TOP Susan-to Mary-SIKA what-ACC draw-not-PAST Q asked
'Ken asked (Susan) [what only Mary drew].'

---

[12] Note that Morita's observation is different from Tomioka's (2007) observation that an intervention effect is weakened or obviated in *matrix wh-questions* when the intervenor and the in-situ wh-phrase are located within an embedded *declarative* clause, as shown in (i).

(i) a. ?(?)Kimi-wa [John-sika nani-o yom-anakat-ta to] omotteiru no?
you-TOP John-SIKA what-ACC read-not-PAST REP think Q
'What do you think that only John read?'

b. Kimi-wa [John-ka Bill-ga nani-o yonda to] omotteiru no?
you-TOP John-or Bill-NOM what-ACC read REP think Q
'What do you think that John or Bill read?' (Tomioka 2007: 1573)

I will leave open how the proposal to be made in this chapter captures this observation. However, in fn.21, I will briefly comment on Tomioka's analysis of intervention effects in Japanese.

Crucially, (23) is judged well-formed, similarly to (22a) but unlike (22b). This observation thus indicates that Morita's conclusion that the obviation of intervention effects in embedded wh-questions depends on the semantic class of the selecting verb does not hold up.

Note here that the only difference between (22b) and (23) is whether the complementizer *to* is present (22b) or absent (23). Furthermore, recall that no intervention effect takes place in the embedded wh-question in (22a), where *to* cannot be added (see (21a)). Given these considerations, I (tentatively) argue that the crucial factor that determines the presence or absence of intervention effects in embedded wh-questions is whether they are followed by the complementizer *to*, or with the terminology introduced in Section 3.2, whether they are realized as (standard) *embedded wh-questions* or to-*embedded wh-questions*.

Now, let us dig more into the generalization proposed above that intervention effects are obviated in (standard) embedded wh-questions. Recall that in Section 3.2.2, I have posited two subclasses of them, namely embedded CQ wh-questions and embedded non-CQ wh-questions. Notice here that in the above examples where an obviation of intervention effects is observed (see (22a) and (23)), all wh-questions are embedded CQ wh-quesitons, with the matrix predicates (i.e. *sitteita* 'knew', *sirabeta* 'investigated' and *tazuneta* 'asked') being CQ-predicates, i.e. question-selecting predicates that can select concealed questions, as (24) shows.

(24) Ken-wa [sono taikai-no tyanpion]-o {sitteita / sirabeta / tazuneta}.
Ken-TOP that tournament-GEN champion-ACC knew / investigated / asked
'Ken {knew / investigated / asked} the champion of the tournament.'
≈ 'Ken {knew / investigated / asked} who the champion of the tournament is.'

The question to ask, then, is whether intervention effects are obviated even in embedded non-CQ wh-questions. To examine this question, observe (25), which contains an embedded non-CQ wh-question, selected by non-CQ-predicates (i.e. question-selecting predicates that cannot select concealed questions) such as *sinpaisiteiru* 'is anxious', *gironsita* 'discussed' and *kaisetusita* 'explained' (see Section 3.2.2).

(25) ?*Ken-wa [Mary-sika nani-o kak-anakat-ta ka] {sinpaisiteiru / gironsita /
Ken-TOP Mary-SIKA what-ACC draw-not-PAST Q is.anxious / discussed /
kaisetusita}.
explained
'Ken {is anxious / discussed / explained} [what only Mary drew].'

In the embedded non-CQ wh-question in (25), an intervenor is placed above the in-situ wh-phrase, and thus an intervention effect is expected to appear. This expectation is borne out, as its degradedness indicates. The above observations thus lead to the conclusion that the obviation of intervention effects is not observed in all embedded wh-questions; it is observed in embedded CQ wh-questions (e.g. (22a), (23)) but not in embedded non-CQ wh-questions (e.g. (25)).

To recap, this subsection has argued against Morita's (2019) generalization regarding the presence/absence of intervention effects in Japanese wh-questions, and has established the new generalization which is summarized in (26), with the (new) relevant examples given in (27-28).[13]

[13] The three examples in (27) become acceptable if the wh-phrase is replaced with a non-wh item, rendering the relevant wh-questions *yes*/*no*-/*whether*-questions, as shown in (ia-c). (I assume that (ib) with *kaisetusita* 'explained' is more marginal just because it is harder to imagine an appropriate context.)

(i) a. Ken-sika ringo-o kai-masen desita ka.
Ken-SIKA apple-ACC buy-not.POL COP.POL.PAST Q
'Did only Ken buy apples?'

b. Mai-wa [Ken-sika ringo-o kawa-nakat-ta ka] {sinpaisiteiru / gironsita / ?kaisetusita}.
Mai-TOP Ken-SIKA apple-ACC buy-not-PAST Q is.anxious / discussed / explained
'Mai {is anxious / discussed / explained} [whether only Ken bought apples].'

c. Mai-wa [Ken-sika ringo-o kawa-nakat-ta ka to] {tazuneta / sinpaisiteiru}.
Mai-TOP Ken-SIKA apple-ACC buy-not-PAST Q REP asked / is.anxious
Lit. 'Ken {asked / is anxious} [that whether only Mai bought apples].'

This indicates that the wh-phrase is crucially involved in the degradedness of (27a-c).

(26) *An asymmetry regarding intervention effects*

a. Intervention effects appear in matrix wh-questions, embedded non-CQ wh-questions, and *to*-embedded wh-questions (see (27)).

b. Intervention effects are obviated in embedded CQ wh-questions (see (28)).

(27) a. Matrix wh-questions:

?*Ken-sika nani-o {kaw-anakat-ta desu ka / kai-masen desita ka}.
Ken-SIKA what-ACC buy-not-PAST COP.POL Q buy-not.POL COP.POL.PAST Q
'What did only Ken buy?' (= (17a))

b. Embedded non-CQ wh-questions:

?*Mai-wa [Ken-sika nani-o kawa-nakat-ta ka] {sinpaisiteiru / gironsita /
Mai-TOP Ken-SIKA what-ACC buy-not-PAST Q is.anxious / discussed /
kaisetusita}.
explained
'Mai {is anxious (about) / discussed / explained} [what only Ken bought.]'

c. *To*-embedded wh-questions:

?*Mai-wa [Ken-sika nani-o kawa-nakat-ta ka to] {tazuneta / sinpaisiteiru}.
Mai-TOP Ken-SIKA what-ACC buy-not-PAST Q REP asked / is.anxious
Lit. 'Mai {asked / is anxious} [that what only Ken bought].'

(28) Embedded CQ wh-questions:

Mai-wa [Ken-sika nani-o kawa-nakat-ta ka] {sitteiru / akasita / tazuneta}.
Mai-TOP Ken-SIKA what-ACC buy-not-PAST Q know / revealed / asked
'Mai {knows / revealed / asked} [what only Ken bought].'

3.3.1.4 A brief remark on Beck's (2006) observation

Before closing this subsection, it is worth introducing Beck's (2006) data from German (29b), where an intervention effect does not appear despite the fact that the focused phrase, normally an intervenor, occurs above the in-situ wh-phrase in the embedded wh-question (compare it with its matrix counterpart in (29a), which is degraded due to an intervention effect).

(29) a. ??Wen hat LUISE <u>wo</u> gesehen?
who has Luise where seen
'Where did LUISE see who?'

b. Ich habe mich (nur) gefragt, wen LUISE <u>wo</u> gesehen hat.
I have myself only asked who Luise where seen has
'I (only) wondered where LUISE saw who.'

(Beck 2006: 32)

However, Beck points out that for (29b) to be grammatical, the focus status of *LUISE* must be interpreted with respect to the matrix clause, rather than the embedded clause. She attributes the lack of an intervention effect in (29b) to this interpretational factor.

Given that, one might wonder if the obviation of intervention effects in embedded CQ wh-questions, as in (28), can be attributed to the same reason why (29b) does not show an intervention effect, i.e. the embedded intervenor being construed with respect to the matrix clause.[14] However, I argue that this cannot be the reason (or, at least, cannot be the only reason) for the obviation of intervention effects under consideration based on the following two observations, which show that an intervention effect is absent in embedded CQ wh-questions even if an intervenor is interpreted with respect to the embedded clause.

The first case is (28), where the NPI *Ken-sika* is an intervenor. Note that in (28), both the NPI *Ken-sika* and the negation are placed within the embedded clause. Given that the scope of an NPI is determined by the location of negation, this guarantees that in (28) the scope of the NPI does

[14] Under this view, the same option, i.e. the matrix scope interpretation of an embedded intervenor, would somehow be unavailable to embedded non-CQ wh-questions (e.g. (27b)) and *to*-embedded wh-questions (e.g. (27c)).

not go beyond the embedded clause. Therefore, the lack of an intervention effect in (28) is different from that in (29b) regarding its nature.

The second case involves another intervenor, a disjunction *X-ka Y(-ka)* ‘X or Y’. Note first of all that embedded CQ wh-questions do not give rise to an intervention effect even with a disjunction as an intervenor, as shown in (30).

(30) Mai-wa [Ken-ka Kooji(-ka)-ga nani-o tabeta ka] {sitteiru / akasita / tazuneta}.
Mai-TOP Ken-or Koji-or-NOM what-ACC ate Q know / revealed / asked
‘Mai {knows / revealed / asked} [what Ken or Koji ate].’

In Japanese, the matrix scope reading of an embedded disjunction is usually unavailable, as exemplified in (31), where a disjunction occurs in an embedded *whether*-question.

(31) Mai-wa [Ken-ga susi-ka raamen(-ka)-o tabeta ka] sitteiru.
Mai-TOP Ken-NOM sushi-or ramen-or-ACC ate Q know
(i) ✓ ‘Mai knows whether Ken ate either sushi or ramen.’
(ii) * ‘Mai knows either whether Ken ate sushi or whether Ken ate ramen.’

However, some native speakers (including the author) at least marginally allow the embedded disjunction in (31) to take a matrix scope with a special prosody (such as putting a strong prosodic emphasis on the disjunction and adding a long pause between the disjunction and the immediately following element, i.e. *tabeta* ‘ate’); this is shown by (32), where the capitals and the comma indicate a prosodic emphasis and a pause respectively, and “%” means that that reading is at least marginally possible for some native speakers.

(32) Mai-wa [Ken-ga SUSI-KA RAAMEN(-KA)-O, tabeta ka] sitteiru.
Mai-TOP Ken-NOM sushi-or ramen-or-ACC ate Q know

(i) ✓ ‘Mai knows whether Ken ate either sushi or ramen.’

(ii) % ‘Mai knows either whether Ken ate sushi or whether Ken ate ramen.’

The crucial fact, then, is that for those speakers who accept the reading (ii) in (32), (30) is acceptable even without such prosodic amendment on the embedded disjunction. This indicates that they do not find an intervention effect in (30) even when the disjunction does not take a matrix scope.

Taking these observations into consideration, I assume that in the examples in the following discussion where an intervention effect is obviated in an embedded CQ wh-question, the intervenor is construed with respect to the embedded clause, not to the matrix clause.

3.3.2 Second asymmetry: Cleft wh-questions

This subsection turns to the second asymmetry regarding Japanese wh-questions, which pertains to *clefts* like (33b), whose basic counterpart is (33a).[15]

(33) a. Ken-ga hondana-ni kabin-o oita.
Ken-NOM bookshelf-on vase-ACC put
‘Ken put a vase on the bookshelf.’

b. [Ken-ga e_i kabin-o oita no]-wa hondana-ni_i da.
Ken-NOM vase-ACC put C-TOP bookshelf-on COP
‘It is on the bookshelf that Ken put a vase.’

---

[15] Note that clefts like (33b) must be distinguished from *pseudoclefts* like (i); the two superficially differ only in whether a Case particle or a postposition is attached to the focus phrase, but they syntactically behave differently (e.g. Hoji 1987, Hiraiwa and Ishihara 2002, 2012). Pseudoclefts will be discussed in Section 3.4.2.3.1.

(i) [Ken-ga e_i kabin-o oita no]-wa $hondana_i$ da.
Ken-NOM vase-ACC put NMLZ-TOP bookshelf COP
‘(The place) Where Ken put a vase is the bookshelf.’

Note also that, since many native speakers judge as marginal or ungrammatical clefts where a nominative or an accusative Case particle is attached to the focus phrase, in what follows I will mainly use clefts with a focus phrase followed by *ni* ‘to/on/in’ (or the dative Case particle).

As (33b) shows, in Japanese clefts, the presupposed clause is headed by the complementizer *no*, and the focus (or clefted) phrase is followed by the copula *da* (or *desu*; see, e.g., (34a)).

Let us now consider *cleft wh-questions*, i.e. clefts used as wh-questions with their focus position occupied by a wh-phrase, in terms of the four classes of wh-questions introduced in Section 3.2. The relevant paradigm is given below:

(34) a. Matrix wh-questions
[Ken-ga e_i kabin-o oita no]-wa doko-ni$_i$ desu ka.
Ken-NOM vase-ACC put C-TOP where-on COP.POL Q
'Where is it that Ken put a vase?'

b. Embedded non-CQ wh-questions
Mai-wa [[Ken-ga e_i kabin-o oita no]-wa doko-ni$_i$ (da) ka] {gironsita /
Mai-TOP Ken-NOM vase-ACC put C-TOP where-on COP Q discussed
?kaisetusita / kansin-ga-aru}.
explained / is.interested
'Mai {discussed / explained / is interested in} [where it is that Ken put a vase].'

c. *To*-embedded wh-questions
Mai-wa [[Ken-ga e_i kabin-o oita no]-wa doko-ni$_i$ (da) ka to] {tazuneta /
Mai-TOP Ken-NOM vase-ACC put C-TOP where-on COP Q REP asked
?sinpaisiteiru}.
is.anxious
Lit. 'Mai {asked / is anxious} [that where it is that Ken put a vase].'

(35) Embedded CQ wh-questions
?*Mai-wa [[Ken-ga e_i kabin-o oita no]-wa doko-ni$_i$ (da) ka] {sitteiru / akasita /
Mai-TOP Ken-NOM vase-ACC put C-TOP where-on COP Q know / revealed /
tazuneta}.
asked
'Mai {knows / revealed / asked} [where it is that Ken put a vase].'

On the one hand, (34) shows that cleft wh-questions can be realized as matrix wh-questions, embedded non-CQ wh-questions, and *to*-embedded wh-questions.[16] On the other hand, (35) indicates that forming a cleft wh-questions as an embedded CQ wh-question leads to degradation.[17] This asymmetry is summarized in (36).

(36) *An asymmetry regarding cleft formation*

a. Forming cleft wh-questions as matrix wh-questions, embedded non-CQ wh-questions and *to*-embedded wh-questions leads to well-formedness (see (34)).

b. Forming cleft wh-questions as embedded CQ wh-questions leads to degradation (see (35)).

3.3.3 Distribution of the two asymmetries

To conclude this section, let us consider what implications obtain from the two asymmetries discussed above. The summaries of those asymmetries are repeated below:

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[16] If a non-CQ predicate *sinpaisiteiru* 'is anxious' selects a cleft wh-question (i.e. an embedded non-CQ wh-question), the sentence is degraded, in contrast with (34b), as shown in (i).

(i) ?*Mai-wa [[Ken-ga *e*$_i$ kabin-o oita no]-wa doko-ni$_i$ (da) ka] sinpaisiteiru.
Mai-TOP Ken-NOM vase-ACC put C-TOP where-on COP Q is.anxious
'Mai is anxious [where it is that Ken put a vase].'

Recall, however, that that predicate is generally incompatible with a copula wh-question (see fn.6); the relevant example is repeated below from Section 3.2.

(ii) ??Ken-wa [sono taikai-no tyanpion-ga dare (da) ka] sinpaisiteiru.
Ken-TOP that tournament-GEN champion-NOM who COP Q is.anxious
'Ken is anxious [who the champion of that tournament is].'

The marginality of (i), then, can be ascribed to the reason why (ii) is degraded, given that a cleft wh-question is one type of a copula wh-question. Hence, the marginality of (i) does not constitute a counterexample to the generalization in (36a).

[17] (35) becomes acceptable if the wh-phrase is replaced with a non-wh item, making the embedded wh-question a *whether*-question, as shown below:

(i) Mai-wa [[Ken-ga *e*$_i$ kabin-o oita no]-wa <u>hondana-ni$_i$</u> (da) ka] {sitteiru / (?)akasita / tazuneta}.
Mai-TOP Ken-NOM vase-ACC put C-TOP bookshelf-on COP Q know / revealed / asked
'Mai {knows / revealed / asked} [whether it is on the bookshelf that Mai put a vase].'

This indicates that the presence of the wh-phrase matters for the degradedness of (35).

(26) *An asymmetry regarding intervention effects*

a. Intervention effects appear in matrix wh-questions, embedded non-CQ wh-questions, and *to*-embedded wh-questions (see (27)).

b. Intervention effects are obviated in embedded CQ wh-questions (see (28)).

(36) *An asymmetry regarding cleft formation*

a. Forming cleft wh-questions as matrix wh-questions, embedded non-CQ wh-questions and *to*-embedded wh-questions leads to well-formedness (see (34)).

b. Forming cleft wh-questions as embedded CQ wh-questions leads to degradation (see (35)).

Crucially, by comparing (26) and (36), it turns out that the four types of wh-questions are divided into two groups with respect to the two asymmetries under consideration; matrix wh-questions, embedded non-CQ wh-questions and *to*-embedded wh-questions pattern together, as opposed to embedded CQ wh-questions. More specifically, the wh-questions in the former group give rise to intervention effects and are compatible with the cleft structure, whereas the latter wh-questions obviate intervention effects and are incompatible with the cleft structure. This consideration then gives rise to the following question: what is the (syntactic) property that makes the demarcation in question? The next section will present a proposal about that syntactic property and illustrate how that proposal accounts for the two asymmetries in a uniform manner.

3.4 Proposal: (Non-)involvement of covert wh-movement

The last section has observed two asymmetries in Japanese wh-questions, one concerning intervention effects and the other concerning cleft formation. As a result, it has crucially turned out that with respect to these asymmetries, there is a boundary that divides the four types of

Japanese wh-questions into two groups; matrix wh-questions, embedded non-CQ wh-questions, and *to*-embedded wh-questions pattern together, while embedded CQ wh-questions behave differently. To capture this demarcation, I now propose that Japanese wh-questions are derived in a non-uniform manner, depending on their types. More specifically, I propose (37).

(37) a. Matrix wh-questions, embedded non-CQ wh-questions, and *to*-embedded wh-questions do not involve covert wh-movement.

b. Embedded CQ wh-questions involve covert wh-movement.

The rest of this section will be devoted to proving the validity of this proposal, by demonstrating that it gives a uniform account of the two asymmetries under consideration.

3.4.1 Capturing the first asymmetry: Intervention effects

I will begin with the asymmetry regarding intervention effects. The crux of this asymmetry and relevant examples are repeated below:

(26) *An asymmetry regarding intervention effects*

a. Intervention effects appear in matrix wh-questions, embedded non-CQ wh-questions, and *to*-embedded wh-questions (see (27)).

b. Intervention effects are obviated in embedded CQ wh-questions (see (28)).

(27) a. Matrix wh-questions:

?*$\boxed{\text{Ken-sika}}$	<u>nani-o</u>	{kaw-anakat-ta	desu	ka / kai-masen	desita	ka}.	
Ken-SIKA	what-ACC	buy-not-PAST	COP.POL	Q	buy-not.POL	COP.POL.PAST	Q

'What did only Ken buy?' (= (17a))

b. Embedded non-CQ wh-questions:

?*Mai-wa [Ken-sika nani-o kawa-nakat-ta ka] {sinpaisiteiru / gironsita /
Mai-TOP Ken-SIKA what-ACC buy-not-PAST Q is.anxious / discussed /
kaisetusita}.
explained

'Mai {is anxious (about) / discussed / explained} [what only Ken bought.]'

c. *To*-embedded wh-questions:

?*Mai-wa [Ken-sika nani-o kawa-nakat-ta ka to] {tazuneta / sinpaisiteiru}.
Mai-TOP Ken-SIKA what-ACC buy-not-PAST Q REP asked / is.anxious

Lit. 'Mai {asked / is anxious} [that what only Ken bought].'

(28) Embedded CQ wh-questions:

Mai-wa [Ken-sika nani-o kawa-nakat-ta ka] {sitteiru / akasita / tazuneta}.
Mai-TOP Ken-SIKA what-ACC buy-not-PAST Q know / revealed / asked

'Mai {knows / revealed / asked} [what only Ken bought].'

I will first introduce a widely accepted view regarding the obviation of intervention effects. I will then show that with this view, the proposal in (37) gives a straightforward account of the asymmetry in question (i.e. (26)). In addition, I will also discuss a scope property of embedded CQ wh-questions where intervention effects are obviated, and show that it lends further support to the current proposal.

3.4.1.1 A widely accepted view

(38) gives a recent standard view regarding the obviation of intervention effects.

(38) Intervention effects do not appear when a wh-phrase is located above an intervenor *overtly or covertly*.

According to (38), intervention effects are obviated when a wh-phrase is located *overtly* above an intervenor. This is quite obvious. For example, scrambling an in-situ wh-phrase over an intervenor prevents an intervention effect from arising, as noted in Section 3.3.1; compare, e.g., (39b) with (39a).

(39) a. ?*Ken-sika nani-o {kaw-anakat-ta desu ka / kai-masen desita ka}.
Ken-SIKA what-ACC buy-not-PAST COP.POL Q buy-not.POL COP.POL.PAST Q
'What did only Ken buy?'

b. Nani-o$_i$ Ken-sika t_i {kaw-anakat-ta desu ka / kai-masen desita ka}.
what-ACC Ken-SIKA buy-not-PAST COP.POL Q buy-not.POL COP.POL.PAST Q

Crucially, (38) also states that a wh-phrase appearing even *covertly* above an intervenor obviates an intervention effect. Of relevance for the current discussion is this aspect of (38).[18]

The general view in (38) derives from the analyses of intervention effects proposed in the literature. One such analysis is provided in Pesetsky (2000). He observes that in English, intervention effects occur in a restricted range of multiple wh-questions, namely those where the Superiority Condition is violated. Though such a violation usually leads to unacceptability, it does not with D-linked wh-phrases, such as *which X* (Pesestky 1987), as (40b) shows.[19]

(40) a. Which girl$_i$ did Mary introduce t_i to which boy? (Superiority-obeying)

b. Which boy$_i$ did Mary introduce which girl to t_i? (Superiority-violating)

[18] Demirok (2017) capitalizes on this aspect of (38) to argue for the presence/absence of covert wh-movement in Turkish and Tsez.

[19] D(iscourse)-linked wh-phrases refer to wh-phrases whose value lies in a limited set of elements that both the speaker and the hearer have in mind based on the context they share (e.g. Pesetsky 1987).

Importantly, if the subject *Mary* in (40) is replaced with an intervenor *only Mary*, (40b) becomes degraded, i.e. there is an intervention effect, while (40a) is still grammatical, as shown in (41).

(41) a. Which $girl_i$ did only Mary introduce t_i to which boy? (Superiority-obeying)

b. ??Which boy_i did only Mary introduce which girl to t_i? (Superiority-violating)

Pesetsky (2000) takes Superiority to be an indicator of covert (phrasal) wh-movement. According to this view, in Superiority-obeying wh-questions like (40a), an in-situ wh-phrase undergoes (or at least can undergo) covert wh-movement, whereas in Superiority-violating wh-questions like (40b), an in-situ wh-phrase does not undergo covert wh-movement and stays in its original position.[20] Building on this view and the observation in (41), Pesetsky (2000) presents the generalization that an intervention effect appears when an in-situ wh-phrase (overtly and) covertly stays in-situ (as in (41b)), while it does not when a wh-phrase undergoes overt or covert wh-movement (as in (41a)). (38) then obtains from this generalization.

While Pesetsky (2000) puts forth a syntactic generalization regarding intervention effects, the proposals by Beck (2006) and Kotek (2017) can be taken to be semantic explanations for that generalization. What the two proposals have in common is the claim that the LF configuration where a wh-phrase is located below an intervenor gives rise to a semantic problem, thus leading to an intervention effect. For Beck (2006), that configuration makes the semantic value of the whole structure undefined due to the focus-sensitive operator ~ (Rooth 1985, 1992) being located above a wh-phrase; see Kotek (2017) for a different semantic issue. Both Beck (2006) and Kotek (2017) argue that if a wh-phrase moves (e.g. scrambling, wh-movement) at least covertly to build the LF configuration where it is located above an intervenor, the semantic problems they note get

[20] Pesestky (2000) argues that cases like (40b) involve feature movement, instead of cover wh-movement.

resolved, thus resulting in intended semantic interpretations. The standard view in (38) thus follows as well from Beck's (2006) and Kotek's (2017) semantic analyses of intervention effects.

To recap, I have briefly shown that the widely accepted view regarding the obviation of intervention effects in (38) follows from the analyses of intervention effects proposed in the literature, including Pesetsky (2000), Beck (2006) and Kotek (2017), thus establishing its validity. In the following discussion, I will adopt this general view, being agnostic about which particular analysis should be adopted for intervention effects in Japanese.[21]

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[21] Tomioka's (2007) pragmatic analysis of intervention effects in Japanese does not take covert structures into consideration and thus does not predict that an in-situ wh-phrase being covertly located above a wh-phrase would obviate intervention effects, being incompatible with the general view in (38). To understand his analysis, consider how it accounts for the basic contrast in intervention effects between (ia) and (ib).

(i) a. ?*Mai-sika nani-o kai-masen desita ka?
Mai-SIKA what-ACC buy-not.POL COP.POL.PAST Q
'What did only Mai buy?'

b. Nani-o$_i$ Mai-sika t_i kai-masen desita ka?
what-ACC Mai-SIKA buy-not.POL COP.POL.PAST Q

His analysis builds on the information-structural assumption that a sentence is partitioned into two parts: *focus* (i.e. new information) and *ground* (i.e. old information). In the case of wh-questions, the wh-phrase serves as a focus, while the other part is a ground. Tomioka assumes that in Japanese, there are two ways for an element to count as part of a ground; one is to mark it with the topic particle *-wa* and put it in the sentence-initial position (so that it serves as a pure topic, not a contrastive one), while the other is to put it in the prosodically reduced part. Then, when an intervenor appears in a wh-question, it must become part of a ground by one of these strategies. Concerning the first strategy, he observes that intervenors in Japanese are "anti-topic items", i.e. elements that cannot be topicalized, thus arguing that the first strategy is not available for intervenors. As for the second strategy, an intervenor is hard to interpret as part of a ground if it appears before a wh-element, where it is likely to get a secondary stress (Nagahara 1994). This accounts for the degraded status of (ia); the intervenor, preceding the wh-phrase, cannot be readily interpreted as part of the ground. However, scrambling a wh-phrase over an intervenor enables the latter to be part of the ground, since in Japanese the string following a wh-phrase is prosodically reduced (post-focal reduction; see Chapter 2). This captures the acceptability of (ib); the intervenor can count as part of the ground since it follows a wh-phrase and is thus in the prosodically reduced part. Further, Tomioka argues that when an intervenor is an embedded subject, it can be construed as part of the ground even if it occurs before a wh-phrase. This captures why (ii) is better than (ia).

(ii) ?(?)Kimi-wa [John-sika nani-o yom-anakat-ta to] omotteiru no?
you-TOP John-SIKA what-ACC read-not-PAST REP think Q
'What do you think that only John read?' (Tomioka 2007: 1573)

Notice, however, that with his analysis, it is hard to capture the observation in Section 3.3.1 that intervention effects appear in embedded non-CQ wh-questions and *to*-embedded wh-questions, but are obviated in embedded CQ wh-questions; the relevant examples are repeated below:

(iii) a. Embedded non-CQ wh-questions:
?*Mai-wa [Ken-sika nani-o kawa-nakat-ta ka] {sinpaisiteiru / gironsita / kaisetusita}.
Mai-TOP Ken-SIKA what-ACC buy-not-PAST Q is.anxious / discussed / explained
'Mai {is anxious (about) / discussed / explained} [what only Ken bought].' (= (27b))

b. *To*-embedded wh-questions:
?*Mai-wa [Ken-sika nani-o kawa-nakat-ta ka to] {tazuneta / sinpaisitieru}.
Mai-TOP Ken-SIKA what-ACC buy-not-PAST Q REP asked / is.anxious

3.4.1.2 Turning back to the asymmetry

Now that the idea in (38) has been motivated, let us turn back to the asymmetry regarding intervention effects. Three relevant ingredients, the gist of the asymmetry regarding intervention effects (26), the current proposal (37), and the widely accepted view regarding the obviation of intervention effects (38), are repeated below:

(26) *An asymmetry regarding intervention effects*

a. Intervention effects appear in matrix wh-questions, embedded non-CQ wh-questions, and *to*-embedded wh-questions (see (27)).

b. Intervention effects are obviated in embedded CQ wh-questions (see (28)).

(37) a. In Japanese, matrix wh-questions, embedded non-CQ wh-questions, and *to*-embedded wh-questions do not involve covert wh-movement.

b. In Japanese, embedded CQ wh-questions involve covert wh-movement.

(38) Intervention effects do not appear when a wh-phrase is located above an intervenor *overtly or covertly*.

Let us now consider how the proposal in (37), supplemented by the view in (38), gives an account of the asymmetry in (26). According to (37a), in matrix wh-questions, embedded non-CQ wh-questions, and *to*-embedded wh-questions, wh-phrases do not undergo covert wh-movement,

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Lit. 'Mai {asked / is anxious} [that what only Ken bought].' (= (27c))

(iv) Embedded CQ wh-questions:
Mai-wa [Ken-sika nani-o kawa-nakat-ta ka] {sitteiru / akasita / tazuneta}.
Mai-TOP Ken-SIKA what-ACC buy-not-PAST Q know / revealed / asked
'Mai {knows / revealed / asked} [what only Ken bought].' (= (28))

In these, the intervenors appear as the subject of the embedded clause, hence Tomioka's analysis predicts all these examples to be better than the basic cases of intervention effects, contrary to fact. For this reason, this chapter puts aside Tomioka's pragmatic analysis, instead adopting the more standard view in (38). However, I leave open how the (relative) acceptability of (ii) can be captured by an analysis different from Tomioka's.

staying in-situ even covertly. This means that if a wh-phrase is located below an intervenor in those wh-questions in overt syntax, it stays in that position even covertly, never being located above that intervenor. Hence, it follows from (38) that an intervention effect cannot be voided in those three types of wh-questions, leading to (26a). According to (37b), in embedded CQ wh-questions, wh-phrases undergo covert wh-movement. As a result, even if a wh-phrase is overtly positioned below an intervenor in those wh-questions, it is covertly located above that intervenor by virtue of covert wh-movement. According to (38), this configuration obviates an intervention effect; hence, (26b). This way, the proposal in (37), along with the general view in (38), captures the asymmetry regarding intervention effects in (26).

3.4.1.3 Additional support: Inverse scope readings

Before turning to the second asymmetry, I here provide additional support for the current proposal that pertains to the obviation of intervention effects. More specifically, I will observe that in embedded CQ wh-questions where a wh-phrase occurs below an intervenor but an intervention effect is obviated, that wh-phrase exhibits an inverse scope interpretation with respect to that intervenor. Consider, e.g., (42), which involves three intervenors, the NPI *dare-mo* '(not) anyone' (42a), the universal quantifier *da're-mo* 'everyone' (42b) and the disjunction '*Ken-ka Kooji(-ka)*' 'Ken or Koji' (42c).[22] (Concerning the NPI *dare-mo* '(not) anyone', I here follow Shimoyama 2011 and assume that it is in fact a universal quantifier that takes wide scope over negation.)

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[22] Here I will not take into consideration the narrow scope readings of the wh-phrases, or the reconstructed scope readings in (42), instead focusing on the inverse scope readings. However, one relevant note is in order. The judgement "*" on the second reading of (42a,b) is intended to indicate that pair-list readings are unavailable to them. However, functional readings are available to them. For example, one of the intended readings of (ia) (= (42b)) can be paraphrased as (ib) (i.e. a functional reading) but not as (ic) (i.e. a pair-list reading).

(i) a. Mai-wa [da're-mo-ga <u>nani</u>-o katta ka] sitteiru.
Mai-TOP who-MO-NOM what-ACC bought Q know
'Mai knows what everyone bought.'

(42) a. Mai-wa [$\boxed{\text{dare-mo}}$ nani-o kawa-nakat-ta ka] {sitteiru / akasita / tazuneta}.
Mai-TOP who-MO what-ACC buy-not-PAST Q know / revealed / asked
'Mai {knows / revealed / asked} [what no one bought(/what everyone did not buy)].'
- ✓: wh>∀ (e.g. Mai knows what *x* is s.t. everyone did not buy *x*.)
- * :∀>wh (e.g. Mai knows, for every *y*, what is *x* s.t. *y* did not buy *x*.) (See fn.22.)

b. Mai-wa [$\boxed{\text{da're-mo}}$-ga nani-o katta ka] {sitteiru / akasita / tazuneta}.
Mai-TOP who-MO-NOM what-ACC bought Q know / revealed / asked
'Mai {knows / revealed / asked} [what everyone bought].'
- ✓: wh>∀ (e.g. Mai knows what *x* is s.t. everyone bought *x*.)
- * :∀>wh (e.g. Mai knows, for every *y*, what is *x* s.t. *y* bought *x*.) (See fn.22.)

c. Mai-wa [$\boxed{\text{Ken-ka Kooji(-ka)}}$-ga nani-o katta ka] {sitteiru / akasita / tazuneta}.
Mai-TOP Ken-or Koji-or-NOM what-ACC bought Q know / revealed / asked
'Mai {knows / revealed / asked} [what Ken or Koji bought].'
- ✓: wh > or (e.g. Mai knows what *x* is s.t. Ken or Koji bought *x*.)
- ??: or > wh (e.g. Mai knows, for *y* s.t. *y* is Ken or Koji, what is *x* s.t. *y* bought *x*.)

The crucial point in (42) is that the wide scope readings of the wh-phrases with respect to the intervenors are available, despite the fact that the former elements are apparently located below the latter ones. Note further that the scope reading patterns observed in (42) obtain as well with the corresponding matrix wh-questions where a wh-phrase is scrambled over an intervenor to obviate an intervention effect; compare (43a-c) with (42a-c) respectively.[23]

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b. Mai-wa [da're-mo$_i$-ga zibun$_i$-no hon-o katta koto]-o sitteiru.
Mai-TOP who-MO-NOM self-GEN book-ACC bought fact-ACC know
'Mai knows (the fact) that everyone bought his own book.'

c. Mai-wa [Ken-ga hon-o, Kooji-ga CD-o, Hiro-ga pen-o katta koto]-o sitteiru.
Mai-TOP Ken-NOM book-ACC Koji-NOM CD-ACC Hiro-NOM pen-ACC bought fact-ACC know
'Mai knows (the fact) that Ken bought a book, Koji a CD and Hiro a pen.

I will leave it aside how to explain this fact, given that it is not relevant to the current discussion.

[23] In (43a,b), the judgement "*" on the second reading indicates the unavailability of pair-list readings. However, functional readings are available, as in the case of (42a,b) (see fn.22). For instance, the matrix question (43b), repeated as (iA), can be answered with (iBI) (i.e. a functional reading) but not with (iBII) (i.e. a pair-list reading).

(43) a. Nani-o$_i$ $\boxed{\text{dare-mo}}$ t_i kai-masen desita ka.

what-ACC who-MO buy-noy.POL COP.POL.PAST Q

'What did no one buy(/What did everyone not buy)?'

- ✓: wh > ∀ (What is *x* s.t. everyone did not buy *x*?)
- * :∀ > wh (For every *y*, there is *x* s.t. *y* did not buy *x*; what is that *x*?) (See fn.23.)

b. Nani-o$_i$ $\boxed{\text{da're-mo}}$-ga t_i kai-masi-ta ka.

what-ACC who-MO-NOM buy-POL-PAST Q

'What did everyone buy?'

- ✓: wh > ∀ (What is *x* s.t. everyone bought *x*?)
- * :∀ > wh (For every *y*, there is *x* s.t. *y* bought *x*; what is that *x*?) (See fn.23.)

c. Nani-o$_i$ $\boxed{\text{Ken-ka Kooji(-ka)}}$-ga t_i kai-masi-ta ka.

what-ACC Ken-or Koji-or-NOM buy-POL-PAST Q

'What did Ken or Koji buy?'

- ✓: wh > or (What is *x* s.t. Ken or Koji bought *x*?)
- * : or > wh (For *y* s.t. *y* is Ken or Koji, what is *x* s.t. *y* bought *x*?)

The scope reading patterns observed in (42) (in particular, the inverse scope readings), then, straightforwardly follow from the current proposal (37), or more precisely, (37b). According to it, in the embedded CQ wh-questions in (42), the wh-phrases undergo covert wh-movement and are thus covertly located above the intervenors, creating the configuration similar to the overt structure

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(i) A: Nani-o$_i$ $\boxed{\text{da're-mo}}_j$-ga t_i kai-masi-ta ka.
what-ACC who-MO-NOM buy-POL-PAST Q
'What did everyone buy?'

B: I. Zibun$_j$-no hon-o kai-masi-ta.
self-GEN book-ACC buy-POL-PAST
'(They) bought their own book.'

II. #Ken-wa hon-o, Kooji-wa CD-o, Hiro-wa pen-o kai-masi-ta.
Ken-TOP book-ACC Koji-TOP CD-ACC Hiro-TOP pen-ACC buy-POL-PAST
'Ken bought a book, Koji a CD, and Hiro a pen.'.

in (43) and thus allowing for the scope readings parallel to (43). The above observation thus lends additional support to the proposal in (37) (or (37b), in particular).

3.4.2 Capturing the second asymmetry: Cleft wh-questions

This subsection turns to the second asymmetry regarding Japanese wh-questions, which concerns cleft formation. The summary of this asymmetry (36) and the relevant data (34-35) are repeated below:

(36) *An asymmetry regarding cleft formation*

a. Forming cleft wh-questions as matrix wh-questions, embedded non-CQ wh-questions and *to*-embedded wh-questions leads to well-formedness (see (34)).

b. Forming cleft wh-questions as embedded CQ wh-questions leads to degradation (see (35)).

(34) a. Matrix wh-questions

[Ken-ga *e*$_i$ kabin-o oita no]-wa doko-ni$_i$ desu ka.

Ken-NOM vase-ACC put C-TOP where-on COP.POL Q

'Where is it that Ken put a vase?'

b. Embedded non-CQ wh-questions

Mai-wa [[Ken-ga *e*$_i$ kabin-o oita no]-wa doko-ni$_i$ (da) ka] {gironsita /

Mai-TOP Ken-NOM vase-ACC put C-TOP where-on COP Q discussed

?kaisetusita / kansin-ga-aru}.

explained / is.interested

'Mai {discussed / explained / is interested in} [where it is that Ken put a vase].'

c. *To*-embedded wh-questions

Mai-wa [[Ken-ga *e*$_i$ kabin-o oita no]-wa doko-ni$_i$ (da) ka to] {tazuneta /

Mai-TOP Ken-NOM vase-ACC put C-TOP where-on COP Q REP asked

?sinpaisiteiru}.

is.anxious

Lit. 'Mai {asked / is anxious} [that where it is that Ken put a vase].'

(35) Embedded CQ wh-questions

?*Mai-wa [[Ken-ga e_i kabin-o oita no]-wa doko-ni$_i$ (da) ka] {sitteiru / akasita /

Mai-TOP Ken-NOM vase-ACC put C-TOP where-on COP Q know / revealed /

tazuneta}.

asked

'Mai {knows / revealed / asked} [where it is that Ken put a vase].'

I will discuss below how the asymmetry in question (i.e. (36)) can be accounted for by the current proposal in (37), repeated below:

(37) a. In Japanese, matrix wh-questions, embedded non-CQ wh-questions, and *to*-embedded wh-questions do not involve covert wh-movement.

b. In Japanese, embedded CQ wh-questions involve covert wh-movement.

In particular, I will focus on the degradedness of (35) and pursue an account of it, i.e. the reason why forming cleft wh-questions as embedded CQ wh-questions causes degradation, in contrast with the other types of wh-questions (see (34)). For that purpose, I will adopt Hiraiwa and Ishihara's (2002, 2012) analysis of clefts in Japanese, and show that the proposal in (37b) gives an account of the degradedness under consideration in terms of freezing effects. It will be further shown that the proposed account of the degradedness of (35) is compatible with the other relevant data, i.e. (34), given the proposal in (37a).

3.4.2.1 Hiraiwa and Ishihara (2002, 2012)

I begin this subsection with an overview of Hiraiwa and Ishihara's (2002, 2012) (H&I, henceforth) analysis of Japanese clefts, which constitutes the basis for the following discussion.[24] They argue that Japanese clefts are derived through movement of the focus (or clefted) phrase, i.e. focus movement. On their analysis, the cleft in (44a) is derived from (44b), which they call an *in-situ focus construction*.

(44) a. [Ken-ga *e*$_i$ kabin-o oita no]-wa hondana-ni$_i$ da.
Ken-NOM vase-ACC put C-TOP bookshelf-on COP
'It is on the bookshelf that Ken put a vase.'

b. Ken-ga HONDANA-ni kabin-o oita no da.
Ken-NOM bookshelf-on vase-ACC put C COP
'It is on the bookshelf that Ken put a vase.'

In (44b), a regular declarative sentence is followed by the sequence of complementizer *no* + copula *da*, and the focus phrase *HONDANA-ni* 'on the bookshelf' is pronounced with phonological emphasis (represented as capitals in (44b)). Assuming the cartographic CP structure proposed by Rizzi (1997) in (45), H&I claim that (44b) has the structure in (46), where the complementizer *no* and the copula *da* head Fin(ite)P and Foc(us)P, respectively.

(45) [$_{\text{ForceP}}$ Force [$_{\text{Top(ic)P}}$ Top [$_{\text{Foc(us)P}}$ Foc [$_{\text{(TopP)}}$ (Top) [$_{\text{Fin(ite)P}}$ Fin [$_{\text{TP}}$ …]]]]]]

(46) [$_{\text{FocP}}$ [$_{\text{FinP}}$ [$_{\text{TP}}$ Ken-ga HONDANA-ni kabin-o oita] no] da]
Ken-NOM bookshelf-on vase-ACC put C COP

[24] Among previous studies that follow H&I's analysis are Nishigauchi and Fujii (2006), Takahashi (2006), and Hasegawa (2011). Also, Watanabe (2003) suggests a similar analysis of Japanese clefts.

Building on (46), H&I argue that the cleft in (44a) is derived by applying two movements to (44b): first, the focus phrase moves to Spec,FocP (i.e. focus movement) (47a), and then, the remnant FinP moves to Spec,Top(ic)P above FocP (i.e. topicalization) (47b).[25]

(47) a. focus movement of *hondana-ni* 'on the bookshelf' to Spec,FocP

[$_{\text{FocP}}$ hondana-ni$_i$ [$_{\text{FinP}}$ [$_{\text{TP}}$ Ken-ga *t*$_i$ kabin-o oita] no] da]

bookshelf-on Ken-NOM vase-ACC put C COP

b. remnant movement of FinP to Spec,TopP

[$_{\text{TopP}}$ [$_{\text{FinP}}$ Ken-ga *t*$_i$ kabin-o oita no]$_j$-wa [$_{\text{FocP}}$ hondana-ni$_i$ *t*$_j$ da]]

Ken-NOM vase-ACC put C-TOP bookshelf-on COP

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[25] The copula in in-situ focus constructions and clefts in Japanese can inflect for tense, as shown in (i).

(i) a. Mai-wa HONDANA-ni kabin-o oita no {da / dat-ta}.

Mai-TOP bookshelf-on vase-ACC put C COP / COP-PAST

Lit. 'It {is/was} on the bookshelf that Mai put a vase.'

b. [Mai-ga *e*$_i$ kabin-o oita no]-wa hondana-ni$_i$ {da / dat-ta}.

Mai-NOM vase-ACC put C-TOP bookshelf-on COP / COP-PAST

Lit. 'It {is/was} on the bookshelf that Mai put a vase.'

This might suggest that the copula here is associated with TP, which would be problematic for H&I's claim that it heads FocP, located above TP. However, Hasegawa (2011) and Hiraiwa and Ishihara (2012) maintain this view by observing that *dat-ta* (copula *da* + past tense *ta*) in (i) does not indicate tense. Instead, it indicates that "the speaker remembers or reaffirms what the speaker has previously acknowledged, which is the content of the *no*-clause" (Hasegawa 2011: 27). In this sense, *dat-ta* in (i) "is not a real tensed predicate but more like a modal particle" (Hiraiwa and Ishihara 2012: 155).

Indeed, a similar view has been put forth for the copula *be* in English specificational pseudoclefts (SPCs). They have been argued to have no independent temporal value, in that its only possible forms are *is* and *was*, the use of which depends on the tense value of the wh-clause (Akmajian 1970, Higgins 1973), as in (ii).

(ii) a. [What John is] is very tall.

b. [What John was] was very tall.

c. *[What John will be] will be very tall.

d. [What John will be] is very tall.

e. *[What John is being] is being very tall.

f. [What John is being] is very tall. (Bošković 1997a: 266)

Based on this, it has been argued that *be* in English SPCs is not associated with TP. Thus, Bošković (1997a) proposes that English SPCs have a VP-size structure, lacking TP. Note also that Boeckx (2007) argues that *be* in English SPCs is the head of TopP, with the wh-clause and the element following *be* located in Spec,TopP and Spec,FocP respectively, as in the structure of (ia) in (iii). (His analysis of English SPCs is quite similar to H&I's analysis of Japanese clefts.)

(iii) [$_{\text{TopP}}$ [what$_i$ John is *t*$_i$]$_j$ [$_{\text{Top}}$ is] [$_{\text{FocP}}$ [very tall]$_i$ [$_{\text{IP}}$ [$_{\text{VP}}$ *t*$_j$]]]]

These proposals regarding English SPCc have in common that the copula verb *be* is not associated with TP. Given that, it is not unreasonable to assume that the same holds for the relevant element in (i), as H&I propose.

Of importance in H&I's analysis of Japanese clefts is that they are derived through the focus movement of the focus phrase. This point will be crucial for the discussion in Section 3.4.2.2. Given that, in the rest of this sub-subsection, I will show a few arguments that uphold this particular aspect of H&I's analysis.

First, the involvement of movement in the derivation of clefts is confirmed by the fact that the focus phrase is sensitive to islands, as observed in the literature (e.g. Hoji 1987, H&I); while the focus phrase can be associated with a gap position in the presupposed clause across a clause boundary, it cannot if it appears in an island, as the contrast in (48) shows.

(48) a. [Ken-ga [Mai-ga e_i kabin-o oita to] omotteiru no]-wa hondana-ni_i da.
Ken-NOM Mai-NOM vase-ACC put REP think C-TOP bookshelf-on COP
'It is on the bookshelf that Ken thinks that Mai put a vase.'

b. *[Ken-ga [[e_i e_j kabin-o oita] $hito_i$]-o sitteiru no]-wa hondana-ni_j da.
Ken-NOM vase-ACC put person-ACC know C-TOP bookshelf-on COP
Lit. 'It is on the bookshelf that Ken knows a person who put a vase.'

In addition, Takahashi (2006) observes that weak crossover effects occur in clefts, as in (49).

(49) a. [Mai-ga e_i [[$soitu_j$-ga e_k kaita] $ronbun_k$]-o henkyakusita no]-wa [$dare_j$-ni]$_i$
Mai-NOM that.person-NOM wrote paper-ACC returned C-TOP who-to
desu ka.
COP.POL Q
'To whom is it that Mai returned a paper that he wrote?'

b. ?*[[[e_k $Soitu_j$-o sonkeisiteiru] $gakusee_k$]-ga e_i gakkoo-de atta no]-wa
that.person-ACC respect student-NOM school-at met C-TOP
[$dare_j$-ni]$_i$ desu ka.
who-to COP.POL Q
Lit. 'Who is it that the student that respects him met at school?'

The above two examples structurally differ in whether the gap that corresponds to the clefted phrase (i.e. e_i) c-commands the bound variable *soitu* '(Lit.) that person'; it does in (49a), while it does not in (49b). Given that, the grammaticality contrast between the two follows if those gaps are A'-traces of the clefted phrases; with this assumption, a weak crossover effect occurs in (49b) but not in (49a), given that it arises when the trace of an element that has undergone A'-movement does not c-command its antecedent. Takahashi's (2006) observation thus supports H&I's claim that the focus phrase of clefts undergoes focus movement, which is plausibly one type of A'-movement. (See also Takano 2002 for evidence based on reconstruction effects that the A'-movement involved in the derivation of clefts is not long-distance scrambling.)

One might wonder, however, if the relevant movement actually applies to the focus phrase itself. In fact, some previous studies argue that it is a null operator that moves in the derivation of clefts (e.g. Kuwabara 1996, Matsuda 1998, Koizumi 2000, Kizu 2005). According to Kizu (2005), for example, cleft sentences have the structure in (50), where the focus phrase XP is base-generated in the focus position and the null operator that corresponds to it moves from the original argument position to Spec,CP of the presupposed clause, where the operator is associated with the focus phrase.

(50) $[_{\text{TopP}} [_{\text{CP}} \text{Op}_i [_{\text{TP}} \ldots t_i \ldots]] [_{\text{TP}} \text{(Subject)} [_{\text{VP}} \ldots \text{XP}_i \ldots \textit{da}]]]$

H&I, however, note that under this operator movement analysis, it is unclear how a Case particle/postposition attached to the focus element is licensed. In contrast, their analysis provides a straightforward account of this issue: the Case particle/postposition in question is licensed in the position where the focus phrase originates in the presupposed clause before it undergoes focus movement.

3.4.2.2 Freezing effect account

With the analysis of Japanese clefts illustrated above, I will now provide an account of the degraded status of the embedded CQ cleft wh-question in (35), repeated below:

(35) ?*Mai-wa [[Ken-ga *e*$_i$ kabin-o oita no]-wa doko-ni$_i$ (da) ka] {sitteiru / akasita / tazuneta}.
Mai-TOP Ken-NOM vase-ACC put C-TOP where-on COP Q know / revealed / asked
'Mai {knows / revealed / asked} [where it is that Ken put a vase].'

Before delving into the discussion, I here exclude two potential accounts of the degradedness of (35). First, as shown in (51), it is possible to realize as an embedded CQ wh-question an in-situ focus construction, which can be transformed into a cleft according to H&I (see Section 3.4.2.1).

(51) Mai-wa [Ken-ga(/-wa) doko-ni kabin-o oita no (da) ka] {sitteiru / akasita / tazuneta}.
Mai-TOP Ken-NOM/-TOP where-on vase-ACC put C COP Q know / revealed / asked
'Mai {knows / revealed / asked} [where it is that Ken put a vase].'

This fact indicates that the marginality of (35) cannot be ascribed to the realization of the underlying in-situ focus construction as an embedded CQ wh-question.

Second, in general, topic phrases can appear within embedded questions (see, e.g., Saito 2012, 2013). Of particular relevance here is that they can occur in embedded CQ wh-questions, as in (52). (Note importantly that the embedded topic phrase *Mai-wa* in (52) can be construed as a non-contrastive topic.)

(52) Mai-wa [Ken-wa nani-o katta ka] {sitteiru / akasita / tazuneta}.
Mai-TOP Ken-TOP what-ACC bought Q know / revealed / asked
'Mai {knows / revealed / asked} [what Ken bought].'

Given that, it cannot be that (35) is degraded because a topic phrase occurs in the embedded CQ wh-question.

Ruling out those potential accounts, I now propose an account of the degradedness of (35). This account crucially draws on freezing effects (e.g. Rizzi 2006, Bošković 2008), which are observed with elements that have undergone A'-movement; they are "frozen" at their landing site — movement can no longer be applied to them. This effect is observed in the examples below:

(53) a. Someone thinks that Mary solved every problem. (∃>∀ / (?)∀>∃)
b. Someone thinks that every problem, Mary solved. (∃>∀ / *∀>∃)
(Bošković 2008: 251, citing from Lasnik and Uriagereka 1988)

(54) *What$_i$ do you wonder [$_{CP}$ t_i C [$_{TP}$ John bought t_i (when)]]? (ibid: 256)

The examples in (53) differ regarding their scope interpretations. At least for some speakers, the universal quantifier can take scope over the existential quantifier across the clause boundary in (53a). However, (53b) does not allow the wide scope interpretation of the universal quantifier even for those who accept that interpretation in (53a). Notice that (53b) is different from (53a) in that the universal quantifier undergoes topicalization within the embedded clause. With the assumption that quantifier raising (QR) into the matrix clause is needed for the wide scope of the universal quantifier, the unavailability of its wide scope reading in (53b) can be accounted for in terms of freezing effects; QR cannot apply to the universal quantifier since it has already undergone topic movement. Furthermore, (54) shows that a wh-phrase which has undergone wh-movement (in the embedded clause) cannot undergo further wh-movement. This can also be captured in terms of freezing effects; the wh-phrase is "frozen" in Spec,CP of the embedded question, being incapable of undergoing movement any longer.

With this overview, I now illustrate the account of the degradedness of (35) based on freezing effects, which I will call the *freezing effect account* henceforth. If we follow H&I's analysis of Japanese clefts (Section 3.4.2.1), the structure of the embedded CQ wh-question in (35) can be represented as in (55). As shown there, I assume that the question particle *ka* is the head of ForceP, the topmost projection in the left periphery (see (45)).

(55) (Mai-wa) [$_{\text{ForceP}}$ [$_{\text{TopP}}$ [$_{\text{FinP}}$ Ken-ga t_i kabin-o oita no]$_j$-wa [$_{\text{FocP}}$ doko-ni$_i$ t_j da]] ka]
Mai-TOP Ken-NOM vase-ACC put C-TOP where-on COP Q
({sitteiru / akasita / tazuneta})
know / revealed / asked

Notice in particular that the wh-phrase has been moved to Spec,FocP (i.e. focus movement). What is crucial now is the current proposal (37), or more precisely, (37b), repeated below.

(37) b. In Japanese, embedded CQ wh-questions involve covert wh-movement.

Here, I assume that in cartographic terms, the landing site of wh-movement in Japanese is the specifier of ForceP, which is located above FocP as represented in (55).[26] It then follows that the wh-phrase in (55) (i.e. (35)) must further undergo wh-movement to Spec,ForceP. However, this movement is blocked due to a freezing effect; the wh-phrase is "frozen" in Spec,FocP because it

[26] Note incidentally that Rizzi (1997) argues that the landing site of wh-movement in English embedded wh-questions is Spec,ForceP.

has already undergone focus movement. I argue that this impossibility of wh-movement is what leads to the degradation of (35).

3.4.2.3 Arguments for the freezing effect account

To fortify the freezing effect account proposed above, I will now present two arguments for it. Those arguments concern (i) pseudoclefts and (ii) cleft wh-questions involving the reason wh-phrase *naze* 'why'.

3.4.2.3.1 Pseudoclefts

The first argument builds on the construction that H&I call *pseudoclefts*, examples of which are shown in (56) (cf. fn.15).[27] (In particular, I refer to sentences like (56b) as *pseudocleft wh-questions*.)

(56) a. [Ken-ga e_i kabin-o oita no]-wa hondana$_i$ da.
Ken-NOM vase-ACC put NMLZ-TOP bookshelf COP
'(The place) Where Ken put a vase is the bookshelf.'

b. [Ken-ga e_i kabin-o oita no]-wa doko$_i$ desu ka.
Ken-NOM vase-ACC put NMLZ-TOp where COP.POL Q
'Where is (the place) where Ken put a vase?'

Note that (56a-b) resemble the clefts in (57a-b) respectively.

(57) a. [Ken-ga e_i kabin-o oita no]-wa hondana-ni$_i$ da.
Ken-NOM vase-ACC put C-TOP bookshelf-on COP

[27] Following H&I, I assume that *no* in pseudoclefts is a (pro)nominal element, rather than a complementizer, and gloss it as NMLZ, which stands for a "nominalizer". Accordingly, I translate pseudoclefts differently from clefts (e.g. (56) vs. (57)). This assumption will not matter for the current discussion.

‘It is on the bookshelf that Ken put a vase.’ (= (44a))

b. [Ken-ga e_i kabin-o oita no]-wa doko-ni$_i$ desu ka.
Ken-NOM vase-ACC put C-TOP where-on COP.POL Q
‘Where is it that Ken put a vase?’ (= (34a))

Superficially, pseudoclefts differ from clefts only in that no Case or postposition is attached to their clefted phrase; compare (56) with (57). It has been argued, however, that these two similar constructions have different structures and derivations, building on the observation that they show different syntactic properties (e.g. Hoji 1987, H&I). Particularly relevant to the current discussion is the fact that pseudoclefts do not show an island effect, while clefts do (see Section 3.4.2.1); see the contrast between the cleft in (58a) and the pseudocleft in (58b).

(58) a. *[Ken-ga [[e_i e_j kabin-o oita] hito$_i$]-o sitteiru no]-wa sono hondana-ni$_j$ da.
Ken-NOM vase-ACC put person-ACC know C-TOP that bookshelf-on COP
Lit. ‘It is on that bookshelf that Ken knows a person who put a vase.’

b. [Ken-ga [[e_i e_j kabin-o oita] hito$_i$]-o sitteiru no]-wa sono hondana$_j$ da.
Ken-NOM vase-ACC put person-ACC know C-TOP that bookshelf COP
Lit. ‘(The place) Where Ken knows a person who put a vase is that bookshelf.’

This indicates that pseudoclefts do not involve (focus) movement of their focus phrase, unlike clefts. It has been argued, instead, that their focus phrase is base-generated in the original position (while the gap position is occupied by a null pronoun that corresponds to the focus phrase). In terms of the freezing effect account, it is then predicted that no degradedness will arise even if a pseudocleft wh-question is realized as an embedded CQ wh-question; applying wh-movement to its focus phrase should not give rise to a freezing effect since it has not undergone any movement.

This prediction is borne out, as (59b) shows; compare it with (59a) (= (35)), which involves a cleft wh-question formed as an embedded CQ wh-question.

(59) a. ?*Mai-wa [[Ken-ga e_i kabin-o oita no]-wa doko-ni$_i$ (da) ka] {sitteiru / akasita / tazuneta}.
Mai-TOP Ken-NOM vase-ACC put C-TOP where-on COP Q know / revealed / asked
'Mai {knows / revealed / asked} [where it is that Ken put a vase].' (= (35))

b. ?*Mai-wa [[Ken-ga e_i kabin-o oita no]-wa doko$_i$ (da) ka] {sitteiru / akasita / tazuneta}.
Mai-TOP Ken-NOM vase-ACC put C-TOP where COP Q know / revealed / asked
Lit. 'Mai {knows / revealed / asked} [where (the place) where Ken put a vase is].'

The above contrast thus lends support to the claim that a freezing effect is involved in (59a) (= (35)).

3.4.2.3.2 Cleft *why*-questions

The second argument for the freezing effect account concerns cleft wh-questions whose focus position is occupied by *naze* 'why', like (60), which I call *cleft* why-*questions*.[28]

(60) [Ken-ga hondana-ni kabin-o oita no]-wa naze desu ka?
Ken-NOM bookshelf-on vase-ACC put C-TOP why COP.POL Q
'Why is it that Ken put a vase on the bookshelf?'

[28] I thank Kensuke Takita for inspiring me to consider cleft *why*-questions.

Crucially, cleft *why*-questions are well-formed even as embedded CQ wh-questions; compare (61b) with (61a) (= (35)).

(61) a. ?*Mai-wa [[Ken-ga e_i kabin-o oita no]-wa doko-ni$_i$ (da) ka] {sitteiru / akasita /
Mai-TOP Ken-NOM vase-ACC put C-TOP where-on COP Q know / revealed /
tazuneta}.
asked
'Mai {knows / revealed / asked} [where it is that Ken put a vase].' (= (35))

b. Mai-wa [[Ken-ga hondana-ni kabin-o oita no]-wa naze (da) ka] {sitteiru /
Mai-TOP Ken-NOM bookshelf-on vase-ACC put C-TOP why COP Q know
akasita / tazuneta}.
Revealed / asked
'Mai {know / revealed / asked} [why it is that Ken put a vase on the bookshelf].'

Given that the only difference between (61a) and (61b) concerns the wh-phrase (i.e., *where* vs. *why*), the amelioration must be ascribed to some property of *naze* 'why'. In fact, it has been cross-linguistically observed that reason wh-phrases (like *why* in English) do not pattern with other wh-phrases in many respects (e.g. Rizzi 1990, 2001, Ko 2005, Stepanov and Tsai 2008). Below, I present two examples of the peculiarity of reason wh-phrases, from French and Italian (after which I will also discuss a case from Japanese).

First, French reason wh-phrase *pourquoi* 'why' shows a peculiar property with respect to in-situ wh-questions (see, e.g., Rizzi 1990). One notable characteristic of French (matrix) wh-questions is that wh-phrases need not undergo wh-movement and thus can stay in-situ, as shown below:

(62) a. Il a parlé de <u>quoi</u>?
he has spoken of what
Lit. 'He spoke about what?'

b. Il a parlé <u>comment</u>?
he has spoken how
Lit. 'He spoke how?' (Rizzi 1990: 47)

However, *pourquoi* 'why' resists surfacing in a low position (as in (62)) and necessarily appears in a left peripheral position; witness the contrast in (63).[29]

(63) a. ?*Il a parlé <u>pourquoi</u>?
he has spoken why
Lit. 'He spoke why?'

b. <u>Pourquoi</u> a-t-il parlé?
why has.he spoken
'Why did he spoke?' (ibid: 47)

Second, Rizzi (2001) observes that Italian reason wh-phrase *perché* 'why' is distinct from other wh-phrases regarding compatibility with a fronted focus phrase. More specifically, *perché* 'why' can co-occur with a fronted focus phrase with the former preceding the latter, whereas other wh-phrases cannot with either order; observe the contrast in (64), for example.[30]

(64) a. *{Che cosa A GIANNI / A GIANNI che cosa} hanno detto (non a Piero)?
what to Gianni to Gianni what have said not to Piero
Lit. '{What TO GAINNI / TO GIANNI what} they have said (not to Piero)?'
(Rizzi 2001: 291)

b. {Perché QUESTO / *QUESTO perché} avremmo dovuto dirgli, non
why this this why had should say.him not

[29] Rizzi (1990) also shows that French reason wh-phrase *pourquoi* 'why' behaves differently from other wh-phrases with respect to stylistic inversion.
[30] Rizzi (2001) also observes a peculiar property of Italian reason wh-phrase *perché* 'why' regarding the optionality of subject inversion.

qualsco'alro?

something.else

Lit. '{Why THIS / THIS why} we should have said to him, not something else?'

(cf. ibid: 294)

Such peculiar properties of reason wh-phrases have led researchers to propose that the wh-phrase in question is base-generated in Spec,CP, in contrast to other wh-phrases, which move to Spec,CP from their original position below TP (e.g. Rizzi 1990, 2001, Ko 2005, Stepanov and Tsai 2008). This nicely captures the above data. Concerning French (63), since *pourquoi* originates in Spec,CP, it must precede all elements within TP, which is the case in (63b) but not in (63a). Concerning Italian (64), Rizzi (2001) presents a more specific proposal; he proposes the fine-grained CP structure in (65) and argues that in Italian matrix wh-questions, all wh-phrases except *perché* 'why' move to Spec,FocP, while *perché* externally merges in the specifier of Int(errogative)P, which is located between ForceP and FocP (and does not move further).

(65) ForceP (> TopP*) > IntP (> TopP*) > FocP (> TopP*) > FinP > IP (Rizzi 2001: 289)

In (64a), the wh-phrase *che cosa* and the focused phrase *A GIANNI* compete for the same position, Spec,FocP, hence its ungrammaticality. This issue does not occur in (64b), because *perché* 'why' originates in Spec,IntP, a position different from (and higher than) that of the focused phrase, Spec,FocP.

What about the Japanese reason wh-phrase *naze* 'why', then? In fact, *naze* has also been observed to behave differently from other wh-phrases, and this peculiarity has led researchers to argue that it is base-generated in Spec,CP, like its counterpart in other languages (as seen above) (e.g. Ko 2005, Kuwabara 2008, 2013). Thus, wh-questions with *naze* do not give rise to intervention effects, unlike those with other wh-phrases, even when they are not realized as embedded CQ wh-questions (see Ko 2005 and references therein). Observe, e.g., the contrast between the matrix wh-questions in (66) (involving an object wh-phrase) and those in (67) (involving *naze* 'why'); notice in particular that (67a) is (at least marginally) acceptable even though the NPI *Ken-sika*, which is an intervenor, is located above *naze* 'why'.[31]

(66) a. ?*Ken-sika nani-o kaw-anakat-ta no?
Ken-SIKA what-ACC buy-not-PAST Q
'What did only Ken buy?'

b. Nani-o$_i$ Ken-sika t_i kaw-anakat-ta no?
what-ACC Ken-SIKA buy-not-PAST Q

(based on Ko 2005: 870)

(67) a. (?)Ken-sika naze sono hon-o kaw-anakat-ta no?
Ken-SIKA why that book-ACC buy-not-PAST Q
'Why did only Ken buy that book?'

b. Naze Ken-sika sono hon-o kaw-anakat-ta no?
why Ken-SIKA that book-ACC buy-not-PAST Q

(based on ibid: 872)

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[31] Note that the wh-questions in (66-67) end with the particle *no*, instead of the question particle *ka*. This is because matrix wh-questions with *naze* 'why' must contain *no* in the sentence-final part; observe, e.g., the contrast between (ia) and (ib). (Note also that *no*-ending wh-questions do not require the politeness affix *-mas*, in contrast to matrix *ka*-ending wh-questions, as noted in fn.1.)

(i) a. Ken-wa nani-o {kai-masi-ta ka / katta no}.
Ken-TOP what-ACC buy-POL-PAST Q bought Q
'What did Ken buy?'

b. Ken-wa naze sono hon-o {??kai-masi-ta ka / katta no}.
Ken-TOP why that book-ACC buy-POL-PAST Q bought Q
'Why did Ken buy that book?'

I will not discuss this property here, referring the reader to Kuwabara (2008, 2013) and references therein. Of importance for the current discussion, however, is that the acceptability of (67a) cannot be ascribed to the use of *no*, given that intervention effects generally appear even in *no*-ending wh-questions, as (66a) indicates.

Ko (2005) argues that this special property of *naze* 'why' can be accounted for by assuming that *naze* originates in Spec,CP. Her account draws on Beck and Kim's (1997) approach to intervention effects, according to which intervention effects occur as a result of covert wh-movement crossing an intervenor. Notice, however, that this approach is incompatible with the assumption regarding (the obviation of) intervention effects that this chapter has adopted in Section 3.4.1, i.e. (38), repeated below.

(38) Intervention effects do not appear when a wh-phrase is located above an intervenor *overtly or covertly*.

According to this view, covert wh-movement across an intervenor would prevent intervention effects from occurring. Given that, I will provide below a different account of the lack of intervention effects with *naze* 'why' (i.e. (67a)) which is based on the widely adopted view regarding (the obviation of) intervention effects in (38) and the assumption that *naze* 'why' is base-generated in Spec,CP.[32]

Recall first that the important point regarding (67a) is that it does not induce an intervention effect even though the intervenor *Ken-sika* is located above the wh-phrase *naze* 'why'. Interestingly, a similar pattern is observed when an intervenor appears above a wh-phrase as a result of scrambling; e.g., compare (68a) and (68b).

(68) a. ?*$\boxed{\text{Ken-sika}}$ <u>nani</u>-o kaw-anakat-ta no.
Ken-SIKA what-ACC buy-not-PAST Q
'What did only Ken buy?' (= (66a))

[32] In fact, Ko (2005) notes that her "paper can be made compatible with other approaches [to intervention effects than Beck and Kim (1997)]" (Ko 2005: 871).

b. ?$\boxed{\text{Banana-sika}}_i$ <u>dare</u>-ga *t*i kaw-anakat-ta no.
banana-SIKA who-NOM buy-not-PAST Q
'Who bought only bananas?'

(68a) instantiates the typical configuration where an intervention effect occurs; the subject intervenor *Ken-sika*, being located in the canonical subject position, appears above the object wh-phrase. (68b) is similar to (68a) in that the intervenor surfaces above the wh-phrase. However, it differs from (68a) in that that configuration has resulted from the scrambling of the intervenor; *banana-sika*, which originated in the object position, has been scrambled over the subject wh-phrase, as a result of which the former is located above the latter. Crucially, (68b) does not induce an intervention effect, in contrast to (68a), but similarly to (67a).

Regarding the lack of an intervention effect in (68b), I argue that it can be captured if we adopt the widely accepted view that scrambling (or more precisely, A'-scrambling) can undergo complete reconstruction (e.g. Saito, 1989, Tada 1993, Bošković and Takahashi 1998). To see this, suppose that the scrambled intervenor in (68b) is reconstructed to its original position, i.e. the object position, in LF. According to the view regarding the obviation of intervention effects in (38), this LF structure should not induce an intervention effect, since the reconstructed intervenor *banana-sika* is located below the subject wh-phrase. The acceptability of (68b) then follows.

To bolster this reconstruction-based account, I will now focus on another property of scrambling: (clause-internal) scrambling can serve as A-movement and thus an element that has undergone this type of scrambling can bind anaphors or variables. Compare (69a) and (69b), for example.

(69) a. *[Soko$_i$-no syain]-ga [mit-tu izyoo-no kaisya]$_i$-o tyoosasita.
there-GEN employee-NOM 3-CLS over-GEN company-ACC investigated

Lit. 'Employees there$_i$ investigated [more than three companies]$_i$.'

b. [Mit-tu izyoo-no kaisya]$_i$-o$_j$ [soko$_i$-no syain]-ga t_j tyoosasita.
3-CLS over-GEN company-ACC there-GEN employee-NOM investigated
Lit. '[More than three companies]$_i$, employees there$_i$ investigated.'

In (69a), *soko* 'there' in the subject noun phrase cannot be interpreted as a variable bound by the object noun phrase *mit-tu izyoo-no kaisya* 'more than three companies', since the former is not c-commanded by the latter. However, the interpretation in question becomes available if the object is scrambled over the subject, as shown in (69b). This indicates that the scrambling of the object in (69b) counts as A-movement, which enables it to bind the variable in the subject noun phrase.

If this property of scrambling is taken into consideration, the reconstruction-based account presented above makes the following prediction: if an intervenor undergoes scrambling over a wh-phrase and binds an anaphor or a variable, it cannot be reconstructed and thus an intervention effect should arise, unlike (68b).

To investigate whether this prediction is borne out, I use a universal quantifier *(hotondo) da're-mo* '(almost) everyone', which serves as an intervenor, as shown in (70) (see Section 3.3.1.1).

(70) a. ?? $\boxed{\text{(Hotondo) da're-mo}}$-ga <u>nani</u>-o kai-masi-ta ka.
almost who-MO-NOM what-ACC buy-POL-PAST Q
'What did (almost) everyone buy?'

b. <u>Nani</u>-o$_i$ $\boxed{\text{(hotondo) dar'e-mo}}$-ga t_i kai-masi-ta ka.
what-ACC almost who-MO-NOM buy-POL-PAST Q

More specifically, I will use it as the direct object of a double object construction; (71) shows the two basic sentences for the following discussion. (Note that (71b) does not involve a configuration which gives rise to an intervention effect, since the wh-phrase *doko-ni* 'to where' originates above the intervenor.)

(71) a. Tanaka-sensee{-wa/-ga} MIT-ni (hotondo) da're-mo-o suisensita.
Tanaka-Prof.-TOP/-NOM MIT-to almost who-MO-ACC recommended
'Prof. Tanaka recommended (almost) everyone to MIT.'

b. Tanaka-sensee{-wa/-ga} doko-ni (hotondo) da're-mo-o suisensi-masi-ta ka.
Tanaka-Prof.-TOP/-NOM where-to almost who-MO-ACC recommend-POL-PAST Q
'To where (≈ which university) did Prof. Tanaka recommend (almost) everyone?'

Two notes are in order. First, if the subject in (71a) involves a variable, it can be bound by the direct object *(hotondo) daremo* '(almost) everyone' which has been scrambled to the sentence-initial position; observe the contrast between (72a) and (72b).

(72) a. *Sono-hito$_i$-no sidookyooin-ga MIT-ni (hotondo) da're-mo$_i$-o suisensita.
that-person-GEN advisor-NOM MIT-to almost who-MO-ACC recommended
Lit. 'His$_i$ advisor recommended (almost) everyone$_i$ to MIT.'

b. (Hotondo) da're-mo$_i$-o$_j$ sono-hito$_i$-no sidookyooin-ga MIT-ni t_j suisensita.
almost who-MO-ACC that-person-GEN advisor-NOM MIT-to recommended
Lit. '(Almost) everyone$_i$, his$_i$ advisor recommended to MIT.'

Second, even if the direct object in (71b) is scrambled to the sentence-initial position, no intervention effect arises, as is the case with (68b); this is illustrated in (73).

(73) (?) (Hotondo) da're-mo-o$_i$ Tanaka-sensee{-wa/-ga} doko-ni t_i suisensi-masi-ta
almost who-MO-ACC Tanaka-Prof.-TOP/-NOM where-to recommend-POL-PAST
ka.
Q
'To where (≈ which university) did Prof. Tanaka recommend almost everyone?'

Now, if the reconstruction-based account given above is correct, it is expected that (73) will become degraded if the subject contains a variable bound by the scrambled intervenor; since the scrambled intervenor cannot be reconstructed due to the variable binding, it must be located above the wh-phrase even in LF, thus inducing an intervention effect (according to (38)). This expectation is borne out; observe (74) (in particular, compare it with (73)).

(74) ?? [(Hotondo) da're-mo]$_i$-o$_j$ sono-hito$_i$-no sidookyooin-ga doko-ni t_j
almost who-MO-ACC that-person-GEN advisor-NOM where-to
suisensi-masi-ta ka.
recommend-POL-PAST Q
Lit. '(Almost) everyone$_i$, to where (≈ which university) did his$_i$ advisor recommend?'

This observation thus lends support to the reconstruction-based account, i.e. the claim that no intervention effect arises in (68b) (and (73)) because the scrambled intervenor is reconstructed, thus being covertly located below the wh-phrase.

Taking the account in question into consideration, let us now turn back to (67a), which is our main concern. I will show that the acceptability of (67a) can be accounted for in the same manner as that of (68b), if the reason wh-phrase *naze* 'why' is base-generated in Spec,CP. Under this assumption, the word order in (67a) suggests that the subject intervenor *Ken-sika* has been scrambled over the wh-phrase *naze*; more specifically, (67a) has the structure in (75), where I assume the subject has undergone scrambling from Spec,TP.[33]

(75) [[Ken-sika]$_i$ [$_{CP}$ naze [$_{TP}$ t_i sono hon-o kawa-nakat-ta]]] no
Ken-SIKA why that book-ACC buy-not-PAST Q

[33] Ko (2005) also claims that the subject has been scrambled over *naze* 'why' when the former precedes the latter as in (67a). Based on this claim, she argues that subjects can undergo scrambling, *pace* Saito (1985).

Suppose now that the scrambled intervenor *Ken-sika* is reconstructed to its original position, i.e. Spec,TP, in LF. Note crucially that in this LF structure, the reconstructed intervenor is located below the wh-phrase *naze* 'why', which resides in Spec,CP. Hence, if we follow the standard view regarding the obviation of intervention effects in (38), the LF configuration in question does not induce an intervention effect, hence the grammaticality of (67a).

To recap, I have argued above that the assumption that *naze* 'why' originates in Spec,CP can give an account of the lack of an intervention effect in (67a), with the two independently motivated assumptions, i.e. the widely accepted view regarding the obviation of intervention effects in (38) and the property of scrambling regarding reconstruction. The hypothesis that reason wh-phrases are base-generated in CP is thus motivated for Japanese as well.

With the CP-base-generation approach described above, let us now return to (61), repeated as (76) below, which importantly shows that cleft wh-questions realized as embedded CQ wh-questions are ameliorated with *naze* 'why'.

(76) a. ?*Mai-wa [[Ken-ga e_i kabin-o oita no]-wa doko-ni$_i$ (da) ka] {sitteiru / akasita /
Mai-TOP Ken-NOM vase-ACC put C-TOP where-on COP Q know / revealed /
tazuneta}.
asked
'Mai {knows / revealed / asked} [where it is that Ken put a vase].' (= (35))

b. Mai-wa [[Ken-ga hondana-ni kabin-o oita no]-wa naze (da) ka] {sitteiru /
Mai-TOP Ken-NOM bookshelf-on vase-ACC put C-TOP why COP Q know
akasita / tazuneta}.
Revealed / asked
'Mai {know / revealed / asked} [why it is that Ken put a vase on the bookshelf].'

Under the current assumption, *naze* in the cleft wh-question in (76b) originates in Spec,CP. Given the standard assumption that wh-movement targets Spec,CP, this means that *naze* in (76b) does not undergo (covert) wh-movement. As a result, the derivation of the cleft wh-question in (76b) does not induce a freezing effect. The grammaticality of (76b) is then straightforwardly captured.

Above, I have adopted H&I's analysis of clefts, where the split CP structure plays a crucial role. To be consistent with their analysis, I will now be more specific about the structure of the left periphery by adopting Rizzi's (2001) split CP in (77) (= (65)).

(77) ForceP (> TopP*) > IntP (> TopP*) > FocP (> TopP*) > FinP > IP (Rizzi 2001: 289)

Based on (77), I will assume that *naze* 'why' is base-generated in Spec,IntP and does not undergo wh-movement (see, e.g. Rizzi 2001) even in embedded CQ wh-questions, where I proposed that other wh-phrases undergo covert wh-movement (i.e. (37b)).[34] In other words, *naze* is interpreted in interrogative Spec,IntP. Under the current assumption, the surface structure of the cleft *why*-question in (76b) can be represented as in (78).

(78)	Ken-wa	[$_{ForceP}$ [$_{TopP}$ [$_{FinP}$ Mai-ga	hondana-ni	kabin-o	oita	no]$_i$-wa	[$_{IntP}$ naze
	Ken-TOP	Mai-NOM	bookshelf-on	vase-ACC	put	C-TOP	why
	[$_{FocP}$ t_i (da)]]] ka] {sitteiru / akasita / tazuneta}.						
	COP Q know / revealed / asked						

In (78), *naze* 'why', which originates in Spec,IntP, will not undergo covert wh-movement. Hence, the derivation of the cleft *why*-question in (76b) does not give rise to a freezing effect.

Shlonsky and Soare (2011), however, argue that reason wh-phrases originate in the left periphery and still undergo wh-movement. More specifically, they propose that a reason wh-phrase is base-generated in the specifier of a projection that they call ReasonP, which is located in a lower field of the left periphery (at least higher than the subject position and, possibly, FinP), and then

[34] Kuwabara (2008, 2013) also argues that *naze* 'why' is base-generated in Spec,IntP.

moves to Spec,IntP. Nothing would change under the current analysis even under this kind of an approach to reason wh-phrases. Under such an approach and the current proposal that embedded CQ wh-questions involve covert wh-movement (to Spec,ForceP) (i.e. (37b)), in embedded CQ wh-questions, *naze* 'why' would originate in Spec,IntP and then move to Spec,ForceP in LF. With this view, *naze* in (78) would undergo covert wh-movement to Spec,ForceP. Crucially, this movement would not induce a freezing effect, given that that wh-phrase has otherwise not undergone any movement, being base-generated at Spec,IntP.[35]

Whichever analysis is adopted, the crucial point is that no freezing effect arises in the derivation of the embedded CQ wh-question in (76b). In (76a), by contrast, the wh-phrase in the clefted position, which originates within VP, has undergone focus movement, and thus applying covert wh-movement to it gives rise to a freezing effect, leading to its degradation. Therefore, the contrast in (76) lends support to the freezing effect account, in that it gives an account of the

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[35] In the embedded CQ wh-question in (i), *naze* 'why' appears within the declarative complement clause, and is associated with the reason for Ken's being fired.

(i) Mai-wa [Kooji-ga [Ken-ga naze kubininatta to] omotteiru ka] siranai.
Mai-TOP Koji-NOM Ken-NOM why was.fired REP think Q know.not
'Mai does not know why Koji thinks that Ken was fired.'

Naze in (i) should then be base-generated in the left periphery of the complement declarative clause. With the proposal in (37b), *naze* should then undergo LF wh-movement to the interrogative projection (i.e. Spec,ForceP) of the embedded CQ wh-question.

Recall why the cleft *why*-question in (76b) does not degrade the sentence even though it is realized as an embedded CQ wh-question, under the assumption that *naze* is base-generated in Spec,IntP and then undergoes covert wh-movement to Spec,ForceP (cf. Shlonsky and Soare 2011). *Naze* in (76b) originates in Spec,IntP, which is located *above* FocP (see (77)); it thus does not undergo focus movement, and hence its covert wh-movement does not induce a freezing effect. In contrast with *naze* in (76b), the one in (i) originates in a position *below* FocP of the embedded CQ wh-question. Given that, if the embedded CQ wh-question in (i) were to be transformed into a cleft wh-question, *naze* would undergo focus movement to Spec,FocP. However, it could not subsequently undergo covert wh-movement (to Spec,ForceP) due to a freezing effect. Therefore, it is predicted that the embedded CQ wh-question in (i) cannot be realized as a cleft wh-question, in contrast to (76b). This prediction is borne out; observe (ii).

(ii) Mai-wa [[Kooji-ga e_i [Ken-ga e_j kubininatta to] omotteiru no]-wa naze$_{i/*j}$ (da) ka] siranai.
Mai-TOP Koji-NOM Ken-NOM was.fired REP think C-TOP why COP Q know.not
'Mai does not know [why$_{i/*j}$ it is [that (e_i) Koji thinks [that (e_j) Ken was fired]]].'

Crucially, the cleft *why*-question in (ii) can ask only for the reason for Koji's thinking, not for Ken's being fired, which means that the above prediction is borne out. (See Rizzi 2001 for a similar point regarding Italian.)

contrast in question with a few plausible and independently motivated assumptions regarding *naze* 'why'.

3.4.2.4 The other side of the asymmetry

To recap, in the above discussion, I have shown that the proposal in (37b) captures the observation that forming cleft wh-questions as embedded CQ wh-questions leads to degradation as in (35). The relevant example in (35) and the proposal in (37) are repeated below.

(35) Embedded CQ wh-questions

?*Mai-wa [[Ken-ga *e*i kabin-o oita no]-wa doko-nii (da) ka] {sitteiru / akasita / tazuneta}.

Mai-TOP Ken-NOM vase-ACC put C-TOP where-on COP Q know / revealed / asked

'Mai {knows / revealed / asked} [where it is that Ken put a vase].'

(37) a. In Japanese, matrix wh-questions, embedded non-CQ wh-questions, and *to*-embedded wh-questions do not involve covert wh-movement.

b. In Japanese, embedded CQ wh-questions involve covert wh-movement.

More specifically, I have illustrated that (37b), along with H&I's analysis of Japanese clefts, accounts for the degradation of (35) in terms of freezing effects; in deriving the embedded CQ wh-question in (35), the clefted wh-phrase undergoes focus movement, as argued by H&I, but it cannot subsequently undergo covert wh-movement (see (37b)) due to a freezing effect.

I will close this subsection by confirming that the freezing effect account is compatible with the other side of the asymmetry regarding cleft formation, i.e. the observation that cleft wh-questions can be formed as matrix wh-questions, embedded non-CQ wh-questions, and *to*-embedded wh-questions. The relevant data are repeated below:

(34) a. Matrix wh-questions

[Ken-ga e_i kabin-o oita no]-wa doko-ni$_i$ desu ka.

Ken-NOM vase-ACC put C-TOP where-on COP.POL Q

'Where is it that Ken put a vase?'

b. Embedded non-CQ wh-questions

Mai-wa [[Ken-ga e_i kabin-o oita no]-wa doko-ni$_i$ (da) ka] {gironsita /

Mai-TOP Ken-NOM vase-ACC put C-TOP where-on COP Q discussed

?kaisetusita / kansin-ga-aru}.

explained / is.interested

'Mai {discussed / explained / is interested in} [where it is that Ken put a vase].'

c. *To*-embedded wh-questions

Mai-wa [[Ken-ga e_i kabin-o oita no]-wa doko-ni$_i$ (da) ka to] {tazuneta /

Mai-TOP Ken-NOM vase-ACC put C-TOP where-on COP Q REP asked

?sinpaisiteiru}.

is.anxious

Lit. 'Mai {asked / is anxious} [that where it is that Ken put a vase].'

Given (37a), covert wh-movement does not apply to the clefted wh-phrases in (34), which have undergone focus movement. Therefore, freezing effects do not appear in (34) unlike (35), hence their grammaticality. This consideration, along with the discussion in the preceding subsections, thus leads to the conclusion that the proposal in (37) captures the asymmetry in Japanese wh-questions regarding cleft formation, which is instantiated by (34) and (35).

In summary, this section has demonstrated that the proposal in (37) provides a uniform account of the two asymmetries in Japanese wh-questions established in Section 3.3, namely the observation that Japanese wh-questions show asymmetric behavior with respect to intervention effects and cleft formation. (See also Chapter 5 for a deduction of (37) from independent mechanisms.)

3.5 Conclusion of the chapter

This chapter has revolved around the two hitherto unnoticed asymmetries in Japanese wh-questions that build on a classification consisting of four classes of wh-questions, i.e. matrix wh-questions, embedded CQ wh-questions, embedded non-CQ wh-questions, and *to*-embedded wh-questions. These asymmetries concern intervention effects and cleft formation. Concerning the former, it has been shown that intervention effects are obviated in embedded CQ wh-questions, but appear in the other types of wh-questions (i.e. matrix wh-questions, embedded non-CQ wh-questions and *to*-embedded CQ wh-questions). Concerning the latter, I have shown that forming cleft wh-questions as embedded CQ wh-questions leads to degradation, which is not the case with the other types of wh-questions. Notably, these two asymmetries make the same cut regarding the classes of Japanese wh-questions; matrix wh-questions, embedded non-CQ wh-questions, and *to*-embedded wh-questions pattern together, as opposed to embedded CQ wh-questions.

Building on this dichotomy, I have proposed that embedded CQ wh-questions are derived through covert wh-movement, while the other types of wh-questions do not involve covert wh-movement. I have then demonstrated that this proposal accounts for the above asymmetries in a uniform manner. First, I have shown that the proposal in question gives a straightforward account of the asymmetry regarding intervention effects if we follow the widely accepted view regarding the obviation of intervention effects, i.e. the view that intervention effects do not occur when a wh-phrase is located above an intervenor *overtly or covertly* (e.g. Pesestky 2000, Beck 2006, Kotek 2017). Second, I have also shown that the proposal in question captures the asymmetry regarding cleft formation; in particular, I have argued that assuming the involvement of covert wh-movement in embedded CQ wh-questions enables us to account for the degraded status of cleft wh-questions formed as embedded CQ wh-questions in terms of freezing effects, building on H&I's analysis of Japanese clefts.

Consequently, this section has established the validity of the aforementioned proposal that Japanese wh-questions differ in the (non-)involvement of covert wh-movement. In terms of the goals of this dissertation, the soundness of the proposal in question gives rise to two important implications. First, Japanese wh-questions cannot be analyzed in a uniform manner; instead, they must be derived differently depending on their types, i.e. which syntactic environments they occur in. Second, the current proposal crucially depends on covert wh-movement. Therefore, it provides a novel argument for the necessity of covert wh-movement in analyzing Japanese wh-questions, from a perspective different from Nishigauchi's (1990).

The above discussion, however, has not explained why the proposed asymmetry regarding the (non-)involvement of covert wh-movement in different types of questions holds. Chapter 5 will be devoted to providing an answer to the "why" question. But before that, in the next chapter, I will discuss one consequence of the proposal in this chapter regarding sluicing in Japanese.

Chapter 4

An Excursion: Sluicing in Japanese

4.1 Introduction

Before we explore the origin of the presence/absence of covert wh-movement in Japanese wh-questions, which will be done in Chapter 5, this chapter provides an excursion into a consequence of the discussion so far with respect to when covert wh-movement takes place in Japanese wh-questions regarding sluicing (and other related elliptical constructions) in Japanese. Sluicing constructions refer to (embedded) wh-questions whose propositional portion is elided. In the English example (1a), the underlined sluicing construction has the same interpretation as the embedded wh-question in (1b), suggesting that its propositional part (i.e. *he bought*) is elided.

(1) a. Jack bought something, but I don't know [<u>what</u>].

b. Jack bought something, but I don't know [<u>what he bought</u>].

Japanese has been noted to have a similar construction (Inoue 1978, Takahashi 1994, Fukaya and Hoji 1999, Hiraiwa and Ishihara 2002, 2012, Saito 2004, Abe 2015, 2016, among many others); see, e.g., the underlined part in (2b), whose antecedent is (2a).[1]

(2) a. Ken-wa dokoka-ni ryokooniitta sooda.

Ken-TOP somewhere-to traveled I.hear

'I hear that Ken traveled to somewhere.'

[1] I here treat *ryokooniitta* 'traveled' in (2) as one predicate for expository purposes; more precisely, it can be morphologically analyzed as *ryokoo-ni itta* 'went for a trip'.

b. (Demo,) boku-wa [doko-ni (da) ka] siranai.
but I-TOP where-to COP Q know.not
'(But) I don't know [to where].'

The underlined elliptical clause in (2b), consisting of the wh-phrase *doko-ni* 'to where', the optional copula *da*, and the Q-particle *ka*, has the same interpretation as the embedded wh-question in (3).

(3) *After (2a),*
(Demo,) boku-wa [Ken-ga doko-ni ryokooniitta ka] siranai.
but I-TOP Ken-NOM where-to traveled Q know.not
'(But) I don't know [to where Ken traveled].'

Given its similarity to sluicing in English, following the literature, I will refer to elliptical clauses as in (2b) as sluicing constructions (or, simply, sluicing). In this chapter, I will investigate how to analyze sluicing constructions in Japanese, by taking into consideration the current proposal, developed in Chapter 3, regarding the (non-)involvement of covert wh-movement in Japanese wh-questions. In particular, I will reveal several problems for the previous analyses of sluicing constructions in Japanese, and then propose a novel analysis of them, which crucially involves null operator movement associated with a particular semantic property.

The rest of this chapter is organized as follows: Section 4.2 introduces two representative analyses of Japanese sluicing proposed in the literature: the focus movement/cleft analysis and the in-situ analysis. Both analyses will be shown to be problematic, which will provide motivation for a new analysis of sluicing. To set the grounds for a new analysis, in Section 4.3, I will compare sluicing with three other elliptical constructions in Japanese, namely *ka*(*dooka*)-stripping, *to*-stripping and fragment answers. It will be shown that sluicing parallels *ka*(*dooka*)/*to*-stripping but differs from fragment answers with respect to several syntactic and semantic properties. Based on this, in Section 4.4, I will propose a novel analysis for sluicing (and *ka*(*dooka*)/*to*-stripping). Section 4.5 provides a summary of the chapter.

4.2 Two representative analyses

In this section, I will discuss two representative analyses of Japanese sluicing.

4.2.1 Focus movement/cleft analysis

The first analysis draws on the derivation and/or structure of clefts. Hiraiwa and Ishihara (2002, 2012) (i.e. H&I) argue that the derivation of sluicing constructions in Japanese involves focus movement, which they argue is an important ingredient for deriving clefts. I will briefly review here H&I's analysis of clefts in Japanese, which was discussed in more detail in Chapter 3 (i.e. Section 3.4.2). According to H&I, the cleft in (4a) is derived from the in-situ focus construction in (4b).

(4) a. [Ken-ga e_i kabin-o oita no]-wa hondana-ni$_i$ da.
Ken-NOM vase-ACC put C-TOP bookshelf-on COP
'It is on the bookshelf that Ken put a vase.'

b. Ken-ga HONDANA-ni kabin-o oita no da.
Ken-NOM bookshelf-on vase-ACC put C COP
'It is on the bookshelf that Ken put a vase.'

More specifically, assuming (5a) to be the structure of (4b), H&I argue that (4b) is transformed into (4a) through two movement operations, namely the movement of the focus phrase to

Spec,FocP (i.e. focus movement) and the movement of the remnant FinP to Spec,TopP (i.e. topicalization), as represented in (5b) and (5c) respectively.

(5) a. the structure of (4b)
[$_{FocP}$ [$_{FinP}$ [$_{TP}$ Ken-ga HONDANA-ni kabin-o oita] no] da]
Ken-NOM bookshelf-on vase-ACC put C COP

b. focus movement of *hondana-ni* 'on the bookshelf' to Spec,FocP
[$_{FocP}$ hondana-ni$_{i}$ [$_{FinP}$ [$_{TP}$ Ken-ga *t*$_{i}$ kabin-o oita] no] da]
bookshelf-on Ken-NOM vase-ACC put C COP

c. remnant movement of FinP to Spec,TopP
[$_{TopP}$ [$_{FinP}$ Ken-ga *t*$_{i}$ kabin-o oita no]$_{j}$-wa [$_{FocP}$ hondana-ni$_{i}$ *t*$_{j}$ da]]
Ken-NOM vase-ACC put C-TOP bookshelf-on COP

With this analysis of clefts in mind, I turn to H&I's analysis of sluicing. They argue that sluicing constructions have the structure of in-situ focus constructions (e.g. (5a) for (4b)) as their base structure. For example, the base structure of the sluicing construction in (2b) can be represented as in (6).

(6) [$_{FocP}$ [$_{FinP}$ [$_{TP}$ Ken-ga doko-ni ryokooniitta] no] da] (ka)
Ken-NOM where-to traveled C COP Q
(Lit. 'it is to where that Ken traveled')

The sluicing construction is then derived from (6) as shown in (7a) or (7b), where the remnant (i.e. the element that has remained undeleted), or the wh-phrase, has undergone focus movement to Spec,FocP (cf. (5b)), and FinP has been elided. The only difference between (7a) and (7b) is whether the FinP has undergone topicalization (cf. (5c)), i.e. whether a cleft has been formed. This difference will not be relevant to the following discussion.

(7) a. [$_{\text{ForceP}}$ [$_{\text{FocP}}$ doko-ni$_i$ ~~[$_{\text{FinP}}$ [$_{\text{TP}}$ Ken-ga t_i ryokooniitta] no]~~ (da)] ka]
where-to Ken-NOM traveled C COP Q
'to where (≈ to where Ken traveled)'

b. [$_{\text{ForceP}}$ [$_{\text{TopP}}$ ~~[$_{\text{FinP}}$ Ken-ga t_i ryokooniitta no]$_j$-wa~~ [$_{\text{FocP}}$ doko-n i$_i$ t_j (da)]] ka]
Ken-NOM traveled C-TOP where-to COP Q

I will call this analysis the *focus movement/cleft analysis* (the *FM/C analysis*, henceforth). There are two arguments for the FM/C analysis. First, it can capture the (optional) presence of the copula in sluicing constructions, since it is also contained in their underlying structure, i.e. in-situ focus constructions and/or clefts (see (6-7)). Second, it is observed in the literature that sluicing constructions are sensitive to islands (e.g. Takahashi 1994). Note first that in the antecedent clause of sluicing, the correlate (i.e. the element in the antecedent clause that corresponds to the remnant in the sluicing construction) can appear within a declarative complement clause (at least marginally), as shown in (8).

(8) a. Mai-wa [Ken-ga <u>dokoka</u>-ni ryokooniitta to] syutyoosi-tei-ru sooda.
Mai-TOP Ken-NOM somewhere-to travelled REP claim-PROG-PRES I.hear
'I hear that Mai is claiming that Ken traveled to somewhere.'

b. (?) (Demo,) boku-wa [doko-ni (da) ka](-wa) siranai.
but I-TOP where-to COP Q-TOP know.not
'(But) I don't know [to where].'
≈ '(But) I don't know [to where Mai is claiming that Ken traveled].'

In contrast, sluicing constructions cannot be formed if the correlate occurs within an island; compare (9) with (8), for example.[2,3]

(9) a. Ken-wa [[e_i <u>dokoka-ni</u> ryokooniitta] hito$_i$]-ni atta sooda.
Ken-TOP somewhere-to traveled person-DAT met I.hear
'I hear that Ken met a person who traveled to somewhere.'

b. # (Demo,) boku-wa [doko-ni (da) ka](-wa) siranai.
but I-TOP where-to COP Q-TOP know.not
'(But) I don't know [to where].'
≉ Lit. '(But) I don't know [to where Ken met [a person who traveled (*t*)]].'

This property follows from the FM/C analysis; sluicing constructions are sensitive to islands because they involve focus movement in their derivation (e.g. (7)). (Recall that clefts have also been observed to be island-sensitive, as noted in Chapter 3.)

With this much background, let us now re-examine the FM/C analysis in terms of the proposal presented in the last chapter. Of particular relevance here are the two subclasses of embedded wh-questions, i.e. embedded CQ wh-questions and embedded non-CQ wh-questions (Chapter 3). In the above examples, the sluicing constructions are selected by a CQ-predicate *siranai* 'not know'

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[2] Some speakers find (9b) acceptable. This is not relevant to the current discussion, however, because such speakers interpret it as involving an island-internal sentence in the antecedent clause, rather than the whole antecedent clause (e.g. Merchant 2001, Fukaya 2003, Abe 2016). Thus, they interpret the sluicing construction in (9b) as in (i); since no island is involved in (i), (9b) is grammatical for those speakers.

(i) [sono hito-ga doko-ni ryokooniitta ka]
that person-NOM where-to traveled Q
'where that person traveled to'

Fukaya (2003) and Abe (2016) in fact demonstrate that island effects appear in sluicing constructions even for those speakers who find examples like (9b) acceptable when island-internal interpretations like (i) are blocked by an independent factor.

[3] Unlike sluicing in Japanese, English sluicing is observed to be island-insensitive (e.g. Ross 1969). To capture this, it has been proposed that PF deletion voids island effects (i.e. rescue by PF deletion; e.g. Chomsky 1971, Lasnik 2001, Bošković 2011) (but for different and dissenting views, see Chung et al.1995, Barros et al. 2014, Abe 2015, among others). The unacceptability of (9b) might then suggest that the mechanism in question is unavailable in Japanese. This issue will be left open in this chapter; see Sugawa (2008) for relevant discussion.

and thus they are regarded as embedded CQ wh-questions. (10b) shows another example where a sluicing construction is selected by other CQ-predicates.

(10) a. Ken-wa dokoka-ni ryokooniitta sooda.
Ken-TOP somewhere-to traveled I.hear
'I hear that Ken traveled to somewhere.'

b. Boku-wa [doko-ni (da) ka] {sitteiru / akasita / tazuneta}.
I-TOP where-to COP Q know / revealed / asked
'I {know / revealed / asked} [to where].'
≈ 'I {know / revealed / asked} [to where Ken traveled].'

In addition, (11b) shows that sluicing constructions can also be selected by non-CQ-predicates, thus realized as embedded non-CQ wh-questions.[4]

(11) a. Ken-wa dokoka-ni ryokooniitta sooda.
Ken-TOP somewhere-to traveled I.hear
'I hear that Ken traveled to somewhere.'

b. Boku-wa [doko-ni (da) ka] {sinpaisiteiru / gironsita / kaisetusita}.
I-TOP where-to COP Q am.anxious / discussed / explained
'I {am anxious / discussed / explained} [to where].'
≈ 'I {am anxious / discussed / explained} [to where Ken traveled].'

Bearing these facts in mind, let us evaluate the FM/C analysis with regard to the current proposal that embedded CQ wh-questions involve covert wh-movement while embedded non-CQ wh-questions do not. The FM/C analysis is compatible with the acceptability of (11b); since the

[4] Note incidentally that (11b) is acceptable even with the matrix predicate being *sinpaisiteiru* 'is anxious'. In Chapter 3 (i.e. Section 3.2.2), I noted that if that predicate takes a cleft wh-question (or more generally, a copula wh-question) as its complement, the sentence is degraded. Given that, the acceptability of (11b) with *sinpaisiteiru* 'is anxious' would suggest that it is not the case that sluicing constructions are derived from clefts as in (7b). This is in fact compatible with what I will argue below.

sluicing construction is realized as an embedded non-CQ wh-question, the remnant wh-phrase no longer moves after it has undergone focus movement. However, the acceptability of (10b) (and (2b)) constitutes a problem for the FM/C analysis; given that the sluicing construction in (10b) is an embedded CQ wh-question, the remnant wh-phrase must undergo wh-movement after it has moved to Spec,FocP, which should degrade the sentence due to a freezing effect. This consideration thus suggests that an alternative analysis is required for Japanese sluicing.

4.2.2 In-situ analysis

To explore an alternative analysis, I here introduce another analysis of sluicing constructions (and other elliptical constructions), which will be called the *in-situ analysis* (e.g. Kimura 2010, Kimura and Takahashi 2011, Abe 2015, 2016, Kimura and Natira 2021). This analysis crucially differs from the FM/C analysis in that it does not assume any movement of the remnant. In this section, for expository purposes, I implement the in-situ analysis as follows: (i) sluicing constructions originate from in-situ focus constructions (Kimura and Takahashi 2011), (ii) a focused element has the feature [Focus] (Abe 2015, 2016), and (iii) the deletion applies to FinP except for the element with [Focus], which then becomes the remnant. With this analysis, the sluicing construction in (2b), repeated below, has (12) as its underlying structure; note in particular that the remnant wh-phrase does not undergo movement.

(2) a. Ken-wa dokoka-ni ryokooniitta sooda.
Ken-TOP somewhere-to traveled I.hear
'I hear that Ken traveled to somewhere.'

b. (Demo,) boku-wa [<u>doko-ni (da) ka</u>] siranai.
but I-TOP where-to COP Q know.not
'(But) I don't know [to where].'

(12) [$_{\text{ForceP}}$ [$_{\text{FocP}}$ ~~[$_{\text{FinP}}$ [$_{\text{TP}}$ Ken-ga~~ doko-ni$_{\text{[Focus]}}$ ~~ryokooniitta] no]~~ (da)] ka]
Ken-NOM where-to traveled C COP Q
'to where (≈ to where Ken traveled)'

Crucially, the in-situ analysis can resolve the above-mentioned issue for the FM/C analysis regarding the acceptability of the sluicing construction in (10b). Since the sluicing construction in (10b) is realized as an embedded CQ wh-question, it should involve covert wh-movement, according to the current proposal. With the FM/C analysis, the remnant wh-phrase in (10b) would have to undergo covert wh-movement after it has undergone focus movement, thus inducing a freezing effect. The FM/C analysis is thus incompatible with the acceptability of (10b), as noted above. With the in-situ analysis, however, the remnant wh-phrase has not undergone any movement before it undergoes covert wh-movement, and thus it is correctly predicted that no freezing effect should arise. The interim conclusion, therefore, is that the in-situ analysis fares better than the FM/C analysis, at least so far as the acceptability of (10b) is concerned.

Let us then investigate the validity of the in-situ analysis from other perspectives. There are two arguments for this analysis. First, as is the case with the FM/C analysis, it can capture the (optional) presence of the copula in sluicing constructions, since it assumes in-situ focus constructions, which also contain the copula, as their underlying structure (e.g. (12)). The second argument is more crucial; the in-situ analysis straightforwardly captures the observation that sluicing constructions in Japanese can involve immobile elements as their remnants (Kimura and Takahashi 2011). For instance, Kimura and Takahashi (2011) observe that remnants of sluicing constructions can be small clause predicates (SCPs, henceforth). In (13b), e.g., the sluicing construction contains the SCP *(dorekurai) tuyoku* '(how) strong' as its remnant, due to the fact that the antecedent clause in (13a) involves a small clause (i.e. SC in (13a)).

(13) a. Ken-wa [SC gakusee-o sake-ni (aruteedo) tuyoku] si-tai sooda.
Ken-TOP student-ACC liquor-DAT to.some.degree strong make-want I.hear
Lit. 'I hear that Ken wants to make his students strong for liquor (i.e. able to hold their liquor) (to some degree).'

b. (Demo) boku-wa [dorekurai tuyoku (da) ka] siranai.
but I-TOP how strong COP Q know.not
Lit. '(But) I don't know [how strong].'
≈ Lit. '(But) I don't know [how strong he wants to make his students for liquor].'

(based on Kimura and Takahashi 2011: 152)

Crucially, SCPs are incapable of undergoing movement. Consider the SCP *(kanari) tuyoku* '(quite) strong' in (14a), for example. It cannot undergo scrambling (Kikuchi and Takahashi 1991), as shown in (14b). Furthermore, it cannot be clefted (Kimura and Takahashi 2011), as (14c) shows. This in particular indicates that SCPs cannot undergo focus movement.

(14) a. Ken-wa [SC gakusee-o sake-ni (kanari) tuyoku] sita.
Ken-TOP student-ACC liquor-DAT quite strong made
Lit. 'Ken made his students (quite) strong for liquor (i.e. able to hold their liquor (quite well)).'

b. * (Kanari) tuyoku$_i$ Ken-wa [SC gakusee-o sake-ni *t*$_i$] sita.
quite strong Ken-TOP student-ACC liquor-DAT made

c. *[Ken-ga [SC gakusee-o sake-ni *e*$_i$] sita no]-wa (kanari) tuyoku$_i$ da.
Ken-NOM student-ACC liquor-DAT made C-TOP quite strong COP
Lit. 'It was (quite) strong that Ken made his students for liquor.'

The grammaticality of (13b) thus indicates that sluicing can involve an immobile element. To establish this fact more firmly, I now add two more immobile elements: complement clauses in ECM (i.e. Exceptional Case Marking) constructions and genitive phrases; they are instantiated by the underlined phrases in (15a) and (16a), respectively. That they are immobile elements can be confirmed by the fact that they cannot be scrambled (15/16b) and they cannot be clefted (15/16c).

(15) a. Ken-wa Mai-o$_i$ [*e*$_i$ (kanari) kasikoi to] omotta.
Ken-TOP Mai-ACC quite smart REP thought
'Ken thought Mai to be (quite) smart.'

b. *[*e*$_i$ (kanari) kasikoi to]$_j$ Ken-wa Mai-o$_i$ *t*$_j$ omotta.
quite smart REP Ken-TOP Mai-ACC thought (Kuno 1976)

c. *[Ken-ga Mai-o$_i$ *e*$_j$ omotta no]-wa [*e*$_i$ (kanari) kasikoi to]$_j$ da.
Ken-NOM Mai-ACC thought C-TOP quite smart REP COP
Lit. 'It was to be (quite) smart that Ken thought Mai.'

(16) a. Ken-wa [Mai-no hahaoya]-o sagasi-tei-ru.
Ken-TOP Mai-GEN mother-ACC look.for-PROG-PRES
'Ken is looking for Mai's mother.'

b. *Mai-no$_i$ Ken-wa [*t*$_i$ hahaoya]-o sagasi-tei-ru.
Mai-GEN Ken-TOP mother-ACC look.for-PROG-PRES

c. *[Ken-ga [*e*$_i$ hahaoya]-o sagasi-tei-ru no]-wa Mai-no$_i$ da.
Ken-NOM mother-ACC look.for-PROG-PRES C-TOP Mai-GEN COP
Lit. 'It was Mai's$_i$ that Ken is looking for [*e*$_i$ mother].'

Crucially, sluicing constructions can contain those elements as their remnants, as (17/18b) show.

(17) a. Ken-wa Mai-o$_i$ [*e*$_i$ (aruteedo) kasikoi to] omotta sooda.
Ken-TOP Mai-ACC to.some.degree smart REP thought I.hear
'I hear that Ken thought Mai to be smart (to some degree)'

b. (Demo) boku-wa [[*e*$_i$ dorekurai kasikoi to] (da) ka] siranai.
but I-TOP how smart REP COP Q know.not
Lit. '(But) I don't know [to be how smart].'

≈ ‘(But) I don’t know [how smart Ken though Mai to be].’

(18) a. Ken-wa [dareka-no hahaoya]-o sagasi-tei-ru sooda.
Ken-TOP someone-GEN mother-ACC look.for-PROG-PRES I.hear
‘I hear that Ken is looking for someone’s mother.’

b. (Demo) boku-wa [dare-no (da) ka] siranai.
but I-TOP who-GEN COP Q know.not
Lit. ‘(But) I don’t know [whose].’
≈ ‘(But) I don’t know [whose mother Ken is looking for].’

One note regarding (18b) is in order here. It is important to confirm that the remnant phrase *dare-no* in (18b) is not a pronominal phrase headed by the light noun *no* ‘one’. To appreciate the importance of this, observe (19).

(19) a. Ken-wa [akai kaban]-o katta. Mai-wa [aoi no]-o katta.
Ken-TOP red bag-ACC bought Mai-TOP blue one-ACC bought
‘Ken bought a red bag. Mai bought a blue one.’

b. Ken-wa [Kooji-no kaban]-o karita. Mai-wa [Eri{*-no no / no}]-o karita.
Ken-TOP Koji-GEN bag-ACC borrowed Mai-TOP Eri-GEN one NO-ACC borrowed
‘Ken borrowed Koji’s bag. Mai borrowed Eri’s one.’

In (19a), the light noun *no* ‘one’ appears after the prenominal adjective *aoi* ‘blue’ in place of the same noun as the antecedent *kaban* ‘bag’. If the same strategy is to be used for (19b), the light noun *no* ‘one’ will occur after the genitive Case -*no*, resulting in the sequence of two homophonous morphemes (i.e. *no no*). This sequence is not allowed; instead, only one of the two morphemes *no* appears, as shown in (19b). I here follow Hiraiwa (2016) and assume that in such a case, either the genitive Case -*no* or the light noun *no* undergoes haplology (i.e., is deleted in PF due to the illicit surface form where the two homophonous elements *no* are adjacent to each other), with both elements being syntactically present (e.g. Kuno 1973, Okutsu 1974, Kamio 1983); see his paper for arguments against the NP-deletion analysis for this case. Given that, one might suspect that the remnant wh-phrase in (18b), *dare-no*, is not a genuine genitive phase but involves the light noun *no* 'one' substituted for *hahaoya* 'mother', with one of the two homophonous morphemes *no* undergoing haplology. Of relevance in examining this issue is one unique property of the light noun *no* 'one': when it is used to refer to a person, it necessarily yields a derogative reading (e.g. McGloin 1985). In (20a-b), e.g., the use of the light noun would suggest that the speaker is looking down on Mai's teacher and Eri's mother respectively.

(20) a. Ken-wa [yasasii <u>sensee</u>]-ni osowatta. Mai-wa [mazimena {#<u>no</u> / sensee}-ni
Ken-TOP kind teacher-DAT was.taught Mai-TOP diligent one teacher-DAT
osowatta.
was.taught
'Ken was taught by a kind teacher. Mai was taught by a diligent {one / teacher}.'

b. Ken-wa [Kooji-no <u>hahaoya</u>]-ni atta. Mai-wa [{#Eri <u>no</u> / Eri-no hahaoya}]-ni
Ken-TOP Koji-GEN mother-DAT met Mai-TOP Eri NO Eri-GEN mother-DAT
atta.
met
'Ken met Koji's mother. Mai met Eri's {one / mother}.'

Now, notice crucially that the remnant wh-phrase in (18b) *dare-no* does not give rise to a derogative reading. This then indicates that the remnant wh-phrase under consideration is a pure genitive phrase, rather than a pronominal phrase involving the light noun *no*.

The above observations thus further confirm that immobile elements can occur in sluicing constructions. The in-situ analysis is clearly consistent with this fact; remnants in sluicing constructions can be immobile elements because they need not move in deriving those constructions. In contrast, the immobility of remnants in sluicing constructions is obviously incompatible with the FM/C analysis, since this analysis requires them to undergo (focus) movement. The in-situ analysis thus fares better than FM/C analysis in this respect as well.

However, a serious problem for the in-situ analysis is the island sensitivity of sluicing constructions, which is illustrated by (9b), repeated below.

(9) a. Ken-wa [[e_i <u>dokoka-ni</u> ryokooniitta] hito$_i$]-ni atta sooda.
Ken-TOP somewhere-to traveled person-DAT met I.hear
'I hear that Ken met a person who traveled to somewhere.'

b. # (Demo,) boku-wa [doko-ni (da) ka](-wa) siranai.
but I-TOP where-to COP Q-TOP know.not
'(But) I don't know [to where].'
≉ Lit. '(But) I don't know [to where Ken met [a person who traveled (*t*)]].'

With the in-situ analysis, the remnant phrase does not undergo any movement and is thus predicted to be insensitive to islands, contrary to fact. I therefore conclude that the in-situ analysis cannot be adopted for Japanese sluicing constructions at its face value.

To recap, this section has re-examined two representative analyses for sluicing constructions in Japanese, with the current proposal concerning the (non-)involvement of covert wh-movement in Japanese wh-questions taken into consideration. Concerning the FM/C analysis, while it is advantageous in capturing the island sensitivity of sluicing, it is problematic regarding the acceptability of sluicing constructions formed as embedded CQ wh-questions (see, e.g., (10b)) and the immobility of their remnants. Concerning the in-situ analysis, while it resolves those two issues for the FM/C analysis, it is incompatible with the island sensitivity of sluicing constructions. These considerations thus lead to the conclusion that a new analysis is required for Japanese sluicing constructions.

Against this backdrop, the rest of this chapter aims to provide a new analysis of Japanese sluicing. Notice here that the above discussion has alluded to a direction for that goal; the successful analysis of Japanese sluicing constructions should be the one where (i) movement is involved but (ii) that movement does not apply to the remnants of sluicing. The first point is indicated by their island sensitivity. The second point is suggested by the acceptability of sluicing constructions formed as embedded CQ wh-questions as in (10b), repeated below; if the movement under consideration applied to the remnant in (10b), its subsequent covert wh-movement would induce a freezing effect.[5]

(10) a. Ken-wa dokoka-ni ryokooniitta sooda.
Ken-TOP somewhere-to traveled I.hear
'I hear that Ken traveled to somewhere.'

b. Boku-wa [doko-ni (da) ka] {sitteiru / akasita / tazuneta}.
I-TOP where-to COP Q know / revealed / asked
'I {know / revealed / asked} [to where].'
≈ 'I {know / revealed / asked} [to where Ken traveled].'

At this point, one might wonder if the movement involved in sluicing constructions could be the covert wh-movement itself. If this is the case, it is predicted that island sensitivity will not be observed with sluicing constructions formed as embedded non-CQ wh-questions, since they should not involve covert wh-movement according to the current proposal. To examine this prediction, consider (9) above again. The sluicing construction in (9b) is formed as an embedded CQ wh-question and thus involves covert wh-movement. Under the hypothesis suggested above,

[5] One might claim that the immobility of remnants in sluicing constructions also indicates the second point. With this property, however, we cannot exclude the possibility that some covert movement (distinct from covert wh-movement) applies to those remnants; while this possibility would be compatible with their immobility, it would be incompatible with the acceptability of sluicing constructions realized as embedded CQ wh-questions (e.g. (10b)), since it would be incorrectly predicted that the subsequent covert wh-movement would induce a freezing effect.

it is expected that if the matrix predicate in (9b) is replaced with a non-CQ predicate and thus the sluicing construction is rendered an embedded non-CQ wh-question, (9b) will be grammatical with no island effect. This expectation is not borne out, as (21) shows.

(21) *After (9a),*
 #Boku-wa [doko-ni (da) ka](-wa) {sinpaisiteiru / gironsita / kaisetusita}.
 I-TOP where-to COP Q-TOP am.anxious / discussed / explained
 'I {am anxious / discussed / explained} [to where].'
 ≉ Lit. 'I {am anxious / discussed / explained} [to where Ken met [a person who traveled (*t*)]].'

Hence, the movement involved in sluicing constructions must be independent of covert wh-movement.

Now, in order to propose a new analysis of sluicing constructions following the direction shown above, we need to answer two questions: (i) What induces the movement involved in sluicing? (ii) What kind of movement is it? The following two sections will explore those two questions respectively.

4.3 Driving force for the movement

This section aims to identify the driving force for the movement involved in Japanese sluicing constructions. To achieve this goal, I will proceed with the following two steps: First, I will attempt to find other elliptical constructions that have one of the following traits: (i) those that are similar to sluicing with respect to both the immobility of their remnants and their island sensitivity, and (ii) those that are similar to sluicing with respect to the former property but differ from it regarding the latter. Second, after identifying such elliptical constructions, I will compare the elliptical constructions characterized by (i) and those characterized by (ii) and establish an independent property that distinguishes them. Given that those two types of constructions syntactically differ only in their island (in)sensitivity, that independent property should be related to the movement in sluicing (and other similar constructions). Section 4.3.1 and 4.3.2 will be devoted to each of these two steps.

4.3.1 Other elliptical constructions in Japanese

Here, I introduce three other elliptical constructions, namely ka(dooka)-*stripping* (Saito 2004, Hiraiwa and Ishihara 2012, Abe 2016, among others), to-*stripping* (e.g. Fukaya and Hoji 1999, Fukaya 2003, Hiraiwa and Ishihara 2012), and *fragment answers* (Nishigauchi and Fujii 2006, Abe 2016, Kimura and Narita 2021, among others). They are exemplified in (22-24), respectively.

(22) *Ka*(*dooka*)-stripping

a. Ken-wa dokoka-ni ryokooniitta sooda.
Ken-TOP somewhere-to traveled I.hear
'I hear that Ken traveled to somewhere.'

b. (Demo,) boku-wa [<u>Igirisu-ni (da) ka(dooka)</u>] siranai.
but I-TOP England-to COP whether know.not
Lit. '(But) I don't know [whether to England].'

(23) *To*-stripping

After (22a),

Boku-wa [<u>Igirisu-ni (da) to</u>] omotteiru.
I-TOP England-to COP REP think
Lit. 'I think [that to England].'

(24) Fragment answers

A: Ken-wa doko-ni ryokooniitta no.
Ken-TOP where-to traveled Q
'Where did Ken travel to?'

B: Igirisu-ni desu.
England-to COP.POL
'To England.'

The underlined elliptical clauses in (22b), (23) and (24B) are interpreted as in (25a-c) respectively.

(25) a. *After (22a),*
(Demo,) boku-wa [Ken-ga Igirisu-ni ryokooniitta ka(dooka)] siranai.
but I-TOP Ken-NOM England-to traveled whether know.not
'(But) I don't know [whether Ken traveled to England].'

b. *After (22a),*
Boku-wa [Ken-ga Igirisu-ni ryokooniitta to] omotteiru.
I-TOP Ken-NOM England-to traveled REP think
'I think [that Ken traveled to England].'

c. *After (24A),*
B: Ken-wa Igirisu-ni ryokooniiki-masi-ta.
Ken-TOP England-to traveled-POL-PAST
'Ken traveled to England.'

Note that these constructions are similar to sluicing in their appearances, in that they involve remnants (which are non-wh-phrases, differently from sluicing) and the copula (whether optionally or obligatorily).

In the following discussion, I will investigate the aforementioned two syntactic properties regarding these three elliptical constructions in order to determine how close they are to sluicing with respect to these properties.

4.3.1.1 Immobility of remnants

Consider first the (im)mobility of the remnants. In Section 4.2.2, I introduced three immobile elements: (i) SCPs (i.e. small clause predicates), (ii) complement clauses of ECM constructions and (iii) genitive phrases. Importantly, the following data show that all these immobile elements can appear as remnants of the three elliptical constructions under consideration.

<u>(i) SCPs</u>

(26) *Ka*(*dooka*)-stripping

a. Ken-wa [$_{SC}$ gakusee-o sake-ni <u>(aruteedo) tuyoku</u>] si-tai sooda.
Ken-TOP student-ACC liquor-DAT to.some.degree strong make-want I.hear
Lit. 'I hear that Ken wants to make his students strong for liquor (i.e. able to hold their liquor) (to some degree).'

b. (Demo) boku-wa [<u>kanari tuyoku</u> (da) ka(dooka)] siranai.
but I-TOP how strong COP whether know.not
Lit. '(But) I don't know [whether quite strong].'
≈ Lit. '(But) I don't know [whether he wants to make his students quite strong for liquor].'

(27) *To*-stripping

After (26a),

Boku-wa [<u>kanari tuyoku</u> (da) to] omotteiru.
I-TOP quite strong COP REP think

Lit. 'I think [that quite strong].'

≈ Lit. 'I think [that he wants to make his students quite strong for liquor].'

(28) Fragment answers

A: Ken-wa [$_{SC}$ gakusee-o sake-ni <u>dorekurai tuyoku</u>] si-tai no.

Ken-TOP student-ACC liquor-DAT how strong make-want Q

Lit. 'How strong does Ken want to make his students for liquor?'

B: <u>Kanari tuyoku</u> desu.

quite strong COP.POL

'Quite strong.'

≈ Lit. 'Ken wants to make his students quite strong for liquor.'

<u>(ii) Complement clauses of ECM constructions</u>

(29) *Ka*(*dooka*)-stripping

a. Ken-wa Mai-o$_i$ <u>[*e*$_i$ (aruteedo) kasikoi to]</u> omotta sooda.

Ken-TOP Mai-ACC to.some.degree smart REP thought I.hear

'I hear that Ken thought Mai to be smart (to some degree)'

b. (Demo) boku-wa [<u>[*e*$_i$ kanari kasikoi to]</u> (da) ka(dooka)] siranai.

but I-TOP quite smart REP COP whether know.not

Lit. '(But) I don't know [whether to be quite smart].'

≈ '(But) I don't know [whether Ken thought Mai to be quite smart].'

(30) *To*-stripping

After (29a),

Boku-wa [<u>[*e*$_i$ kanari kasikoi to]</u> (da) to] omotteiru.

I-TOP quite smart REP COP REP think

Lit. 'I think [that to be quite smart].'

≈ 'I think [that Ken thought Mai to be quite smart].'

(31) Fragment answers

A: Ken-wa Mai-o$_i$ [e_i dorekurai kasikoi to] omotta no.
Ken-TOP Mai-ACC how smart REP thought Q
'How smart did Ken think Mai to be?'

B: [e_i kanari kasikoi to] desu.
quite smart REP COP.POL
'To be quite smart.'
≈ 'Ken thought Mai to be quite smart.'

(iii) Genitive phrases[6]

(32) *Ka*(*dooka*)-stripping

a. Ken-wa [dareka-no hahaoya]-o sagasi-tei-ru sooda.
Ken-TOP someone-GEN mother-ACC look.for-PROG-PRES I.hear
'I hear that Ken is looking for someone's mother.'

b. (Demo) boku-wa [Mai-no (da) ka(dooka)] siranai.
but I-TOP Mai-GEN COP whether know.not
Lit. '(But) I don't know [whether Mai's].'
≈ '(But) I don't know [whether Ken is looking for Mai's mother].'

(33) *To*-stripping

After (32a),

Boku-wa [Mai-no (da) to] omotteiru.
I-TOP Mai-GEN COP REP think
Lit. '(But) I think [that Mai's].'
≈ '(But) I think [that Ken is looking for Mai's mother].'

[6] Note that the remnant phrases in (32b), (33) and (34B), which contain *no*, do not yield derogative readings, which indicates that they are not pronominal phrases headed by the light noun *no* 'one' but genuine genitive phrases, as is the case with sluicing constructions (see (18b)).

(34) Fragment answers

A: Ken-wa [<u>dare-no</u> hahaoya]-o sagasi-tei-ru no.
Ken-TOP who-GEN mother-ACC look.for-PROG-PRES Q
'Whose mother is Ken looking for?'

B: <u>Mai-no</u> desu.
Mai-GEN COP.POL
'Mai's.'
≈ 'Ken is looking for Mai's mother.'

The above observations thus indicate that the three elliptical constructions are similar in that their remnants can be immobile elements.

4.3.1.2 Island (in)sensitivity

Let us next consider the second property, namely island (in)sensitivity. We will first discuss *ka*(*dooka*)/*to*-stripping. Specifically, adding to Fukaya and Hoji's (1999) observation that *to*-stripping is sensitive to islands, I will show that *ka*(*dooka*)-stripping is island-sensitive as well. Note first that the correlate of these constructions can appear in declarative complement clauses (at least marginally), as shown in (35-36).

(35) *Ka*(*dooka*)-stripping

a. Mai-wa [Ken-ga <u>dokoka</u>-ni ryokooniitta to] syutyoosi-tei-ru sooda.
Mai-TOP Ken-NOM somewhere-to travelled REP claim-PROG-PRES I.hear
'I hear that Mai is claiming that Ken traveled to somewhere.'

b. (?) (Demo,) boku-wa [Igirisu-ni (da) ka(dooka)](-wa) siranai.
but I-TOP England-to COP whether-TOP know.not
Lit. '(But) I don't know [whether to England].'
≈ '(But) I don't know [whether Mai is claiming that Ken traveled to England].'

(36) *To*-stripping

After (35a),

(?)Boku-wa [Igirisu-ni (da) to] omotteiru.
I-TOP England-to COP REP think
Lit. 'I think [that to England].'
≈ 'I think [that Mai is claiming that Ken traveled to England].'

However, the two constructions cannot be formed if their correlates appear within islands, as (37-38) show.

(37) *Ka*(*dooka*)-stripping

a. Ken-wa [[*e*$_i$ <u>dokoka-ni</u> ryokooniitta] hito$_i$]-ni atta sooda.
Ken-TOP somewhere-to traveled person-DAT met I.hear
'I hear that Ken met a person who traveled to somewhere.'

b. # (Demo,) boku-wa [Igirisu-ni (da) ka(dooka)](-wa) siranai.
but I-TOP England-to COP whether-TOP know.not
'(But) I don't know [whether to England].'
≉ '(But) I don't know [whether Ken met [a person who traveled to England]].'

(38) *To*-stripping

After (37a),

#Boku-wa [Igirisu-ni (da) to] omotteiru.
I-TOP England-to COP REP think
'I think [that to England].'
≉ 'I think [that Ken met [a person who traveled to England]].'

The data presented above thus indicate that the two constructions under consideration are sensitive to islands.

In contrast, fragment answers are known to be insensitive to islands (e.g. Nishigauchi and Fujii 2006, Abe 2016); see (39), for example.

(39) A: Ken-wa [$_{\text{island}}$ e_i doko-ni ryokooniitta hito$_i$]-ni atta no.
Ken-TOP where-to traveled person-DAT met Q
Lit. 'Ken met a person who traveled to where?'
(≈ 'What is the place such that Ken met a person who traveled there?')

B: Igirisu-ni desu.
England-to COP.POL
'To England.'
≈ 'Ken met [a person who traveled to England].'

In (39), the correlate in the antecedent clause (i.e. the wh-phrase in the preceding question (39A)) appears within the island. Despite that, the fragment answer in (39B) is acceptable. This thus indicates that fragment answers are island insensitive.

It can then be concluded that the three elliptical constructions under consideration are different with regard to island sensitivity; *ka*(*dooka*)/*to*-stripping are sensitive to islands, while fragment answers are not.

4.3.1.3 Summary: Comparison with sluicing

To summarize, I have discussed above two syntactic properties of the three elliptical constructions under consideration. First, they are similar in that all of them can involve immobile elements as their remnants. Second, they differ regarding island sensitivity; *ka*(*dooka*)/*to*-stripping constructions are island-sensitive, while fragment answers are not. Recall here that sluicing constructions can contain immobile elements as their remnants and are sensitive to islands, as illustrated in Section 4.2. Given that, it can be concluded that *ka*(*dooka*)/*to*-stripping constructions parallel sluicing with respect to both syntactic properties under consideration, while fragment answers minimally differ from the other elliptical constructions under consideration regarding island (in)sensitivity, being insensitive to islands. This conclusion thus provides a direction for the pursuit of the driving force of the movement involved in sluicing; it should be related to some property of sluicing that is shared with *ka*(*dooka*)/*to*-stripping but not with fragment answers.

Before embarking on the exploration of that property, I note that the two syntactic properties of fragment answers, i.e. the immobility of their remnants and their island insensitivity, are both compatible with the in-situ analysis (see Section 4.2.2). I thus conclude that the in-situ analysis fares well with fragment answers, as argued by Abe (2016) and Kimura and Narita (2021), but *contra* Nishigauchi and Fujii (2006).

4.3.2 Exhaustivity

Based on the direction suggested above, this subsection aims to identify the property that distinguishes sluicing and *ka*(*dooka*)/*to*-stripping from fragment answers. I will show that the property in question concerns exhaustive interpretations. Below, I will investigate the exhaustivity in each of the elliptical constructions under consideration.

4.3.2.1 Exhaustivity in sluicing and *ka*(*dooka*)/*to*-stripping

Let us begin with sluicing. As a baseline, consider first the sequence of three utterances in (40), which involves a non-elliptical embedded wh-question in (40b).

(40) a. Ken-wa mit-tu-no kuni-ni ryokooniitta sooda.
Ken-TOP 3-CLS-GEN country-to traveled I.hear
'I hear that Ken traveled to three countries.'

b. Mai-wa [Ken-ga {doko / dono kuni}-ni ryokooniitta ka] boku-ni
Mai-TOP Ken-NOM where which country-to traveled Q I-to
osie-tekure-ta.
teach-for.me-PAST
'Mai told me [{where/which countries} Ken traveled to].'

c. Zissai, Mai-wa ni-kakoku-sika sira-nakat-ta ndakedo.
actually Mai-TOP 2-CLS-SIKA know-not-PAST though
'Actually, she knew only two countries, though.'

(40c) suggests that Mai told the speaker about only two countries that Ken traveled to, not all the three countries. The felicity of (40c), then, indicates that non-elided embedded wh-question, as in (40b), are compatible with non-exhaustive interpretations.

With this in mind, observe now (41), which differs from (40) only in that (41b) involves a sluicing construction that corresponds to the embedded wh-question in (40b).

(41) a. Ken-wa mit-tu-no kuni-ni ryokooniitta sooda.
Ken-TOP 3-CLS-GEN country-to traveled I.hear
'I hear that Ken traveled to three countries.'

b. Mai-wa [{doko / dono kuni}-ni (da) ka] boku-ni osie-tekure-ta.
Mai-TOP where which country-to COP Q I-to teach-for.me-PAST
'Mai told me [to {where / which countries}].'
≈ 'Mai told me [to {where / which countries} Ken traveled].'

c. #Zissai, Mai-wa ni-kakoku-sika sira-nakat-ta ndakedo.
actually Mai-TOP 2-CLS-SIKA know-not-PAST though
'Actually, she knew only two countries, though.'

Crucially, in the above sequence of utterances, (41c) is infelicitous, as opposed to (40c). This suggests that the sluicing construction in (41b) is incompatible with a non-exhaustive interpretation. In other words, this suggests that sluicing constructions must yield exhaustive interpretations, unlike non-elliptical embedded wh-questions.

Let us next turn to *ka*(*dooka*)-stripping. Observe first the sequence of utterances in (42), which contains a normal *ka*(*dooka*)-question, or *whether*-question, in (42b).

(42) a. Ken-wa mit-tu-no kuni-ni ryokooniitta sooda.
Ken-TOP 3-CLS-GEN country-to traveled I.hear
'I hear that Ken traveled to three countries.'

b. (Demo) boku-wa [Ken-ga {Igirisu-to Furansu-to Supein-ni / Igirisu-ni}
but I-TOP Ken-NOM England-and France-and Spain-to England-to
ryokooniitta ka(dooka)](-wa) siranai.
traveled whether-TOP know.not
'(But) I don't know [whether Ken traveled {to England, France and Spain / to England}].'

(42b) shows that the non-elliptical *ka*(*dooka*)-question can refer to only one of the three countries that Ken traveled to, as well as all those three countries. This fact indicates that non-elliptical *ka*(*dooka*)-clauses are not characterized by exhaustivity.

Let us now observe (43), where the complement clause in (42b) is replaced with the corresponding *ka*(*dooka*)-stripping construction.

(43) *After (42a),*

(Demo) boku-wa [{Igirisu-to Furansu-to Supein-ni / #Igirisu-ni} (da)
but I-TOP England-and France-and Spain-to England-to COP
ka(dooka)](-wa) siranai.
whether-TOP know.not
Lit. '(But) I don't know [whether {to England, France and Spain / to England}].'
≈ '(But) I don't know [whether Ken traveled {to England, France and Spain / to England}].'

Crucially, (43) shows that the *ka*(*dooka*)-stripping construction cannot refer to only one of the countries that Ken traveled to; rather, it must refer to all those three countries. This thus indicates that *ka*(*dooka*)-stripping constructions involve exhaustivity.

The similar diagnostics can be applied to *to*-stripping. Consider first examples in (44) as a baseline.

(44) *After (42a),*

a. Boku-wa [Ken-ga Igirisu-to Furansu-to Supein-ni ryokooniitta to]
   I-TOP Ken-NOM England-and France-and Spain-to traveled REP
   omotteiru.
   think
   'I think [that Ken traveled to England, France and Spain].'

b. (Sukunakutomo) boku-wa [Ken-ga Igirisu-ni ryokooniitta to] omotteiru.
   at.least I-TOP Ken-NOM England-to traveled REP think
   '(At least,) I think [that Ken traveled to England].'

Note that (44b) is felicitous even though the *to*-clause (i.e. the complement clause headed by the reportative complementizer *to*) refers to only one of the three countries that Ken traveled to. This indicates the non-exhaustivity of non-elliptical *to*-clauses.

Now, compare (45) with (44), which involves the *to*-stripping constructions corresponding to the *to*-clauses in (44).

(45) *After (42a),*

a. Boku-wa [Igirisu-to Furansu-to Supein-ni (da) to] omotteiru
   I-TOP England-and France-and Spain-to COP REP think
   Lit. 'I think [that to England, France and Spain].'
   ≈ 'I think [that Ken traveled to England, France and Spain].'

b. # (Sukunakutomo) boku-wa [Igirisu-ni (da) to] omotteiru.
at.least I-TOP England-to COP REP think
Lit. '(At least,) I think [that to England].'
≈ '(At least,) I think [that Ken traveled to England].'

(45) shows that the *to*-stripping construction must refer to all three countries that Ken traveled to, disallowing only one of them to be mentioned. This thus suggests that *to*-stripping constructions must be interpreted exhaustively.

To recap, the above observations have revealed that the three constructions under consideration, namely sluicing, *ka*(*dooka*)-stripping and *to*-stripping, are all characterized by the involvement of exhaustivity.

4.3.2.2 Non-exhaustivity in fragment answers

Then, what about the remaining elliptical construction, i.e. fragment answers? To examine their exhaustivity, consider the conversation in (46), where (46B) involves a non-fragment and a fragment answer.

(46) A: Ken-wa doko-ni ryokooniitta no.
Ken-TOP where-to traveled Q
'Where did Ken travel to?'

B: {Ken-wa Igirisu-ni ryokooniiki-masi-ta. / Igirisu-ni desu.} <u>Ato,</u>
Ken-TOP England-to travel-POL-PAST England-to COP.POL further
<u>Furansu-ni-mo ryokooniitta soodesu</u>.
France-to-also traveled I.hear.POL
'{Ken traveled to England. / To England.} Further, I hear that he also traveled to France.'

In (46B), the non-fragment and the fragment answer are both followed by a supplemental answer to the question (46A), as underlined there. If fragment answers involve exhaustivity, the fragment answer in (46B) must constitute a complete answer to the question (46A), hence it should be inconsistent with the subsequent supplemental answer. This, however, is not the case, as the felicity of (46B) shows. This then indicates that fragment answers can be characterized as being non-exhaustive, differing in this respect from the other elliptical constructions discussed above.

To recap, the above observations have revealed that the four elliptical constructions differ with respect to their exhaustivity; sluicing and *ka*(*dooka*)/*to*-stripping involve exhaustivity, while fragment answers do not. Notice crucially that this demarcation parallels the one regarding the island (in)sensitivity; sluicing and *ka*(*dooka*)/*to*-stripping are sensitive to islands, while fragment answers are not, as discussed in Section 4.3.1. Based on this parallel, I now argue that the driving force of the movement involved in sluicing (and *ka*(*dooka*)/*to*-stripping) constructions is their exhaustivity.

4.4 Null operator movement analysis

Having identified the driving force of the movement in sluicing, I now turn to the second question posed in Section 4.2: What kind of movement is it? Concerning this question, recall that it was shown in Section 4.2 that this movement should not apply to remnants in sluicing constructions, given the acceptability of sluicing constructions realized as embedded CQ wh-questions, as in (10b) repeated below; if the movement under consideration applied to the remnant wh-phrase in (10b), it would be wrongly predicted that a freezing effect would arise, given that embedded CQ wh-questions involve covert wh-movement, as shown in the last chapter.

(10) a. Ken-wa dokoka-ni ryokooniitta sooda.
Ken-TOP somewhere-to traveled I.hear
'I hear that Ken traveled to somewhere.'

b. Boku-wa [doko-ni (da) ka] {sitteiru / akasita / tazuneta}.
I-TOP where-to COP Q know / revealed / asked
'I {know / revealed / asked} [to where].'
≈ 'I {know / revealed / asked} [to where Ken traveled].'

Given this suggestion and the discussion in the last section, I now propose that sluicing constructions involve null operator movement that is associated with exhaustivity. (This proposal also applies to *ka*(*dooka*)/*to*-stripping constructions. However, the following discussion will focus on sluicing constructions.) More specifically, assuming that they have the structure of in-situ focus constructions as their base structure (see Section 4.2), I propose that a null operator with the feature [Exh(austive)] attaches to the relevant element (i.e. the wh-phrase) and undergoes movement to determine the scope of the exhaustive focus. I assume that the null operator moves to Spec,FocP, whose head is also assumed to have the feature [Exh]. Borrowing insights from the in-situ analysis (see Section 4.2.2), I further propose that after the null operator movement, the deletion applies to FinP except for the element that the operator originally attaches to. With this analysis, for example, the sluicing construction in (2b), repeated below, is derived through the steps shown in (47).

(2) a. Ken-wa dokoka-ni ryokooniitta sooda.
Ken-TOP somewhere-to traveled I.hear
'I hear that Ken traveled to somewhere.'

b. (Demo,) boku-wa [doko-ni (da) ka] siranai.
but I-TOP where-to COP Q know.not
'(But) I don't know [to where].'

(47) a. the structure of an in-situ focus construction, where the null operator $\text{Op}_{\text{[Exh]}}$ is attached to the wh-phrase

$[_{\text{FocP}}\ [_{\text{FinP}}\ [_{\text{TP}}$ Ken-ga [$\text{Op}_{\text{[Exh]}}$ doko-ni] ryokooniitta] no] $(\text{da})_{\text{[Exh]}}$] (ka)
Ken-NOM where-to traveled C COP Q

b. the movement of the null operator $\text{Op}_{\text{[Exh]}}$ to Spec,FocP

$[_{\text{FocP}}\ \text{Op}_{\text{[Exh]i}}\ [_{\text{FinP}}\ [_{\text{TP}}$ Ken-ga [t_i doko-ni] ryokooniitta] no] $(\text{da})_{\text{[Exh]}}$] (ka)
Ken-NOM where-to traveled C COP Q

c. the deletion of FinP except for the element that the null operator originally attaches to, i.e. the wh-phrase

$[_{\text{ForceP}}\ [_{\text{FocP}}\ \text{Op}_{\text{[Exh]i}}$ ~~[FinP [TP Ken-ga~~ [t_i doko-ni] ~~ryokooniitta] no]~~ $(\text{da})_{\text{[Exh]}}$] ka]
Ken-NOM where-to traveled C COP Q

Crucially, (47) shows that the remnant itself does not undergo any movement; what moves instead is the null operator that was base-generated attached to the remnant. The current proposal thus correctly predicts that covert wh-movement does not induce a freezing effect and the remnant can be an immobile element.

Let us further consider a case where an island is involved. A relevant example is repeated below:

(9) a. Ken-wa [[e_i <u>dokoka-ni</u> ryokooniitta] hito_i]-ni atta sooda.
Ken-TOP somewhere-to traveled person-DAT met I.hear
'I hear that Ken met a person who traveled to somewhere.'

b. # (Demo,) boku-wa [doko-ni (da) ka](-wa) siranai.
but I-TOP where-to COP Q-TOP know.not
'(But) I don't know [to where].'
≉ Lit. '(But) I don't know [to where Ken met [a person who traveled (t)]].'

With the current analysis, the sluicing construction in (9b) is derived as shown below:

(48) [$_{\text{ForceP}}$ [$_{\text{FocP}}$ Op$_{\text{[Exh]i}}$ ~~[$_{\text{FinP}}$ [$_{\text{TP}}$ Ken-ga [$_{\text{NP}}$ [e_{j}~~ [t_{i} doko-ni] ~~ryokooniitta] hito$_{\text{j}}$]-ni~~
Ken-NOM where-to traveled person-DAT
~~atta] no]~~ (da)$_{\text{[Exh]}}$] ka]
met C COP Q

Note in particular that in (48), the null operator is base-generated attached to an element that is located within the island. Consequently, the null operator movement crosses an island, thus inducing an island effect. The island-sensitivity of sluicing thus follows from the proposed analysis.

Furthermore, the null operator movement analysis captures the fact that (9b) is rendered acceptable if the sluicing construction contains as its remnant the whole island that contains the wh-phrase (i.e. the complex NP), rather than only that wh-phrase, as (49b) shows (see also Takahashi 1994: fn.7).

(49) a. Ken-wa [[e_{i} <u>dokoka-ni</u> ryokooniitta] hito$_{\text{i}}$]-ni atta sooda.
Ken-TOP somewhere-to traveled person-DAT met I.hear
'I hear that Ken met a person who traveled to somewhere.'

b. (Demo,) boku-wa [[[e_{i} <u>doko-ni</u> ryokooniitta] hito$_{\text{i}}$]-ni (da) ka] siranai.
but I-TOP where-to traveled person-DAT COP Q know.not
Lit. '(But) I don't know [[a person who traveled to where]].'
≈ Lit. '(But) I don't know [to where Ken met [a person who traveled (t)]].'

Under the current analysis, the sluicing construction in (49b) is derived as in (50).

(50) [$_{\text{ForceP}}$ [$_{\text{FocP}}$ Op$_{\text{[Exh]i}}$ ~~[$_{\text{FinP}}$ [$_{\text{TP}}$ Ken-ga~~ [t_{i} [$_{\text{NP}}$ [e_{j} doko-ni ryokooniitta] hito$_{\text{j}}$]]-ni
Ken-NOM where-to traveled person-DAT
~~atta] no]~~ (da)$_{\text{[Exh]}}$] ka]
met C COP Q

Notice crucially that in (50), the null operator is base-generated attached to the complex NP, rather than the wh-phrase within the complex NP; this is why the complex NP remains unelided in (49b). Its movement hence does not cross an island, thus giving rise to no island effect in (49b).

4.5 Summary of the chapter

As one consequence of the proposal presented in Chapter 3 regarding the (non-)involvement of covert wh-movement in Japanese wh-questions, this chapter has discussed how to analyze sluicing in Japanese. I started by discussing two representative analyses of Japanese sluicing, namely the FM/C (i.e. focus movement/cleft) analysis and the in-situ analysis, in terms of the proposal in question. It was first pointed out that the FM/C analysis, which assumes focus movement of the remnant of sluicing, makes an incorrect prediction regarding the acceptability of sluicing constructions formed as embedded CQ wh-questions: given that embedded CQ wh-questions involve covert wh-movement, as shown in the previous chapter, the FM/C analysis wrongly predicts such sluicing constructions to give rise to freezing effects, with covert wh-movement being applied to the remnant wh-phrase that has undergone focus movement. Then, as a potential alternative analysis for sluicing constructions, I introduced the in-situ analysis, where the remnant wh-phrase stays in-situ without undergoing any movement. I have shown that this analysis resolves the problem for the FM/C analysis mentioned above; it correctly predicts sluicing constructions formed as embedded CQ wh-questions to induce no freezing effect, since the remnant wh-phrase has not undergone any movement before covert wh-movement applies to it. The in-situ analysis is also advantageous in that it can capture the immobility of remnants in sluicing, i.e. the fact that sluicing constructions can contain as their remnants items that cannot be moved, such as small clause predicates, complement clauses of ECM constructions, and genitive phrases. However, the in-situ analysis is incompatible with one argument for the FM/C analysis, namely the island sensitivity of sluicing constructions, i.e. the fact that the correlate cannot appear in an island. This consideration has thus revealed that a new analysis is in order for Japanese sluicing constructions. It also suggested that the analysis must involve movement which does not apply to their remnants, given (i) their island sensitivity and (ii) the fact that sluicing constructions formed as embedded CQ wh-questions are grammatical, with no freezing effect induced.

To propose such an analysis, I considered three other elliptical constructions in Japanese, i.e. *ka*(*dooka*)-stripping, *to*-stripping and fragment answers, and compared them with sluicing constructions. The comparison has revealed that *ka*(*dooka*)/*to*-stripping are similar to sluicing regarding the immobility of remnants and island sensitivity, while fragment answers are similar to sluicing regarding the immobility of remnants but differ from it regarding island (in)sensitivity. In addition, I have shown that *ka*(*dooka*)/*to*-stripping parallel sluicing in that they are characterized by exhaustivity, while fragment answers are not.

With these observations, I have proposed that the driving force for the movement in sluicing (and *ka*(*dooka*)/*to*-stripping) is exhaustivity. Building on this, I have proposed that sluicing constructions involve null operator movement associated with exhaustivity. With another proposal that deletion applies to FinP except for the element that the null operator originally attaches to, I have shown that the null operator movement analysis captures all of the following crucial facts regarding Japanese sluicing constructions: (i) they do not induce a freezing effect when formed as embedded CQ wh-questions, (ii) they can involve immobile elements as their remnants and (iii) they are sensitive to islands.

The excursion is now over; in the next chapter, we will turn back to the issue that was left unresolved in the end of Chapter 3, namely we will explore what the (non-)involvement of covert wh-movement in Japanese wh-questions, which the discussion in Chapter 3 established, originates from.

Chapter 5

Origin of Covert Wh-Movement: Nominality and Selection

5.1 Introduction

In Chapter 3, I have proposed (1) regarding Japanese wh-questions and have shown that it gives an account of their asymmetric behavior with respect to intervention effects and cleft formation.

(1) a. Matrix wh-questions, embedded non-CQ wh-questions, and *to*-embedded wh-questions do not involve covert wh-movement.

b. Embedded CQ wh-questions involve covert wh-movement.

However, that chapter has left one significant question unresolved: why does (1) hold, or more specifically, why is it that Japanese wh-questions differ regarding the (non-)involvement of covert wh-movement depending on their types? This chapter will be devoted to the pursuit of an answer to this question.

The asymmetry regarding wh-movement shown in (1), which I will dub the *Wh-Movement Asymmetry*, should be related to other different properties of embedded CQ wh-questions and the other types of wh-questions. In particular, the involvement of covert wh-movement in embedded CQ wh-questions should be ascribed to a property of such questions which is not shared by the other types of wh-questions. Following this line of reasoning, the current chapter aims to give an account of the Wh-Movement Asymmetry by discussing two distinct properties of embedded CQ wh-questions, which will be briefly introduced below.

First, I will focus on a distinct "internal" property of embedded CQ wh-questions, i.e. their nominality. It has been recently argued by Tomioka (2020) that embedded (CQ) wh-questions in

Japanese have nominal structures and that they are interpreted as concealed questions as a result of that. I will hypothesize that embedded CQ wh-questions have the structure of free relatives and explore the possibility of ascribing the involvement of covert wh-movement in such wh-questions to the establishment of the structure of free relatives, building on the syntactic analysis of free relatives proposed in Cecchetto and Donati's works (Cecchetto and Donati 2010, 2015, Donati and Cecchetto 2011). This exploration will uncover a significant fact regarding the nominality of embedded CQ wh-questions; it will show that such wh-questions do not always show nominality, its distribution being restricted.

Second, I will zoom in on a distinct "external" property of embedded CQ wh-questions, namely their selectors, CQ-predicates. I will show that CQ-predicates can select a wider range of syntactic categories as questions compared with selectors of other types of wh-questions. I will then propose a syntactic analysis of Japanese wh-questions that syntactically implements the selectional property under consideration. The analysis will be based on the selection mechanism proposed by Donati and Cecchetto (2011). I will show that the proposed analysis captures two significant properties of Japanese wh-questions that this dissertation is concerned with, namely the Wh-Movement Asymmetry and the nominal property of embedded CQ wh-questions.

The rest of this chapter is organized as follows: In Section 5.2, I will pursue the possibility of accounting for the Wh-Movement Asymmetry in terms of the nominal structure of embedded CQ wh-questions. This will be implemented in terms of a free relative analysis of embedded CQ wh-questions, the hypothesis being that covert wh-movement is induced to make them free relatives. The hypothesis is based on the nominal nature of embedded CQ wh-questions. However, the discussion in this section will reveal that embedded CQ wh-questions do not always exhibit nominality. Building on the discussion in Section 5.2, Section 5.3 will propose an analysis of

Japanese wh-questions that can capture the Wh-Movement Asymmetry and the restricted nominality of embedded CQ wh-questions. This will be done by focusing on a distinct external property of embedded CQ wh-questions, namely the selecting predicates, and the mechanism of selection, where the labeling of syntactic objects will play a special role. Section 5.4 provides a summary of this chapter.

5.2 "Internal" property of embedded CQ wh-questions: Nominality

This section will seek an explanation for the Wh-Movement Asymmetry by appealing to a distinct "internal" property of embedded CQ wh-questions, namely their nominal nature, which was recently discussed by Tomioka (2020). I will start with an overview of Tomioka (2020).

5.2.1 Tomioka (2020): Embedded wh-questions as nominal phrases

Discussing a particular phenomenon regarding Quantificational Variability Effects, Tomioka (2020) presents a novel proposal that embedded (CQ) wh-questions in Japanese are nominal phrases. Specifically, he argues that Japanese embedded wh-questions have the structure of internally-headed relative clauses. While his proposal regarding the nominal structure of embedded CQ wh-questions will be adopted below, I will depart from his analysis of such questions, for reasons to be explained below.

5.2.1.1 Quantificational Variability Effects and wh-classifier matching

Usually, embedded wh-questions are exhaustively interpreted; see, for example, (2).

(2) David knows who attended the meeting.
≈ <u>For *all* of those who attended the meeting,</u> David knows that they attended the meeting.

(Tomioka 2020: 122)

However, their exhaustivity can be altered by an adverbial phrase that belongs to the matrix clause, as exemplified in (3).

(3) a. For the most part, David knows who attended the meeting.
≈ For the *most* of those who attended the meeting, David knows that they attended the meeting.
b. David in part knows who attended the meeting.
≈ For *some* of those who attended the meeting, David knows that they attended the meeting.

(ibid: 122)

This phenomenon is known as *Quantificational Variability Effects* (QVEs, henceforth) (Berman 1989, 1991, Lahiri 2002, Beck and Sharvit 2002, among others). While the above examples are from English, QVEs are observed in Japanese as well, as in (4).

(4) Mari-wa [dare-ga ukatta ka] daitai sitteiru.
Mari-TOP who-NOM passed Q mostly know
'For the most part, Mari knows who passed.'
≈ 'For most of the people who passed, Mari knows that they passed.' (ibid: 123)

In Japanese, QVE-inducing adverbial phrases can be quantifiers involving a (numeral) classifier, and interestingly, that classifier must "match" the wh-phrase in the embedded wh-question; more precisely, it must be the one used to count the same kind of referents as (the value of) the wh-phrase. Observe (5).[1]

[1] Tomioka (2020) notes that when a quantifier with a classifier is used as a QVE-inducing adverbial phrase, the sentence sounds a little odd if that quantifier is not followed by a particle such as an approximating particle (e.g. -

(5) Mari-wa [**dare**-ga ukatta ka] <u>san-**nin**(-gurai)</u> sitteiru.
Mari-TOP who-NOM passed Q 3-CLS-about know
Lit. 'For (about) three (people), Mari knows who passed.'
≈ 'For (about) three of the people who passed, Mari knows that they passed.' (ibid: 124)

In (5), the QVE-inducing quantifier involves the classifier -*nin*, which is used for counting people. Note that the use of this classifier is in accord with the use of the wh-phrase *dare* 'who' (whose value refers to people) in the embedded wh-question. I will call this phenomenon *wh-classifier matching*. This matching is possible with various patterns. For example, it is possible regardless of whether the wh-phrase in the embedded wh-question is an argument (see, e.g., (5) for the subject wh-phrase and (6a) for the object wh-phrase) or an adjunct (see, e.g., (6b)). (Later, however, we will observe that wh-classifier matching is impossible with some wh-phrases.) Also, the matching is possible with any kind of a classifier (e.g. -*nin* for people (5), -*satu* for books (6a), and -*kakoku* for countries (6b)).

(6) a. Mari-wa [doroboo-ga **dono hon**-o nusunda ka] <u>zyus-**satu**(-kurai)</u>
Mari-TOP thief-NOM which book-ACC stole Q 10-CLS-about
age-rare-ru.
list-can-PRES
Lit. 'For (about) ten (books), Mari can tell which books the thief has stolen.'
≈ 'For about ten of the book that the thief has stolen, Mari can tell which ones they are.'
(based on ibid: 124)

b. [Sono kaisya-ga **dono kuni**-kara wain-o yunyuusi-tei-ru ka]
that firm-NOM which country-from wine-ACC import-PROG-PRES Q

---

gurai 'about') and a focus particle (e.g. -*mo* 'as many/much as'). However, relevant sentences sound perfect to the author even without such a particle. I thus add parentheses to the approximating particle -*gurai*/-*hodo* 'about' in the relevant examples.

san-**kakoku**(-hodo) osiete hosii.
3-CLS-about tell want
Lit. 'For (about) three (countries), I want you to tell me which countries that firm imports wine from.'
≈ 'I want you to tell me (the names of) three of the countries from which that firm imports wine.'

(ibid: 125)

Since it was first noted by Kitagawa (2009), this interesting phenomenon has not been paid attention to in the relevant theoretical literature on Japanese before Tomioka (2020). Against this background, Tomioka (2020) aims to give an account of wh-classifier matching. One might attempt to ascribe this phenomenon to the fact that in Japanese, a quantifier (with a classifier) can appear outside the domain of the noun phrase which that quantifier is associated with (i.e. a floated quantifier); see, e.g., (7). (Note also that in (7), the classifier *-nin* (used to count people) "matches" with the noun phrase *gakusee* 'student'.)

(7) **Gakusee**-ga san-**nin**(-gurai) ukatta.
student-NOM 3-CLS-about passed
'(About) three students have passed.'

More specifically, one might claim that the quantifier used in wh-classifier matching is a floated quantifier such as the one in (7) and is thus directly associated with the wh-phrase in the embedded wh-question. However, Tomioka rules out this possibility. For example, under the hypothesis in question, it is expected that a floated quantifier in the matrix clause can be directly associated with an element in a declarative complement clause as well. However, this expectation is not borne out, as shown in (8a), where the quantifier in the matrix clause is intended to be associated with the subject in the declarative complement clause *gakusee* 'student'. (Compare (8a) with its grammatical counterpart in (8b)).

(8) a. *Mari-wa [**gakusee**-ga ukatta to] san-**nin**(-gurai) omotteiru.
Mari-TOP student-NOM passed REP 3-CLS-about think
Intended: 'Mari thinks that three students passed.'

b. Mari-wa [**gakusee**-ga san-**nin**(-gurai) ukatta to] omotteiru.
Mari-TOP student-NOM 3-CLS-about passed REP think
'Mari thinks that three students passed.'

I refer the reader to Tomioka (2020) for other arguments against the potential account of wh-classifier matching based on a direct association between a floated quantifier and a wh-phrase.

In pursuing an alternative account, Tomioka points out the fact that a floated quantifier can be associated with a concealed question, as shown in (9).

(9) a. Kana-wa [uinburudon-no kako-no **syoosya**]-o juu-**nin**(-gurai) sitteiru.
Kana-TOP Winbledon-GEN past-GEN winner-ACC 10-CLS-about know
'Kana knows (about) ten of the past Wimbledon champions.'

b. Kei-wa [yooroppa-no **syuto**]-o jut-**tosi**(-hodo) age-rare-ru.
Kei-TOP Europe-GEN capital-ACC 10-CLS-about list-can-PRES
'Kei can list (about) ten European capitals.'

(ibid: 129)

Inspired by this fact, Tomioka proposes that embedded wh-questions in Japanese are actually nominal phrases and are interpreted as concealed questions.[2] With this proposal, for example, (5),

[2] Tomioka (2020) does not discuss whether embedded (CQ) wh-questions in Japanese *always* have nominal structures. He seems to leave this as an open issue, as suggested by the following statement: "[t]he main claim is that embedded questions in Japanese are or at least can be nominalized" (Tomioka 2020: 121, with an underline added by the author). This issue will be discussed in Section 5.2.2.2.

repeated as (10a), is interpreted in the same way as (10b), which contains a (genuine) concealed question.[3]

(10) a. Mari-wa [**dare**-ga ukatta ka] san-**nin**(-gurai) sitteiru.
Mari-TOP who-NOM passed Q 3-CLS-about know
Lit. 'For (about) three (people), Mari knows who passed.'
≈ 'For (about) three of the people who passed, Mari knows that they passed.'
(ibid: 124)

b. Mari-wa [[e_i ukatta] **hito**$_i$]-o san-**nin**(-gurai) sitteiru.
Mari-TOP passed person-ACC 3-CLS-about know
'Mari knows about three of those who passed the exam.' (ibid: 128)

With this proposal, Tomioka argues, wh-classifier matching can be ascribed to the fact that concealed questions can be associated with floated quantifiers that involve a corresponding classifier (e.g. the parallel between (10a) and (10b)), whatever analysis accounts for this (Tomioka leaves this issue open).

Tomioka (2020) discusses only embedded CQ wh-questions (as suggested by the above examples, where all the matrix predicates are CQ-predicates), not mentioning other types of wh-questions. However, his proposal regarding the nominal structure of wh-questions should not apply to the latter types of wh-questions; if it applied, those wh-questions would not be interpreted as questions, since they are not selected by CQ predicates. Therefore, I maintain that only embedded CQ wh-questions can be characterized as nominal, while the other types of wh-questions cannot be.

[3] The following statement by Tomioka (2020) might help understand his proposal: "An embedded question looks like nothing but an interrogative sentence, but its structure is actually nominal, and the nominal structure is, in turn, interpreted as a concealed question (back to question meaning). In this sense, Japanese embedded questions are doubly concealed." (Tomioka 2020: 129)

5.2.1.2 Embedded wh-questions as internally-headed relative clauses

What kind of nominal structure do embedded CQ wh-questions have, then? Concerning this question, Tomioka (2020) argues that they have the structure of internally-headed relative clauses (IHRCs) such as in (11).

(11) Koji-ga [Mari-ga <u>keeki</u>-o reezooko-ni irete-oi-ta no]-o tabeta.
Koji-NOM Mari-NOM cake-ACC fridge-in place-put-PAST C-ACC ate
'Koji ate the cake that Mari put in the fridge.

(based on ibid: 135)

In (11), the matrix object can be intuitively interpreted as the headed relative clause *the cake that Mai put in the fridge*, but the head *keeki* 'cake' is located within the embedded clause (headed by *no*) as underlined. Hence, their name, IHRCs.

What motivates Tomioka to argue that embedded CQ wh-questions structurally parallel IHRCs is the fact that the head of IHRCs appears clause-internally, as noted above. In the case of embedded CQ wh-questions, the best candidate for the head of their nominal structure should be their wh-phrase, which occurs clause-internally, as does the head of IHRCs. For example, if the embedded CQ wh-question in (12a) is to be interpreted as a concealed question like the one in (12b) (i.e. *ukatta hito* 'the person(s) who passed'), the wh-phrase *dare* 'who' should somehow play the role of the head of that concealed question, *hito* 'person', given that both elements are semantically related to people.

(12) a. Mari-wa [<u>dare</u>-ga ukatta ka] sitteiru.
Mari-TOP who-NOM passed Q know
'Mari knows [who passed].'

b. Mari-wa [[e_i ukatta] <u>hito</u>$_i$]-o sitteiru.
Mari-TOP passed person-ACC know
'Mari knows the person(s) who passed.'

One piece of empirical evidence that he presents for his claim concerns the fact that IHRCs can be followed by a noun phrase with the demonstrative *sono* 'that' which corresponds to their head, as in (13).

(13) Koji-ga [Mari-ga keeki-o reezooko-ni irete-oi-ta] <u>sono keeki</u>-o tabeta.
Koji-NOM Mari-NOM cake-ACC fridge-in place-put-PAST that cake-ACC ate
Lit. 'Koji ate that cake that Mari put a cake in the fridge.'

In a similar vein, embedded wh-questions can be followed by a noun phrase with *sono* 'that' which corresponds to an individual that constitutes the answer to the wh-phrase, as exemplified in (14) (see, e.g., Eguchi 1990, 1992 and Yamaizumi 2008 for discussion of this construction).[4]

(14) Keesatu-wa [dare-ga hooseki-o ubatta ka] <u>sono hannin</u>-o sitteiru.
police-TOP who-NOM jewel-ACC stole Q that culprit-ACC know
Lit. 'The police know who stole jewels, that culprit.'
(ibid: 133)

Tomioka claims that this similarity between IHRCs and embedded CQ wh-questions serves as a piece of empirical evidence for his IHRC-based analysis of embedded CQ wh-questions.

Though it may seem attractive to ascribe the nominal nature of embedded CQ wh-questions to the structure of IHRCs, I will not adopt that idea here for the following two reasons. First, the

[4] The following note from Tomioka (2020) may help to understand the interpretation of (14): "Descriptively speaking, the DPs that come after the embedded interrogative CPs pick up the referents that correspond to the individuals that constitute the answers to the wh-phrases" (Tomioka 2020: 134).

empirical evidence that he gives for the structural parallel between IHRCs and embedded CQ wh-questions, i.e. the fact that they are similar in that both can be followed by a noun phrase with the demonstrative *sono* 'that' as in (13) and (14), is not without problems. If the similarity in question serves as a piece of evidence for the structural parallel of those embedded clauses, it is expected that the constructions exemplified by (13) and (14) would show the same behavior. This expectation is not completely borne out, however; for example, an IHRC cannot be detached from the following *sono*-phrase, while an embedded CQ wh-question can be; observe the contrast between (15b) and (16b) (whose basic counterparts are (15a) and (16a) respectively).

(15) a. Kooji-wa Mika-ni [Mari-ga keeki-o reezooko-ni irete-oi-ta]
Koji-TOP Mika-to Mari-NOM cake-ACC fridge-in place-put-PAST
sono keeki-o ageta.
that cake-ACC gave
Lit. 'Koji gave Mika that cake that Mari put a cake in the fridge.'

b. *Kooji-wa [Mari-ga keeki-o reezooko-ni irete-oi-ta]$_i$ Mika-ni t_i
Koji-TOP Mari-NOM cake-ACC fridge-in place-put-PAST Mika-to
sono keeki-o ageta.
that cake-ACC gave

(16) a. Kooji-wa Mika-ni [dare-ga hooseki-o ubatta ka] sono hannin-o osieta.
Koji-TOP Mika-to who-NOM jewel-ACC stole Q that culprit-ACC taught
Lit. 'Koji told Mika who stole the jewels, that culprit.'

b. Kooji-wa [dare-ga hooseki-o ubatta ka]$_i$ Mika-ni t_i sono hannin-o osieta.
Koji-TOP who-NOM jewel-ACC stole Q Mika-to that culprit-ACC taught

Given that, the similarity between (13) and (14) does not provide strong support for the IHRC-based analysis of embedded CQ wh-questions, unless the contrast between (15b) and (16b) can be attributed to an independent factor, which is unclear.

Second, it is not clear how the IHRC-based analysis would derive the nominal interpretation of embedded CQ wh-questions. To lay out this problem, I first note that Tomioka's (2020) specific analysis cannot be adopted here. He proposes to analyze embedded CQ wh-questions with Shimoyama's (1999) analysis of IHRCs, according to which an IHRC moves to the sentence-initial position in LF to be interpreted as a sentence coordinated to the matrix clause, and an E-type pronoun that originates with the IHRC is interpreted to refer to a contextually salient entity in the event denoted by the coordinated IHRC. Under this analysis, in particular, the nominal interpretation of an embedded CQ wh-question is ascribed to the E-type pronoun that refers to the value of the wh-phrase in that wh-question (interpreted as a coordinated clause). His proposal thus apparently succeeds in assigning nominal interpretations to embedded CQ wh-questions. However, his analysis is not compatible with this section's aim to link the nominal structure of embedded CQ wh-questions to the involvement of covert wh-movement in those questions (which was empirically supported by the discussion in Chapter 3); Shimoyama's (1999) analysis of IHRCs, which Tomioka's proposal draws on, does not assume any movement in IHRCs. Furthermore, the analysis that derives IHRCs without any movement is in fact problematic given that those clauses have been observed to be sensitive to islands (e.g. Watanabe 1992, Grosu and Hoshi 2016). For these reasons, I will not adopt Tomioka's (2020) IHRC-based analysis. Instead, I will now consider the possibility of applying another analysis of IHRCs that assumes movement in those clauses, where the above problems do not arise. More specifically, I will focus on the analysis that assumes movement of the internal head in IHRCs (e.g. *keeki* 'cake' in (11)), given that I am now aiming to ascribe the covert wh-movement in embedded CQ wh-questions to the derivation of IHRCs, and that the wh-phrase in those questions is compared to the internal head of IHRCs as noted above.[5]

[5] Another type of analysis of IHRCs assumes that they involve movement but it is an operator that moves; see Landman (2016).

This type of analysis is proposed by Itō (1986), Watanabe (1992) and Erlewine and Gould (2016); with technical differences aside, they argue that the head covertly raises out of the IHRC to the matrix clause to serve as (part of) an argument of the matrix predicate. However, this cannot be straightforwardly extended to embedded wh-questions; it seems strange to assume that the wh-phrase in an embedded wh-question undergoes raising to serve as an argument of the matrix predicate. One potentially viable way to apply the analysis under consideration to embedded CQ wh-questions is to assume that a wh-phrase covertly moves to merge with the Q-particle *ka*, resulting in the form of an indefinite "wh+*ka*" (e.g. *dare-ka* 'someone'), which induces nominality. With this analysis, for example, the embedded CQ wh-question in (12a) would covertly have the structure like (17).

(17) [[~~dare~~-ga ukatta] dare-ka]
who-NOM passed who-Q

Given that, the wh-question in (12a) would be expected to have the same interpretation as in (18), i.e. the indefinite *dareka* 'someone' modified by the preceding relative clause.

(18) [e_i ukatta] $dareka_i$
passed someone
'someone who passed'

Note, however, that (18) cannot be interpreted as a concealed question when it is selected by the predicate *sitteiru* 'know'. E.g., (19) cannot be interpreted in the same manner as (12a); instead, it is interpreted only as *Mari is acquainted with someone who passed*, as indicated in the translation.

(19) Mari-wa [[e_i ukatta] $dareka_i$]-o sitteiru.
Mari-TOP passed someone-ACC know

‘Mari knows someone who passed.’

≉ ‘Mari knows who passed.’ /

≈ ‘Mari is acquainted with someone who passed.’

This analysis is thus incompatible with Tomioka’s (2020) claim that embedded CQ wh-questions serve as nominal phrases interpreted as concealed questions. With this consideration, I conclude that it is hard to extend even the analysis of IHRCs involving movement of the internal head to embedded CQ wh-questions.

For the above two reasons, I will depart from the IHRC-based analysis for embedded CQ wh-questions; instead, below I will proceed on the assumption that embedded CQ wh-questions have the structure of free relatives.

5.2.2 Linking covert wh-movement to nominality: Free relatives

Building on Tomioka’s (2020) proposal illustrated above that embedded CQ wh-questions in Japanese are nominal phrases, this subsection explores the possibility of analyzing those wh-questions as free relatives. Free relatives refer to wh-clauses that are interpreted like relative clauses. They typically appear in positions where DPs occur, being interpreted like nominal phrases. In (20b), e.g., the free relative *what you read* appears in the object position of the verb *read*, where a DP otherwise occurs (see (20a)), and is construed roughly as the headed relative clause *the thing that you read*.

(20) a. I read [$_{DP}$ the book].

b. I read [what you read].

I will adopt the syntactic analysis of free relatives proposed by Cecchetto and Donati (Cecchetto and Donati 2010, 2015, Donati and Cecchetto 2011; henceforth, C&D), and hypothesize that covert wh-movement takes place in embedded CQ wh-questions to form the structure of free relatives. The scrutiny of this hypothesis, however, will reveal a problem with it.

5.2.2.1 Cecchetto and Donati's analysis of free relatives

Here, I will give a brief overview of C&D's analysis of free relatives, which will be adopted in the following discussion. It is standardly assumed that when a new syntactic object is formed by movement, the target of the movement always projects, labeling the resulting constituent, while the moved element never projects. C&D argue, however, that this is not always the case. Thus, they argue that in free relatives, the wh-clauses are labeled (i.e. projected) by the moved wh-element; consider (21), where (21b) is the structure of the wh-clause (21a), with its label being undetermined.[6]

(21) a. what you read

b. [$_?$ what [$_C$ C you read ~~what~~]]

C&D claim that there are two options to label the wh-clause in (21b). With one option, the target of the wh-movement, i.e. C, projects, as standardly assumed. This makes the wh-clause in (21a) a wh-question, as exemplified in (22a). With the other option, crucially, the moved wh-word *what* is what provides the label. Following C&D, I assume that what projects in this case is its category

[6] In the following discussion, I will make a distinction between the following three terms: (i) (*single*) *wh-words* refer to elements that consist of only a single wh-related item (e.g. *who*, *dare* 'who'), (ii) (*complex*) *wh-phrases* refer to complex phrases containing wh-words (e.g. *whose book*, *dare-no hon* 'whose book'), and (iii) *wh-elements* refer to both/either of them.

feature, namely D. As a result, the wh-clause serves as a nominal phrase (or a DP), i.e. a free relative, as in (22b).

(22) a. I wonder [$_{C(P)}$ what [$_{C}$ C you read]].

b. I read [$_{D(P)}$ what [$_{C}$ C you read]].

5.2.2.2 Free-Relative-Building Movement Account

With C&D's analysis of free relatives in mind, I return to embedded CQ wh-questions in Japanese. Recall that Tomioka (2020) argues that those wh-questions have nominal structures and are interpreted as concealed questions, as discussed in Section 5.2.1. For instance, the embedded CQ wh-question in (23a) is interpreted in the manner parallel to the concealed question in (23b) due to its nominal structure.

(23) a. Mai-wa [Ken-ga nani-o katta ka] sitteiru.
Mai-TOP Ken-NOM what-ACC bought Q know
'Mai knows [what Ken bought].'

b. Mai-wa [[Ken-ga *e*$_{i}$ katta] mono$_{i}$]-o sitteiru.
Mai-TOP Ken-NOM bought thing-ACC know
'Mai knows the thing that Ken bought.'

Building on this proposal, I now make a crucial assumption that embedded CQ wh-questions have the structure of free relatives, or in terms of C&D's analysis, they are labeled as D. This assumption is summarized in (24), which I term *Embedded CQ Wh-questions as Free Relatives*. (In the following discussion, however, (24) will be reconsidered and revised.)

(24) *Embedded CQ Wh-questions as Free Relatives*

Embedded CQ wh-questions have the structure of free relatives, i.e. they are labeled as D.

Recall here that according to C&D, free relatives must involve wh-movement, since this enables the wh-element to label the wh-clause with its categorial feature D (see Section 5.2.2.1). Given that, it follows from the assumption in (24) that embedded CQ wh-questions must involve wh-movement. This thus provides a straightforward account of the involvement of covert wh-movement in embedded CQ wh-questions; those wh-questions are free relatives and free relatives must involve wh-movement. I will refer to this account as *Free-Relative-Building Movement Account* (the *FRBM Account*, henceforth), which is summarized in (25).

(25) *Free-Relative-Building Movement (FRBM) Account*

Covert wh-movement occurs in embedded CQ wh-questions to render their structure that of free relatives, i.e. to label them as D.

Consider, e.g., how the embedded CQ wh-question in (23a) is analyzed. With the current setup, that wh-question has the structure in (26).[7]

(26) [$_D$ nani$_D$ [$_C$ ka$_C$ Ken katta ~~nani~~]] (word order irrelevant)
what Q Ken bought what

In (26), categorial features of lexical items are represented by subscripts. Following C&D, I assume that the wh-word *nani* 'what' has the category feature D. In addition, I assume that the question particle *ka* has the category feature C (or Force, if we follow the assumption in Section 3.4.2; it will not matter which option is taken for the current discussion). The crucial part of (26) is that the wh-word *nani* undergoes (covert) movement; according to C&D's theory, this movement

[7] In considering syntactic derivations involving covert wh-movement, I am agnostic about how covert wh-movement should be implemented. The current analysis is compatible with the model where covert wh-movement is preceded by overt wh-movement (as in Huang 1982a,b and Nishigauchi 1990) and the one where cyclic Spell-out is assumed, with traditional overt and covert movement taking place in the same cycle (Chomsky 2000), including the version in Nissenbaum (2000), where the two take place on the same cycle but differ in the timing of transfer to the interfaces.

enables the wh-word to project and label the wh-clause as D, thus rendering it nominal, i.e. a free relative, as indicated in (26).[8] Importantly, I assume that movement of wh-elements in embedded CQ wh-question, as in (26), occurs for the purpose of labeling, i.e. to make the relevant object nominal, rather than that C serves as a probe and induces that movement (as suggested in (25)).

The FRBM Account/(25) also captures the other side of the Wh-Movement Asymmetry, namely the claim that wh-movement does not occur in the other types of wh-questions: matrix wh-questions, embedded non-CQ wh-questions, and *to*-embedded wh-questions. Since it is assumed, as noted above, that this movement takes place for the purpose of labeling, it would make those wh-questions free relatives, or nominal, labeling them as D. However, they could then not be construed as questions, since they are not selected by CQ-predicates.[9] The wh-questions under consideration thus lack covert wh-movement, not being labeled as D. Instead, I assume that what projects in those wh-questions is C, as standardly assumed. For instance, the embedded non-CQ wh-question in (27) has the structure in (28), which does not involve wh-movement and is labeled as C.

(27) Mai-wa [Ken-ga nani-o katta ka] sinpaisiteiru.
Mai-TOP Ken-NOM what-ACC bought Q is.anxious
'Mai is anxious [what Ken bought].'

(28) $[_{C}$ ka$_{C}$ Ken katta nani$_{D}$] (word order irrelevant)
Q Ken bought what

[8] Donati and Cecchetto (2011) note that "to the best of [their] knowledge, while there are in-situ interrogatives, there are no in-situ free relatives" (Donati and Cecchetto 2011: 545). However, Polinsky (2015) observes that Tsez has wh-in-situ free relatives.

[9] E.g., matrix free relatives could not be interpreted as questions, and the same issue would arise with *to*-embedded wh-questions since they are paraphrases/reports of matrix questions. Note also that non-CQ predicates must select CP (as discussed in more detail below).

So far, so good; the FRBM Account/(25), which can be taken to be a consequence of the assumption in (24), appears to nicely capture the Wh-Movement Asymmetry. In the rest of this subsection, I will scrutinize the validity of (24-25). Consider first (24), repeated below.

(24) *Embedded CQ Wh-questions as Free Relatives*

Embedded CQ wh-questions have the structure of free relatives, i.e. they are labeled as D.

According to this assumption, *all* embedded CQ wh-questions would be interpreted as nominal, or free relatives. I will now examine whether this actually holds.

In what follows, I will draw on what Tomioka (2020) argues shows the nominality of embedded wh-questions in Japanese, namely wh-classifier matching (see Section 5.2.1), which is exemplified by (29a) (= (10a)).

(29) a. Mari-wa [**dare**-ga ukatta ka] san-**nin**(-gurai) sitteiru.
Mari-TOP who-NOM passed Q 3-CLS-about know
Lit. 'For (about) three (people), Mari knows who passed.'
≈ 'For (about) three of the people who passed, Mari knows that they passed.'
(Tomioka 2020: 124)

b. Mari-wa [[e_i ukatta] **hito**$_i$]-o san-**nin**(-gurai) sitteiru.
Mari-TOP passed person-ACC 3-CLS-about know
'Mari knows about three of those who passed the exam.' (ibid: 128)

Recall that Tomioka argues that embedded wh-questions in Japanese have nominal structures and are construed as concealed questions, thus ascribing the possibility of the wh-classifier matching in (29a) to the acceptability of (29b) (= (10b)), where the corresponding concealed question is associated with the same (floated) quantifier. Now, if (24) is correct, it is predicted that wh-classifier matching would be possible with *all* embedded CQ wh-questions.

However, this prediction is not borne out. Below, I will illustrate two cases that are not compatible with the prediction in question. The first case concerns embedded CQ wh-questions that involve an adjunct wh-word such as *doo* 'how' and *naze* 'why', like those in (30).

(30) a. Mai-wa [Ken-ga doo kinko-o aketa ka] sitteiru.
Mai-TOP Ken-NOM how safe-ACC opened Q know
'Mai knows [how Ken opened the safe].'

b. Mai-wa [Ken-ga naze kaisya-o yameta ka] sitteiru.
Mai-TOP Ken-NOM why company-ACC quit Q know
'Mai knows [why Ken quit the company].'

If (24) holds and thus the embedded CQ wh-questions in (30a-b) have nominal structures, they should be interpreted like the concealed questions in (31a-b) respectively, which are headed by *hoohoo* 'way' and *riyuu* 'reason'.

(31) a. Mai-wa [[Ken-ga *e*$_i$ kinko-o aketa] hoohoo$_i$]-o sitteiru.
Mai-TOP Ken-NOM safe-ACC opened way-ACC know
'Mai knows the way Ken opened the safe.'

b. Mai-wa [[*e*$_i$ Ken-ga kaisya-o yameta] riyuu$_i$]-o sitteiru.
Mai-TOP Ken-NOM company-ACC quit reason-ACC know
'Mai knows the reason why Ken quit the company.'

Those concealed questions, or more generally, the nouns *hoohoo* 'way' and *riyuu* 'reason', can be associated with quantifiers with the classifier -*tu*, which can be used to count abstract things, as (32) shows.

(32) a. Mai-wa [[Ken-ga *e*$_i$ kinko-o aketa] **hoohoo**$_i$]-o mit-**tu**(-gurai) sitteiru.
Mai-TOP Ken-NOM safe-ACC opened way-ACC 3-CLS-about know
'Mai knows three ways Ken opened the safe.'

b. Mai-wa [[e_i Ken-ga kaisya-o yameta] **riyuu**$_i$]-o <u>mit-**tu**(-gurai)</u> sitteiru.
Mai-TOP Ken-NOM company-ACC quit reason-ACC 3-CLS-about know
'Mai knows three reasons why Ken quit the company.'

Now, if the embedded CQ wh-questions in (30) have nominal structures, it is predicted that wh-classifier matching would be possible with them, to the extent that the classifier involved is *-tu*. However, this prediction is not borne out, as (33) shows.

(33) a. *Mai-wa [Ken-ga <u>**doo**</u> kinko-o aketa ka] <u>mit-**tu**(-gurai)</u> sitteiru.
Mai-TOP Ken-NOM how safe-ACC opened Q 3-CLS-about know
Lit. 'For (about) three (ways), Mai knows how Ken opened the safe.'
≈ 'For (about) thee of the ways Ken opened the safe, Mai knows that Ken opened the safe in those ways.'

b. *Mai-wa [Ken-ga <u>**naze**</u> kaisya-o yameta ka] <u>mit-**tu**(-gurai)</u> sitteiru.
Mai-TOP Ken-NOM why company-ACC quit Q 3-CLS-about know
Lit. 'For (about) three (reasons), Mai knows why Ken quit the company.'
≈ 'For (about) three of the reasons Ken quit the company, Mai knows that Ken quit the company for those reasons.'

This observation thus indicates that embedded CQ wh-questions with an adjunct wh-word such as *doo* 'how' and *naze* 'why' do not have a nominal structure.[10]

[10] Note that wh-words that correspond to *how* and *why* can introduce "adjunct" free relatives in some languages; (i) gives an English example that involves an adjunct free relative introduced by *how*, while (ii) gives a Romanian example containing an adjunct free relative with *de ce* 'why'. Those adjunct free relatives are interpreted like *in the way in which* … and *for the reason why* … respectively. (Note incidentally that free relatives introduced by a wh-word corresponding to *why* are cross-linguistically quite rare; such free relatives are observed in Romanian (Caponigro and Fălăuș 2023), Teramano (Mantenuto and Caponigro 2021), and some Mesoamerican languages (Caponigro et al. 2021).)

(i) I did it [how you did it].
≈ I did it [in the way in which you did it]. (based on Caponigro 2003: 12)

(ii) Ana a plecat din țară [de ce a plecat și Maria].
Ana has left from country why has left also Maria
'Ana left the country for the reason(s) why Maria left.' (Caponigro and Fălăuș 2023: 412)

The second relevant case concerns embedded CQ wh-questions that involve a complex wh-phrase such as *dare-no hon* 'whose book', like (34).

(34) Mai-wa [Ken-ga dare-no hon-o katta ka] sitteiru.
Mai-TOP Ken-NOM who-GEN book-ACC bought Q know
'Mai knows [whose book Ken bought].'

If (24) holds, it is expected that the embedded CQ wh-question in (34) would have the nominal structure labeled as D and thus wh-classifier matching would be possible in (34), with the classifier being the one for people -*nin* (given the wh-word *dare* 'who') or the one for books -*satu* (given the whole complex wh-phrase *dare-no hon* 'whose book'). However, this expectation is not borne out. Observe (35).

(35) Mai-wa [Ken-ga **dare**-no hon-o katta ka] {??san-**nin**(-gurai) /
Mai-TOP Ken-NOM who-GEN book-ACC bought Q 3-CLS-about
?*san-satu(-gurai)} sitteiru.
3-CLS-about know
Lit. 'For (about) three (authors), Mai knows whose book Ken bought.'
≈ 'For (about) three of the authors whose books Ken bought, Mai knows that Ken bought those authors' books.' /
Lit. 'For (about) three (books), Mai knows whose book Ken bought.'
≈ 'For (about) three of the books of some authors that Ken bought, Mai knows that Ken bought those books.

In light of this, one might consider the possibility that the embedded CQ wh-questions in (30) do have the structure of free relatives but serve as adjunct free relatives. Importantly, this possibility may be compatible with the ungrammaticality of (33) if it is plausibly assumed that those wh-questions are labeled not as D but as another category such as ADV(erb); wh-classifier matching is compatible with nominal phrases but not with adverbial ones. However, this possibility cannot be adopted, since the wh-clauses in (30) do not have the interpretation of adjunct free relatives, or more specifically, the sentences in (30a-b) are not construed as *Mai knows (something) in the way in which Ken opened the safe* and *Mai knows (something) for the reason why Ken quit the company* respectively.

This observation thus shows that embedded CQ wh-questions with a complex wh-phrase such as *dare-no hon* 'whose book' do not have nominal structures.[11]

To recap, the above two observations indicate that it is not the case that *all* embedded CQ wh-questions have the structure of free relatives, being labeled as D. They thus constitute a serious problem for the assumption in (24).

I will now discuss how this issue can be resolved under the current hypothesis. Consider first how the embedded CQ wh-questions with an adjunct wh-word like (30) and those with a complex wh-phrase like (34), which do not have a nominal structure, are labeled. Following the standard assumption that interrogative clauses are syntactically CP, I assume that those wh-quesitons are labeled as C, i.e. they serve as (genuine) interrogative clauses. With this assumption, I now revise, or loosen, the assumption in (24) and submit (36) as a "weaker" version.

(36) *Embedded CQ Wh-questions as D(P) or C(P)*

Embedded CQ wh-questions have *either* the structure of free relatives, i.e. are labeled as D, *or* have that of interrogative clauses, i.e. are labeled as C.

This revision thus accommodates the grammaticality of (30) and (34), where the embedded CQ wh-questions should not be labeled as D but as C.

How can embedded CQ wh-questions be labeled as C, then? Recall first that it was crucially assumed in this subsection that in embedded CQ wh-questions, wh-movement takes place for the purpose of labeling, i.e. in order to label those questions as D. It should then follow that embedded CQ wh-questions labeled as C do not involve covert wh-movement. Recall also that I have argued

[11] There is one exception to this generalization; wh-classifier matching is possible with embedded CQ wh-questions containing a complex wh-phrase *dono X* 'which X', suggesting that those questions have a nominal structure. See (6), for example. This exceptional case will be discussed in Section 5.3.3.2.2.

above that wh-questions that do not (or more precisely, cannot) involve covert wh-movement (i.e. matrix wh-questions, embedded non-CQ wh-questions, *to*-embedded wh-questions) are labeled as C. E.g., observe again the embedded non-CQ wh-question in (27) and its structure in (28), repeated below as (37) and (38) respectively.

(37) Mai-wa [Ken-ga nani-o katta ka] sinpaisiteiru.
Mai-TOP Ken-NOM what-ACC bought Q is.anxious
'Mai is anxious [what Ken bought].'

(38) [$_{C}$ ka$_{C}$ Ken katta nani$_{D}$] (word order irrelevant)
Q Ken bought what

Crucially, the embedded non-CQ wh-question in (37) does not involve wh-movement, as (38) shows. Hence, it cannot be labeled as D. Instead, as represented in (38), what projects is C, i.e. the category feature of the Q-particle *ka*, as standardly assumed. Based on those considerations, it follows that when embedded CQ wh-questions are labeled as C, they do not involve covert wh-movement and their label is provided by the category feature of the Q-particle *ka*.

To recap, under the assumption in (36), there should be two different derivations for embedded CQ wh-questions, and they result in different labels for those wh-questions. With one derivation, embedded CQ wh-questions involve covert wh-movement, which results in them being labeled as D (i.e. being free relatives). With the other derivation, those wh-questions do not involve covert wh-movement, which results in them being labeled as C (i.e. being interrogative clauses). It is then predicted that only the second derivation should be available for the wh-questions in (30) and (34), given that they must be labeled not as D but as C. That is, those wh-questions are predicted to involve no wh-movement.

At this point, one might notice that this prediction seems to be incongruent with the proposal that I presented in Chapter 3, namely that embedded CQ wh-questions (in general) involve covert wh-movement. Notice, however, that the arguments for this proposal given in Chapter 3 did not take into consideration embedded CQ wh-questions such as in (30) and (34), except for those with the adjunct wh-word *naze* 'why', which turned out to involve an interfering factor (Section 3.4.2.3.2; see also relevant discussion below). That is, I did not examine in Chapter 3 embedded CQ wh-questions with the adjunct wh-word *doo* 'how' and with a complex wh-phrase like *dare-no hon* 'whose book'. Hence, the prediction in question is worth examining and might require us to revise the proposal that embedded CQ wh-questions (in general) involve covert wh-movement.

Let us now examine the prediction under consideration, i.e. whether the embedded CQ wh-questions in (30) and (34) involve covert wh-movement. For the purpose of this investigation, I will rely on the diagnostics for wh-movement from Chapter 3, i.e. (i) obviation of intervention effects and (ii) formation of cleft wh-questions. Recall the crux of these diagnostics. Concerning the former, if no intervention effect arises with an intervenor located above a wh-element, that indicates that covert wh-movement is involved, given the widely accepted view that intervention effects are obviated if a wh-element is placed above an intervenor, overtly or covertly (see Section 3.4.1). E.g., the acceptability contrast between (39a) and (39b) suggests that matrix wh-questions do not involve covert wh-movement, while (at least some) embedded CQ wh-questions do.

(39) a. Matrix wh-questions:

?*Ken-sika nani-o {kaw-anakat-ta desu ka / kai-masen desita ka}.
Ken-SIKA what-ACC buy-not-PAST COP.POL Q buy-not.POL COP.POL.PAST Q
'What did only Ken buy?'

b. Embedded CQ wh-questions:

Mai-wa [Ken-sika nani-o kawa-nakat-ta ka] {sitteiru / akasita / tazuneta}.

Mai-TOP Ken-SIKA what-ACC buy-not-PAST Q know / revealed / asked

'Mai {knows / revealed / asked} [what only Ken bought].'

Concerning the latter, if transforming a wh-question into a cleft leads to degradation, that wh-question should involve covert wh-movement; since the derivation of clefts involves (focus) movement of the focus phrase according to Hiraiwa and Ishihara (2002, 2012), applying covert wh-movement to the wh-element in the focus position would give rise to a freezing effect, leading to degradation (see Section 3.4.2). For instance, the following examples show that forming a cleft wh-question as a matrix wh-question is perfectly possible (40a), while building a cleft wh-question as an embedded CQ cleft wh-question leads to degradation (40b). This contrast also suggests that covert wh-movement does not occur in matrix wh-questions, while it does in (at least some) embedded CQ wh-questions.

(40) a. Matrix wh-questions

[Ken-ga e_i kabin-o oita no]-wa doko-ni$_i$ desu ka.

Ken-NOM vase-ACC put C-TOP where-on COP.POL Q

'Where is it that Ken put a vase?'

b. Embedded CQ wh-questions

?*Mai-wa [[Ken-ga e_i kabin-o oita no]-wa doko-ni$_i$ (da) ka] {sitteiru / akasita /

Mai-TOP Ken-NOM vase-ACC put C-TOP where-on COP Q know / revealed /

tazuneta}.

asked

'Mai {knows / revealed / asked} [where it is that Ken put a vase].'

Taking advantage of these diagnostics, let us now examine whether the embedded CQ wh-questions in (30) and (34) involve covert wh-movement.

A note regarding embedded CQ wh-questions with *naze* 'why' like (30b), repeated below as (41), is first in order.

(41) Mai-wa [Ken-ga naze kaisya-o yameta ka] sitteiru.
Mai-TOP Ken-NOM why company-ACC quit Q know
'Mai knows [why Ken quit the company].'

As discussed in Chapter 3 (i.e. Section 3.4.2.3.2), wh-questions involving *naze* 'why' show peculiar patterns with respect to the aforementioned diagnostics. Concerning intervention effects, such effects do not arise in wh-questions with *naze* 'why' even if those wh-questions are matrix wh-questions. Witness, e.g., (42), where *naze* 'why' appears above the intervenor *Ken-sika* in a matrix wh-questions. (In particular, compare (42) with (39a)).

(42) (?)Ken-sika naze sono hon-o kaw-anakat-ta no.
Ken-SIKA why that book-ACC buy-not-PAST Q
'Why did only Mai buy that book?'

Concerning cleft formation, it was observed that cleft wh-questions with *naze* 'why' can be realized as embedded CQ wh-questions in contrast to other wh-words. The relevant example is shown in (43). (In particular, compare (43) with (40b)).

(43) Mai-wa [[Ken-ga hondana-ni kabin-o oita no]-wa naze (da) ka] {sitteiru /
Mai-TOP Ken-NOM bookshelf-on vase-ACC put C-TOP why COP Q know
akasita / tazuneta}.
revealed / asked
'Mai {know / revealed / asked} [why it is that Ken put a vase on the bookshelf].'

In Chapter 3 (i.e. Section 3.4.2.3.2), I have shown that the reason for the peculiar behavior of *naze* 'why', as shown above, is that *naze* is base-generated in the left periphery (see also Rizzi 1990, 2001, Ko 2005, Stepanov and Tsai 2008, Kuwabara 2008, 2013). This is an interfering factor that prevents using the two diagnostics for covert wh-movement under consideration to test whether covert wh-movement is involved in embedded CQ wh-questions with *naze* 'why' like (41) (see Section 3.4.2.3.2 for a more detailed discussion). For this reason, in the following discussion, I will not deal with such wh-questions, and instead, treat embedded CQ wh-questions with *doo* 'how', like (30a), as the representative example of those with an adjunct wh-word.

Let us then begin with (30a), repeated below as (44), which involves the adjunct wh-word *doo* 'how'.

(44) Mai-wa [Ken-ga doo kinko-o aketa ka] sitteiru.
Mai-TOP Ken-NOM how safe-ACC opened Q know
'Mai knows [how Ken opened the safe].'

I have to note first, however, that the second diagnostic, namely forming cleft wh-questions, cannot apply to this case, since *doo* 'how' is generally incompatible with the focus position of cleft wh-questions. For example, (45) shows that *doo* 'how' is not compatible with matrix cleft wh-questions, which are acceptable with other wh-words (see, e.g., (40a)).

(45) ?*[Ken-ga e_i kinko-o aketa no]-wa doo_i desu ka.
Ken-NOM safe-ACC opened C-TOP how COP.POL Q
Intended. 'How is it that Ken opened the safe?'

Hence, for (44), I focus only on the first diagnostic, i.e. the obviation of intervention effects. Now, observe (46), which shows that with *doo* 'how', an intervention effect arises in matrix wh-questions (46a), while it is obviated in embedded CQ wh-questions (46b).[12]

(46) a. ?*$\boxed{\text{Dare-mo}}$ <u>doo</u> sono kinko-o {ake-nakat-ta desu ka / ake-masen
who-MO how that safe-ACC open-not-PAST COP.POL Q / open-not.POL
desita ka}.
COP.POL.PAST Q
'How did no one open that safe? (≈ What is the way such that no one opened that safe in that way?)'

b. (?)Mai-wa [$\boxed{\text{dare-mo}}$ <u>doo</u> sono kinko-o ake-nakat-ta ka] sitteiru.
Mai-TOP who-MO how that safe-ACC open-not-PAST Q know
'Mai knows [how no one opened that safe (≈ what the way is such that no one opened that safe in that way)].'

---

[12] In (46), I use the NPI *dare-mo* (… NEG) 'no one' as the intervenor, instead of another NPI *X-sika* (… NEG) 'only X', which is used as an intervenor in most of the relevant examples in this thesis. I do not use *X-sika* in (46) because this NPI is generally incompatible with the adjunct wh-word *doo* 'how'. Note first that if the intervenor in (46b) is replaced with *Ken-sika* 'only Ken', the sentence is degraded, as (i) shows.

(i) ?*Mai-wa [$\boxed{\text{Ken-sika}}$ <u>doo</u> sono kinko-o ake-nakat-ta ka] sitteiru.
Mai-TOP Ken-SIKA how that safe-ACC open-not-PAST Q know
'Mai knows [how only Ken opened that safe (≈ what the way is such that only Ken opened that safe in that way)].'

This degradedness might be taken to indicate that the wh-word *doo* 'how' does not undergo wh-movement in the embedded CQ wh-question. However, I argue that (i) is degraded because the co-occurrence of the NPI *X-sika* with the wh-word *doo* 'how' generally degrades the sentence. This is demonstrated by the matrix wh-question in (iib), which shows that an intervention effect is not obviated even if the *sika*-phrase is scrambled over *doo* 'how'; if the NPI *dare-mo* is used in the same configuration, an intervention effect is obviated, as shown in (iiib).

(ii) a. ?*$\boxed{\text{Ken-sika}}$ <u>doo</u> sono kinko-o ake-nakat-ta desu ka.
Ken-SIKA how that safe-ACC open-not-PAST COP.POL Q
'How did only Ken open that safe? (≈ What is the way such that only Ken opened that safe in that way?)''

b. ?*<u>Doo</u>$_i$ $\boxed{\text{Ken-sika}}$ t_i sono kinko-o ake-nakat-ta desu ka.
how Ken-SIKA that safe-ACC open-not-PAST COP.POL Q

(iii) a. ?*$\boxed{\text{Dare-mo}}$ <u>doo</u> sono kinko-o ake-nakat-ta desu ka.
who-MO how that safe-ACC open-not-PAST COP.POL Q
'How did no one open that safe? (≈ What is the way such that no one opened that safe in that way?)'
(= (46a))

b. (?)<u>Doo</u>$_i$ $\boxed{\text{dare-mo}}$ t_i sono kinko-o ake-nakat-ta desu ka.
how who-MO that safe-ACC open-not-PAST COP.POL Q

This observation thus suggests that in the embedded CQ wh-question (44), the adjunct wh-word *doo* 'how' undergoes covert wh-movement, even though that wh-question should be labeled as C, not D.

Let us next turn to (34), repeated below as (47), whose embedded CQ wh-question contains the complex wh-phrase *dare-no hon* 'whose book'.

(47) Mai-wa [Ken-ga dare-no hon-o katta ka] sitteiru.
Mai-TOP Ken-NOM who-GEN book-ACC bought Q know
'Mai knows [whose book Ken bought].'

First, consider it in terms of the first diagnostic, i.e. obviation of intervention effects; consider (48).

(48) a. ?*Ken-sika dare-no hon-o {kawa-nakat-ta desu ka / kai-masen
Ken-SIKA who-GEN book-ACC buy-not-PAST COP.POL Q buy-not.POL
desita ka}.
COP.POL.PAST Q
'Whose book did only Ken buy?'

b. ?Mai-wa [Ken-sika dare-no hon-o kaw-anakat-ta ka] sitteiru.
Mai-TOP Ken-SIKA who-GEN book-ACC buy-not-PAST Q know
'Mai knows [whose book only Ken bought].'

(48a) shows that the matrix wh-question with the complex wh-phrase *dare-no hon* 'whose book' exhibits an intervention effect. However, this effect is obviated if the same wh-question is realized as an embedded CQ wh-question, as shown by (48b).[13] This indicates that the wh-phrase under consideration undergoes covert wh-movement in the embedded CQ wh-question in (47).

[13] I suggest that the slight marginality of (48b) arises because it is hard to imagine a context where the intended reading of (48b) is true. This suggestion is confirmed by the observation that an intervention effect is more clearly obviated in (ib) than (48b), even though it involves a complex wh-phrase (i.e. *dare-no ronbun* 'whose paper') as does (48b); notice in particular that the situation where the intended reading of (ib) holds is more easily imaginable compared with (48b).

Next, consider (47) in terms of the second diagnostic, i.e. formation of cleft wh-questions. Witness the relevant examples in (49).

(49) a. [Ken-ga e_i kabin-o oita no]-wa dare-no tukue-ni desu ka.
Ken-NOM vase-ACC put C-TOP who-GEN desk-on COP.POL Q
'On whose desk is it that Ken put a vase?'

b. ?*Mai-wa [[Ken-ga e_i kabin-o oita no]-wa dare-no tukue-ni$_i$ (da) ka]
Mai-TOP Ken-NOM vase-ACC put C-TOP who-GEN desk-on COP Q
sitteiru.
know
Lit. 'Mai knows [on which desk it is that Ken put a vase].'

(49a) shows that the matrix cleft wh-question can be formed with the complex wh-phrase *dare-no tukue* 'whose desk'. However, realizing the same cleft wh-question as an embedded CQ wh-question yields degradation, as shown by (49b). This observation indicates that a complex wh-phrase undergoes covert wh-movement in embedded CQ wh-questions.

The above observations thus indicate that the embedded CQ wh-question with the adjunct wh-word *doo* 'how' in (44)(/(30a)) and that with the complex wh-phrase *dare-no hon* 'whose book' in (47)(/(34)) involve covert wh-movement, even though they are not labeled as D but as C. Therefore, the prediction under consideration (i.e. that the embedded CQ wh-questions in (44) and (47) do not involve covert wh-movement, given that they should be labeled by C, not D) is not borne out.

(Consider, e.g., the situation where graduate students of a syntax course were required to write a review about one of the specified papers.)

(i) a. ?*Ken-sika dare-no ronbun-o {rebyuusi-nakat-ta desu ka / rebyuusi-masen desita
Ken-SIKA who-GEN paper-ACC review-not-PAST COP.POL Q review-not.POL COP.POL.PAST
ka}.
Q
'Whose paper did only Ken review?'

b. Mai-wa [Ken-sika dare-no ronbun-o rebyuusi-nakat-ta ka] sitteiru.
Mai-TOP Ken-SIKA who-GEN paper-ACC review-not-PAST Q know
'Mai knows [whose paper only Ken reviewed].'

Now, let us recapitulate what this subsection has elucidated concerning embedded CQ wh-questions. This is summarized in (50).

(50) a. Embedded CQ wh-questions are labeled either as D (i.e. free relatives) or as C (i.e. interrogative clauses) (i.e. (36)).

b. Embedded CQ wh-questions involve covert wh-movement regardless of their label; in particular, they involve covert wh-movement even if they are labeled as C.

Let us then evaluate these findings with respect to the FRBM Account in (25), repeated below.

(25) *Free-Relative-Building Movement (FRBM) Account*
Covert wh-movement occurs in embedded CQ wh-questions to render their structure that of free relatives, i.e. to label them as D.

A prediction that the hypothetical account in (25) makes is that covert wh-movement would not occur if embedded CQ wh-questions are labeled not as D but C, since it is induced in order to label them as D. However, this prediction is crucially incompatible with our finding (50b); wh-movement takes place in embedded CQ wh-questions even when they have C as their label. Therefore, I conclude that the potential account of the Wh-Movement Asymmetry in (25), or our attempt to attribute the involvement of covert wh-movement in embedded CQ wh-questions to their nominal structure, must be given up.[14]

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[14] There is another potential way of linking the nominality of embedded CQ wh-questions to the involvement of covert wh-movement in those questions that builds on Berman's (1991) analysis of wh-questions. According to him, wh-clauses themselves denote (open) propositions, with wh-elements serving as indefinites. Under this view, for example, when the verb *know* takes a wh-clause as its complement, it denotes a proposition, rather than a question; this is compatible with the fact that *know* can take both declaratives, which obviously denote propositions, and interrogatives as its complement. Concerning verbs that can take only interrogatives as their complement like *ask*, however, Berman suggests that wh-clauses selected by those verbs involve a question operator, as a result of which those wh-clauses denote questions.

Consider then a CQ-predicate in Japanese *sitteiru* 'know'. According to Berman, wh-clauses selected by this verb denote propositions. Consider, e.g., the wh-clause, or the embedded CQ wh-question, selected by *sitteiru* in (i).

5.2.3 Toward a satisfactory analysis of Japanese wh-questions

To recap, this section has pursued an account of the Wh-Movement Asymmetry in terms of an "internal" property of embedded CQ wh-questions, namely their nominal structure (Tomioka 2020). More specifically, adopting C&D's analysis of free relatives, I have explored the possibility of identifying the establishment of the structure of free relatives as the trigger of covert wh-movement in embedded CQ wh-questions. However, it has been revealed that embedded CQ wh-questions do not necessarily show nominality (i.e. they are labeled either as D or C), and that they involve covert wh-movement even when they do not show nominality (i.e. even when they are not labeled as D but as C). These observations have forced us to discard the possibility that covert wh-movement in embedded CQ wh-questions occurs to render them nominal, namely the FRBM Account in (25).

(i) Mai-wa [Ken-ga nani-o katta ka] sitteiru.
Mai-TOP Ken-NOM what-ACC bought Q know
'Mai knows what Ken bought.'

If we assume that the wh-clause in (i) involves covert wh-movement, its structure can be represented as in (ii).

(ii) sitteiru [nani [$_{\text{Proposition}}$ ka Ken katta nani]] (word order irrelevant)
know what Q Ken bought what

Given Berman's claim that wh-elements are interpreted as indefinites, the complement of *sitteiru* can be interpreted as a nominal phrase where the propositional clause serves as a relative clause that modifies its head, i.e. the indefinite *nani* 'what'. Being selected by a CQ-predicate, that nominal phrase is then interpreted as a concealed question. This way, covert wh-movement in embedded CQ wh-questions can be tied to the nominality of those wh-questions.

However, this analysis makes a wrong prediction regarding another CQ-predicate, *tazuneta* 'asked'. Under Berman's view, wh-clauses selected by this predicate denote questions (by containing a question operator), rather than propositions. Consider then the wh-clause selected by *tazuneta* in (iii).

(iii) Mai-wa [Ken-ga nani-o katta ka] tazuneta.
Mai-TOP Ken-NOM what-ACC bought Q asked
'Mai asked what Ken bought.'

Suppose that the wh-clause in (iii) involves covert wh-movement. Then, its structure can be represented as in (iv).

(iv) tazuneta [nani [$_{\text{Question}}$ ka Ken katta nani]] (word order irrelevant)
asked what Q Ken bought what

The complement phrase in (iv) crucially differs from that in (ii) in that the clause following *nani* 'what' is a question, rather than a proposition. That clause then cannot serve as a relative clause to modify the indefinite *nani*. It is hence predicted that the wh-clause in (iii) cannot involve wh-movement. However, wh-clauses selected by *tazuneta* 'ask', being embedded CQ wh-questions, do involve covert wh-movement, as shown in Chapter 3 based on obviation of intervention effects and cleft formation. In addition to this issue, the analysis under consideration (like the FRBM Account) cannot explain why covert wh-movement occurs in embedded CQ wh-questions with an adjunct wh-word or with a complex wh-phrase even though they do not show nominality, as discussed in the main text. Given those issues, I will not adopt this alternative analysis in this dissertation.

Even though our attempt has turned out to be in vain, this does not mean that it was meaningless. In particular, the discussion in this section is significant in that it has uncovered new facts about a peculiar property of embedded (CQ) wh-questions in Japanese that has not been seriously discussed until recently, namely their nominal nature (Tomioka 2020). More specifically, this section has revealed that those wh-questions do not always exhibit nominality; embedded CQ wh-questions with an adjunct wh-word such as *doo* 'how' and *naze* 'why' or a complex wh-phrase such as *dare-no X* 'whose X' cannot be construed as nominal, as indicated by the impossibility of wh-classifier matching with them (see (33) and (35)). Therefore, a satisfactory analysis of Japanese wh-questions must be able to account for this restricted distribution of the nominality of embedded CQ wh-questions. Given that, the next section will aim to propose an analysis of Japanese wh-questions which is satisfactory in the sense that it gives answers to the two relevant significant issues, namely the Wh-Movement Asymmetry and the nominality of embedded CQ wh-questions with its restricted distribution.

5.3 "External" property of embedded CQ wh-questions: Selection

The main goal of this section is to pursue an analysis of wh-questions in Japanese that gives an answer to the question that the last section has left unresolved, namely what the Wh-Movement Asymmetry follows from. Another goal is to give an account of the nominality of embedded CQ wh-questions and its distribution, which has been explored to some extent in the last section. In a nutshell, this section will aim to propose an analysis of Japanese wh-questions that can give answers to all the questions in (51) and (52).

(51) *Wh-Movement Asymmetry*

a. Why is it that embedded CQ wh-questions involve covert wh-movement?

b. Why is it that the other types of wh-questions (i.e. matrix wh-questions, embedded non-CQ wh-questions and *to*-embedded wh-questions) do not involve covert wh-movement?

(52) *Nominality of embedded CQ wh-questions and its distribution*

a. How is the nominality of embedded CQ wh-questions derived?

b. Why is it that embedded CQ wh-questions with an adjunct wh-word such as *doo* 'how' and *naze* 'why' do not show nominality?

c. Why is it that embedded CQ wh-questions with a complex wh-phrase, such as *dare-no hon* 'whose book', do not show nominality?

While the last section has zoomed in on a distinct "internal" property of embedded CQ wh-questions, this section will direct the focus to their distinct "external" property, i.e. the fact that they are selected by CQ-predicates, which are characterized differently from the selectors of other types of wh-questions in terms of their selectional properties. To syntactically implement this property, I will adopt a syntactic selection mechanism proposed by Donati and Cecchetto (2011). I will then propose an analysis of Japanese wh-questions that incorporates that selection mechanism, and discuss how it is relevant to the above-mentioned two issues, i.e. (51) and (52).

5.3.1 Syntactic categories selected as questions

I start this section by establishing a distinct "external" property of embedded CQ wh-questions, or how those wh-questions selectionally differ from the other types of wh-questions. In particular, I will compare different types of wh-questions in terms of what syntactic categories their selectors can select. Since selectional characteristics of those wh-questions have been descriptively introduced in Chapter 3 (i.e. Section 3.2), the following discussion can be taken as a reinterpretation of (part of) that description from a more syntactic perspective. Incidentally, I will not deal with matrix wh-questions below, since they are not selected by any element.

Let us first consider embedded CQ wh-questions. As mentioned above, what selects them is CQ-predicates. Some examples of CQ-predicates are listed in (53).

(53) an incomplete list of CQ-predicates:
akasu 'reveal', *kiku* 'ask', *kirokusuru* 'record', *kotaeru* 'answer', *oboeteiru* 'remember', *osieru* 'teach/tell', *sitteiru* 'know', *tazuneru* 'ask', etc.

As defined in Chapter 3 (i.e. Section 3.2.2), CQ-predicates refer to question-selecting predicates that can take a concealed question as their complement. This means that they select either clausal elements or nominal elements as questions, as exemplified in (54).

(54) a. Mai-wa [Ken-ga nani-o katta ka] {sitteiru / akasita / tazuneta}.
Mai-TOP Ken-NOM what-ACC bought Q know / revealed / asked
'Mai {knows / revealed / asked} [what Ken bought].'

b. Mai-wa [Itaria-no syuto]-o {sitteiru / akasita / tazuneta}.
Mai-TOP Italy-GEN capital-ACC know / revealed / asked
'Mai {knows / revealed / asked} the capital of Italy.'
≈ 'Mai {knows / revealed / asked} [what the capital of Italy is].'

To paraphrase this property from a syntactic perspective, I here follow the standard assumption that clausal and nominal elements correspond to CP and DP respectively. Then, the selectional property of CQ-predicates can be stated as follows: they can select CP and DP as questions.[15]

[15] Note that I here put aside the possibility that question clauses could have D(P) as their syntactic category, as in C&D's analysis; I merely follow the run-of-the-mill syntactic assumption that those clauses correspond to CP.

What about embedded non-CQ wh-questions? They are selected by non-CQ-predicates, examples of which are shown in (55).

(55) an incomplete list of non-CQ-predicates:
gironsuru 'discuss', *kaisetusuru* 'explain', *kansatusuru* 'observe', *kansin-ga aru* 'be interested', *sinpaisiteiru* 'be anxious', *situmonsuru* 'question', etc.

They crucially differ from CQ-predicates in that they cannot select concealed questions. That is, they can select only clausal elements, but not nominal ones, as questions, as (56) shows. (What is important for (56b) is that while it is grammatical (but infelicitous with *gironsita* 'discussed'), its nominal complement can never be construed as a (concealed) question, as the translation suggests.)

(56) a. Mai-wa [Ken-ga nani-o katta ka] {sinpaisiteiru / gironsita / kaisetusita}.
Mai-TOP Ken-NOM what-ACC bought Q is.anxious / discussed / explained
'Mai {is anxious / discussed / explained} [what Ken bought].'

b. Mai-wa [Itaria-no syuto]-o {sinpaisiteiru / #gironsita / kaisetusita}.
Mai-TOP Italy-GEN capital-ACC is.anxious / discussed / explained
'Mai {is anxious about / discussed / explained} the capital of Italy.'
≉ 'Mai {is anxious about / discussed / explained} [what the capital of Italy is].'

From a syntactic perspective, then, it can be stated that non-CQ predicates can select only CP as questions, but not DP.

Let us finally consider *to*-embedded CQ wh-questions. Observe first (57), which contains this type of a wh-question.

(57) Mai-wa [[Ken-ga nani-o katta ka] to] {tazuneta / sinpaisiteiru}.
Mai-TOP Ken-NOM what-ACC bought Q REP asked / is.anxious
Lit. 'Mai {asked / is anxious} [that [what Ken bought]].'

Our only concern here is the (local) selectional relationship between the question clause headed by *ka* and the reportative complementizer *to*; I leave it open how the matrix predicate is involved with selection (e.g. whether it locally selects the *to*-headed clause, or it non-locally selects the *ka*-headed wh-clause, or both; see Ishii 2013 for relevant discussion).

With this in mind, let us now consider what the complementizer *to* can select as a question. (57) shows that it can select clausal elements as questions. However, it cannot select concealed questions; in (58), for example, the nominal element that is apparently selected by *to* is intended to be interpreted as a (concealed) question, but that expected reading is not available (even though the matrix predicate is a CQ-predicate *tazuneta* 'asked').

(58) *Mai-wa [[Itaria-no syuto] to] {tazuneta / sinpaisiteiru}.
Mai-TOP Italia-GEN capital REP asked / is.anxious
Lit. 'Mai {asked / is anxious} [that [the capital of Italy]].'
≉ Lit. 'Mai {asked / is anxious} [that [what the capital of Italy is]].'

Hence, in syntactic terms, the complementizer *to* can select only CP as questions, but not DP.

Now, to compare the three wh-questions in terms of their external/selectional property, observe (59), which summarizes what syntactic categories the selectors of those wh-questions can select as questions.

(59) a. embedded CQ wh-questions:
- selector: CQ-predicates
- selectable categories as questions: <u>CP, DP</u>

b. embedded non-CQ wh-questions:
- selector: non-CQ-predicates
- selectable categories as questions: <u>CP</u>

c. *to*-embedded wh-questions:
- selector: reportative complementizer *to*
- selectable categories as questions: <u>CP</u>

Notice that (59) shows that embedded CQ wh-questions differ from the other types of wh-questions in terms of "selectable categories as questions"; the selector of the former wh-questions can select a wider range of syntactic categories (i.e. CP and DP) as questions than the selector of the latter wh-questions can (i.e. only CP). (Another way to look at this is that non-CQ predicates and the complementizer *to* are more selectionally restricted.) The "external" property of embedded CQ wh-questions is thus differentiated from that of the other types of wh-questions.

The next task, then, is to syntactically implement this property, given the current goal of presenting a syntactic analysis of Japanese wh-questions that captures the Wh-Movement Asymmetry (as well as the nominality of embedded CQ wh-questions). What I will adopt for that purpose is the syntactic mechanism of selection proposed by Donati and Cecchetto (2011); the next subsection will provide an overview of it.

5.3.2 Syntactic mechanism of selection: Donati and Cecchetto (2011)

Donati and Cecchetto's (2011) approach to selection is based on the intuition that selection involves an element still located in the numeration and a syntactic object already formed in the computational space. They propose a theoretical implementation of this idea, according to which a selectional feature of an element in the numeration probes into the syntactic object to find its goal, i.e. the categorial feature that will satisfy its selectional requirement.[16] As an example, let us consider how the lexical item *think* in the numeration selects the syntactic object *that Mary will leave* under the mechanism in question; see the representation in (60).

(60) Numeration: {…, think^{C}, …}
Syntax: [$_{C}$ that Mary will leave] (Donati and Cecchetto 2011: 546)

[16] Donati and Cecchetto (2011) note that a similar idea is proposed by Rizzi (2008).

The verb *think* in the numeration has the selectional feature C, represented by the superscript in (60). This feature probes into the syntactic object and finds C as its goal, represented by the shading. Since the categorial feature C is on the root node, the verb *think* then successfully merges with the syntactic object, satisfying its selectional requirement, as shown in (61).

(61) [think$^{\cancel{C}}$ [$_C$ that Mary will leave]] (ibid: 546)

Since this mechanism in a sense involves the establishment of a probe-goal/Agree relation across representations (i.e. between the numeration and the syntax), I will refer to this syntactic mechanism of selection as the *Cross-Representational Selection Mechanism* (henceforth, *CRSM*).

They further argue that the CRSM triggers movement. In particular, they argue that the mechanism under consideration explains how the head raising is triggered in relative clauses under the head-raising analysis of those clauses (e.g. Vergnaud 1974, Kayne 1994, Bianchi 1999, Bhatt 2002). According to that analysis, the relative clause in (62a), for example, is derived as in (62b), where the bare noun *man* originates in the relative clause and raises out of it, and then the determiner merges with the raised head.

(62) a. the man that will come

b. the man [that ~~man~~ will come]

Donati and Cecchetto explain how the head raising in (62b) is motivated under the CRSM. Consider (63), one derivational point of (62a).

(63) Numeration: {…, the^N, …}

Syntax: [$_C$ that man$_N$ will come] (ibid: 546)

In (63), according to the CRSM, the selectional feature of the lexical item *the* in the numeration, namely N, probes into the syntactic object. It finds its goal *man* below the root node, unlike (60), where the probe finds the root node C itself as its goal.[17] They propose that this probe-goal match (i.e. Agree) triggers movement of *man*, as (64) shows.

(64) [$_?$ man$_N$ [$_C$ that ~~man~~ will come]]

What plays a crucial role at this point is C&D's proposal that moved elements can project, which was introduced in Section 5.2.2.1. Consider, for example, the wh-clause and its structure, where *what* has undergone wh-movement, in (65).

(65) a. what you read

b. [$_?$ what$_D$ [$_C$ C you read ~~what~~]]

According to the proposal in question, the structure in (65) can be projected not only by C (as standardly assumed) but also by the moved wh-word, labeling the structure as its categorial feature D. In particular, the wh-clause is interpreted as a free relative when labeled as D, as shown in (66).

(66) I read [$_{D(P)}$ what$_D$ [$_C$ C you read]].

Baded on this proposal, Donati and Cecchetto argue that the structure in (64) is labeled as N by the moved lexical element *man*, as represented in (67).

(67) [$_N$ man [$_C$ that ~~man~~ will come]]

[17] Notice that this probe-goal match relation itself does not meet the selectional requirement of the determiner *the*; instead, that requirement is satisfied when *the* merges with the syntactic object which is labeled as N as a result of the noun *man* undergoing movement, as will be discussed later (see (67-68)).

The resulting syntactic object, labeled as N, satisfies the selectional requirement of the lexical item *the*, after *the* is merged into the structure, as in (68).

(68) the$^{\cancel{N}}$ [$_N$ man [$_C$ that ~~man~~ will come]] (ibid: 546)

Recall here that what we are aiming to do is to syntactically implement the distinctness of the external/selectional property of embedded CQ wh-questions, as established in Section 5.3.1. The CRSM thus fares well with this aim, given that it implements selection syntactically, i.e. through a syntactic mechanism.

5.3.3 CRSM-based analysis of Japanese wh-questions

In this subsection, I will extend the CRSM outlined above to selection of questions in Japanese, and propose a CRSM-based analysis of wh-questions in Japanese. In particular, we will see how that analysis can provide answers to the Wh-Movement Asymmetry in (51), as well as the questions regarding the nominality of embedded CQ wh-questions in (52), repeated below.

(51) *Wh-Movement Asymmetry*

a. Why is it that embedded CQ wh-questions involve covert wh-movement?

b. Why is it that the other types of wh-questions (i.e. matrix wh-questions, embedded non-CQ wh-questions and *to*-embedded wh-questions) do not involve covert wh-movement?

(52) *Nominality of embedded CQ wh-questions and its distribution*

a. How is the nominality of embedded CQ wh-questions derived?

b. Why is it that embedded CQ wh-questions with an adjunct wh-word such as *doo* ‘how’ and *naze* ‘why’ do not show nominality?

c. Why is it that embedded CQ wh-questions with a complex wh-phrase, such as *dare-no hon* 'whose book', do not show nominality?

5.3.3.1 Assumptions about categorial and selectional features

To propose a CRSM-based analysis of Japanese wh-questions, I will first make some assumptions concerning categorial and selectional features of the relevant elements.

First, I assume (69) regarding categorial features of the Q-particle *ka*, single nominal wh-words, and single adjunct wh-words.

(69) a. The Q-particle *ka* has the categorial feature C_{wh}.

b. Single nominal wh-words, such as *nani* 'what' and *dare* 'who', have the categorial feature D_{wh}.

c. Single adjunct wh-words, such as *doo* 'how' and *naze* 'why', have the categorial feature ADV_{wh}.

Notice in particular that those categorial features involve the subscript "$_{wh}$": C_{wh}, D_{wh} and ADV_{wh}. They can be viewed as subcategories of C, D and ADV, respectively. The assumption of this subscript is motivated by the intuition that all lexical items noted in (69) are related to wh-questions.[18]

[18] It may be said that the current assumption about subcategorization of syntactic categories follows the spirit of *l-selection*, proposed by Pesetsky (1992) (see also Bošković 1997b); "[l]-selection makes reference to *subcategories* of syntactic categories — in the limiting case, to individual words and, additionally (perhaps) to features like [+finite]" (Pesestky 1992: 10). For example, while both nouns *love* and *desire* can co-occur with a PP, they differ regarding which individual preposition is used; *love* allows either *for* or *of*, while *desire* allows only *for*. In addition, while the verbs *like* and *enjoy* can both take CP complements, they are different in the specific types of those complements; *like* takes either an infinitival or a gerund, whereas *enjoy* allows only the latter, as (i) shows.

(i) a. She liked {to hear / hearing} the concerto.

b. She enjoyed {*to hear / hearing} the concerto. (Pesestky 1992: 11)

These examples are instances of l-selection, in that specific lexical information of the complements (e.g. individual prepositions, infinitivals vs. gerunds) is involved in selection. The current assumption is similar to l-selection in that the categorial features under consideration involve particular lexical information that pertains to wh-questions. Given that, the current assumption could be taken as an attempt to implement the idea of l-selection in terms of the CRSM.

Second, I assume (70) regarding the structure of concealed questions.

(70) Concealed questions are headed by D_{wh}.

This assumption is based on the intuition that concealed questions are nominal (hence, D) but are interpreted as wh-questions (hence, the subscript "$_{wh}$").

Finally, I assume (71) regarding the selectional features of CQ-predicates, non-CQ-predicates and the reportative complementizer *to*.

(71) a. CQ-predicates have a selectional feature X_{wh}, where X is any category.
b. Non-CQ-predicates have a selectional feature C_{wh}.
c. The reportative complementizer *to* has a selectional feature C.

This assumption can be taken to be a syntactic implementation of the observation about external/selecitonal properties of wh-questions made in Section 5.3.1. Witness again the summary of that observation in (59), repeated below as (72).

(72) a. embedded CQ wh-questions: - selector: CQ-predicates
- selectable categories as questions: <u>CP, DP</u>
b. embedded non-CQ wh-questions: - selector: non-CQ-predicates
- selectable categories as questions: <u>CP</u>
c. *to*-embedded wh-questions: - selector: reportative complementizer *to*
- selectable categories as questions: <u>CP</u>

What (72) shows is that the selector of embedded CQ wh-questions, i.e. CQ-predicates, can select a wider range of syntactic categories (i.e. CP and DP) than the selectors of the other types of wh-questions can (i.e. only CP). (71) syntactically captures this fact. On the one hand, (71a) amounts to saying that CQ-predicates can select any syntactic category, as long as it is subcategorized by the subscript "$_{\text{wh}}$". On the other hand, according to (71b-c), the other selectors (i.e. non-CQ-predicates and the complementizer *to*) can select only one category, namely C($_{\text{wh}}$). This way, the assumption in (71) reflects the distinctness of the external property of embedded CQ wh-questions indicated in (72).[19] Note further that in (71), the selectional feature of the reportative complementizer *to*, C, is not subcategorized by the subscript "$_{\text{wh}}$", in contrast to that of non-CQ-predicates, C_{wh}. This is based on the fact that *to* can select not only question clauses (e.g. *to*-embedded wh-questions) but also sentences with various clause types, such as declarative clauses and even imperative clauses (e.g. Schwager 2006, Saito 2010, 2012, 2013, 2015, Kaufmann 2012), as noted in Chapter 3 (i.e. Section 3.2.1). The relevant examples from that section are repeated below.[20]

(73) a. Mai$_i$-wa [[Ken-ga kanozyo$_i$-no heya-o soojisuru] to] omotteiru.
Mai-TOP Ken-NOM she-GEN room-ACC clean REP think
'Mai thinks [that Ken will clean her room].'

b. Mai-wa [[Ken-ga nani-o katta ka] to] tazuneta.
Mai-TOP Ken-NOM what-ACC bought Q REP asked
Lit. 'Mai asked [that what Ken bought].'

c. Mai$_i$-wa Ken-ni [[kanozyo$_i$-no heya-o soojisi-ro] to] itta.
Mai-TOP Ken-to she-GEN room-ACC clean-IMP REP said
Lit. 'Mai said to Ken [that clean her room].'

[19] One might wonder if there are other categories subcategorized by "$_{\text{wh}}$", such as P_{wh}, which would wrongly predict that PPs could serve as questions. I suggest that only CPs and DPs can be interpreted as questions (cf. Grimshaw's 1979 notion of s-selection).

[20] Recall that, as discussed in Section 3.2.1, the complement clauses in (73) cannot be regarded as direct quotes. In (73a,c), the embedded clauses contain a pronominal element *kanozyo* 'she', which refers to the matrix subject *Mai*. This reading would not be possible if those embedded clauses were a direct quote. In (73b), the embedded wh-question does not include the politeness marker *-mas*, which is an obligatory element for matrix wh-questions (Miyagawa 1987; see Section 3.2.1), and is thus not a direct quote of a matrix wh-question.

Given this fact and the standard assumption that clause types are associated with C, the selectional feature of the complementizer *to* should be less restricted than C_{wh}, hence C, which allows it to select clauses of any clause type.

5.3.3.2 Implementing the CRSM-based analysis

With the above assumptions, I will now illustrate how a CRSM-based analysis of wh-questions (including concealed questions) in Japanese works. More specifically, below I will demonstrate how that analysis derives each type of wh-questions, and show how it can provide answers to the questions in (51) and (52) above. The following discussion will be divided into three parts: analyses of (i) concealed questions, (ii) embedded CQ wh-questions (with a single nominal wh-word, a single adjunct wh-word, and a complex wh-phrase), and (iii) the other types of wh-questions (i.e. matrix wh-questions, embedded non-CQ wh-questions and *to*-embedded wh-questions).

5.3.3.2.1 Concealed questions

I will first describe a CRSM-based analysis of concealed questions, which will constitute a baseline case for the following demonstrations. Consider (74), where the concealed question *Itaria-no syuto* 'the capital of Italy' is selected by the CQ-predicate *sitteiru* 'know'.

(74) Mai-wa [Itaria-no syuto]-o sitteiru.
Mai-TOP Italy-GEN capital-ACC know
'Mai knows the capital of Italy.'
≈ 'Mai knows [what the capital of Italy is].'

Among the assumptions in Section 5.3.3.1, relevant to the current analysis are (70) and (71a), repeated below.

(70) Concealed questions are headed by D_{wh}.

(71) a. CQ-predicates have a selectional feature X_{wh}, where X is any category.

With those assumptions, one derivational point of (74) can be represented as in (75), where the head of the concealed question D_{wh} provides the label of it.

(75) Numeration: {…, sitteiru^{Xwh}, …}
know
Syntax: [$_{Dwh}$ D_{wh} [Itaria syuto]] (word order irrelevant)
Italy capital

With the CRSM, the selectional feature X_{wh} in the numeration probes into the syntactic object, i.e. the concealed question. Since X in X_{wh} refers to any category, it finds D_{wh} as its goal, as the shading in (76) shows.

(76) Numeration: {…, sitteiru^{Xwh}, …}
know
Syntax: [$_{Dwh}$ D_{wh} [Itaria syuto]] (word order irrelevant)
Italy capital

Since D_{wh} is on the root node, *sitteiru* 'know' then successfully merges with the concealed question, satisfying its selectional requirement, as shown in (77).

(77) sitteiru$_{\cancel{Xwh}}$ [$_{Dwh}$ D_{wh} [Itaria syuto]] (word order irrelevant)
know Italy capital

5.3.3.2.2 Embedded CQ wh-questions

Next, I will describe our main concern, i.e. a CRSM-based analysis of embedded CQ wh-questions. I will show how the analysis in question accounts for the involvement of covert wh-movement in those wh-questions (i.e. (51a)) as well as their nominality, with its restricted distribution (i.e. (52)). In particular, I will present a crucial proposal regarding Agree between a selectional and a categorial feature. The discussion will be split into three subcases: embedded CQ wh-questions with (i) a single nominal wh-word, (ii) a single adjunct wh-word, and (iii) a complex wh-phrase.

<u>[i] Embedded CQ wh-questions with a single nominal wh-word</u>

Let us first consider the derivation of (78), whose embedded CQ wh-question contains a single nominal wh-word *nani* 'what'.

(78) Mai-wa [Ken-ga <u>nani</u>-o katta ka] sitteiru.
Mai-TOP Ken-NOM what-ACC bought Q know
'Mai knows [what Ken bought].'

Among the assumptions made in Section 5.3.3.1, the ones relevant to the current discussion are (69a,b) and (71a), repeated below:

(69) a. The Q-particle *ka* has the categorial feature C_{wh}.
b. Single nominal wh-words, such as *nani* 'what' and *dare* 'who', have the categorial feature D_{wh}.

(71) a. CQ-predicates have a selectional feature X_{wh}, where X is any category.

With those assumptions, one derivational point of (78) can be represented as in (79), where the structure is labeled by C_{wh}, the categorial feature of the Q-particle *ka*.

(79) Numeration: {…, sitteiru$^{X_{wh}}$, …}
know
Syntax: [$_{C_{wh}}$ ka$_{C_{wh}}$ [Ken katta nani$_{D_{wh}}$]] (word order irrelevant)
Q Ken bought what

With the CRSM, the selectional feature X_{wh} in the numeration probes into the syntactic object and finds the root node C_{wh} as its goal, as represented by the shading in (80).

(80) Numeration: {…, sitteiru$^{X_{wh}}$, …}
know
Syntax: [$_{C_{wh}}$ ka$_{C_{wh}}$ [Ken katta nani$_{D_{wh}}$]] (word order irrelevant)
Q Ken bought what

Now, if we followed Donati and Cecchetto (2011) regarding their proposal of the CRSM, the probing by X_{wh} would stop at the point when the probe finds C_{wh} as its (most local) goal. Since C_{wh} is on the root node, the verb *sitteiru* 'know' successfully merges with the syntactic object, satisfying its selectional requirement, as shown in (81).

(81) sitteiru$^{\cancel{X_{wh}}}$ [$_{\mathbf{C_{wh}}}$ ka$_{C_{wh}}$ [Ken katta nani$_{D_{wh}}$]]
know Q Ken bought what

Notice, however, that this derivation is not what we want to obtain in two respects. First, in (81), the wh-word *nani* 'what' does not undergo movement; we thus could not capture the proposal in Chapter 3 that covert wh-movement occurs in embedded CQ wh-questions. Second, with (81), it would be predicted that embedded CQ wh-questions are *always* labeled as C_{wh}. Hence, we could not capture the nominality of embedded CQ wh-questions, as discussed in the last section. Given all this, in the embedded CQ wh-question in (78), we want the wh-word *nani* 'what' to undergo movement and label the structure as D_{wh} by virtue of its categorial feature, thus rendering it nominal (i.e. a free relative).

Against this background, I now propose that a selectional feature as a probe can undergo Multiple Agree (see Hiraiwa 2001, 2005 regarding Multiple Agree), that is, it probes all the way down into the syntactic object where it can find more than one relevant feature as its goal.

Let us now reconsider the derivation of the embedded CQ wh-question in (78), beginning with the derivational step in (79). With the current proposal, when the selectional feature X_{wh} probes into the syntactic object, it first finds C_{wh} as its goal, but it keeps on probing, and then finds D_{wh} as another goal, as represented in (82).

(82) Numeration: {…, sitteiru^{Xwh}, …}
know

Syntax: [$_{Cwh}$ ka$_{Cwh}$ [Ken katta nani$_{Dwh}$]] (word order irrelevant)
Q Ken bought what

As discussed above, Agree by a selectional feature triggers movement (see Section 5.3.2). The wh-word *nani* 'what' then undergoes movement, resulting in (83).[21]

[21] One might wonder whether Agree between the selectional feature X_{wh} and (the categorial feature of) the wh-element, and the resulting covert movement of the latter, violate the Phase Impenetrability Condition (PIC). According to the PIC, syntactic operations such as agreement and movement cannot apply across a phase boundary; an element must move out of a phase to its edge to undergo an operation induced by another element above that phase. The PIC thus deduces successive cyclic movement. However, it has been argued in the literature that the PIC is not the locality condition for syntax, but its effect (i.e. successive cyclic movement) essentially follows from interactions between phases and (multiple) Spell-Out (e.g. Stjepanović and Takahashi 2001, Bošković 2005, Fox and Pesetsky 2005). The gist of Bošković's (2005) idea, for example, goes as follows: under the assumption of multiple Spell-Out, when a phasal domain undergoes Spell-Out and is sent to the phonology, its word order is fixed there, and thus an operation cannot apply to any element inside that domain if it alters the word order; to make such an operation possible, an element which that operation will apply to must escape out of the phase before it undergoes Spell-Out, hence successive cyclic movement. With such a view, he argues that syntactic operations that do not affect word order, such

(83) Numeration: {…, sitteiru$^{X_{wh}}$, …}
know

Syntax: [? nani$_{Dwh}$ [$_{Cwh}$ ka$_{Cwh}$ [Ken katta ~~nani~~]] (word order irrelevant)
what Q Ken bought what

The current proposal thus gives an answer to the question in (51a), repeated below:

(51) a. Why is it that embedded CQ wh-questions involve covert wh-movement?

More specifically, the wh-movement under consideration is triggered by (Multiple) Agree with the selectional feature X_{wh} of the CQ-predicate in the numeration.

Consider now how the structure in (83) is labeled. I claim that it can be labeled in two ways. First, it can be labeled by the categorial feature of the target of the wh-movement in (83), namely C_{wh}, if we follow the standard assumption that the target of movement, or a root node, can project and provide the label. Second, it can be labeled by D_{wh}, the categorial feature of *nani* 'what', given C&D's proposal that a moved element can project (see Section 5.2.2.1 and Section 5.3.2). The observation in Section 5.2.2.2 that embedded CQ wh-questions can be labeled either as D or C (i.e. (36)) thus follows. Those two possibilities can then be represented as in (84), where the CQ-predicate *sitteiru* 'know' has undergone merge as well.

(84) a. sitteiru$^{\cancel{X_{wh}}}$ [$_{\mathbf{Cwh}}$ nani$_{Dwh}$ [$_{\mathbf{Cwh}}$ ka$_{Cwh}$ Ken katta ~~nani~~] (word order irrelevant)
know what Q Ken bought what

b. sitteiru$^{\cancel{X_{wh}}}$ [$_{\mathbf{Dwh}}$ nani$_{\mathbf{Dwh}}$ [$_{Cwh}$ ka$_{Cwh}$ Ken katta ~~nani~~] (word order irrelevant)
know what Q Ken bought what

as Agree and covert movement, can apply across phase boundaries. I adopt this view and hence do not take phases into consideration when I deal with syntactic operations in this section, given that they do not affect word order.

In particular, (84b) shows that the embedded CQ wh-question can be derived as D_{wh}, or as nominal. Hence, the CRSM-based analysis, supplemented by the above proposal, can provide an answer to the question in (52a), repeated below:

(52) a. How is the nominality of embedded CQ wh-questions derived?

<u>[ii] Embedded CQ wh-questions with a single adjunct wh-word</u>

Let us next consider embedded CQ wh-questions with a single adjunct wh-word such as *doo* 'how' and *naze* 'why'. Here, as a case study, I will illustrate how the CRSM-based analysis derives the embedded CQ wh-question in (85), which contains *doo* 'how' (see fn.24 for a remark on embedded CQ wh-questions with *naze* 'why').

(85) Mai-wa [Ken-ga <u>doo</u> kaetta ka] sitteiru.
Mai-TOP Ken-NOM how returned Q know
'Mai knows [how Ken went home].'

The relevant assumptions from Section 5.3.3.1, (69a,c) and (71a), are repeated below:

(69) a. The Q-particle *ka* has the categorial feature C_{wh}.
c. Single adjunct wh-words, such as *doo* 'how' and *naze* 'why', have the categorial feature ADV_{wh}.

(71) a. CQ-predicates have a selectional feature X_{wh}, where X is any category.

With those assumptions, observe (86), one derivational point of (85).

(86) Numeration: {…, sitteiru^{Xwh}, …}
know
Syntax: [$_{Cwh}$ ka$_{Cwh}$ Ken kaetta doo$_{ADVwh}$] (word order irrelevant)
Q Ken returned how

In (86), the selectional feature X_{wh} starts probing into the syntactic object. With the proposal made above, X_{wh} probes all the way down into the syntactic object and finds two goals, C_{wh} and ADV_{wh} (i.e. Multiple Agree), as represented by the shading in (87).

(87) Numeration: {…, sitteiru^{Xwh}, …}
know
Syntax: [$_{Cwh}$ ka$_{Cwh}$ Ken kaetta doo$_{ADVwh}$] (word order irrelevant)
Q Ken returned how

Agree between X_{wh} and ADV_{wh} then triggers movement of the adjunct wh-word *doo* 'how', as discussed above; see (88).

(88) Numeration: {…, sitteiru^{Xwh}, …}
know
Syntax: [$_{?}$ doo$_{ADVwh}$ [$_{Cwh}$ ka$_{Cwh}$ Ken kaetta ~~doo~~]] (word order irrelevant)
how Q Ken returned how

As in the previous case, the structure in (88) can be labeled in two ways; it will be labeled either by C_{wh}, the categorial feature of the target of the wh-movement, or by ADV_{wh}, the categorial feature of the moved wh-word *doo* 'how'. (89) shows these two possible structures (where *sitteiru* 'know' has merged as well).

(89) a. sitteiru$^{\cancel{Xwh}}$ [**Cwh** doo$_{ADVwh}$ [**Cwh** ka$_{Cwh}$ Ken kaetta ~~doo~~] (word order irrelevant)
know how Q Ken returned how

b. sitteiru$^{\cancel{Xwh}}$ [**ADVwh** doo$_{\mathbf{ADVwh}}$ [Cwh ka$_{Cwh}$ Ken kaetta ~~doo~~] (word order irrelevant)
know how Q Ken returned how

Notice here that in (89b), the matrix predicate *sitteiru* 'know' takes the syntactic object labeled as ADV$_{wh}$, or intuitively speaking, an adverbial phrase, as its complement. However, adverbial phrases, being adjuncts, should not be taken by predicates as their complement by definition.[22] If this indeed holds, we can assume that the structure in (89b) is ruled out for an independent reason, even though it is in principle possible to derive it syntactically. The only possible structure for the embedded CQ wh-question in (85) would then be (89a), where the embedded CQ wh-question is labeled as C$_{wh}$.[23]

Recall now that it was shown in Section 5.2.2.2 that embedded CQ wh-questions with *doo* 'how' do not exhibit nominality, as indicated by the fact that wh-classifier matching (Tomioka 2020) is not possible with them, as in (90) (= (33a)).

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[22] Relevant here may be the fact that some predicates must co-occur with a certain adverbial phrase. In English, for example, the verb *treat* (with a specific lexical meaning) requires a manner adverb such as *badly*; without it, the sentence would be unacceptable, as (i) shows.

(i) Mary treated Peter *(badly).

If we are not dealing here with complementation, and such adverbial phrases are actually not selected by predicates, we can ascribe their obligatoriness to the (lexical) semantics of the predicate, rather than syntactic/selectional factors.

[23] In contrast to the current analysis of embedded CQ wh-questions with *doo* 'how' in Japanese, free relatives with *how* in English can be interpreted not only as adjunctive (meaning *in the way* ...; see fn.10) but also as nominal (meaning *the way* ...). Thus, they can appear in the complement position of transitive verbs as in (ia); compare it with (ib), which involves an adjunct free relative with *how*.

(i) a. Lily loathes [how all thieves work] — secretly.
≈ Lily loathes [the way that all thieves work] — secretly.
b. Jack works [how all thieves work] — secretly.
≈ Jack works [in the way that all thieves work] — secretly. (Caponigro and Pearl 2009: 156)

The fact that free relatives with *how* can be interpreted as nominal as in (ia) suggests that the wh-word *how* can involve the categorial feature D, or more generally, that *how* can have a nominal property. In fact, Caponigro and Pearl (2009) propose to analyze *how* as an NP which a null preposition takes as its complement. Regarding Japanese, I assume that *doo* 'how' has only an adjunctive/adverbial property, or the categorial feature ADV, lacking an option to have a nominal property, or the categorial feature D, in contrast to English *how*. (I also assume that the unavailable wh-clause labeled as ADV in (89b) has the reading of an adjunct free relative, being interpreted as *in the way Ken went home*.)

(90) *Mai-wa [Ken-ga **doo** kinko-o aketa ka] mit-**tu**(-gurai) sitteiru.
Mai-TOP Ken-NOM how safe-ACC opened Q 3-CLS-about know
Lit. 'For (about) three (ways), Mai knows how Ken opened the safe.'
≈ 'For (about) thee of the ways Ken opened the safe, Mai knows that Ken opened the safe in those ways.'

The current analysis captures this observation: they do not show nominality since they can be labeled only as C_{wh} as discussed above, not as D_{wh}. This then gives an answer to the question (52b), repeated below.[24]

(52) b. Why is it that embedded CQ wh-questions with an adjunct wh-word such as *doo* 'how' and *naze* 'why' do not show nominality?

[iii] Embedded CQ wh-questions with a complex wh-phrase

Let us finally discuss (91), where the embedded CQ wh-question contains the complex wh-phrase *dare-no hon* 'whose book'.

(91) Mai-wa [Ken-ga dare-no hon-o katta ka] sitteiru.
Mai-TOP Ken-NOM who-GEN book-ACC bought Q know
'Mai knows [whose book Ken bought].'

The assumptions in Section 5.3.3.1 that are relevant to this case are shown below:

(69) a. The Q-particle *ka* has the categorial feature C_{wh}.
b. Single nominal wh-words, such as *nani* 'what' and *dare* 'who', have the categorial feature D_{wh}.

[24] Embedded CQ wh-questions with *naze* 'why' can be treated in a similar way as those with *doo* 'how'.

(71) a. CQ-predicates have a selectional feature X_{wh}, where X is any category.

Based on those assumptions, (92) represents one derivational point of (91).

(92) Numeration: {…, sitteiru^{Xwh}, …}
know
Syntax: [$_{Cwh}$ ka$_{Cwh}$ Ken katta [$_{Dwh}$ dare$_{Dwh}$ hon]] (word order irrelevant)
Q Ken bought who book

Here, I assume that the complex wh-phrase *dare-no hon* 'whose book' is labeled by the categorial feature of *dare* 'who' D_{wh}, without considering how this labeling is realized. The selectional feature X_{wh} then probes into the syntactic object and finds two goals, C_{wh} and D_{wh} (of the complex wh-phrase), as indicated by the shading in (93). (One might wonder whether the probing by X_{wh} finds the categorial feature D_{wh} of the single wh-word *dare* 'who', which would then undergo movement, being extracted out of the complex wh-phrase. I discuss this possibility below.)

(93) Numeration: {…, sitteiru^{Xwh}, …}
know
Syntax: [$_{Cwh}$ ka$_{Cwh}$ Ken katta [$_{Dwh}$ dare$_{Dwh}$ hon]] (word order irrelevant)
Q Ken bought who book

As a result of this Agree, the complex wh-phrase *dare-no hon* 'whose book' undergoes movement, resulting in (94).

(94) Numeration: {…, sitteiru^{Xwh}, …}
know
Syntax: [$_?$ [$_{Dwh}$ dare$_{Dwh}$ hon] [$_{Cwh}$ ka$_{Cwh}$ Ken katta ~~[dare hon]~~]
who book Q Ken bought who book
(word order irrelevant)

Let us now consider how the syntactic object in (94) will be labeled. Note first that C_{wh} can project, given that it is the target of the wh-movement. What about D_{wh}, the categorial feature of the complex wh-phrase *dare-no hon* 'whose book'? Of crucial relevance here is the assumption that it is only a *head* (or a lexical item), not a *phrase* (or a syntactic object), that can project and provide the label when it merges with a syntactic object (see Chomsky 2013, 2015). Consider this assumption in terms of free relatives. For example, C&D observe that while wh-clauses with a single wh-word (e.g. *what you read*) can count as both interrogative clauses and free relatives, wh-clauses with a complex wh-phrase (e.g. *what book you read*) can serve only as interrogative clauses, not as free relatives; observe the contrast between (95) and (96).[25]

(95) a. I wonder [what you read].
b. I read [what you read].

(96) a. I wonder [what book you read].
b. *I read [what book you read].

This difference follows from the aforementioned standard assumption, along with C&D's proposal that moved elements can project. Regarding the wh-clause in (95), what moves is a single wh-word, or a *head*, and thus it can project, rendering the wh-clause a free relative by labeling it as D. Regarding the wh-clause in (96), however, what moves is a complex wh-*phrase*, which cannot project. As a result, the wh-clause in (96) cannot serve as a free relative, or cannot be labeled as D.

[25] Caponigro (2019, 2023) is skeptical about C&D's generalization that complex wh-phrases cannot be involved in free relatives. For example, he notes that (i), where the wh-clause must be interpreted as a free relative, is acceptable in some varieties of American English.
(i) He read [what books she read]. (Caponigro 2019: 360)
Concerning (i), Donati et al. (2022) point out that for wh-clauses with "*what* + *NP*" as in (i) to be interpreted as nominal, the noun following *what* must take a plural form; compare (i) with (96b). With this peculiarity, Donati et al. (2022) argue that the wh-clause in (i) is not a free relative but rather a headed relative clause, where *what* serves as a special kind of a determiner (i.e. D) that takes a plural (or mass) noun like *books* as its complement.

At any rate, the upshot of the above discussion is that only a head or the target/root node can project.

Bearing this point in mind, let us now return to the structure in (94). Note crucially that what has undergone movement in (94) is *dare-no hon* 'whose book', which is a complex wh-*phrase*, not a *head*. Hence, given the assumption in question, *dare-no hon* 'whose book' cannot project, and accordingly, the structure in (94) cannot be labeled as its categorial feature D_{wh}. As a result, the structure in (94) can be labeled only as C_{wh}, as represented in (97) (which involves merge of *sitteiru* 'know' as well).

(97) sitteiru^~~Xwh~~ [**Cwh** [Dwh dare hon] [**Cwh** ka Ken katta ~~[dare hon]~~]]
know who book Q Ken bought who book (word order irrelevant)

Recall here that as observed in Sec.5.2.2.2, embedded CQ wh-questions with a complex wh-phrase cannot be interpreted as nominal, given that wh-classifier matching is not possible with them, as shown in (98) (= (35)).

(98) Mai-wa [Ken-ga **dare**-no hon-o katta ka] {??san-**nin**(-gurai) /
Mai-TOP Ken-NOM who-GEN book-ACC bought Q 3-CLS-about
?*san-satu(-gurai)} sitteiru.
3-CLS-about know
Lit. 'For (about) three (authors), Mai knows whose book Ken bought.'
≈ 'For (about) three of the authors whose books Ken bought, Mai knows that Ken bought those authors' books.' /
Lit. 'For (about) three (books), Mai knows whose book Ken bought.'
≈ 'For (about) three of the books of some authors that Ken bought, Mai knows that Ken bought those books.

This observation then follows from the current analysis, according to which those embedded wh-questions cannot be labeled as D_{wh}, being always labeled as C_{wh}, as (97) shows. The CRSM-based analysis thus provides an answer to the question (52c), repeated below:

(52) c. Why is it that embedded CQ wh-questions with a complex wh-phrase, such as *dare-no hon* 'whose book', do not show nominality?

Before closing the discussion regarding the current subcase, two notes are in order. First, as mentioned above, a possibility to consider is that the derivation of the wh-clause in (91) may involve Agree between the selectional feature X_{wh} and the categorial feature D_{wh} of the single wh-word *dare* 'who', as a result of which the wh-word would undergo movement, being extracted out of the complex wh-phrase. This possibility would allow for the wh-clause to be labeled as D_{wh} by the moved single wh-word. However, this is inconsistent with the observation that wh-classifier matching is not possible with the wh-clause in (91), as shown by (98). Hence, I assume that the alternative, subextraction derivation under consideration is not available. Notice, however, that it is not the case that such a derivation is not allowed in general; some languages can form free relatives with complex wh-phrases as long as a single wh-word within them undergoes movement, i.e. as long as the relevant subextraction is allowed. Observe (99-100), for example.

(99) German: *was für Bücher* 'what kind of books'

a. ??Seine Mutter kauft, **was für Bücher** auch immer Johann gerade liest.
his mother buys what for books also ever Johann currently reads
Lit. 'His mother buys what kind of books Johann also currently reads.'

b. Seine Mutter kauft, **was** auch immer Johann **für Bücher** gerade liest.
his mother buys what also ever Johann for books currently reads
Lit. 'His mother buys what kind Johann also currently reads of books.'

(Cecchetto and Donati 2010: 273)

(100) Italian: *quanto arroganti* 'how arrogant'

a. ??Detesto **quanto arroganti** sono.
I-hate how arrogant they-are
Lit. 'I hate how arrogant they are.'

b. Detesto **quanto** sono **arroganti**.
I-hate how they-are arrogant
Lit. 'I hate how they are arrogant.' (ibid: 273)

In the above a-examples, the wh-clauses involve movement of a complex wh-phrase and thus cannot serve as free relatives. In the b-examples, however, what undergoes movement is a single wh-word, which enables the wh-clauses to be construed as free relatives, being labeled as D. I leave open why a derivation like the one for the free relatives in the above b-examples is unavailable for the wh-clause in (91), merely noting that this kind of subextractions show a great deal of crosslinguistic variation.

Second, there is one exception to the generalization suggested in (52c), i.e. the one that embedded CQ wh-questions with a complex wh-phrase do not show nominality: embedded CQ wh-questions with *dono X* 'which X' can be interpreted as nominal. For example, (101) shows that the embedded CQ wh-question involving the complex wh-phrase *dono hon* 'which book' is compatible with wh-classifier matching, which indicates the nominality of that wh-question (see also (6), which shows the same point).

(101) Mai-wa [Ken-ga **dono hon**-o katta ka] san-**satu**(-gurai) sitteiru.
Mai-TOP Ken-NOM which book-ACC bought Q 3-CLS-about know
Lit. 'For (about) three (books), Mai knows which books Ken bought.'
≈ 'For (about) three books (within a certain domain of books) that Ken bought, Mai knows that Ken bought those books.'

This fact suggests that embedded CQ wh-questions with the complex wh-phrase *dono X* 'which X' can be labeled as D_{wh}, in contrast to those with another complex wh-phrase *dare-no X* 'whose X' (see the above discussion, in particular (98)). I will leave open how to account for this contrast (e.g. (98) vs. (101)), merely noting two possibilities. The first possibility is that in the case of *dono X* 'which X', the single wh-phrase *do*(*no*) 'which' can undergo subextraction, i.e. covert movement out of the complex wh-phrase, as in (99b) and (100b); this option should then be unavailable for *dare* 'who' in *dare-no X* 'whose X', as assumed above. The second possibility is that the complex wh-phrase *dono X* 'which X' can be reanalyzed as a single wh-word, unlike *dare-no X* 'whose X'; as a result, its movement would enable it to label the wh-clause as D_{wh}.

5.3.3.2.3 The other types of wh-questions

I will now discuss the remaining three types of wh-questions, i.e. matrix wh-questions, embedded non-CQ wh-questions and *to*-embedded CQ wh-questions. In particular, it will be shown how the CRSM based analysis of those wh-questions captures the absence of covert wh-movement in them (i.e. (51b)).

<u>[i] Matrix wh-questions</u>

Consider first the matrix wh-question in (102).

(102) Ken-wa nani-o kai-masi-ta ka.
Ken-TOP what-ACC buy-POL-PAST Q
'What did Ken buy?'

The assumptions in Section 5.3.3.1 that are relevant to this case are shown below:

(69) a. The Q-particle *ka* has the categorial feature C_{wh}.

b. Single nominal wh-words, such as *nani* 'what' and *dare* 'who', have the categorial feature D_{wh}.

With those assumptions, the structure of (102) can be represented as in (103).

(103) [$_{Cwh}$ ka$_{Cwh}$ Ken kai-masi-ta nani$_{Dwh}$] (word order irrelevant)
Q Ken buy-POL-PAST what

Obviously, matrix wh-questions differ from the other kinds of embedded wh-questions in that they are not selected by any item. In terms of the CRSM-based analysis, this means that in the derivation of matrix wh-questions, there is no selectional feature that undergoes Agree with a wh-element, which would induce its movement. It then follows that matrix wh-questions do not involve wh-movement. Eventually, they are labeled by the categorial feature of the Q-particle *ka*, C_{wh}, as (103) shows.

[ii] Embedded non-CQ wh-questions

Let us next consider the embedded non-CQ wh-question in (104), where the matrix predicate is a non-CQ-predicate *sinpaisiteiru* 'is anxious'.

(104) Mai-wa [Ken-ga nani-o katta ka] sinpaisiteiru.
Mai-TOP Ken-NOM what-ACC bought Q is.anxious
'Mai is anxious [what Ken bought].'

The assumptions in Section 5.3.3.1 that are relevant to the following discussion are (69a-b) and (71b), repeated below.

(69) a. The Q-particle *ka* has the categorial feature C_{wh}.

b. Single nominal wh-words, such as *nani* 'what' and *dare* 'who', have the categorial feature D_{wh}.

(71) b. Non-CQ-predicates have a selectional feature C_{wh}.

Then, one derivational point of (104) can be represented as in (105).

(105) Numeration: {…, sinpaisiteiru^{Cwh}, …}
is.anxious
Syntax: [$_{Cwh}$ ka$_{Cwh}$ Ken katta nani$_{Dwh}$] (word order irrelevant)
Q Ken bought what

Crucially, the relevant selectional feature in the numeration in (105) is C_{wh}, rather than X_{wh}, which CQ-predicates have. Therefore, when that selectional feature probes into the syntactic object, it can find only the categorial feature of the root node, C_{wh}, as indicated by the shading in (106).

(106) Numeration: {…, sinpaisiteiru^{Cwh}, …}
is.anxious
Syntax: [$_{Cwh}$ ka$_{Cwh}$ Ken katta nani$_{Dwh}$] (word order irrelevant)
Q Ken bought what

Since the categorial feature of the wh-word *nani* 'what', D_{wh}, does not undergo Agree with the seletional feature, unlike the case of embedded CQ wh-questions (see Section 5.3.3.2.2), the wh-word does not undergo movement and stays in-situ. After probing by C_{wh} in (106), the non-CQ predicate *sinpaisiteiru* 'is worried' undergoes merge with the relevant syntactic object, with the selectional requirement of that predicate being satisfied, as shown in (107).

(107) sinpaisiteiru~~Cwh~~ [$_{\text{Cwh}}$ ka Ken katta nani] (word order irrelevant)
is.anxious Q Ken bought what

The CRSM-based analysis can thus account for the claim that covert wh-movement does not take place in embedded non-CQ wh-questions.

[iii] *To*-embedded wh-questions

The final subcase to be discussed is *to*-embedded wh-questions, such as the one in (108).

(108) Mai-wa [[Ken-ga nani-o katta ka] to] tazuneta.
Mai-TOP Ken-NOM what-ACC bought Q REP asked
Lit. 'Mai asked [that what Ken bought].'

Among the assumptions in Section 5.3.3.1, relevant to the current discussion are (69a,b) and (71c), repeated below:

(69) a. The Q-particle *ka* has the categorial feature C_{wh}.
b. Single nominal wh-words, such as *nani* 'what' and *dare* 'who', have the categorial feature D_{wh}.

(71) c. The reportative complementizer *to* has a selectional feature C.

As in Section 5.3.1, our only concern here is the (local) selectional relationship between the reportative complementizer *to* and its complement wh-clause; I will put aside what selectional relationship the matrix predicate holds.

With this in mind, let us now consider (109), one derivational point of (108).

(109) Numeration: {…, to^{C}, …}
REP

Syntax: [$_{Cwh}$ ka$_{Cwh}$ Ken katta nani$_{Dwh}$] (word order irrelevant)
Q Ken bought what

In (109), the selectional feature of *to* in the numeration, C, probes into the syntactic object, and it finds the root label C_{wh} as its goal; it can serve as the goal for the selectional feature C, given that it is a subcategory of C. This probing is represented in (110).

(110) Numeration: {…, to^{C}, …}
REP

Syntax: [$_{Cwh}$ ka$_{Cwh}$ Ken katta nani$_{Dwh}$] (word order irrelevant)
Q Ken bought what

Note that C_{wh} is the only goal for the selectional feature C; obviously, the categorial feature of the wh-word *nani* 'what', D_{wh}, cannot qualify as its goal. Hence, the wh-word does not undergo movement, staying in-situ. Finally, the reportative complementizer *to* merges with the syntactic object in (110), resulting in (111).

(111) $\text{to}^{\cancel{C}}$ [$_{Cwh}$ ka$_{Cwh}$ Ken katta nani$_{Dwh}$] (word order irrelevant)
REP Q Ken bought what

Thus, the lack of covert wh-movement with *to*-embedded CQ wh-questions can be captured by the CRSM-based analysis as well.

To recap, the above discussion has shown that the CRSM-based analysis provides an account of the proposal that the three types of wh-questions discussed above do not involve covert wh-movement, in contrast to embedded CQ wh-questions. In other words, the current analysis provides an answer to the final question, (51b), repeated below.[26,27]

(51) b. Why is it that the other types of wh-questions (i.e. matrix wh-quesitons, embedded non-CQ wh-questions and *to*-embedded wh-questions) do not involve covert wh-movement?

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[26] As discussed so far, according to the CRSM-based analysis, wh-movement which can make a wh-clause a free relative is induced by Agree between the selectional feature of a CQ-predicate X_{wh} and the categorial feature of a single nominal wh-word D_{wh}. This means that, without additional assumptions, wh-clauses cannot be rendered free relatives if they are selected by an element other than CQ predicates, such as non-CQ predicates and the complementizer *to*. This is compatible with the fact that in Japanese, wh-clauses cannot be interpreted as free relatives when they appear in the complement position of non-question-selecting transitive verbs such as *kau* 'buy' and *yomu* 'read', given that those verbs arguably do not have a selectional feature X_{wh}; if those verbs take a wh-clause as their complement, this leads to ungrammaticality.

[27] Note that in all cases in this subsection where Multiple Agree was involved, one goal was the categorial feature D_{wh} or ADV_{wh}, while the other was C_{wh}. One might then wonder how the CRSM-based analysis works if the two potential goals are both $C_{(wh)}$. Here I will discuss (i) as a case study. (Note that when an embedded question headed by the Q-particle *ka* involves no wh-element, it is interpreted as a *whether*-question, as noted in Chapter 2.)

(i) Mai-wa [[Ken-ga [Kooji-ga MIT-ni ukatta ka] sitteiru ka] to] tazuenta.
Mai-TOP Ken-NOM Koji-NOM MIT-GEN passed Q know Q REP asked
Lit. 'Mai asked [that [whether Ken knows [whether Koji passed MIT]]].'

Consider the derivational point of (i) in (ii), where the selectional feature of the reportative complementizer *to*, i.e. C, is about to probe into the syntactic object.

(ii) Numeration: $\{\ldots, \text{to}^{C}_{\text{REP}}, \ldots\}$
Syntax: $[_{Cwh}$ ka$_{Cwh}$ … $[_{Cwh}$ ka$_{Cwh}$ …] …] (word order irrelevant)

The structure in (ii) involves two potential goals for the probe *to*, i.e. two instances of the categorial feature C_{wh}. Under the CRSM-based analysis, if *to* undergoes Agree with both features, it is predicted that the deeply embedded *whether*-question (i.e. *whether Koji passed MIT*) will undergo (covert) movement, resulting in the structure in (iii). (Note that in (iii), the complement clause of *to* is projected by the target of the movement; the moved *whether*-clause cannot project since it is not a head.)

(iii) $\text{to}^{\cancel{C}}$ $[_{Cwh}$ $[_{Cwh}$ ka$_{Cwh}$ … $]_i$ $[_{Cwh}$ ka$_{Cwh}$ … t_i …]]]

I now examine whether (iii) actually holds based on the formation of clefts, which is used as a diagnostic of covert (wh-)movement in this dissertation (Chapter 3). If the deeply embedded *whether*-question in (i) undergoes movement as in (iii), it is predicted that the *whether*-question selected by *to* in (i) (i.e. *whether Ken knows* ...) cannot be transformed into a cleft with the deeply embedded *whether*-question being in the focus position; in such a cleft, the *whether*-question has undergone focus movement (Hiraiwa and Ishihara 2002, 2012) and thus it could not undergo the movement under consideration due to a freezing effect. This prediction is not borne out; (iv), which involves the cleft under consideration, does not sound unacceptable, even though the judgement is a bit subtle.

(iv) ?Mai-wa [[[Ken-ga e_i sitteiru no]-wa [Kooji-ga MIT-ni ukatta ka]$_i$ (da) ka] to] tazuneta.
Mai-TOP Ken-NOM know C-TOP Koji-NOM MIT-GEN passed Q COP Q REP asked
Lit. 'Mai asked [that [whether it is [whether Koji passed MIT] that Ken knows]].'

I take this to indicate that *to* in (i) does not Agree with C_{wh} of the deeply embedded *whether*-question, which thus does not undergo movement. This then suggests that a selectional feature can undergo Multiple Agree only when the relevant goals are different kinds of categorial features (e.g. D_{wh} and C_{wh}). I leave for future study what this could follow from.

To conclude, in this section, I have proposed a CRSM-based analysis of Japanese wh-questions, which implements the fact that the selectors of embedded CQ wh-questions can select more kinds of syntactic categories as questions than those of the other types of wh-questions. It has been demonstrated that the proposed analysis gives an account of the two significant issues regarding Japanese wh-questions, namely the Wh-Movement Asymmetry (see (51)) and the nominality of embedded CQ wh-questions with its restricted distribution (see (52)).

5.4 Summary of the chapter

The main goal of this chapter was to pursue the origin of the Wh-Movement Asymmetry, i.e. the asymmetry regarding the (non-)involvement of covert wh-movement in Japanese wh-questions, which has been established in Chapter 3. More specifically, in this chapter, I aimed to give answers to the following two questions: (i) Why is it that embedded CQ wh-questions involve covert wh-movement? (ii) Why is it that the other types of wh-questions (i.e. matrix wh-questions, embedded non-CQ wh-questions and *to*-embedded wh-questions) do not involve covert wh-movement? To answer these questions, I have focused on two properties of embedded CQ wh-questions that distinguish them from the other types of wh-questions.

The first property of embedded CQ wh-questions that I have highlighted is their "internal" property, i.e. their nominality. Tomioka (2020) argues that Japanese embedded (CQ) wh-questions have nominal structures and thus are interpreted as concealed questions. He bases this claim on the phenomenon that I called wh-classifier matching, i.e. the fact that when an adverbial expression that induces a Quantificational Variability Effect involves a (numeral) classifier, that classifier must be the one used to count the referent of the value of the wh-element in the embedded wh-question. Tomioka argues that if embedded (CQ) wh-questions are assumed to be nominal, then the phenomenon in question can be ascribed to the fact that concealed questions can be associated with floated quantifiers with a classifier that "match" with them. Regarding how the nominality of embedded CQ wh-questions is derived, he proposes that those wh-questions have the structure of internally-headed relative clauses (IHRCs). However, I have shown that this analysis is problematic in that (i) empirically the two constructions do not completely parallel and (ii) it is unclear how nominal interpretations of embedded CQ wh-questions can be derived if they are analyzed as IHRCs. I have thus departed from Tomioka's analysis and instead assumed that embedded CQ wh-questions have the structure of free relatives. I have further adopted the syntactic analysis of free relatives proposed in Cecchetto and Donati (2010, 2015) and Donati and Cecchetto (2011), according to which in free relatives a moved wh-word projects and labels the wh-clause as D. Based on those assumptions, I have discussed the possibility of linking the involvement of covert wh-movement in embedded CQ wh-questions to their structure as free relatives, given that free relatives require wh-movement. The investigation of this possibility has crucially revealed that embedded CQ wh-questions do not always show nominality; in particular, such wh-questions do not exhibit nominality when they involve an adjunct wh-word (e.g. *doo* 'how' and *naze* 'why') or a complex wh-phrase (e.g. *dare-no hon* 'whose book'). It has thus turned out that a satisfactory syntactic analysis of Japanese wh-questions must be able to capture not only the Wh-Movement Asymmetry but also the nominality of embedded CQ wh-questions along with its restricted distribution (e.g. Why do such wh-questions with an adjunctive wh-word or a complex wh-phrase do not exhibit nominality?).

The second property of embedded CQ wh-questions that I have discussed is an "external" property, namely the fact that they are selected by CQ-predicates (i.e. predicates that can select concealed questions). CQ-predicates differ from the selectors of the other types of wh-questions in that the former can select more kinds of syntactic categories (i.e. CP and DP) as questions than the latter can (i.e. only DP; another way to look at this is that the latter have selectional restrictions). To syntactically implement this, I have adopted the syntactic selection mechanism proposed by Donati and Cecchetto (2011), according to which a selectional feature in the numeration probes into the syntactic object to find a categorial feature which it undergoes Agree with, calling this mechanism the Cross-Representational Selection Mechanism (CRSM). I have then proposed a CRSM-based analysis of Japanese wh-questions, whose crucial parts are (i) the assumptions about selectional and categorial features of the relevant elements (e.g. (non-)CQ-predicates, concealed questions, nominal/adjunct wh-words) and (ii) the proposal that a selectional feature as a probe can undergo Multiple Agree, which can induce movement. I have demonstrated that the CRSM-based analysis gives an account of the Wh-Movement Asymmetry, as well as the nominality of embedded CQ wh-questions with its restricted distribution. In this sense, the CRSM-based analysis can be regarded as a satisfactory analysis of wh-questions in Japanese.

Chapter 6

Some Consequences for Covert Wh-Movement in Japanese

6.1 Introduction

Thus far, through investigating asymmetries in Japanese wh-questions regarding intervention effects and cleft formation, I have argued that covert wh-movement is involved in embedded CQ wh-questions, while it is not in the other types of wh-questions (Chapter 3). Furthermore, I have proposed a CRSM-based analysis of Japanese wh-questions, which I have demonstrated can capture the asymmetry regarding the (non-)involvement of covert wh-movement, as well as the nominal character of embedded CQ wh-questions (Chapter 5). In this chapter, I will examine some consequences of the current discussion regarding certain debated properties of covert wh-movement in Japanese, namely island sensitivity and large-scale pied-piping. Regarding these two properties, Nishigauchi (1990) argues that covert wh-movement is sensitive to islands and that covert large-scale pied-piping is available in Japanese with respect to wh-movement. However, as discussed in Chapter 2, his arguments for those two properties have been challenged in the literature and thus have turned out not to be as solid as they were originally thought to be. Against this background, I will explore the two properties of covert wh-movement in Japanese under consideration from new perspectives, namely in terms of the nominal structure of wh-questions, the obviation of intervention effects and the formation of cleft wh-quesiotns, which I have discussed in the previous chapters. As a result, I will argue that (i) covert wh-movement is sensitive to islands and that (ii) covert large-scale pied-piping is available for Japanese wh-questions, thus defending Nishigauch's (1990) original claims.

The rest of this chapter is structured as follows: Section 6.2 will give a brief reminder of Chapter 2, establishing that it is still an open issue whether covert wh-movement is sensitive to islands and whether covert large-scale pied-piping is available for Japanese wh-questions. In Section 6.3 and Section 6.4, I will discuss the first and the second issue respectively, building on the discussion in the previous chapters. Section 6.5 gives a summary of this chapter.

6.2 A brief reminder of Chapter 2

In Chapter 2, I have given an overview of Nishigauchi's (1990) argument for the covert wh-movement analysis in Japanese. Specifically, he observes an asymmetry in island effects in Japanese wh-questions; they are not sensitive to complex NP islands (Kuno 1973), while they are sensitive to wh-islands, as (1) and (2) indicate. (What matters in (2) is the unavailability of the second reading.)

(1) Ken-wa [[*e*$_i$ nani-o kaita] hito$_i$]-o home-masi-ta ka.
Ken-TOP what-ACC wrote person-ACC praise-POL-PAST Q
Lit. 'Ken praised [the person who wrote what]?'
(≈ 'What is the thing such that Ken praised the person who wrote it?')

(2) Mai-wa [Ken-ga nani-o katta ka] sittei-mas-u ka.
Mai-TOP Ken-NOM what-ACC bought Q know-POL-PRES Q
(i) ✓'Does Mai know [what Ken bought]?'
(ii) * Intended: 'What did Mai know [whether Ken bought (*t*)]?'
(≈ 'For which *x*, *x* a thing, does Mai know [whether Ken bought *x*]?')
(based on Nishigauchi 1990: 30)

Taking island effects to be an indicator of movement, Nishigauchi claims that the wh-island effect in (2) suggests the existence of wh-movement, and thus argues for covert wh-movement in

Japanese wh-questions. Importantly, he argues that covert wh-movement is sensitive to islands, departing from the claim in preceding works that covert wh-movement is insensitive to islands (e.g. Huang 1982b, Lasnik and Saito 1984). Regarding the asymmetry in islands observed in the contrast between (1) and (2), Nishigauchi proposes covert large-scale pied-piping. In (1), what undergoes covert wh-movement is the whole complex NP that contains the wh-word *nani* 'what', rather than that wh-word itself, as a result of the wh-word pied-piping the complex NP. More specifically, according to Nishigauchi's analysis, (1) has the LF structure in (3), where the wh-element *nani* 'what' moves to the edge of the complex NP, where its [+WH] feature is percolated up to the complex NP through CP, and as a result the whole complex NP undergoes wh-movement.

(3) [$_{CP}$ [$_{NP[+WH]}$ [$_{CP[+WH]}$ nani-o$_{j[+WH]}$ [$_{TP}$ e_i t_j kaita]] hito$_i$]-o$_k$ [$_{TP}$ Ken-wa t_k home-masi-ta]
what-ACC wrote person-ACC Ken-TOP praise-POL-PAST
ka] (in LF)
Q

In this derivation, the wh-word does not cross the complex NP island and thus no island effect arises. While Nishigauchi does not discuss the case of wh-islands, we have seen that within his overall approach, (1) can still be ruled out given that in contrast to complex NP islands, covert large-scale pied-piping is not available to wh-islands due to the inapplicability of feature percolation; see Chapter 2 for relevant discussion. As a result, the wh-word *nani* 'what' in (2) has to undergo covert wh-movement, which, however, crosses the wh-island, thus leading to an island effect. To the extent that this short summary is concerned, the main claims by Nishigauchi (1990) can be summarized as follows:

(4) a. Japanese wh-questions involve covert wh-movement.
b. Covert wh-movement is sensitive to islands.

c. Covert large-scale pied-piping is available to Japanese with respect to wh-movement.

However, as discussed in Chapter 2, his arguments that support (4) have been challenged in subsequent works. In particular, (i) Shimoyama (2006) argues that the island asymmetry follows from her semantic analysis of Japanese wh-questions based on Hamblin (1973) without assuming covert wh-movement, and (ii) it has been argued in the literature that wh-island effects in Japanese are merely an illusion, since they are not observed if the prosodic facet of Japanese wh-questions is taken into consideration (e.g. Deguchi and Kitagawa 2002, Ishihara 2003, Kitagawa 2005, 2017). Given that, the three claims in (4) seem to have lost the strong empirical backbone.

As discussed thus far, however, this dissertation has bolstered (4a) with novel arguments, with a proviso that covert wh-movement is involved only in embedded CQ wh-questions. Then, the question that immediately arises is, what about (4b) and (4c)? Since Nishigauchi's arguments for them, i.e. wh-island effects and the island asymmetry noted above, have proved not to be fully reliable as discussed above, they should be examined through new lenses. Against this backdrop, in the remainder of this chapter, I will discuss those issues, (4b) and (4c), in terms of the three phenomena that have been discussed in the previous chapters, i.e. the nominality of embedded CQ wh-questions, the obviation of intervention effects, and the formation of cleft wh-questions.

6.3 Island sensitivity of covert wh-movement

This subsection investigates the island (in)sensitivity of covert wh-movement. This investigation builds on the nominal character of embedded CQ wh-questions that we have discussed in the previous chapter, instead of wh-islands (which is what Nishigauchi 1990 discussed). Recall, first of all, that Tomioka (2020) proposes that embedded (CQ) wh-questions in Japanese involve nominal structures, which are eventually interpreted as concealed questions. The motivation for this proposal is to ascribe wh-classifier matching as in (5a) to the fact that concealed questions can be associated with a (floated) quantifier with a classifier that they "match", as in (5b).

(5) a. Mari-wa [**dare**-ga ukatta ka] san-**nin**(-gurai) sitteiru.
Mari-TOP who-NOM passed Q 3-CLS-about know
Lit. 'For (about) three (people), Mari knows who passed.'
≈ 'For (about) three of the people who passed, Mari knows that they passed.'
(Tomioka 2020: 124)

b. Mari-wa [[e_i ukatta] **hito**$_i$]-o san-**nin**(-gurai) sitteiru.
Mari-TOP passed person-ACC 3-CLS-about know
'Mari knows about three of those who passed the exam.' (ibid: 128)

I have observed, however, that some embedded CQ wh-questions do not show nominality, assuming that wh-classifier matching qualifies as a diagnostic of their nominal structure. Thus, I have shown that embedded CQ wh-questions with a complex wh-phrase such as *dare-no X* 'whose X' are incompatible with wh-classifier matching, as in (6).

(6) Mai-wa [Ken-ga **dare**-no hon-o katta ka] {??san-**nin**(-gurai) /
Mai-TOP Ken-NOM who-GEN book-ACC bought Q 3-CLS-about
?*san-satu(-gurai)} sitteiru.
3-CLS-about know
Lit. 'For (about) three (authors), Mai knows whose book Ken bought.'
≈ 'For (about) three of the authors whose books Ken bought, Mai knows that Ken bought those authors' books.' /
Lit. 'For (about) three (books), Mai knows whose book Ken bought.'
≈ 'For (about) three of the books of some authors that Ken bought, Mai knows that Ken bought those books.

I will give below a brief review of the account of the difference regarding the nominality between (5a) and (6) presented in the last chapter, to which I refer the reader for a more detailed analysis. The account crucially assumes C&D's (i.e. Ceccetto and Donati's) analysis of free relatives (Ceccetto and Donati 2010, 2015, Donati and Cecchetto 2011). C&D propose that a moved element can project. With this proposal, the wh-clause *what you read* can be projected not only by C but also by the categorial feature of the wh-word *who*, i.e. D, as in (7): when D projects, the wh-clause serves as a free relative, as (7b) shows.

(7) a. I wonder [$_{C(P)}$ what [$_{C}$ C you read ~~what~~]].

b. I read [$_{D(P)}$ what [$_{C}$ C you read ~~what~~]].

It further follows from their analysis that wh-clauses cannot qualify as free relatives if they involve a complex wh-phrase, instead of a single wh-word (as in (7)), given the standard assumption that it is only a head, not a phrase, that can project when it merges with a syntactic object. Thus, the wh-clause with a complex wh-word, *whose book you read*, can be projected only by C and thus can serve only as an interrogative clause, not as a free relative, as shown in (8).

(8) a. I wonder [$_{C}$ [$_{D}$ what book] [$_{C}$ C you read ~~what book~~]].

b. *I read [what book you read].

The difference between (5a) and (6) then follows this analysis of fee relatives, along with the proposal that embedded CQ wh-questions involve covert wh-movement (see Chapter 3). In the embedded CQ wh-question in (5a), the single wh-word *dare* 'who' undergoes movement to create the constituent {*dare*, C}. In this configuration, not only C but also D, the categorial feature of *dare* 'who', can project, given that *dare* is a lexical item, i.e. a head.[1] When D projects, the embedded wh-clause in (5a) will serve as a free relative, hence its nominality. In the embedded CQ wh-question in (6), what undergoes movement is the complex wh-phrase *dare-no hon* 'whose book'. The resulting constituent thus can only be labeled by C, not by the categorial feature of *dare*(*-no hon*), i.e. D. The wh-clause in (6) thus cannot serve as a free relative, hence the absence of nominality.

With this background in mind, let us now examine the island (in)sensitivity of covert wh-movement in terms of the nominality of embedded CQ wh-questions, or wh-classifier matching. Before observing core data, I will first verify that wh-classifier matching is not constrained by long-distance dependencies between wh-words and the Q-particle (which they are associated with). Consider (9).

(9) Mai-wa [Kooji-ga [Ken-ga dare-o hometa to] omotteiru ka] sitteiru.
Mai-TOP Koji-NOM Ken-NOM who-ACC praised REP think Q know
'Mai knows [who Koji thinks [that Ken praised]].'

In the embedded CQ wh-question in (9), the wh-word *dare* 'who' occurs in the declarative complement clause of the embedded predicate. Given that declarative complement clauses do not constitute islands, *dare* 'who' in (9) should be able to undergo covert wh-movement across the clause boundary, regardless of whether covert wh-movement is sensitive to islands or not. This wh-movement will enable the embedded wh-question to be labeled as D. It is hence expected that wh-classifier matching will be possible with (9). This expectation is borne out, as shown in (10).

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[1] The analysis proposed in the last chapter assumes the categorial features C_{wh} and D_{wh}, rather than simple C and D. However, I here adopt the latter categories in order to simplify the discussion.

(10) Mai-wa [Kooji-ga [Ken-ga dare-o hometa to] omotteiru ka] <u>san-**nin**(-gurai)</u>
Mai-TOP Koji-NOM Ken-NOM who-ACC praised REP think Q 3-CLS-about
sitteiru.
know
Lit. 'For (about) three (people), Mai knows [who Koji thinks [that Ken praised]].'
≈ 'For (about) three of the people that Koji thinks that Ken praised, Mai knows that Koji thinks that Ken praised them.'

Let us now consider the core case, where an embedded CQ wh-question involves a complex NP that contains a wh-word, as in (11).

(11) Mai-wa [Ken-ga [[dare-ga e_i kaita] hon_i]-o katta ka] sitteiru.
Mai-TOP Ken-NOM who-NOM wrote book-ACC bought Q know
Lit. 'Mai knows [Ken bought [books that <u>who</u> wrote]].'
(≈ 'Mai knows [who the person is such that Ken bought books that (s)he wrote].')

Suppose first that covert wh-movement is insensitive to islands. With this assumption, the wh-word *dare* 'who' in the complex NP in (11) would successfully undergo wh-movement across the complex NP island. This movement would create the constituent {*dare*, C}. In this configuration, the categorial feature of the wh-word D can project and thus form the free relative structure, giving rise to nominality. Therefore, it is predicted that (11) would be compatible with wh-classifier matching. Suppose, however, that covert wh-movement is sensitive to islands. In this case, the covert wh-movement of the wh-word *dare* 'who' could not cross the complex NP island due to an island effect. Hence, the configuration that is necessary to render the wh-clause a free relative, namely {*dare*, C} could not result. Consequently, the wh-clause in (11) would not obtain nominality.[2] Therefore, it is predicted that wh-classifier matching would be impossible with the wh-clause in (11).

To examine which prediction is correct, consider (12).

(12) Mai-wa [Ken-ga [[**dare**-ga e_i kaita] hon$_i$]-o katta ka] {*san-**nin**(-gurai) /
Mai-TOP Ken-NOM who-NOM wrote book-ACC bought Q 3-CLS-about
*san-satu(-gurai)} sitteiru.
3-CLS-about know
Lit. 'For (about) three (authors), Mai knows [Ken bought [books that who wrote]].'
≈ 'For (about) three of the authors such that Ken bought books that they wrote, Mai knows that Ken bought books that those authors wrote.'
Lit. 'For (about) three (books), Mai knows [Ken bought [books that who wrote]].'
≈ 'For (about) three of the books such that Ken bought them and some author wrote them, Mai knows that Ken bought those books.'

(12) shows that (11) is incompatible with wh-classifier matching, whichever classifier, -*nin* (for people) or -*satu* (for books), is used. This means that the latter prediction is borne out. Therefore, the current discussion indicates that covert wh-movement is sensitive to islands, as Nishigauchi (1990) originally argued.

6.4 Covert large-scale pied-piping

This section turns to the second issue, namely that of whether covert large-scale pied-piping is available for Japanese wh-questions. One should bear in mind here that large-scale pied-piping is overtly observed with respect to wh-movement in languages such as Basque (Ortiz de Urbina 1989,

[2] At this point, one might wonder why (11) is acceptable in the first place if covert wh-movement of *dare* 'who' is blocked by the island. This issue will be discussed in the next section.

1990) and Imbabura Quechua (Cole 1982), as noted in Chapter 2. (13) gives relevant examples, where the wh-element has undergone movement inside the complex NP and then pied-pipes it to Spec,CP of the matrix clause.

(13) a. Basque:

[Nork	idatzi	zuen	liburua]	irakurri	du	Peruk.
what.ERG	write	AUX	book	read	AUX	Peter.ERG

Lit. '[The book that who wrote] did Peter read?'

(≈ 'Who is the person such that Peter read the book that (s)he wrote?')

(Ortiz de Urbina 1990: 249)

b. Imbabura Quechua:

[Ima-ta	randi-shka	runa-ta	taj]	riku-rka-ngui.
what-ACC	buy-NMLZ	man-ACC	Q	see-PAST-2P

Lit. '[The man that bought what] did you see?'

(≈ 'What is the thing such that you saw the man that bought it?') (Cole 1982: 24)

Given that the mechanism is clearly in principle available, it is not unreasonable to posit its covert counterpart for wh-in-situ languages such as Japanese. The question, then, is whether it is actually available in Japanese.

Let us begin the discussion by reconsidering the data in (11). Note that (11) involves an embedded CQ wh-question where the wh-word *nani* 'what' appears in the complex NP island. In the last section, I have argued that covert wh-movement is sensitive to islands, by showing that (11) is incompatible with wh-classifier matching (see (12)). With this claim, then, the question that immediately arises is how covert wh-movement is realized in (11). It should not be the case that the wh-word *nani* 'what' itself undergoes movement; this would induce an island effect. An island effect, however, will not occur in (11), if the whole complex NP undergoes covert wh-movement;

the wh-word will then not cross the complex NP island. The current discussion therefore suggests that covert large-scale pied-piping is available for Japanese wh-questions.[3]

Below, I will provide another argument for the availability of covert large-scale pied-piping for Japanese wh-questions, by discussing obviation of intervention effects (Chapter 3). In the following discussion, I will use as intervenors not only the NPI *X-sika* 'only X', but also the NPI wh-*mo* (e.g. *dare-mo* '(not) anyone') and the disjunction *X ka Y (ka)* 'X or Y', for a reason that will be uncovered below. The fact that they induce intervention effects can be confirmed by the following examples repeated from Chapter 3 (i.e. Section 3.3.1), where those items appearing above wh-elements induce degradation.

(14) a. NPI: *X-sika* (… NEG) 'only X'
?*[Ken-sika] <u>nani-o</u> {kaw-anakat-ta desu ka / kai-masen desita ka}.
Ken-SIKA what-ACC buy-not-PAST COP.POL Q buy-not.POL COP.POL.PAST Q
'What did only Ken buy?'

b. NPI: wh-*mo* (… NEG) (e.g. *dare-mo* (who-MO) … NEG 'no one …')
??[Dare-mo] <u>nani-o</u> {kaw-anakat-ta desu ka / kai-masen desita ka}.
who-MO what-ACC buy-not-PAST COP.POL Q buy-not.POL COP.POL.PAST Q
'What did no one buy?'

c. Disjunctions: *X ka Y (ka)* 'X or Y'
??[Ken-ka Kooji(-ka)]-ga <u>nani</u>-o kai-masi-ta ka.
Ken-or Koji-or-NOM what-ACC buy-POL-PAST Q
'What did Ken or Koji buy?'

[3] Since what undergoes movement to Spec,CP in (11) is a complex wh-phrase, i.e. the complex NP containing a wh-word, the embedded CQ wh-question in (11) cannot obtain nominality, as indicated in (12), for the same reason the one in (6) (with the complex wh-phrase *dare-no hon* 'whose book') cannot.

As observed in Chapter 3 (i.e. Section 3.3.1 and 3.4.1), those effects are obviated in embedded CQ wh-questions; compare (15) with the above examples.

(15) a. Mai-wa [Ken-sika nani-o kawa-nakat-ta ka] sitteiru.
Mai-TOP Ken-SIKA what-ACC buy-not-PAST Q know
'Mai knows [what only Ken bought].'

b. Mai-wa [dare-mo nani-o kawa-nakat-ta ka] sitteiru.
Mai-TOP who-MO what-ACC buy-not-PAST Q know
'Mai knows [what no one bought].'

c. Mai-wa [Ken-ka Kooji(-ka)-ga nani-o katta ka] sitteiru.
Mai-TOP Ken-or Koji-or-NOM what-ACC bought Q know
'Mai knows [what Ken or Koji bought].'

I have argued that the obviation of intervention effects in (15) can be accounted for by the proposal that embedded CQ wh-questions involve covert wh-movement, if we adopt the widely accepted view that intervention effects are obviated when a wh-element is located above an intervenor, overtly or covertly.

With these points in mind, let us now turn our eyes to the case where islands are involved. Before examining the crucial case, however, it is important to confirm whether long-distance dependencies between wh-words and the Q-particle matter for the obviation of intervention effects in embedded CQ wh-questions. For this purpose, observe the examples in (16).

(16) a. *Mai-wa [Ken-sika [dare-ga butyoo-o hihansita to] iw-anakat-ta ka]
Mai-TOP Ken-SIKA who-NOM manager-ACC criticized REP say-not-PAST Q
sitteiru.
know
'Mai knows [who only Ken said [(*t*) criticized the manager]].'

b. ?Mai-wa [dare-mo [dare-ga butyoo-o hihansita to] iw-anakat-ta ka]
Mai-TOP who-MO who-NOM manager-ACC criticized REP say-not-PAST Q
sitteiru.
know
'Mai knows [who no one said [(*t*) criticized the manager]].'

c. ?Mai-wa [Ken-ka Kooji(-ka)-ga [dare-ga butyoo-o hihansita to] itta ka]
Mai-TOP Ken-or Koji-or-Nom who-NOM manager-ACC criticized REP said Q
sitteiru.
know
'Mai knows [who Ken or Koji said [(*t*) criticized the manager]].

In the embedded CQ wh-questions of the above examples, the wh-words occur within the declarative complement clauses and the intervenors appear above those wh-words. With this configuration in mind, notice first the acceptability of (16b,c), where the NPI *dare-mo* '(not) anyone' and the disjunction *Ken-ka Kooji(-ka)* 'Ken or Koji' appear as intervenors. This means that intervention effects are obviated in those examples. In contrast, however, the unacceptability is observed with (16a), which involves the NPI *Ken-sika* as an intervenor. This suggests that the obviation of intervention effects is not involved in (16a). With these observations, I conclude that whatever the reason is, long-distance dependencies between wh-elements and the Q-particle matter for the obviation of intervention effects if the intervenor is the NPI *X-sika*, but not if other intervenors are involved.

With this conclusion in mind, let us now observe the crucial data in (17), whose embedded CQ wh-questions involve complex NPs that contain a wh-word.

(17) a. *Mai-wa [$\boxed{\text{Ken-sika}}$ [[<u>dare</u>-ga e_i tukutta] keeki$_i$]-o tabe-nakat-ta ka] sitteiru.
Mai-TOP Ken-SIKA who-NOM made cake-ACC eat-not-PAST Q know
Lit. 'Mai knows [only Ken ate [the cake that <u>who</u> made]].'

b. ?Mai-wa [$\boxed{\text{dare-mo}}$ [[<u>dare</u>-ga e_i tukutta] keeki$_i$]-o tabe-nakat-ta ka] sitteiru.
Mai-TOP who-MO who-NOM made cake-ACC eat-not-PAST Q know
Lit. 'Mai knows [no one ate [the cake that <u>who</u> made]].'
(≈ 'Mai knows [who the person is such that no one ate the cake that (s)he made.]')

c. ?Mai-wa [$\boxed{\text{Ken-ka Kooji(-ka)}}$-ga [[<u>dare</u>-ga e_i tukutta] keeki$_i$]-o tabeta ka]
Mai-TOP Ken-or Koji-or-NOM who-NOM made cake-ACC ate Q
sitteiru.
know
Lit. 'Mai knows [Ken or Koji ate [the cake that <u>who</u> made]].'
(≈ 'Mai knows [who the person is such that Ken or Koji ate the cake that (s)he made.]')

Note, first of all, that the unacceptability of (17a) should be attributed to the above observation that intervention effects cannot be obviated if an intervenor is the NPI *X-sika* and a wh-word forms a long-distance dependency with the Q-particle (see (16a)). Its unacceptability is thus irrelevant to islands. For this reason, let us focus on the other two data patterns (17b,c). Crucially, they are judged to be acceptable, meaning that intervention effects are obviated there. This then suggests that the wh-word *nani* is covertly located above the intervenors as a result of movement. Recall, however, that the discussion in the last section has shown that covert wh-movement is sensitive to islands. Hence, it should not be the case that what undergoes covert wh-movement in (17b,c) is the wh-word *dare* 'who' itself; if it did, it would give rise to an island effect. Now, if the wh-word pied-pipes the whole complex NP, hence the latter undergoes covert wh-movement (i.e. covert large-scale pied-piping), the acceptability of (17b,c) will follow: (i) no island effect arises since the wh-word does not cross the complex NP island, and (ii) intervention effects are obviated because the wh-word is covertly located above the intervenors as a result of the movement of the complex NP that contains it. This then suggests that covert large-scale pied-piping is available for Japanese wh-questions.[4]

Before concluding this section, I will briefly discuss covert large-scale pied-piping in Japanese wh-questions in terms of the formation of cleft wh-questions. Recall first that in Chapter 3 (i.e. Section 3.3.2 and 3.4.2), it has been observed that forming cleft wh-questions as embedded CQ wh-questions gives rise to degradation, as shown in (18b); compare (18b) with the cleft wh-question formed as a matrix wh-question in (18a), which is well-formed.

(18) a. [Ken-ga e_i kabin-o oita no]-wa doko-ni$_i$ desu ka.
Ken-NOM vase-ACC put C-TOP where-on COP.POL Q
'Where is it that Ken put a vase?'

b. ?*Mai-wa [[Ken-ga e_i kabin-o oita no]-wa doko-ni$_i$ (da) ka] sitteiru.
Mai-TOP Ken-NOM vase-ACC put C-TOP where-on COP Q know
'Mai knows [where it is that Ken put a vase].'

To account for the degradation of (18b) based on the proposal that covert wh-movement occurs in embedded CQ wh-questions, I have adopted the syntactic analysis of Japanese clefts proposed by Hiraiiwa and Ishihara (2002, 2012), which crucially involves focus movement of the focus phrase. I have then argued that the degradation of (18b) is ascribed to a freezing effect; the wh-phrase cannot undergo covert wh-movement (to Spec,ForceP) since it has undergone focus movement, being "frozen" in the focus position (i.e. Spec,FocP).

[4] Note that the other intervenor that this dissertation draws on, i.e. the universal quantifier wh-*mo* (e.g. *da're-mo* 'everyone'), patterns with the NPI wh-*mo* and the disjunction *X-ka Y(-ka)* with respect to the data discussed in this section.

With this point in mind, consider (19), where the cleft wh-question is formed as an embedded CQ wh-question and its focus phrase is a complex NP that contains a wh-word *dare* 'who'.

(19) ?*Mai-wa [[Ken-ga e_i kabin-o oita no]-wa [[dare-ga e_i tukutta] tukue]$_i$-ni (da) ka]
Mai-TOP Ken-NOM vase-ACC put C-TOP who-NOM made desk-on COP Q
sitteiru.
know
Lit. 'Mai knows [it is [on the desk that <u>who</u> made] that Ken put a vase].'

Notice that (19) is judged to be degraded, like (18b). Suppose then that the degradation of (19) arises for the same reason as that of (18b), i.e. due to a freezing effect. Under this assumption, the element that is supposed to undergo wh-movement in (19) should be the whole complex NP containing a wh-word *dare* 'who'; since that complex NP has undergone focus movement, applying covert wh-movement to it would give rise to a freezing effect, thus degradaing (19). This account of the degradation of (19) would then lend support to the availability of covert large-scale pied-piping in Japanese wh-questions.

Note, however, that this is not the only possible account of the degradedness of (19); even if the pied-piping does not occur, (19) would be degraded because the covert wh-movement of the wh-word *dare* 'who' would cross the complex NP island, thus inducing an island effect. the degradation of (19) then serves as evidence either for the availability of covert large-scale pied-piping in Japanese wh-questions or for the island sensitivity of covert wh-movement.[5]

[5] One might wonder whether the long-distance dependency between the wh-word and the Q-particle has anything to do with the degradation of (19). The best way to test this possibility would be to investigate what would result if the focus phrase in (19) is replaced with a declarative clause containing a wh-element. However, an issue here is that Japanese clefts in general cannot have a declarative complement clause in their focus position, as (i) shows.

(i) *[Mai-ga e_i omotteiru no]-wa [Ken-ga hon-o katta to]$_i$ da.
Mai-NOM think C-TOP Ken-NOM book-ACC bought REP COP
Lit. 'It is that Ken bought a book that Mai thinks.'

6.5 Summary of the chapter

In this chapter, I have discussed what the findings and proposals in the previous chapters can tell us about the following two claims by Nishigauchi (1990): (i) Covert wh-movement is sensitive to islands. (ii) Covert large-scale pied-piping is available in Japanese with respect to wh-movement. This discussion was motivated by the fact that his arguments for these claims have been challenged in the literature, as discussed in Chapter 2.

Concerning the first claim, I have investigated the case where an embedded CQ wh-question involves a single (nominal) wh-word in a complex NP island. In particular, I have shown that wh-classifier matching is not possible with such an embedded CQ wh-question, indicating that it does not show nominality. I took this observation to mean that the wh-word cannot cross the complex NP island, given that the nominality of embedded CQ wh-questions indicates movement of the wh-word. Based on this consideration, I have argued that covert wh-movement in Japanese is island-sensitive.

Concerning the second claim, I have first considered the same type of embedded CQ wh-question as above, i.e. the one where a wh-element appears within a complex NP island. Building on the claim corroborated by the preceding discussion that covert wh-movement is sensitive to islands, I have argued that the acceptability of such an embedded CQ wh-question indicates that the whole complex NP, rather than the wh-element itself, undergoes covert wh-movement by being pied-piped by the wh-element, thus inducing no island effect. In addition, I have observed that intervention effects are obviated in embedded CQ wh-questions even when a wh-element is involved in a complex NP island. Given that the obviation of intervention effects indicates involvement of covert wh-movement (Chapter 3) and that covert wh-movement is island-sensitive as argued in the preceding discussion, I have argued that in the case under consideration, the wh-element pied-pipes the whole complex NP island regarding the covert wh-movement and thus no island effect arises. These considerations thus indicate that covert large-scale pied-piping is available in Japanese with respect to wh-movement. Finally, I have discussed the fact that forming cleft wh-questions as embedded wh-questions leads to degradation even when the focus phrase is a complex NP that contains a wh-element. I have shown that that degradation arises either because covert wh-movement cannot apply to the whole complex NP containing a wh-element due to a freezing effect or because covert wh-movement of a wh-element crosses a complex NP island, inducing an island effect. The degradation in question thus serves as evidence either for the availability of covert large-scale pied-piping in Japanese wh-questions or for the island sensitivity of covert wh-movement.

In a nutshell, this chapter has shown that Nishigauchi's (1990) original claims regarding the island-sensitivity of covert wh-movement and the availability of covert large-scale pied-piping in Japanese wh-questions can be "reinvigorated" on the basis of the findings and proposals of this thesis.

Chapter 7

Conclusion

In this dissertation, I have explored the syntax of Japanese wh-questions from new perspectives, in order to shed new light on the contentious issue regarding how Japanese wh-questions should be syntactically analyzed. The exploration has built on the new observation that Japanese wh-questions show asymmetric syntactic behavior with respect to intervention effects and cleft formation. Based on this, I have argued that Japanese wh-questions are derived in a non-uniform manner depending on their types. More specifically, I have argued that embedded CQ wh-questions (i.e. embedded wh-questions selected by predicates that can select concealed questions) are derived through covert wh-movement, while other types of wh-questions do not involve covert wh-movement. This claim lends support to the necessity of covert wh-movement in analyzing (a subset of) Japanese wh-questions. I have further aimed to explain why Japanese wh-questions differ in the (non-)involvement of covert wh-movement (i.e. the Wh-Movement Asymmetry). In pursuing the answer to this question, I have paid special attention to selectional properties of Japanese wh-questions. I have then proposed a syntactic analysis of Japanese wh-questions that implements those properties and demonstrated that the proposed analysis gives an account of the Wh-Movement Asymmetry. I have further shown that the analysis in question can also capture the nominal nature of embedded CQ wh-questions. The discussion in this dissertation has resulted in a number of consequences regarding various mechanisms and phenomena, including sluicing constructions and properties of covert wh-movement.

In Chapter 2, I have presented a brief background regarding one aspect of the syntax of Japanese wh-questions, namely covert wh-movement. Covert wh-movement has been mainly motivated by Nishigauchi's (1990) claim that Japanese wh-questions show wh-island effects (unlike other island effects, such as complex NP constraints); based on this claim, he argued that Japanese wh-questions are derived through covert wh-movement, and that covert wh-movement is sensitive to islands (*pace* Huang 1982b and Lasnik and Saito 1984). However, his argument has been challenged in terms of, e.g., semantics (e.g. Shimoyama 2006) and the syntax-phonology interface (e.g. Deguchi and Kitagawa 2002, Ishihara 2003, Kitagawa 2005, 2017). His argument for covert wh-movement in Japanese wh-questions has thus proven not to be so strong as originally thought. This background has thus motivated the discussion in the subsequent chapters.

In Chapter 3, I have discussed asymmetric syntactic behavior of Japanese wh-questions that has gone unnoticed. I have first introduced a classification of Japanese wh-questions that consists of four classes: (i) matrix wh-questions, (ii) embedded CQ wh-questions (see above), (iii) embedded non-CQ wh-questions (i.e. embedded wh-questions selected by predicates that cannot select concealed questions), and (iv) *to*-embedded wh-questions (i.e. embedded wh-questions followed by the reportative complementizer *to*). Building on this classification, I have shown that wh-questions in Japanese exhibit asymmetric behavior with respect to intervention effects and cleft formation. Crucially, these two asymmetries make the same cut; matrix wh-questions, embedded non-CQ wh-questions and *to*-embedded wh-questions pattern together, as opposed to embedded CQ wh-questions. To capture this demarcation, I have proposed that Japanese wh-questions are derived non-uniformly depending on their types. More specifically, I have proposed that embedded CQ wh-questions are derived through covert wh-movement, while the other types of wh-questions do not involve covert wh-movement. I have shown that this proposal gives a uniform account of the above-mentioned asymmetric syntactic behavior of Japanese wh-questions. The proposal has two significant implications regarding the syntax of Japanese wh-questions: (i) Japanese wh-questions must be non-uniformly analyzed, being derived differently depending on their types. (ii) Covert wh-movement is necessary in analyzing (at least a subset of) Japanese wh-questions.

In Chapter 4, I have discussed one consequence of the proposal in Chapter 3, regarding sluicing in Japanese. I have first reconsidered in terms of the current proposal two representative analyses of Japanese sluicing, i.e. the focus movement/cleft analysis and the in-situ analysis, and shown that both face problems. In particular, this investigation has revealed that Japanese sluicing requires a new analysis that involves movement that does not apply to the remnant of sluicing. To propose such an analysis, I have introduced three other elliptical constructions, namely *ka*(*dooka*)-stripping, *to*-stripping and fragment answers, and compared them with sluicing in terms of several relevant properties. Based on this comparison, I have argued that sluicing (and *ka*(*dooka*)/*to*-stripping) involve movement of a null operator associated with exhaustivity. Then, I have proposed that in sluicing (and *ka*(*dooka*)/*to*-stripping) constructions, deletion applies to the propositional portion except for the element that the null operator in question originally attaches to.

In Chapter 5, I have sought to deduce the Wh-Movement Asymmetry, i.e. to answer the following questions: (i) Why is it that embedded CQ wh-questions involve covert wh-movement? (ii) Why is it that the other types of wh-questions do not involve covert wh-movement? To this end, I have paid attention to two properties of Japanese wh-questions. First, I have focused on Tomioka's (2020) claim that Japanese embedded (CQ) wh-questions have nominal structures. In particular, I have explored the possibility that embedded CQ wh-questions have the structure of free relatives under Ceccheto and Donati's analysis of free relatives (Cecchetto and Donati 2010, 2015, Donati and Cecchetto 2011), where a free relative obtains when a wh-clause is projected by the moved wh-element and labeled as its categorial feature D. Based on this approach, I have examined the possibility of linking the involvement of covert wh-movement in embedded CQ wh-questions to the structure of free relatives. This investigation has revealed the following two facts regarding the nominality of embedded CQ wh-questions: (i) Embedded CQ wh-questions do not always have the nominal structure, being labeled either as D or C; in particular, I have shown that embedded CQ wh-questions containing an adjunct wh-word (e.g. *doo* 'how', *naze* 'why') or a complex wh-phrase (e.g. *dare-no hon* 'whose book') are always labeled as C. (ii) Covert wh-movement is involved in embedded CQ wh-questions regardless of whether they are labeled as D or C. Then, I have turned to another property of Japanese wh-questions, namely the fact that the selectors of embedded CQ wh-questions (i.e. CQ-predicates) can select a wider range of syntactic categories (i.e. CP/interrogative clauses and DP/concealed questions) than those of embedded non-CQ questions (i.e. non-CQ-predicates) and *to*-embedded CQ wh-questions (i.e. the reportative complementizer *to*) can (i.e. only CP; in other words, the latter two have selectional restrictions). To implement this selectional property, I have adopted the syntactic selection mechanism proposed by Donati and Cecchetto (2011), which I called the Cross-Representational Selection Mechanism (CRSM). In this mechanism, a selectional feature of an element in the numeration probes into the syntactic object and undergoes Agree with a matching categorial feature, which can then trigger movement of the latter, with the moving element projecting. I have then proposed a CRSM-based analysis of Japanese wh-questions and demonstrated that it captures both the Wh-Movement Asymmetry and the nominality of embedded CQ wh-questions, including its restricted distribution.

In Chapter 6, I have discussed consequences of the discussion in the previous chapters, concerning two properties of covert wh-movement in Japanese argued for by Nishigauchi (1990), namely the island sensitivity of covert wh-movement and the availability of covert large-scale pied-piping with respect to wh-movement. I have discussed these properties in terms of the nominal structure of embedded CQ wh-questions, the obviation of intervention effects, and the formation of cleft wh-questions. More specifically, I have observed the following facts: (i) Embedded CQ wh-questions do not show nominality when a wh-word appears in an island. (ii) Intervention effects are obviated in embedded CQ wh-questions even when a wh-element appears in an island. (iii) Forming cleft wh-questions as embedded CQ wh-questions leads to degradation even when the focus phrase constitutes an island that contains a wh-element. Given the discussion in the preceding chapters, these observations provide clear evidence that covert wh-movement is sensitive to islands and that covert large-scale pied-piping is available in Japanese with respect to wh-movement. They have thus provided new arguments for Nishigauchi's (1990) claims from completely new perspectives.

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